The Riemann–von Mangoldt clock and the zeros of the Riemann zeta function: an exact defect identity, prime cutoffs and a computable corrected clock

Rotkiv Aynäsac

Abstract

Let γ_k be the ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, and let τ_k be the midpoint Riemann–von Mangoldt clock, $N_0(\tau_k) = k - \frac{1}{2}$, where N_0 is the smooth part of the Riemann–von Mangoldt formula. *Known:* $\delta_k = \tau_k - \gamma_k$ is bounded, $\limsup_k |\delta_k| \le 0.6331$ (Bellotti–Wong), its mean square over $T < \gamma_k \le 2T$ is $\sim 2\log\log T/\log^2 T$, and $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ is asymptotically normal with variance $\frac{1}{2\pi^2}\log\log T$ (Selberg, Korolev): the clock is close to the zeros in height but misplaces almost every zero by an unbounded number of positions. *New, unconditional:* (A) an exact identity turns sums of $g(k - \frac{1}{2} - F(\gamma_k))$ over consecutive ordinates into dF-integrals of $g(N - F)$, for every continuous g and every C^1 model F of the counting function; (B) for the clock equation corrected by the Euler phase S_X of the primes $p \le X$, the mean square of the ordinal defect $k - \frac{1}{2} - N_0(\gamma_k) - S_X(\gamma_k)$ is $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log\frac{\log T}{\log X}})$ uniformly for $2 \le X \le T^{1/2}$, also with a cutoff depending on the index, so the error in the position of the zeros becomes bounded in mean square exactly when $\log X \asymp \log T$; (C) an exact finite-height accounting of the computable one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$; (D) with $X = T^{1/\sqrt{\log\log T}}$ the computable clock satisfies $\langle(\tau_k^X - \gamma_k)^2\rangle = (\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$, smaller than the mean square of the bare clock by a factor $\sim \log\log\log T/(2\log\log T)$, a second Newton step with cutoff $T^{1/\sqrt{\log\log\log T}}$ replaces $\log\log\log T$ by $\log\log\log\log T$, and the iteration continues with constants uniform in the number of steps: after $\log^* T + O(1)$ Newton steps, with cutoffs depending only on T, the corrected clock locates the ordinates with a mean-square error of $O(1)$ mean spacings, using no ordinate in its construction; the proof rests on an exact moment recursion, on a multiplicity bound for the corrected clock points obtained from Fujii's moments of $S(t+h) - S(t)$, and on a sampling inequality for maximal functions; (E) $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_t |S(t)|/\log t$ exactly, and $\limsup_k |\tau_k - \gamma_k| < 0.60832$. We also prove the Landau-type formula on the critical line needed for the prime responses of s_k, that the cross term of the Euler phase at the clock points vanishes for $X \le T^{1/4-\varepsilon}$, and an exact identity expressing the twisted integral $\frac{1}{T}\int_T^{2T} p^{i(t+G(t)/\mathcal{L}(t))}dt$ through the displacements $\tau_k - \gamma_k$ sampled at the clock, with a resonance at $p \approx t/2\pi$ of amplitude $2\pi\sqrt{p}/T$. The prime phases at the clock points are almost orthonormal for all $p \le T^{1-\varepsilon}$, which gives a spectral energy inequality relating the mean-square error of any computable approximation to its prime responses; the response of the moving primes $p = T^\theta$ is shown to be carried by the transport and the residual of the Newton steps, not by the prime polynomials themselves. A conditional law for primes $p = T^\theta$ and a determinantal model are kept separate. All observables are computed on the first 2 001 052 zeros of the Riemann zeta function.

Keywords: Riemann zeta function, zeros of the Riemann zeta function, Riemann–von Mangoldt formula, argument of the Riemann zeta function, truncated Euler product
2020 MSC: 11M06, 11M26, 11Y35

Contents

1 Introduction **3**
1.1 The Riemann–von Mangoldt clock and the ordinates . . . 3
1.2 What is known and only used . . . 4
1.3 What this paper adds . . . 5
1.4 Organization . . . 7

2 Notation **7**
2.1 List of symbols . . . 7
2.2 Counting functions . . . 9
2.3 The RvM clock and the displacements . . . 10
2.4 Euler phases and the corrected clock . . . 10

3 External results **11**

4 Theorem A: an exact defect identity at the zeros **15**

5 Theorem B: how much prime information the clock needs **17**
5.1 Lemmas on prime polynomials . . . 19
5.2 Proof of Theorem 5.1 . . . 20
5.3 A cutoff varying with the index . . . 21

6 Theorem C: exact accounting for the one-step clock **23**

7 Theorem D: a computable clock that approaches the zeros **25**

8 Iterating the Newton step **28**
8.1 The exact recursion . . . 28
8.2 Moments of order $2r$ of the defect . . . 29
8.3 Multiplicity of the one-step clock points . . . 29
8.4 The second step . . . 31
8.5 All further steps: the exact identity, the local deformation, and the multiplicity 32
8.6 The transport identity in a prime frequency . . . 36

9 Theorem E: the pointwise envelope **43**

10 Computations **46**
10.1 Data and methods . . . 46
10.2 The clock and the ordinates . . . 46
10.3 Direct defects (Theorem B) . . . 46
10.4 Exact accounting (Theorem C) . . . 46

10.5 The one-step clock against the cutoff . . . 50
10.6 Transport and the Newton term (Theorem D) . . . 50
10.7 Onset of the cross term . . . 52

11 Prime responses at the zeros and at the clock points 52
11.1 Fixed and slowly moving frequencies . . . 52
11.2 The response at the ordinates and its Fourier reduction . . . 55
11.3 The kernel $\mathfrak{K}_\theta$ . . . 56
11.4 The arithmetic of the Newton nonlinearity . . . 57
11.5 The moving-prime law under Hypotheses U and W . . . 62

12 Heuristic support and numerical evidence 63
12.1 A determinantal model whose connected kernel is $\mathfrak{K}_\theta$ . . . 63
12.2 Numerical evidence . . . 64

13 Conclusions 67
13.1 What was known . . . 67
13.2 What this paper adds . . . 67
13.3 Status of the proofs and pending verification . . . 69
13.4 What is missing for a complete prediction of the zeros from the clock . . . 70
13.5 Why a finer knowledge of $\tau_k - \gamma_k$ matters . . . 71
13.6 Future work . . . 71

Appendix A Programs 74
Appendix A.1 `make_inputs_v14.py` . . . 75
Appendix A.2 `opc_analysis_v14.py` . . . 77
Appendix A.3 `opc_checks_v14.py` . . . 84
Appendix A.4 `opc_computable_response_v14.py` . . . 88
Appendix A.5 `opc_envelope_check_v14.py` . . . 89
Appendix A.6 `opc_envelope_coefficient_v14.py` . . . 90
Appendix A.7 `opc_extensions_v14.py` . . . 92
Appendix A.8 `opc_resonance_v14.py` . . . 95
Appendix A.9 `opc_transport_v14.py` . . . 96
Appendix A.10 `opc_twist_v14.py` . . . 98
Appendix A.11 `opc_validate_v14.py` . . . 99
Appendix A.12 `tools/make_tips_v14.py` . . . 100
Appendix A.13 `tools/replace_tips_v14.py` . . . 103

1. Introduction

1.1. The Riemann–von Mangoldt clock and the ordinates

Let $\zeta(s)$ be the Riemann zeta function, let $N(T)$ be the number of its nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity, and list their ordinates as $\gamma_1 \le \gamma_2 \le \cdots$,

with multiplicity. Away from ordinates the Riemann–von Mangoldt formula reads [33, §§9.3–9.4]

$$N(T) = N_0(T) + \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2}+iT\Big) + O\Big(\frac{1}{T}\Big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.1}$$

the argument of the Riemann zeta function being obtained by continuous variation along $2 \to 2+iT \to \frac{1}{2}+iT$ from $\arg\zeta(2) = 0$. The *midpoint Riemann–von Mangoldt clock* (RvM clock) is the sequence τ_k defined by

$$N_0(\tau_k) = k - \frac{1}{2}, \qquad \tau_k = \frac{2\pi\,(k-\frac{11}{8})}{W_0\Big((k-\frac{11}{8})/e\Big)} \qquad (k \geq 1), \tag{1.2}$$

where W_0 is the principal branch of the Lambert function; the Lambert form appears in the work of França and LeClair [9, 10]. The clock is explicit and lies, in phase, half way between consecutive Gram points. This paper studies its relation to the ordinates γ_k of the zeros of the Riemann zeta function: in which sense the clock describes them, and how much prime information has to be added to it. The relation is measured in units of zeros by the ordinal displacement and by its prime-corrected version, the *ordinal defect*,

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad u_k(X) = s_k - S_X(\gamma_k), \qquad S_X(t) = -\frac{1}{\pi}\sum_{p\leq X}\arg\Big(1-p^{-1/2-it}\Big). \tag{1.3}$$

$u_k(X)$ measures how well the true ordinate satisfies the corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$; $|u_k(X)| < \frac{1}{2}$ means that the corrected count assigns to γ_k its correct index.

1.2. What is known and only used

The following facts are quoted with references in Section 3 and are not reproved.

- The displacement $\delta_k = \tau_k - \gamma_k$ is bounded, and $\limsup_k |\delta_k| \leq 2\pi \cdot 0.10076 = 0.6331\ldots$; this follows at once from the explicit bound of Bellotti and Wong for $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ [2], which improves those of Trudgian [34] and of Hasanalizade, Shen and Wong [15].

- The ordinal displacement s_k differs by less than 1 from minus the integer displacement of γ_k with respect to the Gram intervals, whose moments are those of a normal variable with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae as proved by Korolev [20, 19]. Hence $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ and $\langle \delta_k^2\rangle \sim 2\log\log T/\log^2 T$ over $T < \gamma_k \leq 2T$: the bare clock is close to the zeros in height but misplaces almost all of them by an unbounded number of positions.

- In t-mean, $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$ is asymptotically normal (Selberg [28], Tsang [35]), and Selberg's approximation of $S(t)$ by prime sums up to a small power of T holds in $L^4(T, 2T)$ [28, 29], in the form proved by Fujii [12, §3]. These concern t-averages, not the ordinates.

- Landau's formula for $\sum_{0<\gamma\leq t} n^\rho$ [21], in Gonek's uniform version [13], and Selberg's zero-density estimate [33, Thm. 9.19(C)].

- LeClair [22, Sect. 3] truncates the Euler product at a number of primes that tends to infinity with the height, and the truncated product is used in the equation of the individual zero with a fixed number of primes (10^6 for the 10^{100}-th zero, [11, §XXI, Note added 7]).

1.3. What this paper adds

Let $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$ and let $\langle\cdot\rangle_{\mathcal{I}_T}$ be the mean over $\mathcal{I}_T$. Theorems A–E are unconditional. We have not found them in the literature; absence from a search is weak evidence and does not establish priority.

A. Exact defect identity (Theorems 4.1 and 4.2). For every real C^1 model F of the counting function, every continuous g with primitive $\mathcal{G}$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$, the sum of $\mathcal{G}(u_k + \frac{1}{2}) - \mathcal{G}(u_k - \frac{1}{2})$ over consecutive ordinates equals $\int g(N - F)\, dF$ up to two boundary terms. For $g(x) = x^{2j}$ this gives $\sum u_k^{2j}$ ($\frac{1}{12}$ per zero when $j = 1$); for $g(x) = e^{i\xi x}$ it identifies the characteristic function of the empirical law of u_k with a t-integral. No hypothesis on the zeros and no monotonicity of F is used.

B. How much prime information the clock needs (Theorems 5.1 and 5.12). Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2} \log\frac{1}{c} + O\Big(1 + \sqrt{\log(1/c)}\Big), \tag{1.4}$$

and the same holds with the cutoff $X_k = \max(2, (\tau_k/2\pi)^c)$ varying with the index. A fixed set of primes leaves a mean-square defect $\sim (2\pi^2)^{-1} \log\log T$; primes up to T^c keep it bounded; boundedness holds if and only if $\liminf \log X/\log T > 0$ (Corollary 5.4). Thus the growth of the number of primes asked for in [22] is necessary but not sufficient.

C. Exact accounting for the computable clock (Theorem 6.1). For the one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, the mean-square gain over any finite set of zeros equals a diagonal prime-response term minus a cross term, with no independence assumption; the residual of τ_k^X differs from $u_k(X)$ by an explicit transport term and by the exact nonnegative term $\chi_k = \frac{\gamma_k}{2\pi}(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k})$.

D. A computable clock that approaches the zeros by an order of magnitude (Theorem 7.3). For $2 \le X \le T^{1/2}$ the residual $r_k(X) = \mathcal{L}(\tau_k)(\tau_k^X - \gamma_k)$ satisfies $\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{2\pi^2}\log\frac{1}{c} + O((1 + \sqrt{\log(1/c)})(1 + c^2\log\log T))$; the term $c^2\log\log T$ is the transport between τ_k and γ_k (Lemma 7.2). With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ read off the clock itself,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} \big(\tau_k^X - \gamma_k\big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}. \tag{1.5}$$

A second Newton step from τ_k^X, with cutoff $T^{1/\sqrt{\log\log\log T}}$, has mean-square error $(\log\log\log\log T + O(\sqrt{\log\log\log\log T}))/\log^2 T$ (Theorem 8.5), and the m-th step, with the cutoffs (8.11) depending only on T, has error $(H_m + O(\sqrt{H_m} + 1))/\log^2 T$ with absolute constants, where H_m is an iterated logarithm of T (Theorem 8.10); after $m^* = \log^* T + O(1)$ steps the error is $O(1)/\log^2 T$, that is, $O(1)$ mean spacings in mean square. The proof rests on an exact moment recursion, on a multiplicity bound for the corrected clock points of every level (Theorem 8.9, from Fujii's moments of $S(t+h) - S(t)$), and on a sampling inequality for maximal functions at points of bounded

multiplicity. The second clock preserves the local order and spacing of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices (Proposition 8.7).

E. The pointwise envelope (Proposition 9.1 and Theorem 9.3). $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_{t\to\infty} |S(t)|/\log t$, with equality, also for the corrected clocks with $X \le (\log T)^{2-\varepsilon}$; and

$$\limsup_{k\to\infty} |\tau_k - \gamma_k| \le \frac{1}{\log 2} \int_0^1 \frac{\mathfrak{e}(\sigma)\, d\sigma}{\sqrt{\sigma(2-\sigma)}} < 0.60832 \tag{1.6}$$

for the exponent profile $\mathfrak{e}$ given by Bourgain's exponent pair and the functional equation, against 0.6331 before.

Main finding. The Euler phase of the primes up to a power of the height turns the clock from an approximation in height into an approximation in position. For the bare clock the error in units of zeros, $s_k = \mathcal{L}(\xi_k)\delta_k$, has mean square $\sim \frac{1}{2\pi^2}\log\log T$. With $X = T^c$ the error in units of zeros of the corrected clock equation has bounded mean square (Theorem B), so outside a proportion $\ll (1 + \log(1/c))/A^2$ of the zeros it is at most A; and the computable clock, which uses only τ_k and the primes, keeps the gain up to the transport between τ_k and γ_k (Theorem D). With $X = T^{1/\sqrt{\log\log T}}$,

$$\frac{\langle(\tau_k^X - \gamma_k)^2\rangle_{\mathcal{I}_T}}{\langle(\tau_k - \gamma_k)^2\rangle_{\mathcal{I}_T}} \sim \frac{\log\log\log T}{2\log\log T} \longrightarrow 0, \tag{1.7}$$

and in units of zeros the mean square drops from $\frac{1}{2\pi^2}\log\log T$ to $\frac{1}{4\pi^2}\log\log\log T$. The gain is a statement in mean square; for the worst zero, Proposition 9.1 shows that no cutoff $X \le (\log T)^{2-\varepsilon}$ changes $\limsup |\tau_k - \gamma_k|$, and the improvement (1.6) comes from the exponents of the Riemann zeta function, not from the primes.

Further unconditional results (Section 11). The symmetry $\rho \mapsto 1 - \bar\rho$ of the zeros and Selberg's density estimate give the Landau-type formula $\sum_{0<\gamma\le t} n^{i\gamma} = -\frac{t}{2\pi}\frac{\Lambda(n)}{\sqrt{n}} + O(\frac{t\log^2 n}{\log t} + \sqrt{n}\log^2 t)$ uniformly for $n \le t^{1/8}$ (Theorem 11.1); hence the prime responses of s_k tend to 1 uniformly for primes up to $(\log T)^{2-\varepsilon}$, and the mean-square gain of the full Euler phase at the clock points is $(2\lambda - \lambda^2)v_X + o(1)$ for $X \le (\log T)^{2-\varepsilon}$. The cross term at the clock points is $O(T^{-\varepsilon/2})$ for $X \le T^{1/4-\varepsilon}$; numerically it switches on when X^2 reaches the height in the scale $\tau/2\pi$, through prime pairs $pq \approx \tau_k/2\pi$, and Propositions 8.15 and 8.16 prove the mechanism: the clock sum $\sum(\tau_{k+1} - \tau_k)p^{i\tau_{k+1/2}}$ has a stationary point at $\tau = 2\pi p$ with value exactly $2\pi\sqrt{p}$, and the cross term of the primes is, up to error terms, $\frac{1}{2\pi^2 \#I}\sum \log(pq)$ over the pairs with $2\pi pq$ in the window. The transport identity (Theorem 8.12) writes the integral $\frac{1}{T}\int p^{i(t+G/\mathcal{L})}dt$ exactly as a sum over the displacements δ_k at the clock points; it characterizes the response at the clock, not at the ordinates. *Conditional results* (Section 11.5) are rigorous implications from two named hypotheses: the moving-prime law $1 - \Phi(\theta)$ for $p = T^\theta$, resting on an unconditional reduction to a pair correlation shifted by a prime frequency. *Heuristic support* (Section 12) is kept apart.

1.4. Organization

Section 2 lists the notation and Section 3 the external results. Sections 4–9 prove Theorems A–E; Section 8 iterates the Newton step, uniformly in the number of steps, and proves the transport identity in a prime frequency. Section 10 computes the observables on the first 2 001 052 zeros of the Riemann zeta function. Section 11 contains the prime responses, unconditional and conditional, and Section 12 the heuristic support. Section 13 separates what is known from what is new and discusses the use of a finer knowledge of $\tau_k - \gamma_k$. All programs and outputs are in a public repository (Data availability).

2. Notation

2.1. List of symbols

Every symbol has one meaning throughout. The letter μ is not used (it is reserved for the Möbius function), π is only the number 3.14159…, and prime counts are written $\#\{p \le y\}$. Dummy variables of integration (t, y) are local.

symbol	meaning
$\zeta(s)$	the Riemann zeta function
$\rho_k = \beta_k + i\gamma_k$	nontrivial zeros of the Riemann zeta function, $0 < \gamma_1 \le \gamma_2 \le \cdots$ with multiplicity
$\nu(\gamma)$	number of zeros of the Riemann zeta function with ordinate γ (multiplicities included)
$N(t)$, $N(t^{\pm})$	number of zeros with $0 < \gamma \le t$; one-sided limits
ϑ, Θ_0, ϖ	Riemann–Siegel theta function; $\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}$; $\varpi = \vartheta - \Theta_0$
g_m	Gram points, $\vartheta(g_m) = m\pi$
N_0, $\mathcal{L}$	$N_0 = \Theta_0/\pi + 1$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$
S, S^*, G	$S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$; midpoint value at ordinates; $G = N - N_0 = S + \varpi/\pi$
$\mathcal{I}_T$, $\langle\cdot\rangle_I$	$\{k : T < \gamma_k \le 2T\}$; mean over k in a finite index set I
τ_k, τ_x, W_0	RvM clock (2.5); the same for real $x \ge 1$; principal Lambert function
s_k, δ_k, ξ_k	$s_k = k - \frac{1}{2} - N_0(\gamma_k)$; $\delta_k = \tau_k - \gamma_k$; mean-value point in $s_k = \mathcal{L}(\xi_k)\delta_k$
Δ_k	integer displacement of γ_k with respect to the Gram intervals (Theorem 3.4)
$\mathfrak{d}_k$	$s_k + \Delta_k$, $\lvert\mathfrak{d}_k\rvert < 1$ (Remark 3.22)
$\mathfrak{s}_T$	$(\frac{1}{2\pi^2}\log\log T)^{1/2}$ (Remark 3.22)
Λ, a_n	von Mangoldt function; $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$
A, A_X, $\mathcal{P}$	set of prime powers with $\sum_{n\in A}\lvert a_n\rvert < \infty$; $A_X = \{p^m : p \le X\}$; finite set of primes
S_A, S_X, $S_{\mathcal{P}}$	$\sum_{n\in A} a_n \sin(t\log n)$; S_{A_X}; $S_{\{p^m : p\in\mathcal{P}\}}$
v_A, v_X, $v_{\mathcal{P}}$	$\frac{1}{2}\sum_{n\in A} a_n^2$; v_{A_X}; $v_{\{p^m : p\in\mathcal{P}\}}$
σ_X, M_X	bounds for $\lvert S_X\rvert$ and $\lvert S_X'\rvert$, (2.11)
F, F_X	a real C^1 model of N; $F_X = N_0 + S_X$
u_k, $u_k(X)$	$k - \frac{1}{2} - F(\gamma_k)$; $k - \frac{1}{2} - F_X(\gamma_k)$

symbol	meaning
Ψ_j, κ_j	sawtooth polynomial (4.2); $\kappa_j = \frac{(2j)!}{j!(2\pi)^{2j}}$
X, c, C, T_0	prime cutoff; $c = \log X/\log T$; absolute constants
Y, Y_0, S_Y^{pr}	polynomial length; $Y_0 = T^{1/160}$; $S_Y^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p<Y} p^{-1/2}\sin(t\log p)$
$\mathcal{J}$, V, $\mathcal{W}$, $\mathcal{W}_1$	primes between X and Y_0; $V = \sum_{p\in\mathcal{J}} p^{-1}$; $\mathcal{W} = S_{Y_0}^{\mathrm{pr}} - S_X$; its part over $\mathcal{J}$ (5.4)
E_0, E_X, $\mathcal{E}_2$	$S - S_{Y_0}^{\mathrm{pr}}$; $S - S_X$; $\int_T^{2T} E_X^2\,dt$
z_n, $\mathcal{D}$, $\mathcal{D}_1$, $\mathcal{D}_2$, η_n, η_n'	coefficients and Dirichlet polynomials in Lemmas 5.9–5.10
τ_k^X, δ_k^X, $r_k(X)$, χ_k	one-step clock; $\tau_k^X - \gamma_k$; $\mathcal{L}(\tau_k)\delta_k^X$; second-order error (6.3)
$\mathcal{R}_n^\gamma(I)$, $\mathcal{R}_n^\tau(I)$	prime-power responses at the zeros and at the clock points (6.1)
$\mathrm{Cr}_A(I)$, $D_X(I)$	cross term (6.2); diagonal term (6.5)
λ	real multiplier of an Euler phase
ς_n, $\mathbf{Q}$, $\mathbf{b}$, $\boldsymbol{\lambda}$, $\Upsilon_{\boldsymbol{\lambda}}$	$\varsigma_n(k) = \sin(\tau_k\log n)$; Gram matrix; projection vector; coefficients; trigonometric correction (Remark 6.3)
$\mathrm{BW}(t)$	$0.10076\log t + 0.24460\log\log t + 8.08344$ (Bellotti–Wong)
t_k^X	root of $F_X(t) = k - \frac{1}{2}$ (Remark 5.7)
ℓ_n	$\log n$
ψ, t_1, t_2	test function and endpoints in Lemma 4.7
L_T, $\mathcal{N}_T$, θ	$\log(T/2\pi)$; $TL_T/2\pi$; $\log p/L_T$
$\Phi(\theta)$	$(1+\theta)\log(1+\theta) + (1-\theta)\log(1-\theta)$
$\mathcal{A}_{T,p}$, $\mathcal{S}_{T,p}$	one-particle term and shifted correlation (11.5)
$\mathcal{H}_n(t)$	$\int_{14}^t S(y)n^{iy}\,dy$ (proof of Theorem 11.3)
f_T, $\mathfrak{m}_T$, Z_T, $\mathcal{B}_T$	$S\mathbf{1}_{[T,2T]}$; its distributional derivative; Fourier transform of $\mathfrak{m}_T$; scaled bilinear integral (11.11)
ω, ω_0, α	frequency variables; $\alpha = \omega/L_T$
$\mathfrak{K}_\theta$, h	kernel (11.12); $h(q) = \max(1-\lvert q\rvert, 0)$
R_0, $\mathfrak{n}$, $\Pi_{\mathfrak{n}}$, q, Ξ_q	model: circle length, number of modes, projection, lattice frequency, linear statistic
ι_1, ι_2, m_i	tilt parameters; multiplication operators (Section 12.1)
y_0, w_T	fixed threshold; threshold tending to infinity
K, B	2 001 052, the number of ordinates used; saturation level (Section 10.5)
$\mathfrak{g}$, $\mathfrak{G}$, ω	continuous test function and a primitive (Theorem 4.2); real frequency variable
$\mathfrak{t}_k(X)$, y_k	transport term $S_X(\gamma_k) - S_X(\tau_k)$; one-step correction $S_X(\tau_k)/\mathcal{L}(\tau_k)$
φ, $\mathfrak{h}$, $\mathcal{M}$, $\mathfrak{b}_T$, $\mathcal{U}_k$	cutoff function; $\varphi S_X'$; centred Hardy–Littlewood maximal operator; $2\pi/\log(3T)$; sampling intervals (Lemma 7.2)
X_k, $\tilde{u}_k$, $X_\pm$, $V_\pm$	variable cutoff; its defect; extreme cutoffs in a window; $\sum_{X_-<p\le X_+} p^{-1}$ (Theorem 5.12)
$\mathcal{Q}$, $\mathfrak{w}$	maximal partial prime sum; smooth window (Lemma 5.11)
$\mathcal{I}_T^\tau$, τ_k^{loc}	$\{k : T \le \tau_k < 2T\}$; one-step clock with cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ (Corollary 7.5)
$\mathfrak{a}$, BW_S	$\limsup_{t\to\infty} \lvert S(t)\rvert/\log t$; Bellotti–Wong bound for $\lvert S(t)\rvert$ (Theorem 3.2)

symbol	meaning
$\mathfrak{e}$, (k,l)	exponent profile of the Riemann zeta function; exponent pair (Section 9)
$\sigma_\circ$, ϱ, $\varkappa$, $\sigma^\sharp$, $\delta^\sharp$	centre and radius of the Jensen disc; power in Backlund's function; parameters of Backlund's trick (Theorem 3.21)
$N^+(\sigma,t)$, $\mathcal{K}_n(t)$	number of zeros with $\beta>\sigma$, $0<\gamma\le t$; symmetric remainder (Theorem 11.1)
$x_k^{(j)}$, $e_{j,k}$, $\chi_{j,k}$, $\|\cdot\|_{r,I}$	iterated Newton points (8.1); their errors; curvature terms; discrete norms (Section 8)
H_j, c_j, ρ_k, y_k, $\tilde n_j$	$H_0=\log\log T$, $H_j=2\log(1/c_j)$ and the cutoff law (8.11); $\rho_k=x_k^{(1)}$; $y_k=S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$; multiplicity functions (Lemma 8.3, Theorem 8.9)
$A_j(r)$, $\mathfrak{k}_j$, m^*, $C_\star$, H^*	normalized moments of the errors, multiplicity constants, number of steps and constants of Theorem 8.10
$n_\Lambda(u)$, C_1,C_2,C_3	near-ordinate count and constants of Lemma 8.8 and Theorem 8.9
v_p, G, $C_T(y)$, $\mathcal{S}$	normalized sine vectors, their Gram matrix, clock sums $\frac{1}{M}\sum_k e^{i\tau_k\log y}$ and the stationary band (Proposition 8.17)
$\mathcal{A}_p^{(m)}$, $\mathcal{D}_p^{(m)}$, $\mathcal{B}_p^{(m)}$, $\mathcal{E}_p^{(m)}$, $B_k^{(m)}$	computable, diagonal, transport and residual responses, and the transport term (11.20)
P_X, z_q, $A_{\mathbf{j}}$, $\Lambda_{\mathbf{j}}$, $D(\mathbf{j})$, λ_n, Q_T	prime part of S_X, Bessel arguments, coefficients, frequencies and degree of (11.15); $\lambda_n(t)=\log n/\mathcal{L}(t)$; arithmetic polynomial of Proposition 11.10
$I_T(w)$, $J_T(w)$, v_k, w, $\mathfrak{w}_k$, $\widetilde{\mathcal{R}}_p^\tau$	twisted integral; clock sum; $\tau_{k+1/2}$; real frequency; weights and weighted response (Section 8.6)
$\mathfrak{f}_\varkappa$, $\mathcal{Z}_\varkappa$, $F_\circ$	Backlund's auxiliary function, its zero sum and the majorant (Theorem 3.21)

2.2. Counting functions

Let $\vartheta(t)=\Im\log\Gamma(\frac14+\frac{it}{2})-\frac{t}{2}\log\pi$ for $t>0$. For $t>0$ not an ordinate [33, Thms. 9.3–9.4],

$$N(t)=\frac{1}{\pi}\vartheta(t)+1+S(t),\qquad S(t)=\frac{1}{\pi}\arg\zeta\Big(\tfrac12+it\Big)=O(\log t),\tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 to $2+it$ and then horizontally to $\frac12+it$. At an ordinate we keep the right-continuous N and define S by (2.1); $S^*(\gamma)=\frac12(S(\gamma^-)+S(\gamma^+))$. Put

$$\Theta_0(t)=\frac{t}{2}\log\frac{t}{2\pi e}-\frac{\pi}{8},\qquad N_0(t)=\frac{1}{\pi}\Theta_0(t)+1,\qquad \varpi(t)=\vartheta(t)-\Theta_0(t).\tag{2.2}$$

By Stirling's formula $\varpi(t)=\frac{1}{48t}+O(t^{-3})$ and $\varpi'(t)=O(t^{-2})$; in particular $|\varpi(t)|/\pi<\frac{1}{100}$ for $t\ge14$. For all $t>0$,

$$N(t)=N_0(t)+S(t)+\frac{\varpi(t)}{\pi},\qquad \mathcal{L}(t)=N_0'(t)=\frac{1}{2\pi}\log\frac{t}{2\pi}.\tag{2.3}$$

Since $N(2T)-N(T)=N_0(2T)-N_0(T)+O(\log T)$,

$$\#\mathcal{I}_T=T\mathcal{L}(T)+O(T).\tag{2.4}$$

2.3. The RvM clock and the displacements

N_0 is C^∞ and strictly increasing on $(2\pi,\infty)$, with $N_0(2\pi)=-\frac{1}{8}$ and $N_0(t)\to\infty$. Hence for every real $x\ge 1$ the equation $N_0(t)=x-\frac{1}{2}$ has exactly one solution $\tau_x>2\pi$. Writing $t=2\pi y$, the equation is $y\log(y/e)=x-\frac{11}{8}\ge-\frac{3}{8}$; its root $y>1$ is $y=(x-\frac{11}{8})/W_0((x-\frac{11}{8})/e)$, with $y=e$ if $x=\frac{11}{8}$ [6]. Thus

$$\tau_x=\frac{2\pi\,(x-\frac{11}{8})}{W_0\Big((x-\frac{11}{8})/e\Big)},\qquad \frac{d\tau_x}{dx}=\frac{1}{\mathcal{L}(\tau_x)}=\frac{2\pi}{\log(\tau_x/2\pi)},\tag{2.5}$$

the derivative by implicit differentiation. Since $\Theta_0(\tau_k)=\pi(k-\frac{3}{2})$ and $\varpi=O(1/t)$, τ_k lies in phase half way between g_{k-2} and g_{k-1}, up to $O(\tau_k^{-1}(\log\tau_k)^{-1})$.

Definition 2.1. For $k\ge 1$, $s_k=k-\frac{1}{2}-N_0(\gamma_k)$ and $\delta_k=\tau_k-\gamma_k$.

Since $\gamma_1>14>2\pi$ and $\tau_k>2\pi$, the mean value theorem applied to $s_k=N_0(\tau_k)-N_0(\gamma_k)$ gives

$$s_k=\mathcal{L}(\xi_k)\,\delta_k\qquad\text{for some }\xi_k\text{ between }\gamma_k\text{ and }\tau_k.\tag{2.6}$$

Lemma 2.2. *For every $k\ge 1$,*

$$N(\gamma_k^-)-N_0(\gamma_k)+\frac{1}{2}\le s_k\le N(\gamma_k^+)-N_0(\gamma_k)-\frac{1}{2},\tag{2.7}$$

and

$$\Big|s_k-\frac{\varpi(\gamma_k)}{\pi}\Big|\le\max\Big(|S(\gamma_k^-)|,|S(\gamma_k^+)|\Big)-\frac{1}{2}.\tag{2.8}$$

Proof. The indices j with $\gamma_j=\gamma_k$ are exactly $N(\gamma_k^-)+1,\dots,N(\gamma_k^+)$, so $N(\gamma_k^-)+1\le k\le N(\gamma_k^+)$; subtract $\frac{1}{2}+N_0(\gamma_k)$. By (2.3) at $\gamma_k^\pm$ and continuity of N_0 and ϖ, (2.7) reads $L+\frac{1}{2}\le x\le U-\frac{1}{2}$ with $x=s_k-\varpi(\gamma_k)/\pi$, $L=S(\gamma_k^-)$, $U=S(\gamma_k^+)$, and $U-L=\nu(\gamma_k)\ge 1$. Put $m=\max(|L|,|U|)$. Then $x\le U-\frac{1}{2}\le m-\frac{1}{2}$ and $x\ge L+\frac{1}{2}\ge -m+\frac{1}{2}$, which is (2.8). □

2.4. Euler phases and the corrected clock

For $n\ge 2$ let $a_n=-\Lambda(n)/(\pi\sqrt{n}\log n)$. For a set A of prime powers with $\sum_{n\in A}|a_n|<\infty$,

$$S_A(t)=\sum_{n\in A}a_n\sin(t\log n),\qquad v_A=\frac{1}{2}\sum_{n\in A}a_n^2;\tag{2.9}$$

the series converges absolutely and uniformly in t, so S_A is continuous and bounded. For $X\ge 2$, $\sum_{n\in A_X}|a_n|\le\frac{1}{\pi}\sum_{p\le X}\sum_{m\ge1}\frac{1}{mp^{m/2}}<\infty$ and, since $\arg(1-z)=-\Im\sum_{m\ge1}z^m/m$ for $|z|<1$ with the principal argument,

$$S_X(t)=-\frac{1}{\pi}\sum_{p\le X}\arg\Big(1-p^{-1/2-it}\Big)=-\frac{1}{\pi}\sum_{p\le X}\sum_{m\ge1}\frac{\sin(mt\log p)}{m\,p^{m/2}}=S_{A_X}(t).\tag{2.10}$$

The sum of principal arguments is a continuous argument of the finite Euler product, not necessarily its principal argument. The series of derivatives converges uniformly, because $\sum_{p\le X}\log p\sum_{m\ge1}p^{-m/2}<\infty$; hence $S_X\in C^\infty(\mathbb{R})$, its derivatives may be taken termwise, and

$$|S_X(t)|\le\sigma_X:=\frac{1}{\pi}\sum_{p\le X}\sum_{m\ge1}\frac{1}{m\,p^{m/2}},\qquad |S_X'(t)|\le M_X:=\frac{1}{\pi}\sum_{p\le X}\frac{\log p}{\sqrt{p}-1},\tag{2.11}$$

with $\sigma_X\ll\sqrt{X}/\log X$ and $M_X\ll\sqrt{X}$ by the prime number theorem.

Definition 2.3. For $X\ge2$ let $F_X=N_0+S_X$ and $u_k(X)=k-\frac12-F_X(\gamma_k)=s_k-S_X(\gamma_k)$. The one-step clock and its residual are

$$\tau_k^X=\tau_k-\frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)},\qquad \delta_k^X=\tau_k^X-\gamma_k,\qquad r_k(X)=\mathcal{L}(\tau_k)\,\delta_k^X.\tag{2.12}$$

Since $F_X(\tau_k)-(k-\frac12)=S_X(\tau_k)$, τ_k^X is the first step of $t\mapsto t-(F_X(t)-k+\frac12)/\mathcal{L}(t)$ from $t=\tau_k$. At a simple ordinate, (2.3) gives $N_0(\gamma_k)+S^*(\gamma_k)+\varpi(\gamma_k)/\pi=k-\frac12$, so $u_k(X)=S^*(\gamma_k)-S_X(\gamma_k)+\varpi(\gamma_k)/\pi$ is the error of the prime approximation of the argument of the Riemann zeta function at the zero.

3. External results

The results of this section are known; they are used, not reproved.

Theorem 3.1 ([33, Thms. 9.4, 9.9 and §9.12], [24])**.** *$S(t)=O(\log t)$; $\gamma_{k+1}-\gamma_k$ is bounded; and $S_1(t)=\int_0^t S(y)\,dy=O(\log t)$.*

Theorem 3.2 (Bellotti–Wong [2, Thm. 1.1, Cor. 1.5 and (1.2)])**.** *For $T\ge e$, $\bigl|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}\bigr|\le\mathrm{BW}(T)$ and $|S(T)|\le\mathrm{BW}_S(T)$, where*

$$\mathrm{BW}_S(t)=0.10076\log t+\min\{0.24460\log\log t+7.20844,\ 1.68845\log\log t+1.50956\}.\tag{3.1}$$

Moreover $|S(T)|\le2.5167$ for $0\le T\le30\,610\,046\,000$.

Earlier explicit versions are due to Trudgian [34] and to Hasanalizade, Shen and Wong [15]. The last bound rests on Platt's rigorous isolation of the zeros up to that height [2, (1.2)].

Theorem 3.3 (Selberg [28]; see [12, p. 225])**.** *For every integer $j\ge1$, $\int_T^{2T}S(t)^{2j}\,dt=\kappa_j\,T(\log\log T)^j+O_j\bigl(T(\log\log T)^{j-1/2}\bigr)$.*

Theorem 3.4 (Selberg; Korolev [20], [19, Thm. 4.2])**.** *Let Δ_k be the integer such that $g_{k+\Delta_k-2}<\gamma_k\le g_{k+\Delta_k-1}$ (in [19] the Gram points are $t_m=g_{m-1}$ and $t_{m-1}<\gamma_k\le t_m$ with $m=k+\Delta_k$), and let $0<\varepsilon_0<10^{-3}$. For every fixed $j\ge1$, as $n_0\to\infty$, uniformly for $n_0^{27/82+0.9\varepsilon_0}\le h_0\le n_0^{27/82+\varepsilon_0}$,*

$$\sum_{n_0<k\le n_0+h_0}\Delta_k^{2j}=\kappa_j h_0(\log\log n_0)^j\bigl(1+o(1)\bigr),\qquad \Bigl|\sum_{n_0<k\le n_0+h_0}\Delta_k^{2j-1}\Bigr|\ll_j h_0(\log\log n_0)^{j-1}.\tag{3.2}$$

Theorem 3.5 (Selberg's approximation in the form of Fujii [12, §3, pp. 232–233]). *For every integer $k \ge 1$, $\int_T^{2T} \Big(S(t) - S_{Y_k}^{\mathrm{pr}}(t)\Big)^{2k} dt \le (Ck^2)^{2k}\, T$ with $Y_k = T^{1/(80k)}$ and an absolute constant C. In particular $\int_T^{2T} (S - S_{Y_0}^{\mathrm{pr}})^4 \ll T$ with $Y_0 = Y_2 = T^{1/160}$.*

In [12, §3] Selberg's difference between $\pi^{-1} \arg L(\frac12 + it, \chi)$ and a prime sum is written as five explicit terms, each bounded on p. 233 in $L^{2k}(T, T+H)$ by $\ll k^2 H^{1/2k}$ under the hypotheses of Fujii's Main Theorem, which restrict q, H and h but not k. For the principal character (so that the L-function is the Riemann zeta function), $H = T$ and Fujii's parameter $a_2 = \frac14$, the prime sum runs over $p < x^3$ with $x = T^{a_2/60k}$, that is $p < T^{1/(80k)} = Y_k$; the five terms are bounded by $k^{1/2}$, k^2, k^2, $k^{1/2}$ and $k^{3/2}$ times $H^{1/2k}$, with absolute implied constants, whence the constant $(Ck^2)^{2k}$; and $\frac{1}{\pi}\Im p^{-1/2-it} = -\frac{1}{\pi} p^{-1/2} \sin(t \log p)$. For $k = 2$ this is the form used in Section 5; $k = 4$, with $Y_4 = T^{1/320}$, and the general k are used in Section 8.

Theorem 3.6 (Fujii [12, Main Theorem, p. 225]). *For every integer $k \ge 1$ there is C_k such that, for $T \ge T_0$ and $0 < h \le 1$,*

$$\int_T^{2T} \Big(S(t+h) - S(t)\Big)^{2k} dt \le C_k\, T \Big(\log(2 + h \log T)\Big)^k.$$

Fujii's statement is an asymptotic formula with main term $c_k H (2 \log(2 + h \log T))^k$ and relative error $O((Ak)^{3k} (\log(2 + h \log T))^{-1/2})$ for $T^{1/2+a_2} \le H \le T$; the upper bound with $H = T$ follows for all $0 < h \le 1$. Only $k = 1, 2$ are used (Lemma 8.8).

Theorem 3.7 (Montgomery–Vaughan [26, Cor. 3]). *For complex z_n, $\int_T^{2T} \Big|\sum_{n \le Y} z_n n^{-it}\Big|^2 dt = \sum_{n \le Y} |z_n|^2 \Big(T + O(n)\Big)$ with an absolute implied constant.*

(Replace z_n by $z_n n^{-iT}$ to pass from $[0, T]$ to $[T, 2T]$.)

Theorem 3.8 (Mertens [25]; see [32, Ch. I.1, Thms. 1.8 and 1.10]). *For $y \ge 2$, $\sum_{p \le y} \frac{\log p}{p} = \log y + O(1)$ and $\sum_{p \le y} \frac{1}{p} = \log\log y + O(1)$.*

Theorem 3.9 (Kusmin–Landau [14, Thm. 2.1]). *Let Ω be real with continuous monotone derivative on $[y_1, y_2]$ and $\|\Omega'\| \ge 1/H_{\mathrm{KL}} > 0$ there, where $\|\cdot\|$ is the distance to the nearest integer. Then $\sum_{y_1 < n \le y_2} e^{2\pi i \Omega(n)} \ll H_{\mathrm{KL}}$.*

Theorem 3.10 (Stationary phase [33, Lemma 4.6]). *Let F be real with three derivatives on $[a, b]$, $0 < \lambda_2 \le -F'' \le A\lambda_2$ and $|F'''| \le A\lambda_3$ there, and $F'(c) = 0$ with $a \le c \le b$. Then*

$$\int_a^b e^{iF(x)} dx = \frac{(2\pi)^{1/2} e^{iF(c) - i\pi/4}}{|F''(c)|^{1/2}} + O\Big(\lambda_2^{-4/5} \lambda_3^{1/5}\Big) + O\Big(\min\Big(\frac{1}{|F'(a)|}, \lambda_2^{-1/2}\Big)\Big) + O\Big(\min\Big(\frac{1}{|F'(b)|}, \lambda_2^{-1/2}\Big)\Big),$$

and if F' does not vanish on $[a, b]$ the same holds without the leading term.

Theorem 3.11 (Van der Corput's exponential sum to integral [33, Lemma 4.7]). *Let f be real with a continuous, steadily decreasing derivative on (a, b), $f'(b) = \alpha$, $f'(a) = \beta$. Then for any $0 < \eta < 1$,*

$$\sum_{a < n \le b} e^{2\pi i f(n)} = \sum_{\alpha - \eta < \nu < \beta + \eta} \int_a^b e^{2\pi i (f(x) - \nu x)} dx + O\Big(\log(\beta - \alpha + 2)\Big).$$

Theorem 3.12 (Van der Corput's exponential sum to integral, weighted form [33, Lemma 4.10]). *Let f be real with a continuous, steadily decreasing derivative on (a,b), $f'(b)=\alpha$, $f'(a)=\beta$, and let $\mathfrak{g}>0$ be decreasing with continuous derivative such that $|\mathfrak{g}'|$ is steadily decreasing. Then for any $0<\eta<1$,*

$$\sum_{a<n\le b}\mathfrak{g}(n)e^{2\pi i f(n)}=\sum_{\alpha-\eta<\nu<\beta+\eta}\int_a^b \mathfrak{g}(x)e^{2\pi i(f(x)-\nu x)}dx+O\Big(\mathfrak{g}(a)\log(\beta-\alpha+2)\Big)+O\Big(|\mathfrak{g}'(a)|\Big).$$

Theorem 3.13 (Van der Corput's second-derivative test [33, Thm. 5.9]). *Let f be real and twice differentiable on $[a,b]$, $b\ge a+1$, with $0<\lambda_2\le f''\le h\lambda_2$ (or $\lambda_2\le -f''\le h\lambda_2$) throughout. Then $\sum_{a<n\le b}e^{2\pi i f(n)}=O\Big(h(b-a)\lambda_2^{1/2}\Big)+O\Big(\lambda_2^{-1/2}\Big)$.*

Theorem 3.14 (Landau [21]; uniform version Gonek [13], as stated in [1, (1.1)]). *For integers $n\ge 2$ and real $t\ge 2$, $\sum_{0<\gamma\le t}n^{\rho}=-\frac{t}{2\pi}\Lambda(n)+O\Big(n\log(2nt)\log\log(3n)\Big)$, zeros counted with multiplicity, with an absolute implied constant.*

(Gonek's sum runs over $0<\gamma<t$; the zeros with $\gamma=t$ contribute $O(n\log t)$.)

Theorem 3.15 (Selberg [33, Thm. 9.19(C)]). *Let $N^+(\sigma,t)$ be the number of zeros $\beta+i\gamma$ of the Riemann zeta function with $\beta>\sigma$ and $0<\gamma\le t$, counted with multiplicity. Uniformly for $\frac12\le\sigma\le 1$ and $t\ge 2$, $N^+(\sigma,t)\ll t^{1-\frac14(\sigma-\frac12)}\log t$.*

Theorem 3.16 (Selberg [28]; Tsang [35]; see [7, (1.1)]). *For fixed real $y_1<y_2$, $\frac1T\operatorname{meas}\{t\in[T,2T]: y_1<S(t)/\mathfrak{s}_T\le y_2\}\to\frac{1}{\sqrt{2\pi}}\int_{y_1}^{y_2}e^{-y^2/2}dy$, where $\mathfrak{s}_T=(\frac{1}{2\pi^2}\log\log T)^{1/2}$.*

In [7] the statement is that $\log\zeta(\frac12+it)/\sqrt{\log\log T}$ tends to a standard complex Gaussian for t uniform in $[T,2T]$; its imaginary part is $\pi S(t)/\sqrt{\log\log T}$, of variance $\frac12$.

Theorem 3.17 (Hardy–Littlewood [31, Thm. I.1], as quoted in [18, (1.2)]; Kinnunen [18, Thm. 1.4]). *Let $\mathcal{M}f(x)=\sup_{y>0}\frac{1}{2y}\int_{x-y}^{x+y}|f|$ be the centred maximal function of $f\in L^1_{\mathrm{loc}}(\mathbb{R})$. For $1<p<\infty$ and $f\in L^p(\mathbb{R})$, $\|\mathcal{M}f\|_p\le A_p\|f\|_p$, where A_p is bounded for $p\ge 2$. If f and f' belong to $L^p(\mathbb{R})$, $1<p<\infty$, then $\mathcal{M}f$ has a locally absolutely continuous representative, $(\mathcal{M}f)'\in L^p(\mathbb{R})$, and $|(\mathcal{M}f)'|\le\mathcal{M}(f')$ almost everywhere.*

Theorem 3.18 (Carleson [5], Hunt [16]; on $\mathbb{R}$ see [23]). *For $f\in L^2(\mathbb{R})\cap L^4(\mathbb{R})$ with Fourier transform $\mathcal{F}f(\omega)=\int_{\mathbb{R}}f(t)e^{-i\omega t}dt$, let $\mathcal{C}f(x)=\sup_{\omega_1\in\mathbb{R}}\Big|\frac{1}{2\pi}\int_{-\infty}^{\omega_1}\mathcal{F}f(\omega)e^{i\omega x}d\omega\Big|$. Then $\|\mathcal{C}f\|_4\ll\|f\|_4$.*

In [23] the statement is that $f\mapsto\sup_{\xi\in\mathbb{R}}|\mathcal{H}(e^{i\xi\cdot}f)|$, $\mathcal{H}$ the Hilbert transform, is bounded on $L^p(\mathbb{R})$ for $1<p<\infty$; since the partial Fourier integral up to ω_1 is $e^{i\omega_1 x}\cdot\frac12(\mathrm{Id}\pm i\mathcal{H})(e^{-i\omega_1\cdot}f)(x)$, the operator $\mathcal{C}f$ is at most $\frac12(|f|+\sup_\xi|\mathcal{H}(e^{i\xi\cdot}f)|)$.

Theorem 3.19 ([17, Lemma 2.8 and (7.57)]; Bourgain [4, §5, Thm. 6]). *(a) If (k,l) is an exponent pair, so is $A(\mathsf{k},\mathsf{l})=\Big(\frac{\mathsf{k}}{2\mathsf{k}+2},\frac12+\frac{\mathsf{l}}{2\mathsf{k}+2}\Big)$. (b) If (k,l) is an exponent pair with $\mathsf{l}-\mathsf{k}\ge\frac12$, then $\zeta(\mathsf{l}-\mathsf{k}+it)\ll|t|^{\mathsf{k}}\log|t|$ for $|t|\ge 2$. (c) For every $\varepsilon>0$, $(\frac{13}{84}+\varepsilon,\frac{55}{84}+\varepsilon)$ is an exponent pair.*

For (b): [17, (7.57)] gives $\zeta(\sigma+it)\ll|t|^{(\mathsf{k}+\mathsf{l}-\sigma)/2}\log|t|$ for $\frac12\le\sigma\le\mathsf{l}-\mathsf{k}$; take $\sigma=\mathsf{l}-\mathsf{k}$.

Theorem 3.20 (Phragmén–Lindelöf principle in a strip [2, Lemma 2.5], [34, Lemma 3]). *Let $\sigma_1 < \sigma_2$ and $e_1 \geq e_2 \geq 0$. If $|\zeta(\sigma_i + it)| \leq C_i |t|^{e_i} \log|t|$ for $|t| \geq 2$ and $i = 1, 2$, then $|\zeta(\sigma+it)| \ll |t|^{e(\sigma)} \log|t|$ uniformly for $\sigma_1 \leq \sigma \leq \sigma_2$ and $|t| \geq 2$, where e is the linear function with $e(\sigma_i) = e_i$.*

(Apply [2, Lemma 2.5] to the entire function $(s-1)\zeta(s)$ of order 1, with exponents $1 + e_i$ and one power of the logarithm on the two lines.)

Theorem 3.21 (Backlund's method [2, Lemmas 2.1, 2.3 and 2.4], after [34, 15]). *Let T be large, $\varkappa \in \mathbb{N}$, $\mathfrak{f}_\varkappa(s) = \frac{1}{2}\Big[((s+iT-1)\zeta(s+iT))^\varkappa + ((s-iT-1)\zeta(s-iT))^\varkappa\Big]$, and for a disc $D(\sigma_\circ, \varrho)$ let $\mathcal{Z}_\varkappa(\sigma_\circ, \varrho) = \frac{1}{\varkappa}\sum_z \log\frac{\varrho}{|z-\sigma_\circ|}$ over the zeros z of $\mathfrak{f}_\varkappa$ in $D(\sigma_\circ, \varrho)$. Let $1 < \sigma_\circ < \varrho$, $\sigma^\sharp = \sigma_\circ + (\sigma_\circ - \frac{1}{2})^2/\varrho$, $\delta^\sharp = 2\sigma_\circ - \sigma^\sharp - \frac{1}{2}$, and assume $0 < \delta^\sharp < \frac{1}{2}$.*

(a) If $F_\circ : [-\pi, \pi] \to \mathbb{R}$ is even and $F_\circ(\theta) \geq \frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ + \varrho e^{i\theta})|$ for all $\varkappa$ and θ, then there are $\varkappa_m \to \infty$ with $\limsup_m \mathcal{Z}_{\varkappa_m}(\sigma_\circ, \varrho) \leq \log\frac{\zeta(\sigma_\circ)}{\sqrt{(\sigma_\circ-1)^2+T^2}\,\zeta(2\sigma_\circ)} + \frac{1}{\pi}\int_0^\pi F_\circ(\theta)\,d\theta$.

(b) For every $\varkappa$, $\Big|\Delta^{1/2}_{\sigma=\sigma^\sharp} \arg\Big((\sigma+iT-1)\zeta(\sigma+iT)\Big)\Big| \leq \frac{\pi\mathcal{Z}_\varkappa(\sigma_\circ,\varrho)}{2\log(\varrho/(\sigma_\circ-\frac{1}{2}))} + \frac{\pi}{100} + \frac{3\pi}{2\varkappa} + \frac{\pi}{4}$.

In (b) the term $\frac{\pi}{100}$ bounds $\frac{1}{2}E(T, \delta^\sharp)$ of [2, Lemma 2.3], by the monotonicity in $\delta^\sharp$ and the bound for $E(T, \frac{1}{2})$ in [2, Lemma 2.4]; (a) is [2, Lemma 2.1], which also requires $\sigma_\circ - \varrho < \frac{1}{2} < 1 < \sigma_\circ < \sigma^\sharp < \sigma_\circ + \varrho$, a consequence of $1 < \sigma_\circ < \varrho$.

Remark 3.22 (Known consequences for the clock). (a) *Moments and law of s_k.* Since $\vartheta(\gamma_k)/\pi \in (k + \Delta_k - 2, k + \Delta_k - 1]$ and $\vartheta/\pi = N_0 - 1 + \varpi/\pi$, one has $s_k + \Delta_k = k + \Delta_k - \frac{1}{2} - N_0(\gamma_k) \in [-\frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi}, \frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi})$, so $s_k = -\Delta_k + \mathfrak{d}_k$ with $|\mathfrak{d}_k| < 1$. The index set $\mathcal{I}_T$ is the integer interval $(N(T), N(2T)]$, of length $\asymp T\log T$. Cover it by consecutive windows $(n_0, n_0 + h_0]$ with $h_0 = \lfloor n_0^{27/82 + 0.95\varepsilon_0}\rfloor$, the last one shifted to end at $N(2T)$, so that it overlaps the previous one in at most $h_0 = O(T^{-1/2}\#\mathcal{I}_T)$ indices; in each window $\log\log n_0 = \log\log T + o(1)$. Summing (3.2) over the windows, and bounding the doubly counted indices by one window sum (for odd powers, by Hölder's inequality and the even moments), gives $\langle\Delta_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j(1 + o(1))$ and $\langle\Delta_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j((\log\log T)^{j-1})$. Expanding $s_k^r = (-\Delta_k + \mathfrak{d}_k)^r$ and bounding $\langle|\Delta_k|^{r-i}\rangle_{\mathcal{I}_T} \leq \langle\Delta_k^{2\lceil r/2\rceil}\rangle_{\mathcal{I}_T}^{(r-i)/2\lceil r/2\rceil} \ll (\log\log T)^{(r-i)/2}$ for $1 \leq i \leq r$, one obtains for every fixed $j \geq 1$

$$\langle s_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j\Big(1 + o(1)\Big), \qquad \langle s_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j\Big((\log\log T)^{j-1}\Big). \tag{3.3}$$

With $\mathfrak{s}_T^2 = \frac{\log\log T}{2\pi^2}$, so that $\kappa_j(\log\log T)^j = \frac{(2j)!}{j!2^j}\mathfrak{s}_T^{2j}$, the normalized moments $\langle(s_k/\mathfrak{s}_T)^r\rangle_{\mathcal{I}_T}$ therefore tend to the moments of a standard normal variable Z for every $r \geq 1$: the even ones by the first relation, the odd ones to 0 because $(\log\log T)^{j-1}/\mathfrak{s}_T^{2j-1} \ll_j (\log\log T)^{-1/2}$. The even moments alone determine only the law of $|s_k|$; it is the odd-moment bound in (3.2) that gives the sign. The normal law is determined by its moments, so the method of moments [3, Thm. 30.2] shows that the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to that of Z. Only consequences for $|s_k|$ are used in this paper: $\langle s_k^2\rangle_{\mathcal{I}_T} \sim \frac{1}{2\pi^2}\log\log T$, $\limsup_k |s_k| = \infty$, and $\#\{k \in \mathcal{I}_T : |s_k| \leq y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$; for these the even moments suffice.

(b) *Bound for δ_k.* Theorem 3.2, applied as $T \to \gamma_k^{\pm}$ (both sides have one-sided limits), gives $|N(\gamma_k^{\pm}) - N_0(\gamma_k) + \frac{7}{8}| \le \mathrm{BW}(\gamma_k)$, and (2.7) gives $-\mathrm{BW}(\gamma_k) - \frac{3}{8} \le s_k \le \mathrm{BW}(\gamma_k) - \frac{11}{8}$. As $\mathcal{L}$ is positive and increasing on $(2\pi, \infty)$ and $s_k = \int_{\gamma_k}^{\tau_k} \mathcal{L}$,

$$-\frac{\mathrm{BW}(\gamma_k) + 3/8}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{\mathrm{BW}(\gamma_k) - 11/8}{\mathcal{L}(\gamma_k)} \qquad (k \ge 1). \tag{3.4}$$

Hence $\delta_k = O(1)$, $\mathcal{L}(\tau_k)/\mathcal{L}(\gamma_k) \to 1$, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$, an immediate consequence of [2]. A limit superior is not a supremum: $|\delta_{34}| = 0.993711$.

4. Theorem A: an exact defect identity at the zeros

Theorem 4.1. *Let $1 \le k_1 < k_2$ be indices, F a real C^1 function on an open interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$ for $k_1 \le k \le k_2$. For every integer $j \ge 1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^{2j} F'(t)\, dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(u_k) + \frac{(\frac{1}{2} + u_{k_1})^{2j+1} + (\frac{1}{2} - u_{k_2})^{2j+1}}{2j+1}, \tag{4.1}$$

where

$$\Psi_j(u) = \frac{(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}}{2j+1} = \sum_{l=0}^{j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1}\, u^{2l} = u^{2j} + \sum_{l<j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1} u^{2l}. \tag{4.2}$$

In particular, since $\Psi_1(u) = u^2 + \frac{1}{12}$,

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^2 F'(t)\, dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2}+u_{k_1})^3 + (\frac{1}{2}-u_{k_2})^3}{3}. \tag{4.3}$$

Proof. Existence. On $[\gamma_{k_1}, \gamma_{k_2}]$ the function N is a step function with finitely many jumps (Theorem 3.1) and F, F' are continuous; the integrand is bounded and piecewise continuous, hence Riemann integrable.

One gap. Fix k with $k_1 \le k < k_2$. If $\gamma_k < \gamma_{k+1}$, then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly k ordinates counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is C^1 with derivative $(k - F(t))^{2j} F'(t)$ (chain rule; no monotonicity of F is needed), so

$$\begin{aligned} \int_{\gamma_k}^{\gamma_{k+1}} \Big(k - F(t)\Big)^{2j} F'(t)\, dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2}+u_k)^{2j+1} + (\frac{1}{2} - u_{k+1})^{2j+1}}{2j+1}, \end{aligned} \tag{4.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. If $\gamma_k = \gamma_{k+1}$, the left side of (4.4) is 0; then $u_{k+1} = u_k + 1$, and the right side equals $((\frac{1}{2}+u_k)^{2j+1} + (-\frac{1}{2} - u_k)^{2j+1})/(2j+1) = 0$. So (4.4) holds for every k, and the values of N at the finitely many ordinates do not affect the integrals.

Summation and boundary terms. Summing (4.4) over $k_1 \le k < k_2$ (a finite sum), each k with $k_1 < k < k_2$ contributes $(\frac{1}{2}+u_k)^{2j+1}$ from the gap starting at γ_k and $(\frac{1}{2}-u_k)^{2j+1}$ from the gap ending at γ_k, that is $\Psi_j(u_k)$; the unmatched terms are $(\frac{1}{2}+u_{k_1})^{2j+1}$ and $(\frac{1}{2}-u_{k_2})^{2j+1}$. In the binomial expansion of $(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}$ the odd powers of u cancel and u^{2l} has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (4.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. $\square$

Theorem 4.2. *Let $k_1 < k_2$, F and u_k be as in Theorem 4.1, and let $\mathfrak{g} : \mathbb{R} \to \mathbb{C}$ be continuous with a primitive $\mathfrak{G}$. Then*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \mathfrak{g}\Big(N(t)-F(t)\Big)F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1}\Big[\mathfrak{G}\Big(u_k+\tfrac{1}{2}\Big)-\mathfrak{G}\Big(u_k-\tfrac{1}{2}\Big)\Big]+\mathfrak{G}\Big(u_{k_1}+\tfrac{1}{2}\Big)-\mathfrak{G}\Big(u_{k_2}-\tfrac{1}{2}\Big). \quad (4.5)$$

Proof. The integrand is bounded and piecewise continuous, as in the proof of Theorem 4.1. If $\gamma_k < \gamma_{k+1}$, then $N = k$ on (γ_k, γ_{k+1}) and $t \mapsto -\mathfrak{G}(k - F(t))$ is C^1 with derivative $\mathfrak{g}(k - F(t))F'(t)$, so the integral over $[\gamma_k, \gamma_{k+1}]$ is $\mathfrak{G}(k - F(\gamma_k)) - \mathfrak{G}(k - F(\gamma_{k+1})) = \mathfrak{G}(u_k + \frac{1}{2}) - \mathfrak{G}(u_{k+1} - \frac{1}{2})$. If $\gamma_k = \gamma_{k+1}$, the integral is 0 and $u_{k+1} = u_k + 1$, so the same formula holds. Summing over $k_1 \le k < k_2$, each k with $k_1 < k < k_2$ contributes $\mathfrak{G}(u_k + \frac{1}{2})$ from the gap that starts at γ_k and $-\mathfrak{G}(u_k - \frac{1}{2})$ from the gap that ends there. □

For $\mathfrak{g}(x) = x^{2j}$ this is (4.1), since $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = \Psi_j(u)$.

Corollary 4.3. *For real $\omega \neq 0$,*

$$\frac{2\sin(\omega/2)}{\omega}\sum_{k=k_1+1}^{k_2-1} e^{i\omega u_k} = \int_{\gamma_{k_1}}^{\gamma_{k_2}} e^{i\omega(N(t)-F(t))}F'(t)\,dt - \frac{e^{i\omega(u_{k_1}+1/2)} - e^{i\omega(u_{k_2}-1/2)}}{i\omega}. \quad (4.6)$$

Proof. Theorem 4.2 with $\mathfrak{g}(x) = e^{i\omega x}$ and $\mathfrak{G}(x) = e^{i\omega x}/(i\omega)$, for which $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = e^{i\omega u} \cdot 2\sin(\omega/2)/\omega$. □

Corollary 4.4 (Signed law of s_k from the t-law of S)**.** *For fixed real η, $\langle e^{i\eta s_k/\mathfrak{s}_T}\rangle_{\mathcal{I}_T} \to e^{-\eta^2/2}$. Hence the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to the standard normal law.*

Proof. Let $\eta \neq 0$, $\omega = \eta/\mathfrak{s}_T$, $k_1 = N(T)$ and $k_2 = N(2T)+1$, so that $\{k_1+1, \dots, k_2-1\} = \mathcal{I}_T$. Apply (4.6) with $F = N_0$, so $u_k = s_k$ and $N - F = G$. By Theorem 3.1 the intervals $[\gamma_{k_1}, T]$ and $[2T, \gamma_{k_2}]$ have bounded length, so the integral equals $\int_T^{2T} e^{i\omega G}\mathcal{L}\,dt + O(\log T)$; the boundary term is $O(1/|\omega|) = O(\mathfrak{s}_T)$; and $2\sin(\omega/2)/\omega = 1 + O(\omega^2)$. Divide by $\#\mathcal{I}_T = \int_T^{2T} \mathcal{L}\,dt + O(\log T)$. On $[T, 2T]$, $|e^{i\omega G} - e^{i\omega S}| \le |\omega||\varpi|/\pi \ll (\mathfrak{s}_T T)^{-1}$ and $\mathcal{L}(t) = \mathcal{L}(T)(1 + O(1/\log T))$, whence

$$\langle e^{i\omega s_k}\rangle_{\mathcal{I}_T} = \frac{1}{T}\int_T^{2T} e^{i\eta S(t)/\mathfrak{s}_T}dt + O\Big(\omega^2 + \frac{1}{\log T}\Big).$$

By Theorem 3.16 and the definition of convergence in distribution, applied to the bounded continuous function $y \mapsto e^{i\eta y}$, the integral tends to $e^{-\eta^2/2}$, and $\omega \to 0$. The last statement is Lévy's continuity theorem [3, Thm. 26.3]. □

Remark 4.5. $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j}dx$ is the mean of x^{2j} over the unit interval centred at u; in particular $\Psi_1(u) = u^2 + \frac{1}{12}$ and $\frac{1}{12} = \int_{-1/2}^{1/2} x^2 dx$. The constant comes from the unit jump of $N - F$ at each zero, through the primitive in Theorem 4.2; when F is not monotone, $N - F$ need not run through an interval of length one between consecutive zeros, and the identities remain exact with the integral counted with sign. With $F = N_0$, Theorem 3.3 and induction on j through the lower-degree terms of Ψ_j recover the even moments in (3.3) without Gram intervals; Corollary 4.4 gives the sign without Korolev's odd moments. The main purpose of Theorem 4.1 is Theorem 5.1, where $F = F_X$ need not be monotone and no Gram-interval structure is available.

Corollary 4.6. *For $X \geq 2$ and $1 \leq k_1 < k_2$,*

$$\sum_{k=k_1+1}^{k_2-1} u_k(X)^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(S - S_X + \frac{\varpi}{\pi}\Big)^2 \big(\mathcal{L} + S_X'\big)dt \\ - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1}(X))^3 + (\frac{1}{2} - u_{k_2}(X))^3}{3}. \tag{4.7}$$

Proof. F_X is C^∞ on $(0,\infty)$ (Section 2.4), $F_X' = \mathcal{L} + S_X'$, and $N - F_X = S - S_X + \varpi/\pi$ by (2.3); apply Theorem 4.1 with $j = 1$. □

Lemma 4.7 (First-moment identity)**.** *Let $2\pi < t_1 < t_2$ be points which are not ordinates and $\psi \in C^1[t_1, t_2]$, real or complex. Then*

$$\sum_{t_1 < \gamma_k \leq t_2} s_k \psi(\gamma_k) = \int_{t_1}^{t_2} G(t)\psi(t)\mathcal{L}(t)\,dt + \frac{1}{2}\Big[\psi\, G^2\Big]_{t_1}^{t_2} - \frac{1}{2}\int_{t_1}^{t_2} G(t)^2\psi'(t)\,dt. \tag{4.8}$$

Proof. On $[t_1, t_2]$ there are finitely many ordinates, none at t_1 or t_2. The function $G = N - N_0$ is right-continuous and of bounded variation, C^∞ with $G' = -\mathcal{L}$ between ordinates, and it jumps by $\nu(\gamma)$ at an ordinate γ. Let $G^*(\gamma) = \frac{1}{2}(G(\gamma^-) + G(\gamma^+))$. The indices with $\gamma_k = \gamma$ are $N(\gamma^-) + 1, \ldots, N(\gamma^-) + \nu(\gamma)$, so $\sum_{\gamma_k=\gamma}(k - \frac{1}{2}) = \nu(\gamma)N(\gamma^-) + \frac{\nu(\gamma)^2}{2} = \nu(\gamma)\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and, N_0 being continuous, $\sum_{\gamma_k=\gamma} s_k = \nu(\gamma)G^*(\gamma)$. The left side of (4.8) therefore equals the finite sum $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma)$ over distinct ordinates. As Lebesgue–Stieltjes measures on $(t_1, t_2]$,

$$d(G^2) = -2G\mathcal{L}\,dt + \sum_{t_1<\gamma<t_2} 2G^*(\gamma)\nu(\gamma)\,\delta_\gamma, \tag{4.9}$$

because $(G^2)' = 2GG' = -2G\mathcal{L}$ between ordinates and $G(\gamma^+)^2 - G(\gamma^-)^2 = (G(\gamma^+) + G(\gamma^-))\nu(\gamma) = 2G^*(\gamma)\nu(\gamma)$. Integrating the bounded Borel function ψ against (4.9) gives $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma) = \frac{1}{2}\int_{(t_1,t_2]} \psi\, d(G^2) + \int_{t_1}^{t_2} G\psi\mathcal{L}\,dt$. Since ψ is C^1 and G^2 is right-continuous of bounded variation and continuous at t_1 and t_2, integration by parts for Lebesgue–Stieltjes integrals [8, §3.5] gives $\int_{(t_1,t_2]} \psi\, d(G^2) = [\psi G^2]_{t_1}^{t_2} - \int_{t_1}^{t_2} G^2\psi'\,dt$. All integrands are bounded and measurable, so all integrals exist. □

5. Theorem B: how much prime information the clock needs

Theorem 5.1. *There are absolute constants T_0 and C such that for $T \geq T_0$ and every X with $2 \leq X \leq T^{1/2}$, writing $c = \log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \leq C\Big(1 + \sqrt{\log(1/c)}\Big). \tag{5.1}$$

Here $\log(1/c) = \log\log T - \log\log X \in [\log 2, \log\log T - \log\log 2]$. For fixed c both the main term and the error are bounded; the theorem does not identify a limit at fixed c. It concerns the defect at the true ordinate and does not select a root of $F_X(t) = k - \frac{1}{2}$.

Corollary 5.2 (fixed cutoff)**.** *For fixed $X \geq 2$, $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\log T + O_X(\sqrt{\log\log T})$, and $\#\{k \in \mathcal{I}_T : |u_k(X)| \leq y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$.*

Proof. The first statement is (5.1). Since $|u_k(X) - s_k| \le \sigma_X$, $|u_k(X)| \le y_0$ implies $|s_k| \le y_0 + \sigma_X$; apply Remark 3.22(a). □

Corollary 5.3 (power cutoff)**.** *For fixed $0 < c \le \frac{1}{2}$ and $X = T^c$,*

$$\limsup_{T\to\infty} \frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} u_k(T^c)^2 \le \frac{1}{2\pi^2} \log\frac{1}{c} + C\Big(1 + \sqrt{\log(1/c)}\Big), \tag{5.2}$$

and $\#\{k \in \mathcal{I}_T : |u_k(T^c)| > w_T\} = o(\#\mathcal{I}_T)$ whenever $w_T \to \infty$. The same holds for $X = N(T)^c$ with fixed $0 < c < \frac{1}{2}$, since then $X = T^{c'}$ with $c' = c + O(\log\log T/\log T)$.

Proof. (5.1) and Chebyshev's inequality. □

Corollary 5.4 (dichotomy)**.** *Let $X = X(T)$ with $2 \le X \le T^{1/2}$. The mean square of $u_k(X)$ over $\mathcal{I}_T$ is bounded as $T \to \infty$ if and only if $\liminf_{T\to\infty} \log X/\log T > 0$. For fixed $X_0 \ge 2$ and $c \in (0, \frac{1}{2}]$,*

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} \Big(u_k(X_0)^2 - u_k(T^c)^2\Big) = \frac{1}{2\pi^2}\log\log T + O\Big(\sqrt{\log\log T}\Big). \tag{5.3}$$

Proof. Both directions follow from (5.1), because $y \mapsto \frac{y}{2\pi^2} \pm C(1+\sqrt{y})$ is bounded on bounded sets and tends to infinity; (5.3) is the difference of the two cases of (5.1). □

Proposition 5.5 (Rounding the corrected count misassigns a positive proportion)**.** *There are absolute constants $c_* > 0$, $\kappa_0 > 0$ and T_0 such that for $T \ge T_0$ and $2 \le X \le T^{c_*}$,*

$$\frac{1}{T}\operatorname{meas}\Big\{t \in [T, 2T] : \Big\lfloor F_X(t) + \tfrac{1}{2}\Big\rfloor \ne N(t)\Big\} \ge \kappa_0.$$

Proof. If $\lfloor F_X(t) + \frac{1}{2}\rfloor = N(t)$ then $|N(t) - F_X(t)| \le \frac{1}{2}$, i.e. $|E| \le \frac{1}{2}$ with $E = E_X + \varpi/\pi$. Let $\mathfrak{m}$ be the measure of $\{t \in [T, 2T] : |E| > \frac{1}{2}\}$. Then $\int_T^{2T} E^2 \le \frac{T}{4} + (\int_T^{2T} E^4)^{1/2}\mathfrak{m}^{1/2}$. By (5.10) and (5.11) (with $|\varpi| \ll 1/T$), $\int_T^{2T} E^2 \ge \frac{T}{2\pi^2}V - CT(1+\sqrt{V})$ and $\int_T^{2T} E^4 \le CT(1+V)^2$, so $\mathfrak{m}/T \ge (\frac{V}{2\pi^2} - C(1+\sqrt{V}) - \frac{1}{4})_+^2/(C(1+V)^2)$, which is $\ge \kappa_0 > 0$ once $V \ge V_0$, that is $c \le c_*$ by Lemma 5.8. □

Rounding the corrected count to the nearest integer therefore does not assign the correct index to the zeros, for small cutoffs, on a set of positive density: the obstruction is the size of $S - S_X$, not the sawtooth $N - N_0$. For c near $\frac{1}{2}$ the question depends on the constant in (5.1) (Question 13.1), and for $c > \frac{1}{2}$ nothing is proved.

Remark 5.6. Since $\log\#\{p \le X\} = \log X + O(\log\log X)$, the condition of Corollary 5.4 says that the logarithm of the number of primes must grow proportionally to $\log T$. A fixed number of primes, as in the computations reported in [11, §XXI, Note added 7], leaves almost all zeros violating the corrected equation by more than any fixed amount; the requirement that the number of primes tend to infinity [22, Sect. 3] is necessary but not sufficient. The range $c > \frac{1}{2}$ is not covered. In Theorem 5.1 the cutoff is constant on each window $\mathcal{I}_T$; Theorem 5.12 below allows the cutoff $X_k = (\tau_k/2\pi)^c$, which varies at every ordinate.

Remark 5.7 (Localization of the ordinates). If $M_X \le \frac{1}{2}\mathcal{L}(T)$, then $F_X' \ge \mathcal{L} - M_X \ge \frac{1}{2}\mathcal{L}(T)$ on $[T, 2T]$, so F_X is strictly increasing there. For $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac{1}{2} \le F_X(2T)$ the equation $F_X(t) = k - \frac{1}{2}$ has exactly one root $t_k^X \in [T, 2T]$ (intermediate value theorem), and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$ gives $|\gamma_k - t_k^X| \le 2|u_k(X)|/\mathcal{L}(T)$ (mean value theorem). As $M_X \asymp \sqrt{X}$, this applies only for $X \ll \log^2 T$, and in that range the localization gains nothing in order (Proposition 7.6). For $X = T^c$, F_X need not be monotone, and Theorem 5.1 controls the equation at the true ordinates without selecting a root; Theorem 7.3 replaces the root by one Newton step.

5.1. Lemmas on prime polynomials

Fix $T \ge T_0$ and $2 \le X \le T^{1/2}$, let $Y_0 = T^{1/160}$, and split $S_X = S_X^{(1)} + S_X^{(2)} + S_X^{(\ge 3)}$ according to $m = 1$, $m = 2$, $m \ge 3$ in (2.10); the three series converge absolutely and uniformly. Let $\mathcal{J}$ be the set of primes in (X, Y_0) if $X < Y_0$ and in $[Y_0, X]$ if $X \ge Y_0$, and put

$$\begin{aligned} \mathcal{W}_1(t) &= -\frac{1}{\pi} \sum_{p \in \mathcal{J}} \frac{\sin(t \log p)}{\sqrt{p}}, \qquad V = \sum_{p \in \mathcal{J}} \frac{1}{p}, \\ \mathcal{W} &= S_{Y_0}^{\mathrm{pr}} - S_X = \pm \mathcal{W}_1 - S_X^{(2)} - S_X^{(\ge 3)}, \end{aligned} \tag{5.4}$$

with $+$ if $X < Y_0$ and $-$ if $X \ge Y_0$. The last identity holds because $S_X^{(1)}$ is the sum of $-\frac{1}{\pi} p^{-1/2} \sin(t \log p)$ over the primes $p \le X$, while $S_{Y_0}^{\mathrm{pr}}$ is the same sum over $p < Y_0$.

Lemma 5.8. $V = \log(1/c) + O(1)$ *uniformly for* $\log 2/\log T \le c \le \frac{1}{2}$.

Proof. By Theorem 3.8, $V = |\log\log Y_0 - \log\log X| + O(1) = |\log \frac{1}{160c}| + O(1)$. For $c \le \frac{1}{160}$ this is $\log \frac{1}{c} - \log 160 + O(1)$; for $\frac{1}{160} < c \le \frac{1}{2}$ both V and $\log \frac{1}{c}$ are $O(1)$. □

Lemma 5.9. *For real* z_n, $2 \le n \le Y$,

$$\int_T^{2T} \Big(\sum_{n \le Y} z_n \sin(t \log n) \Big)^2 dt = \frac{T}{2} \sum_{n \le Y} z_n^2 + O\Big(\sum_{n \le Y} n z_n^2 + \Big(\sum_{n \le Y} |z_n| \Big)^2 \Big), \tag{5.5}$$

and the same with cos *in place of* sin.

Proof. With the finite sum $\mathcal{D}(t) = \sum z_n n^{-it}$, $\sum z_n \sin(t \log n) = -\Im \mathcal{D}$ and $(\Im \mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 - \Re \mathcal{D}^2)$. Theorem 3.7 evaluates $\int |\mathcal{D}|^2$, and $|\int_T^{2T} (n_1 n_2)^{-it} dt| \le 2/\log(n_1 n_2) \le 2/\log 4$ gives $|\int \mathcal{D}^2| \le (2/\log 4)(\sum |z_n|)^2$. For cos use $\Re \mathcal{D}$ and $(\Re \mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 + \Re \mathcal{D}^2)$. □

Lemma 5.10. *With absolute implied constants,*

$$\int_T^{2T} \mathcal{W}^2 \, dt = \frac{T}{2\pi^2} V + O\Big(T(1 + \sqrt{V}) \Big), \tag{5.6}$$

$$\int_T^{2T} \mathcal{W}^4 \, dt \ll T(1 + V)^2, \tag{5.7}$$

$$\int_T^{2T} S_X'(t)^2 \, dt \ll T(1 + \log X)^2. \tag{5.8}$$

Proof. All primes involved are $\le \max(X, Y_0) \le T^{1/2}$. Lemma 5.9 with $z_p = \frac{1}{\pi\sqrt{p}}$ ($p \in \mathcal{J}$) gives $\int \mathcal{W}_1^2 = \frac{T}{2\pi^2}V + O(T^{1/2} + (\sum_{p\le T^{1/2}} p^{-1/2})^2) = \frac{T}{2\pi^2}V + O(T^{1/2})$. For $S_X^{(2)}(t) = -\frac{1}{2\pi}\sum_{p\le X} p^{-1}\sin(t\log p^2)$ the same lemma gives $\int (S_X^{(2)})^2 \ll T\sum_p p^{-2} + \#\{p \le X\} + (\sum_{p\le X} p^{-1})^2 \ll T$. Finally $|S_X^{(\ge 3)}| \le \frac{1}{\pi}\sum_p \sum_{m\ge 3} \frac{1}{mp^{m/2}} \le \frac{1}{\pi}\sum_p \frac{p^{-3/2}}{1-p^{-1/2}} < \infty$, uniformly in X. Expanding $\mathcal{W}^2$ and bounding the cross terms by Cauchy–Schwarz gives (5.6).

For (5.7) use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. Let $\mathcal{D}_1(t) = \sum_{p\in\mathcal{J}} p^{-1/2}p^{-it}$, so that $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$ and $\mathcal{D}_1^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$, $\eta_{p_1p_2} = 2(p_1p_2)^{-1/2}$ for $p_1 < p_2$, all $n \le T$. Theorem 3.7 gives $\int |\mathcal{D}_1|^4 = \sum |\eta_n|^2 (T + O(n)) \ll T\sum |\eta_n|^2 \le T(1+2V^2)$. For $S_X^{(2)}$ let $\mathcal{D}_2(t) = \sum_{p\le X} p^{-1}(p^2)^{-it}$, so that $|S_X^{(2)}| \le |\mathcal{D}_2|/2\pi$ and $\mathcal{D}_2^2 = \sum_n \eta'_n n^{-it}$ over $n = (p_1p_2)^2$ with $|\eta'_n| \le 2(p_1p_2)^{-1}$; then $\sum |\eta'_n|^2 \ll 1$ and $\sum n|\eta'_n|^2 \le 4\#\{p \le X\}^2 \le 4X^2 \le 4T$. $S_X^{(\ge 3)}$ is bounded.

For (5.8), $S'_X(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n)$ (termwise differentiation, Section 2.4). The terms $m = 1$ have $|z_p| = \frac{\log p}{\pi\sqrt{p}}$; Lemma 5.9 and Theorem 3.8 give $\ll T\log X \sum_{p\le X}\frac{\log p}{p} + \sum_{p\le X}\log^2 p + (\sum_{p\le X}\frac{\log p}{\sqrt{p}})^2 \ll T(1+\log X)^2 + X\log X$, and $X \le T^{1/2}$. The terms $m = 2$ have $|z_{p^2}| = \frac{\log p}{\pi p}$ and contribute $\ll T + X\log X + \log^2 X$. The terms $m \ge 3$ are bounded by $\frac{1}{\pi}\sum_p \log p \sum_{m\ge 3} p^{-m/2} < \infty$. Combine with $(y_1+y_2+y_3)^2 \le 3(y_1^2+y_2^2+y_3^2)$. □

5.2. Proof of Theorem 5.1

Proof. Integrability. On $[T, 2T]$, S is bounded (Theorem 3.1) and piecewise continuous, and S_X, S'_X, ϖ are continuous; all integrals below exist.

Second and fourth moments of E_X. Let $E_0 = S - S_{Y_0}^{\mathrm{pr}}$ and $E_X = S - S_X = E_0 + \mathcal{W}$. Theorem 3.5 and Hölder's inequality give

$$\int_T^{2T} E_0^4\,dt \ll T, \qquad \int_T^{2T} E_0^2\,dt \le T^{1/2}\Big(\int_T^{2T} E_0^4\,dt\Big)^{1/2} \ll T. \tag{5.9}$$

Let $\mathcal{E}_2 = \int_T^{2T} E_X^2\,dt$. From (5.6), (5.9) and Cauchy–Schwarz,

$$\mathcal{E}_2 = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.10}$$

and from (5.7), (5.9) and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$,

$$\int_T^{2T} E_X^4\,dt \ll T(1+V)^2. \tag{5.11}$$

Transfer to the ordinates. Let k_1, k_2 be the smallest and largest indices in $\mathcal{I}_T$ ($\mathcal{I}_T \ne \emptyset$ for $T \ge T_0$) and apply (4.7). By Lemma 2.2 and (2.11), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and $u_{k_1}(X)^2 + u_{k_2}(X)^2$ are $O(T^{3/4})$. The intervals $[T, \gamma_{k_1}]$ and $[\gamma_{k_2}, 2T]$ have bounded length (Theorem 3.1), and on them $E_X^2(\mathcal{L} + |S'_X|) \ll T^{1/2}\cdot T^{1/4}$. Since $|\varpi| \ll 1/T$, replacing $(E_X + \varpi/\pi)^2$ by E_X^2 costs $\ll T^{-1}\int_T^{2T}(|E_X|+1)(\mathcal{L}+|S'_X|)\,dt \ll (1+V)^{1/2}\log T$ by Cauchy–Schwarz, (5.10) and (5.8). Hence

$$\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \int_T^{2T} E_X^2\mathcal{L}\,dt + \int_T^{2T} E_X^2 S'_X\,dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{5.12}$$

Main term and derivative term. Since $0 \le \mathcal{L}(t) - \mathcal{L}(T) \le \frac{\log 2}{2\pi}$ on $[T, 2T]$, the first integral is $\mathcal{L}(T)\mathcal{E}_2 + O(\mathcal{E}_2)$. By Cauchy–Schwarz, (5.11) and (5.8),

$$\Big|\int_T^{2T} E_X^2 S_X'\, dt\Big| \le \Big(\int_T^{2T} E_X^4\Big)^{1/2}\Big(\int_T^{2T} S_X'^2\Big)^{1/2} \ll T(1+V)(1+\log X). \tag{5.13}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (5.10),

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\,c\Big), \tag{5.14}$$

because $\log X = c\log T$. By Lemma 5.8, $(1+V)c \ll c(1+\log\frac{1}{c}) \ll 1$ on $(0, \frac12]$; $V \ll \log\log T$ since $c \ge \log 2/\log T$, so $(1+V)/\log T = o(1)$. A second use of Lemma 5.8 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2}\log\frac{1}{c} + O(1)$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$, which proves (5.1). □

5.3. A cutoff varying with the index

Let $\mathcal{I}_T^\tau = \{k : T \le \tau_k < 2T\}$; by (2.5), $\#\mathcal{I}_T^\tau = N_0(2T) - N_0(T) + O(1) \asymp T\log T$.

Lemma 5.11. *Let $T \ge T_0$ and $2 \le X_- < X_+ \le (T/3)^{1/2}$ with $X_+ \le 4X_-$; let $V_\pm = \sum_{X_-<p\le X_+} p^{-1}$ and*

$$\mathcal{Q}(t) = \max_{X_-\le y\le X_+}\Big|\sum_{X_-<p\le y} p^{-1/2-it}\Big|.$$

Then $\mathcal{Q}$ is Lipschitz, $|\mathcal{Q}'(t)| \le \max_{X_-\le y\le X_+}|\sum_{X_-<p\le y}\log p\, p^{-1/2-it}|$ for almost every t, and

$$\int_{T-3}^{2T+3}\mathcal{Q}^4\,dt \ll TV_\pm^2, \qquad \int_{T-3}^{2T+3}|\mathcal{Q}'|^4\,dt \ll T(\log X_+)^4 V_\pm^2, \tag{5.15}$$

$$\langle\mathcal{Q}(\gamma_k)^2\rangle_{\mathcal{I}_T} \ll V_\pm\Big(1 + \frac{\log X_+}{\log T}\sqrt{\log\log T}\Big) + \frac{X_+}{T}, \qquad \langle\mathcal{Q}(\tau_k)^2\rangle_{\mathcal{I}_T^\tau} \ll V_\pm. \tag{5.16}$$

Proof. If $(X_-, X_+]$ contains no prime, $\mathcal{Q} = 0$. Otherwise let $p_1 < \cdots < p_J$ be its primes and $P_j(t) = \sum_{i\le j} p_i^{-1/2-it}$ ($P_0 = 0$), so that $\mathcal{Q} = \max_{0\le j\le J}|P_j|$, a maximum of finitely many Lipschitz functions; hence $\mathcal{Q}$ is Lipschitz and differentiable almost everywhere. At a point t where $\mathcal{Q}$ and all $|P_j|$ are differentiable and $\mathcal{Q}(t) = |P_j(t)|$, the function $\mathcal{Q} - |P_j| \ge 0$ has a minimum, so $\mathcal{Q}'(t) = |P_j|'(t)$, and $\big||P_j|'\big| \le |P_j'|$ wherever $|P_j|$ is differentiable. This gives the bound for $\mathcal{Q}'$.

Window. Fix $\psi \in C_c^\infty(-\frac12, \frac12)$, real, even, nonnegative and not identically 0, let $\check\psi(x) = \int\psi(\omega)e^{i\omega x}d\omega$ and $W = |\check\psi|^2/\check\psi(0)^2$. Then $W \ge 0$ is a Schwartz function with $W(0) = 1$, and $\mathcal{F}W$ is supported in $[-1,1]$. Choose $0 < y_0 \le \frac14$ with $W \ge \frac12$ on $[-y_0, y_0]$ and put $\mathfrak{w}(t) = W(y_0(t - \frac32 T)/(\frac{T}{2} + 3))$. Then $\mathfrak{w} \ge \frac12$ on $[T-3, 2T+3]$, $\|\mathfrak{w}\|_4^4 \asymp T$, and $\mathcal{F}\mathfrak{w}$ and $\mathcal{F}(\mathfrak{w}^4)$ are supported in $|\omega| \le 2y_0/T$ and $|\omega| \le 8y_0/T$.

Proof of (5.15). Let $f_j = \mathfrak{w}P_j$. Then $\mathcal{F}f_j(\omega) = \sum_{i\le j} p_i^{-1/2}\mathcal{F}\mathfrak{w}(\omega + \log p_i)$, a sum of packets supported in $|\omega + \log p_i| \le 2y_0/T$. Since $\log p_{i+1} - \log p_i \ge \log(1 + 1/X_+) \ge (3/T)^{1/2}/2 > 4y_0/T$ for $T \ge T_0$, the packets are disjoint, and f_j is obtained from f_J by removing the packets of $p_{j+1}, \dots, p_J$, which lie below a point ω_j separating them from

the others: $f_j(t) = f_J(t) - \frac{1}{2\pi}\int_{-\infty}^{\omega_j} \mathcal{F}f_J(\omega)e^{i\omega t}d\omega$. Hence $\max_j |f_j| \le |f_J| + \mathcal{C}f_J$, and on $[T-3, 2T+3]$, $\mathcal{Q} \le 2\max_j |f_j|$. By Theorem 3.18,

$$\int_{T-3}^{2T+3} \mathcal{Q}^4 \le 16\int_{\mathbb{R}} \Big(|f_J| + \mathcal{C}f_J\Big)^4 \ll \int_{\mathbb{R}} \mathfrak{w}^4 |P_J|^4\, dt.$$

Write $P_J^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$ and $\eta_{p_i p_{i'}} = 2(p_i p_{i'})^{-1/2}$ for $i < i'$, all $n \le X_+^2 \le T/3$ (unique factorization makes the representation unique). Expanding, $\int \mathfrak{w}^4 |P_J^2|^2 = \sum_{n,n'} \eta_n \eta_{n'} \mathcal{F}(\mathfrak{w}^4)(\log(n/n'))$, and $|\log(n/n')| \ge 1/\max(n,n') \ge 3/T > 8y_0/T$ for $n \ne n'$, so only the diagonal remains: $\int \mathfrak{w}^4 |P_J|^4 = \|\mathfrak{w}\|_4^4 \sum_n \eta_n^2 \ll T\Big(\sum_p p^{-2} + 4V_\pm^2\Big) \ll TV_\pm^2$, because $\sum_{X_-<p\le X_+} p^{-2} \le V_\pm/X_- \le 4V_\pm/X_+ \le 4V_\pm^2$. The same argument with the polynomials $\sum_{i\le j} \log p_i\, p_i^{-1/2-it}$, whose maximum dominates $|\mathcal{Q}'|$, gives the second bound, with η_n multiplied by $\log p_i \log p_{i'} \le (\log X_+)^2$.

At the ordinates. Let $T, 2T$ be non-ordinates (otherwise move them by less than the distance to the nearest ordinate, which changes neither side). As Lebesgue–Stieltjes measures on $(T, 2T]$, $dN = \mathcal{L}\, dt + dG$, and $\mathcal{Q}^2$ is Lipschitz, so integration by parts [8, §3.5] gives

$$\sum_{k\in\mathcal{I}_T} \mathcal{Q}(\gamma_k)^2 = \int_T^{2T} \mathcal{Q}^2 \mathcal{L}\, dt + \Big[\mathcal{Q}^2 G\Big]_T^{2T} - 2\int_T^{2T} G\, \mathcal{Q}\mathcal{Q}'\, dt, \tag{5.17}$$

multiplicities included. By Cauchy–Schwarz and (5.15) the first term is $\ll \log T \cdot T^{1/2}(TV_\pm^2)^{1/2} = TV_\pm \log T$. The boundary term is $\ll X_+ \log T$, since $\mathcal{Q} \le \sum_{p\le X_+} p^{-1/2} \ll X_+^{1/2}$ and $G \ll \log T$. By Hölder's inequality with exponents $2, 4, 4$, Theorem 3.3 ($\int_T^{2T} G^2 \ll T\log\log T$) and (5.15), the last term is $\ll (T\log\log T)^{1/2}(TV_\pm^2)^{1/4}(T(\log X_+)^4 V_\pm^2)^{1/4} = T\sqrt{\log\log T}\,(\log X_+)V_\pm$. Divide by $\#\mathcal{I}_T \gg T\log T$.

At the clock points. For consecutive indices, $\tau_{k+1} - \tau_k = \int_k^{k+1} dx/\mathcal{L}(\tau_x) \ge 2\pi/\log(3T) = \mathfrak{b}_T$ while $\tau_{k+1} \le 3T$, so the intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T^\tau$, are disjoint and contained in $[T, 2T+1]$. For an absolutely continuous g on an interval $\mathcal{U}$ of length $\mathfrak{b}$ and $x \in \mathcal{U}$, $|g(x)| \le \frac{1}{\mathfrak{b}}\int_{\mathcal{U}} |g| + \int_{\mathcal{U}} |g'|$, hence $g(x)^2 \le \frac{2}{\mathfrak{b}}\int_{\mathcal{U}} g^2 + 2\mathfrak{b}\int_{\mathcal{U}} g'^2$. With $g = \mathcal{Q}$ and summing, $\sum_{k\in\mathcal{I}_T^\tau} \mathcal{Q}(\tau_k)^2 \le \frac{2}{\mathfrak{b}_T}\int_T^{2T+1} \mathcal{Q}^2 + 2\mathfrak{b}_T \int_T^{2T+1} \mathcal{Q}'^2 \ll TV_\pm \log T + \frac{T(\log X_+)^2 V_\pm}{\log T}$, by Cauchy–Schwarz and (5.15). Divide by $\#\mathcal{I}_T^\tau \gg T\log T$ and use $\log X_+ \le \log T$. □

Theorem 5.12. *For $\frac{\log 2}{\log T} \le c \le \frac{1}{2}$ let $X_k = \max\{2, (\tau_k/2\pi)^c\}$ and $\tilde{u}_k = s_k - S_{X_k}(\gamma_k)$. Uniformly in c,*

$$\left|\langle \tilde{u}_k^2\rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2}\log\frac{1}{c}\right| \ll 1 + \sqrt{\log(1/c)}. \tag{5.18}$$

For fixed c, $\langle(\tilde{u}_k - u_k(X_-))^2\rangle_{\mathcal{I}_T} \to 0$, with X_- as in the proof.

Proof. For $T \ge T_0$ and $k \in \mathcal{I}_T$, $|\delta_k| \le 1$ by (3.4), so $\tau_k \in [T-1, 2T+1]$ and $X_- \le X_k \le X_+$ with $X_\pm$ the values of $\max\{2, (y/2\pi)^c\}$ at $y = T-1$ and $y = 2T+1$. Then $X_+ \le ((2T+1)/2\pi)^{1/2} \le (T/3)^{1/2}$ and $X_+/X_- \le ((2T+1)/(T-1))^{1/2} \le 2$. The prime powers p^m with $X_- < p \le X_k$ are those in $S_{X_k} - S_{X_-}$, so by (2.10)

$$|S_{X_k}(\gamma_k) - S_{X_-}(\gamma_k)| \le \frac{1}{\pi}\mathcal{Q}(\gamma_k) + \frac{1}{\pi}\sum_{X_-<p\le X_+}\sum_{m\ge 2}\frac{1}{mp^{m/2}} \le \frac{1}{\pi}\mathcal{Q}(\gamma_k) + 1.1\, V_\pm,$$

since the terms $m=1$ are $\frac{1}{\pi}\Im\sum_{X_-<p\le X_k}p^{-1/2-i\gamma_k}$ and $\sum_{m\ge2}p^{-m/2}/m\le p^{-1}(1-2^{-1/2})^{-1}$. By Theorem 3.8, $V_\pm=\log\frac{\log X_+}{\log X_-}+O(\frac{1}{\log X_-})\ll\frac{1}{\log X_-}\ll\frac{1}{1+c\log T}$ (and $V_\pm=0$ if $X_-=X_+=2$). Lemma 5.11 gives

$$\langle(\tilde u_k-u_k(X_-))^2\rangle_{\mathcal{I}_T}\ll V_\pm+\frac{(1+c\log T)\sqrt{\log\log T}}{(1+c\log T)\log T}+T^{-1/2}+V_\pm^2\ll1,$$

which tends to 0 for fixed c. Theorem 5.1 with $X=X_-$, for which $\log(1/c_-)=\log(1/c)+O(1)$ with $c_-=\log X_-/\log T$, gives (5.1) for $u_k(X_-)$; (5.18) follows from $|\langle\tilde u^2\rangle-\langle u^2\rangle|\le 2\langle u^2\rangle^{1/2}\langle(\tilde u-u)^2\rangle^{1/2}+\langle(\tilde u-u)^2\rangle$. □

6. Theorem C: exact accounting for the one-step clock

For a finite nonempty set I of indices and a prime power n, define the response statistics at the zeros and at the clock points,

$$\mathcal{R}_n^\gamma(I)=\frac{2}{a_n}\langle s_k\sin(\gamma_k\log n)\rangle_I,\qquad \mathcal{R}_n^\tau(I)=\frac{2}{a_n}\langle s_k\sin(\tau_k\log n)\rangle_I,\tag{6.1}$$

and, for a set A of prime powers with $\sum_{n\in A}|a_n|<\infty$, the cross term

$$\mathrm{Cr}_A(I)=\langle S_A(\tau_k)^2\rangle_I-v_A.\tag{6.2}$$

For a prime p, $2/a_p=-2\pi\sqrt{p}$. These are finite-window statistics of the zeros of the Riemann zeta function, defined without any hypothesis; Cr_A contains the finite-window diagonal discrepancies and all off-diagonal products and has no fixed sign.

Theorem 6.1 (Finite-window accounting)**.** *Let I be a finite nonempty set of indices and A a set of prime powers with $\sum_{n\in A}|a_n|<\infty$.*

(a) For every k and $X\ge2$,

$$\begin{aligned}&r_k(X)=u_k(X)+\Big(S_X(\gamma_k)-S_X(\tau_k)\Big)+\chi_k,\\&\chi_k=\frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k}-1-\log\frac{\tau_k}{\gamma_k}\Big),\qquad \frac{\delta_k^2}{4\pi\max(\gamma_k,\tau_k)}\le\chi_k\le\frac{\delta_k^2}{4\pi\min(\gamma_k,\tau_k)},\\&|S_X(\gamma_k)-S_X(\tau_k)|\le M_X|\delta_k|.\end{aligned}\tag{6.3}$$

(b) For every real λ,

$$\Big\langle s_k^2-(s_k-\lambda S_A(\tau_k))^2\Big\rangle_I=\sum_{n\in A}a_n^2\Big(\lambda\,\mathcal{R}_n^\tau(I)-\frac{\lambda^2}{2}\Big)-\lambda^2\,\mathrm{Cr}_A(I).\tag{6.4}$$

In particular, for the full local factors,

$$\begin{aligned}&\langle s_k^2-(s_k-S_X(\tau_k))^2\rangle_I=D_X(I)-\mathrm{Cr}_X(I),\\&D_X(I)=\sum_{n\in A_X}a_n^2\Big(\mathcal{R}_n^\tau(I)-\frac12\Big)=2\langle s_kS_X(\tau_k)\rangle_I-v_X,\end{aligned}\tag{6.5}$$

with $\mathrm{Cr}_X=\mathrm{Cr}_{A_X}$.

(c) In units of zeros, the gain of the one-step clock over the bare clock is

$$\begin{aligned}&\Big\langle(\mathcal{L}(\tau_k)\delta_k)^2 - r_k(X)^2\Big\rangle_I = D_X(I) - \mathrm{Cr}_X(I) + 2\langle\chi_k S_X(\tau_k)\rangle_I,\\ &|\langle\chi_k S_X(\tau_k)\rangle_I| \le \sigma_X \max_{k\in I}\chi_k.\end{aligned} \tag{6.6}$$

Proof. (a) By (2.12), $r_k(X) = \mathcal{L}(\tau_k)\delta_k - S_X(\tau_k)$. Put $\chi_k = \mathcal{L}(\tau_k)\delta_k - s_k$; since $s_k = N_0(\tau_k) - N_0(\gamma_k) = \int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$\chi_k = \int_{\gamma_k}^{\tau_k}\Big(\mathcal{L}(\tau_k) - \mathcal{L}(y)\Big)dy = \frac{1}{2\pi}\int_{\gamma_k}^{\tau_k}\log\frac{\tau_k}{y}\,dy = \frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k}\Big),$$

the last step by $\int\log y\,dy = y\log y - y$. If $\gamma_k < \tau_k$, then $\frac{\tau_k-y}{\tau_k} \le \log\frac{\tau_k}{y} \le \frac{\tau_k-y}{y} \le \frac{\tau_k-y}{\gamma_k}$ for $y\in[\gamma_k,\tau_k]$, and integration gives $\frac{\delta_k^2}{4\pi\tau_k} \le \chi_k \le \frac{\delta_k^2}{4\pi\gamma_k}$; if $\tau_k < \gamma_k$, write $\chi_k = \frac{1}{2\pi}\int_{\tau_k}^{\gamma_k}\log\frac{y}{\tau_k}\,dy$ and use $\frac{y-\tau_k}{\gamma_k} \le \frac{y-\tau_k}{y} \le \log\frac{y}{\tau_k} \le \frac{y-\tau_k}{\tau_k}$; if $\tau_k = \gamma_k$, $\chi_k = 0$. Insert $s_k = u_k(X) + S_X(\gamma_k)$. The last bound is the mean value theorem with (2.11).

(b) The series $S_A(\tau_k) = \sum_n a_n\sin(\tau_k\log n)$ converges absolutely, and so does the double series for $S_A(\tau_k)^2$; since I is finite, the mean over I may be exchanged with these sums. Expanding, the left side is $2\lambda\langle s_k S_A(\tau_k)\rangle_I - \lambda^2\langle S_A(\tau_k)^2\rangle_I$, with $\langle s_k S_A(\tau_k)\rangle_I = \sum_n a_n\langle s_k\sin(\tau_k\log n)\rangle_I = \frac{1}{2}\sum_n a_n^2\mathcal{R}_n^\tau(I)$ and $\langle S_A(\tau_k)^2\rangle_I = v_A + \mathrm{Cr}_A(I)$. Taking $\lambda = 1$ and $A = A_X$ gives (6.5).

(c) By (a), $\mathcal{L}(\tau_k)\delta_k = s_k + \chi_k$ and $r_k(X) = s_k + \chi_k - S_X(\tau_k)$, so $(\mathcal{L}(\tau_k)\delta_k)^2 - r_k(X)^2 = 2s_kS_X(\tau_k) - S_X(\tau_k)^2 + 2\chi_kS_X(\tau_k)$; average and use (6.5) and $|S_X| \le \sigma_X$. □

Remark 6.2. (6.5) is an exact sign criterion for the gain of the full Euler phase at the clock points: the gain is positive exactly when the diagonal responses outweigh the cross term. No independence of the phases $\tau_k\log p$ is assumed. A threshold $\mathcal{R}_p^\tau = \frac{1}{2}$ describes a single diagonal prime contribution and does not determine the optimal cutoff when Cr_X and the prime powers matter.

Remark 6.3 (Optimal trigonometric correction). For a finite set A of frequencies let $\varsigma_n(k) = \sin(\tau_k\log n)$, $\mathbf{Q} = (\langle\varsigma_{n_1}\varsigma_{n_2}\rangle_I)_{n_1,n_2\in A}$, $\mathbf{b} = (\langle s_k\varsigma_n(k)\rangle_I)_{n\in A}$ and $\Upsilon_{\boldsymbol{\lambda}} = \sum_n\lambda_n\varsigma_n$. Ordinary least squares gives $\langle(s_k - \Upsilon_{\boldsymbol{\lambda}}(k))^2\rangle_I = \langle s_k^2\rangle_I - 2\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{b} + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{Q}\boldsymbol{\lambda}$, minimized by $\boldsymbol{\lambda} = \mathbf{Q}^\dagger\mathbf{b}$ with gain $\mathbf{b}^{\mathsf{T}}\mathbf{Q}^\dagger\mathbf{b}$; for prime powers $b_n = \frac{1}{2}a_n\mathcal{R}_n^\tau(I)$. The Euler phase is the choice $\lambda_n = a_n$, and (6.4) measures what it loses through $\mathcal{R}_n^\tau \ne 1$ and $\mathrm{Cr}_A \ne 0$.

Remark 6.4 (Clock point against zero). For a prime power n, $\sin(\gamma_k\ell_n) - \sin(\tau_k\ell_n) = \int_0^1\frac{d}{dy}\sin((\tau_k - y\delta_k)\ell_n)\,dy$, whence, exchanging the finite mean and the integral,

$$\mathcal{R}_n^\gamma(I) - \mathcal{R}_n^\tau(I) = -\frac{2\ell_n}{a_n}\int_0^1\Big\langle s_k\delta_k\cos\Big((\tau_k - y\delta_k)\ell_n\Big)\Big\rangle_I dy. \tag{6.7}$$

For fixed n and $I = \mathcal{I}_T$ the right side is $O(\langle|s_k\delta_k|\rangle_{\mathcal{I}_T}) = O(\log\log T/\log T)$ by (2.6) and Remark 3.22; for $\ell_n = \theta L_T$ with fixed θ the phase change $\delta_k\ell_n$ has mean square $\theta^2L_T^2\langle\delta_k^2\rangle_{\mathcal{I}_T} \sim 2\theta^2\log\log T$, so it is not small, the transport integral need not be small, and the computed responses do differ (Section 12); no asymptotic statement about this difference is proved.

Remark 6.5 (Fixed prime sets). For fixed X, $|\delta_k^X - \delta_k| \le \sigma_X/\mathcal{L}(\tau_k) \to 0$ and $|u_k(X) - s_k| \le \sigma_X$. A fixed set of primes changes neither $\limsup_k|\delta_k|$ nor the unboundedness of the defect.

7. Theorem D: a computable clock that approaches the zeros

The one-step clock $\tau_k^X = \tau_k - y_k$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, uses only τ_k and the primes $p \le X$. By Theorem 6.1(a),

$$r_k(X) = u_k(X) + \mathfrak{t}_k(X) + \chi_k, \qquad \mathfrak{t}_k(X) = S_X(\gamma_k) - S_X(\tau_k), \tag{7.1}$$

where Theorem 5.1 controls $u_k(X)$ and $0 \le \chi_k \ll 1/T$ on $\mathcal{I}_T$. This section controls the transport term $\mathfrak{t}_k(X)$ in mean square, with no assumption on the joint behaviour of δ_k and the phases $\tau_k \log p$.

Lemma 7.1. *For $T \ge T_0$ and $2 \le X \le T^{1/2}$,*

$$\int_{T-3}^{2T+3} S_X'(t)^4\,dt \ll T(1+\log X)^4, \qquad \int_{T-3}^{2T+3} S_X''(t)^4\,dt \ll T(1+\log X)^8.$$

Proof. By Section 2.4, the series

$$S_X'(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n), \qquad S_X''(t) = -\sum_{n\in A_X} a_n \log^2 n \sin(t\log n)$$

converge absolutely and uniformly. Split according to $m=1$, $m=2$, $m\ge 3$ in $n=p^m$ and use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. For $\nu \in \{1,2\}$ the terms $m\ge 3$ of the ν-th derivative are bounded by $\frac{1}{\pi}\sum_p \sum_{m\ge 3} \frac{(m\log p)^\nu}{mp^{m/2}} < \infty$, uniformly in X and t. The terms $m=1$ are bounded by $\frac{1}{\pi}|\mathcal{D}_\nu(t)|$ with $\mathcal{D}_\nu(t) = \sum_{p\le X} \log^\nu p\; p^{-1/2-it}$; $\mathcal{D}_\nu^2 = \sum_n \eta_n n^{-it}$ over $n = pp' \le X^2 \le T$ with $\sum_n \eta_n^2 \le 2(\sum_{p\le X}\log^{2\nu} p/p)^2 \ll (1+\log X)^{4\nu}$ (Theorem 3.8 and partial summation), so Theorem 3.7 on an interval of length $T+6$ gives $\int |\mathcal{D}_\nu|^4 = \sum \eta_n^2 (T+6+O(n)) \ll T(1+\log X)^{4\nu}$. The terms $m=2$ are bounded by $\frac{1}{2\pi}|\mathcal{D}_\nu'(t)|$ with $\mathcal{D}_\nu'(t) = \sum_{p\le X}(2\log p)^\nu p^{-1-2it}$; here $\mathcal{D}_\nu'^2 = \sum_n \eta_n' n^{-it}$ over $n = (pp')^2 \le X^4$, with $\sum_n \eta_n'^2 \ll 1$ and $\sum_n n\eta_n'^2 \ll (\sum_{p\le X}\log^{2\nu} p)^2 \ll X^2(1+\log X)^{4\nu} \le T(1+\log X)^{4\nu}$, so again $\int |\mathcal{D}_\nu'|^4 \ll T(1+\log X)^{4\nu}$. □

Lemma 7.2 (Transport)**.** *For $T \ge T_0$ and $2 \le X \le T^{1/2}$,*

$$\left\langle \mathfrak{t}_k(X)^2 \right\rangle_{\mathcal{I}_T} \ll \frac{\log\log T\,(1+\log X)^2}{\log^2 T}. \tag{7.2}$$

Proof. Fourth moment of δ_k. For $T \ge T_0$ and $k \in \mathcal{I}_T$, $|\delta_k| \le 1$ by (3.4); so the point ξ_k of (2.6) is $\ge T-1$ and $\mathcal{L}(\xi_k) \gg \log T$. By (3.3) with $j=2$,

$$\langle \delta_k^4 \rangle_{\mathcal{I}_T} = \langle s_k^4 \mathcal{L}(\xi_k)^{-4} \rangle_{\mathcal{I}_T} \ll \frac{(\log\log T)^2}{\log^4 T}. \tag{7.3}$$

Maximal function. Let $\varphi \in C^\infty(\mathbb{R})$ with $0 \le \varphi \le 1$, $\varphi = 1$ on $[T-2, 2T+2]$, $\varphi = 0$ outside $[T-3, 2T+3]$ and $|\varphi'| \le 2$, and let $\mathfrak{h} = \varphi S_X'$. For $k \in \mathcal{I}_T$ the segment between γ_k and τ_k lies in $[T-1, 2T+1]$, where $\mathfrak{h} = S_X'$, so

$$|\mathfrak{t}_k(X)| = \left| \int_{\tau_k}^{\gamma_k} \mathfrak{h}(y)\,dy \right| \le \int_{\tau_k - |\delta_k|}^{\tau_k + |\delta_k|} |\mathfrak{h}| \le 2|\delta_k|\,\mathcal{M}\mathfrak{h}(\tau_k) \tag{7.4}$$

(trivially if $\delta_k = 0$). The function $\mathfrak{h}$ is C^∞ with compact support. By Theorem 3.17, $g = \mathcal{M}\mathfrak{h}$ is locally absolutely continuous with $\|g\|_4 \ll \|\mathfrak{h}\|_4$ and $\|g'\|_4 \le \|\mathcal{M}(\mathfrak{h}')\|_4 \ll \|\mathfrak{h}'\|_4$; by Lemma 7.1, $\|\mathfrak{h}\|_4^4 \ll T(1+\log X)^4$ and, since $\mathfrak{h}' = \varphi' S_X' + \varphi S_X''$, $\|\mathfrak{h}'\|_4^4 \ll T(1+\log X)^8$.

Sampling at the clock points. As in the proof of Lemma 5.11, the intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T$, are disjoint and contained in $[T-1, 2T+2]$. For $x \in \mathcal{U}$, an interval of length $\mathfrak{b}$, $|g(x)| \le \frac{1}{\mathfrak{b}} \int_{\mathcal{U}} |g| + \int_{\mathcal{U}} |g'|$; by Hölder's inequality and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$, $g(x)^4 \le \frac{8}{\mathfrak{b}} \int_{\mathcal{U}} g^4 + 8\mathfrak{b}^3 \int_{\mathcal{U}} |g'|^4$. Summing over $k \in \mathcal{I}_T$ and dividing by $\#\mathcal{I}_T \gg T \log T$,

$$\langle g(\tau_k)^4 \rangle_{\mathcal{I}_T} \ll \frac{1}{T}\int_{\mathbb{R}} g^4 + \frac{1}{T\log^4 T}\int_{\mathbb{R}} |g'|^4 \ll (1+\log X)^4 + \frac{(1+\log X)^8}{\log^4 T} \ll (1+\log X)^4.$$

Conclusion. By (7.4) and Cauchy–Schwarz, $\langle \mathfrak{t}_k^2 \rangle \le 4\langle \delta_k^2 g(\tau_k)^2 \rangle \le 4\langle \delta_k^4 \rangle^{1/2} \langle g(\tau_k)^4 \rangle^{1/2}$; insert (7.3). □

Theorem 7.3 (Theorem D)**.** *There are absolute constants T_0 and C such that for $T \ge T_0$ and $2 \le X \le T^{1/2}$, with $c = \log X / \log T$,*

$$\Big| \big\langle r_k(X)^2 \big\rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2} \log \frac{1}{c} \Big| \le C\Big(1 + \sqrt{\log(1/c)}\Big)\Big(1 + c^2 \log\log T\Big). \tag{7.5}$$

Proof. By Theorem 6.1(a), $0 \le \chi_k \le \delta_k^2/(4\pi(T-1)) \le 1/T$ on $\mathcal{I}_T$. Since $1 + \log X \le (1 + \frac{1}{\log 2}) c \log T$, Lemma 7.2 gives $\langle \mathfrak{t}_k^2 \rangle \ll c^2 \log\log T$, so $d_k = r_k(X) - u_k(X) = \mathfrak{t}_k + \chi_k$ has $\langle d_k^2 \rangle \ll c^2 \log\log T + T^{-2}$. Theorem 5.1 gives $\langle u_k(X)^2 \rangle \ll 1 + \log(1/c)$. Then $|\langle r_k^2 \rangle - \langle u_k^2 \rangle| \le 2\langle u_k^2 \rangle^{1/2} \langle d_k^2 \rangle^{1/2} + \langle d_k^2 \rangle \ll (1 + \sqrt{\log(1/c)})(c\sqrt{\log\log T} + T^{-1}) + c^2 \log\log T$, which is bounded by the right side of (7.5) because $y \le 1 + y^2$. Combine with (5.1). □

Corollary 7.4. *Let $X = T^{1/\sqrt{\log\log T}}$. Then*

$$\langle r_k(X)^2 \rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2} \log\log\log T + O\Big(\sqrt{\log\log\log T}\Big), \tag{7.6}$$

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} \Big(\tau_k^X - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}, \tag{7.7}$$

while $\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} (\tau_k - \gamma_k)^2 = (2 + o(1)) \log\log T / \log^2 T$; the ratio is asymptotic to $\log\log\log T / (2 \log\log T)$.

Proof. Here $c = (\log\log T)^{-1/2}$, $2 \le X \le T^{1/2}$ for $T \ge T_0$, $\log(1/c) = \frac{1}{2}\log\log\log T$ and $c^2 \log\log T = 1$, so (7.5) gives (7.6). On $\mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$; divide r_k^2 by $\mathcal{L}(\tau_k)^2$. For the bare clock, $\delta_k = s_k / \mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$, and $\langle s_k^2 \rangle \sim \frac{1}{2\pi^2} \log\log T$ by (3.3). □

Corollary 7.5 (A cutoff read off the clock)**.** *For $\tau_k > e^e$ let $X_k^{\mathrm{loc}} = \tau_k^{1/\sqrt{\log\log \tau_k}}$ and $\tau_k^{\mathrm{loc}} = \tau_k - S_{X_k^{\mathrm{loc}}}(\tau_k)/\mathcal{L}(\tau_k)$. Then*

$$\frac{1}{\#\mathcal{I}_T^\tau} \sum_{k \in \mathcal{I}_T^\tau} \Big(\tau_k^{\mathrm{loc}} - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}.$$

Proof. Let $X_0 = T^{1/\sqrt{\log\log T}}$ and $X_0' = (2T)^{1/\sqrt{\log\log 2T}}$. The function $y \mapsto y/\sqrt{\log y}$ is increasing for $y > e^{1/2}$, so $X_0 \le X_k^{\mathrm{loc}} \le X_0'$ for $k \in \mathcal{I}_T^\tau$, and $\log X_0' - \log X_0 \le \log 2/\sqrt{\log\log T} = o(1)$; hence $X_0' \le 2X_0$ and $X_0' \le (T/3)^{1/2}$ for $T \ge T_0$. As in the proof of Theorem 5.12, $|S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k)| \le \frac{1}{\pi}\mathcal{Q}(\tau_k) + 1.1V_\pm$ with $X_\pm$ replaced by X_0, X_0', and Lemma 5.11 gives a mean square over $\mathcal{I}_T^\tau$ that is $\ll V_\pm \ll 1/\log X_0 = \sqrt{\log\log T}/\log T$. The sets $\mathcal{I}_T$ and $\mathcal{I}_T^\tau$ differ by $O(\log T)$ indices, those with γ_k or τ_k within distance 1 of T or $2T$, and for them $|r_k(X_0)| \le |s_k| + \chi_k + \sigma_{X_0} \ll \log T + X_0^{1/2}$ (Lemma 2.2); so the means of $r_k(X_0)^2$ over the two sets differ by $O(T^{-1/2})$, and (7.6) holds over $\mathcal{I}_T^\tau$. Finally $\mathcal{L}(\tau_k)(\tau_k^{\mathrm{loc}} - \gamma_k) = r_k(X_0) - (S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k))$; conclude with Cauchy–Schwarz and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$. □

Proposition 7.6 (The root of the corrected equation in the monotone regime)**.** *Let $X = X(T) \ge 2$ with $M_X \le \frac{1}{2}\mathcal{L}(T)$, and let $\mathcal{I}_T'$ be the set of $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac{1}{2} \le F_X(2T)$, so that the root $t_k^X \in [T, 2T]$ of Remark 5.7 exists. Then $X \ll \log^2 T$, $\#(\mathcal{I}_T \setminus \mathcal{I}_T') \ll \log T$, and*

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T'} \bigl(t_k^X - \gamma_k\bigr)^2 \ge \frac{4}{9\mathcal{L}(2T)^2}\Bigl(\frac{1}{2\pi^2}\log\log T - O\bigl(\sqrt{\log\log T}\bigr)\Bigr) \gg \frac{\log\log T}{\log^2 T}.$$

Proof. $M_X \ge \frac{1}{\pi}\sum_{p \le X} \log p/\sqrt{p} \gg \sqrt{X}$ by Chebyshev's estimate, so $X \ll \mathcal{L}(T)^2 \ll \log^2 T$ and $c = \log X/\log T \le (2\log\log T + O(1))/\log T$. On $[T, 2T]$, $0 < F_X' \le \mathcal{L} + M_X \le \frac{3}{2}\mathcal{L}(2T)$, so $|u_k(X)| = |F_X(t_k^X) - F_X(\gamma_k)| \le \frac{3}{2}\mathcal{L}(2T)|t_k^X - \gamma_k|$ for $k \in \mathcal{I}_T'$. An index $k \in \mathcal{I}_T$ is missing from $\mathcal{I}_T'$ only if $k - \frac{1}{2} < F_X(T)$ or $k - \frac{1}{2} > F_X(2T)$; since $N(T) < k \le N(2T)$ and $|F_X - N| \le |G| + \sigma_X \ll \log T$ at T and $2T$, there are $O(\log T)$ such k, each with $|u_k(X)| \ll \log T$ (Lemma 2.2). Hence $\frac{1}{\#\mathcal{I}_T}\sum_{k \in \mathcal{I}_T'} u_k(X)^2 = \langle u_k(X)^2\rangle_{\mathcal{I}_T} + O(\frac{\log^3 T}{T\log T})$, and Theorem 5.1 with $\log(1/c) \ge \log\log T - \log(2\log\log T + O(1))$ bounds it from below. □

Thus a root of the corrected equation can be selected only where it gains nothing in order: the error in height of t_k^X has mean square of the same order $\log\log T/\log^2 T$ as that of τ_k. Theorem 7.3 avoids roots and takes one Newton step instead.

Remark 7.7 (The Newton term and the restriction on c). Let ξ_k^X be a point between γ_k and τ_k with $\mathfrak{t}_k(X) = -S_X'(\xi_k^X)\delta_k$ (mean value theorem). Since $\delta_k = (r_k(X) + S_X(\tau_k))/\mathcal{L}(\tau_k)$ by (2.12), (7.1) becomes the exact relation

$$r_k(X)\Bigl(1 + \frac{S_X'(\xi_k^X)}{\mathcal{L}(\tau_k)}\Bigr) = u_k(X) + \chi_k - y_k\, S_X'(\xi_k^X). \tag{7.8}$$

The term $y_k S_X'(\xi_k^X)$ is the second-order term of Newton's method. Heuristically, $\langle S_X^2\rangle \approx v_X \approx \frac{1}{2\pi^2}\log\log X$ and $\langle S_X'^2\rangle/\mathcal{L}(T)^2 \approx \sum_{p \le X} \frac{\log^2 p}{2\pi^2 p}/\mathcal{L}(T)^2 \approx c^2$, so this term has mean square of order $c^2\log\log T$ if the two factors decorrelate: the order of Lemma 7.2 is the order of the Newton term, and the lemma cannot be improved by bounds that ignore the correlation between $u_k(X)$ and $\mathfrak{t}_k(X)$. At $T = 2^{19}$ (Table 7) the transport is smaller than this order and anticorrelated with $u_k(X)$, increasingly with c; a proof of this cancellation would allow a fixed c in Corollary 7.4 (Question 13.2). The denominator in (7.8) can vanish: $F_X'(\tau_k) < 0$ at 2.7% of the indices of that window for $c = \frac{1}{2}$.

8. Iterating the Newton step

Theorem 7.3 takes one Newton step from τ_k. Its restriction $c \le (\log\log T)^{-1/2}$ comes from the transport over the interval between τ_k and γ_k, whose length is that of δ_k. After one step the error is smaller, so a second step can use a larger cutoff. This section proves the second step and then every further step, with constants uniform in the number of steps (Section 8.5).

8.1. The exact recursion

For a sequence of cutoffs $X_1, X_2, \dots \ge 2$ define $x_k^{(0)} = \tau_k$ and

$$x_k^{(j)} = x_k^{(j-1)} - \frac{N_0(x_k^{(j-1)}) + S_{X_j}(x_k^{(j-1)}) - (k-\frac12)}{\mathcal{L}(x_k^{(j-1)})}, \qquad e_{j,k} = x_k^{(j)} - \gamma_k, \tag{8.1}$$

so that $x_k^{(1)} = \tau_k^{X_1}$ and $e_{0,k} = \delta_k$. For a finite index set I and $r \ge 1$ let $\|a\|_{r,I} = \langle |a_k|^r \rangle_I^{1/r}$.

Proposition 8.1 (Exact recursion)**.** *Let $j \ge 1$ and suppose $x_k^{(j-1)} > 2\pi$. Then*

$$\mathcal{L}(x_k^{(j-1)})\, e_{j,k} = u_k(X_j) + \Big(S_{X_j}(\gamma_k) - S_{X_j}(x_k^{(j-1)})\Big) + \chi_{j,k}, \qquad 0 \le \chi_{j,k} \le \frac{e_{j-1,k}^2}{4\pi \min(\gamma_k, x_k^{(j-1)})}. \tag{8.2}$$

Let $\varphi_j \in C^\infty(\mathbb{R})$, $0 \le \varphi_j \le 1$, be 1 on an interval containing all segments between γ_k and $x_k^{(j-1)}$, $k \in I$, and $g_j = \mathcal{M}(\varphi_j S'_{X_j})$. Then for every $r \ge 1$,

$$\begin{aligned} &\Big\| \mathcal{L}(x^{(j-1)}) e_j \Big\|_{2r,I} \le \|u(X_j)\|_{2r,I} + 2\,\|e_{j-1}\|_{4r,I}\,\|g_j(x^{(j-1)})\|_{4r,I} + \|\chi_j\|_{2r,I}, \\ &\|\chi_j\|_{2r,I} \le \frac{\|e_{j-1}\|_{4r,I}^2}{4\pi \min_{k\in I} \min(\gamma_k, x_k^{(j-1)})}. \end{aligned} \tag{8.3}$$

Proof. Write $x = x_k^{(j-1)}$. By (8.1), $\mathcal{L}(x)e_{j,k} = \mathcal{L}(x)(x-\gamma_k) - N_0(x) - S_{X_j}(x) + k - \frac12$. As in the proof of Theorem 6.1(a), $N_0(x) - N_0(\gamma_k) = \int_{\gamma_k}^x \mathcal{L} = \mathcal{L}(x)(x-\gamma_k) - \chi_{j,k}$ with $\chi_{j,k} = \int_{\gamma_k}^x (\mathcal{L}(x) - \mathcal{L}(y))\,dy$, which lies between $(x-\gamma_k)^2/(4\pi\max(\gamma_k,x))$ and $(x-\gamma_k)^2/(4\pi\min(\gamma_k,x))$ by the same estimates of log. Hence $\mathcal{L}(x)e_{j,k} = k - \frac12 - N_0(\gamma_k) - S_{X_j}(x) + \chi_{j,k} = u_k(X_j) + S_{X_j}(\gamma_k) - S_{X_j}(x) + \chi_{j,k}$. For (8.3), $|S_{X_j}(\gamma_k) - S_{X_j}(x)| \le \int_{x-|e_{j-1,k}|}^{x+|e_{j-1,k}|} |\varphi_j S'_{X_j}| \le 2|e_{j-1,k}|\, g_j(x)$, and Minkowski's and Hölder's inequalities in $\ell^{2r}(I)$. □

The recursion (8.3) is exact and uses no independence between the errors and the prime phases. It reduces the error of each step to three quantities: the moments of the defect $u_k(X_j)$ at the true ordinates (Theorem 5.1 and its higher-moment versions), the moments of the previous error, and the moments of the maximal function g_j sampled at the computable points $x_k^{(j-1)}$. The last is the only new difficulty: the points $x_k^{(j-1)}$ are not separated like the τ_k.

8.2. Moments of order $2r$ of the defect

Lemma 8.2. *There is an absolute constant C such that for every integer $r \ge 1$, every T with $T^{1/(160r)} \ge 2$ and every $2 \le X \le T^{1/(160r)}$, with V as in Lemma 5.8 computed with $Y_{2r} = T^{1/(160r)}$ in place of Y_0,*

$$\int_T^{2T} E_X^{4r}\,dt \le \Big(C(r^2+\sqrt{r(1+V)})\Big)^{4r} T, \qquad \langle u_k(X)^{2r}\rangle_{\mathcal{I}_T} \le \Big(C(r^2+\sqrt{r(1+V)})\Big)^{2r},$$

and for every $r \ge 1$ and $2 \le X \le T^{1/(2r)}$,

$$\int_{T-3}^{2T+3} |S_X^{(\nu)}(t)|^{4r} dt \le (Cr)^{2r}\, T(1+\log X)^{4r\nu} \qquad (\nu = 1, 2).$$

Proof. The first bound: $E_X = E_0 + \mathcal{W}$ with $E_0 = S - S^{\mathrm{pr}}_{Y_{2r}}$, $\|E_0\|_{L^{4r}(T,2T)} \le C(2r)^2 T^{1/4r}$ by Theorem 3.5 with $k = 2r$, and $\mathcal{W}$ as in Lemma 5.10: $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$, $\mathcal{D}_1^{2r} = \sum_n \eta_n n^{-it}$ over products of $2r$ primes $\le \max(X, Y_{2r})$, so $n \le T^{1/80}$, with $\sum |\eta_n|^2 \le (2r)!\,(V+1)^{2r}$ (each n has at most $(2r)!$ ordered factorizations, so $\sum_n |\eta_n|^2 \le (2r)! \sum_{p_1,\dots,p_{2r}} \prod p_i^{-1}$); Theorem 3.7 gives $\int |\mathcal{D}_1|^{4r} \le 2T \sum |\eta_n|^2$, whence $\|\mathcal{W}_1\|_{L^{4r}} \le C\sqrt{r}\sqrt{1+V}\,T^{1/4r}$ by $((2r)!)^{1/4r} \le \sqrt{2r}$. The terms $m = 2$ and $m \ge 3$ are $O(T^{1/4r})$ in the same norm (products of $2r$ squares of primes, $\sum |\eta_n'|^2 \le (2r)!\,(\sum_p p^{-2})^{2r} \le (2r)!$, and a bounded function). Minkowski's inequality in L^{4r} gives the first bound. The second: Theorem 4.1 with $j = r$ and $F = F_X$, $\Psi_r(u) \ge u^{2r}$, gives $\sum_{k\in\mathcal{I}_T} u_k^{2r} \le \int_T^{2T} (|E_X| + T^{-1})^{2r}(\mathcal{L} + |S_X'|)\,dt + O\Big(4^r T^{1/100} + (2\log T)^{2r+1}\Big)$, the last term bounding the boundary terms and the end pieces, since $|u_k(X)| \le \log T + \sqrt{X}$ (Lemma 2.2 and (2.11)) and $X^{r+1/2} \le T^{1/100}$. With $\mathcal{E}_{4r} = \int_T^{2T} E_X^{4r}$, Hölder's inequality gives $\int_T^{2T} E_X^{2r} \le T^{1/2}\mathcal{E}_{4r}^{1/2}$ and, by (5.8), $\int_T^{2T} E_X^{2r}|S_X'| \le \mathcal{E}_{4r}^{1/2} C_0 T^{1/2}(1+\log X)$; so the integral is at most $(\mathcal{L}(2T) + C_0(1+\log X))T^{1/2}(\mathcal{E}_{4r}^{1/2} + T^{-2r+1/2})$. Dividing by $\#\mathcal{I}_T \ge \frac{1}{2}T\mathcal{L}(T)$ and using $\mathcal{L}(2T) \le 2\mathcal{L}(T)$ and $1 + \log X \le 2\pi\mathcal{L}(T)$, $\langle u_k(X)^{2r}\rangle_{\mathcal{I}_T} \le (4 + 4\pi C_0)T^{-1/2}\mathcal{E}_{4r}^{1/2} + o(1)$, and the first bound gives the second. The third: Lemma 7.1 with products of $2r$ primes, $n \le X^{2r} \le T$, and $\sum \eta_n^2 \le (2r)!\,(\sum_{p\le X} \log^{2\nu} p/p)^{2r} \le (Cr)^{2r}(1+\log X)^{4r\nu}$, by Theorem 3.8 and partial summation; the terms $m = 2$ and $m \ge 3$ as in Lemma 7.1. □

The dependence on r matters in Theorem 8.10, where r grows with the number of Newton steps; for fixed r the three bounds read $\ll_r T(1+V)^{2r}$, $\ll_r (1+V)^r$ and $\ll_r T(1+\log X)^{4r\nu}$.

8.3. Multiplicity of the one-step clock points

Let $X_1 = T^{c_1}$ with $c_1 = (\log\log T)^{-1/2}$, $\rho_k = x_k^{(1)} = \tau_k^{X_1} = \tau_k - y_k$ with $y_k = S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$, $\mathfrak{b}_T = 2\pi/\log(3T)$ as in Lemma 5.11, and

$$\tilde{n}(u) = \#\{k \in \mathcal{I}_T :\ u - \mathfrak{b}_T \le \rho_k \le u\} \qquad (u \in \mathbb{R}).$$

Note that $|y_k| \le \sigma_{X_1}/\mathcal{L}(\tau_k) \ll T^{c_1/2}$, so $\rho_k \in [T - T^{1/2}, 2T + T^{1/2}]$ for $T \ge T_0$.

Lemma 8.3 (Multiplicity)**.** $\displaystyle\int_{\mathbb{R}} \tilde{n}(u)^2\,du = \sum_{j,k\in\mathcal{I}_T} \Big(\mathfrak{b}_T - |\rho_j - \rho_k|\Big)_+ \ll \mathfrak{b}_T\,\#\mathcal{I}_T.$

Proof. The identity: $\tilde{n} = \sum_k \mathbf{1}_{[\rho_k, \rho_k + \mathfrak{b}_T]}$, and the measure of the intersection of two such intervals is $(\mathfrak{b}_T - |\rho_j - \rho_k|)_+$. Write $b = \mathfrak{b}_T$ and $M = \#\mathcal{I}_T$. The pairs with $|j-k| \le 1$ contribute at most $3bM$. Let $j = k+m$ with $m \ge 2$ and $|\rho_{k+m} - \rho_k| \le b$. Since $\tau_{k+m} - \tau_k \ge mb$ (proof of Lemma 5.11), this forces

$$|y_{k+m} - y_k| \ge (m-1)b. \tag{8.4}$$

Two consequences of (8.4) are counted separately; a close pair at distance m satisfies both.

Variation. $y = S_{X_1}/\mathcal{L}$ is C^∞ with $y' = S'_{X_1}/\mathcal{L} - S_{X_1}\mathcal{L}'/\mathcal{L}^2$, and on $J_{k,m} = [\tau_k, \tau_{k+m}]$, of length $|J_{k,m}| \le mb(1+o(1))$,

$$|y_{k+m} - y_k| \le \frac{1+o(1)}{\mathcal{L}(T)} \int_{J_{k,m}} |S'_{X_1}| + \frac{\sigma_{X_1} |J_{k,m}|}{2\pi(T-1)\mathcal{L}(T)^2},$$

and the last term is $\ll T^{c_1/2-1} mb/\log^2 T = o((m-1)b)$ uniformly in $m \ge 2$. As $b\mathcal{L}(T) = 1 - o(1)$, (8.4) gives $\frac{1}{|J_{k,m}|}\int_{J_{k,m}} |S'_{X_1}| \ge \frac{1}{3}\mathcal{L}(T)$ for $T \ge T_0$. Each t lies in at most m of the intervals $J_{k,m}$ (k varying), and $|J_{k,m}| \ge mb$; by Chebyshev's inequality and Jensen's inequality,

$$\#\{k : \text{variation}\} \le \frac{9}{\mathcal{L}(T)^2} \sum_k \frac{1}{|J_{k,m}|} \int_{J_{k,m}} |S'_{X_1}|^2 \le \frac{9}{\mathcal{L}(T)^2\, b} \int_{T-1}^{2T+1} |S'_{X_1}|^2 \ll T \log T\, c_1^2 \asymp M\, c_1^2,$$

by (5.8), uniformly in m.

Size. (8.4) gives $\max(|y_k|, |y_{k+m}|) \ge \frac{m-1}{2} b$, hence $\max(|S_{X_1}(\tau_k)|, |S_{X_1}(\tau_{k+m})|) \ge \frac{m-1}{3}$ for $T \ge T_0$. Sampling at the clock points as in the proof of Lemma 5.11 (Sobolev on the disjoint intervals $[\tau_k, \tau_k + b]$), $\langle S_{X_1}(\tau_k)^4 \rangle_{\mathcal{I}_T} \ll \frac{1}{T}\int S_{X_1}^4 + \frac{1}{T\log^4 T}\int |S'_{X_1}|^4 \ll (1+\log\log T)^2$, the first integral by Theorem 3.7 with products of two primes ($\sum_{p \le X_1} p^{-1} \ll \log\log T$) and the second by Lemma 7.1. By Chebyshev's inequality of order 4, $\#\{k : \text{size}\} \ll M(\log\log T)^2/(m-1)^4$.

Sum. With $H_0 = \log\log T$ and $c_1^2 = H_0^{-1}$, the number of close ordered pairs at distance $m \ge 2$ is $\ll M \min(H_0^{-1}, H_0^2 (m-1)^{-4})$, and

$$\sum_{m \ge 2} \min\Big(H_0^{-1}, \frac{H_0^2}{(m-1)^4}\Big) \le \sum_{m-1 \le H_0^{3/4}} H_0^{-1} + \sum_{m-1 > H_0^{3/4}} \frac{H_0^2}{(m-1)^4} \ll H_0^{-1/4}.$$

Hence the number of pairs (j,k) with $|\rho_j - \rho_k| \le b$ is $\ll M$, and each contributes at most b. □

Lemma 8.4 (Sampling at points of bounded multiplicity)**.** *Let ϱ_k, $k \in \mathcal{I}_T$, be real points of $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ with multiplicity function $\tilde{n}(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le \varrho_k \le u\}$ satisfying $\int_{\mathbb{R}} \tilde{n}^2 \le \kappa\, \mathfrak{b}_T \#\mathcal{I}_T$ for some $\kappa \ge 1$. Let $r \ge 1$ be an integer, $2 \le X \le T^{1/(4r)}$, φ a cutoff equal to 1 on $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ as in Lemma 7.2, and $g = \mathcal{M}(\varphi S'_X)$. Then, with an absolute constant C,*

$$\|g(\varrho_k)\|_{4r, \mathcal{I}_T} \le C\, \kappa^{1/(8r)} \sqrt{r}\, (1 + \log X).$$

In particular $\langle g(\rho_k)^4 \rangle_{\mathcal{I}_T} \ll (1+\log X)^4$ at the one-step clock points ρ_k, for $2 \le X \le T^{1/4}$.

Proof. Let $F = g^{4r}$, locally absolutely continuous with $F' = 4rg^{4r-1}g'$ (Theorem 3.17). For $x \in [u-b, u]$, $F(x) \le \frac{1}{b}\int_{u-b}^{u} F + \int_{u-b}^{u} |F'|$. Averaging over $u \in [\varrho_k, \varrho_k + b]$ and summing over k,

$$\sum_{k\in\mathcal{I}_T} F(\varrho_k) \le \frac{1}{b}\int_{\mathbb{R}} \tilde{n}(u)\Big(\frac{1}{b}\int_{u-b}^{u} F + \int_{u-b}^{u} |F'|\Big)du \le \frac{1}{b}\|\tilde{n}\|_2\big(\|F\|_2 + b\|F'\|_2\big),$$

by Cauchy–Schwarz and because the averaging operator $h \mapsto \frac{1}{b}\int_{u-b}^{u} h$ has norm ≤ 1 on $L^2(\mathbb{R})$. Here $\|F\|_2 = \|g\|_{8r}^{4r} \le (A_{8r}\|\varphi S'_X\|_{8r})^{4r} \le (C\sqrt{r}(1+\log X))^{4r}T^{1/2}$ and $\|F'\|_2 \le 4r\|g\|_{8r}^{4r-1}\|g'\|_{8r} \le 4r(1+\log X)\,(C\sqrt{r}(1+\log X))^{4r}T^{1/2}$, by Theorem 3.17 ($p = 8r$, A_p bounded) and Lemma 8.2 (third bound with $2r$ in place of r, which needs $X^{4r} \le T$). Since $b \cdot 4r(1+\log X) \le 8\pi r\, c\,(1+o(1)) \le 7$ for $c = \log X/\log T \le 1/(4r)$, we get $\sum_k F(\varrho_k) \le 8\,b^{-1}(\kappa b M)^{1/2}T^{1/2}(C\sqrt{r}(1+\log X))^{4r} \le C'\sqrt{\kappa}\,M\,(C\sqrt{r}(1+\log X))^{4r}$, since $b^{-1}T \asymp M$; take the $4r$-th root. For the ρ_k, $\kappa \ll 1$ by Lemma 8.3. □

8.4. The second step

Theorem 8.5 (Second Newton step)**.** *Let $H_0 = \log\log T$, $H_1 = \log H_0$, $H_2 = \log H_1$, $X_1 = T^{H_0^{-1/2}}$, $X_2 = T^{H_1^{-1/2}}$, and let $x_k^{(2)}$ be given by* (8.1)*. Then*

$$\begin{aligned} &\Big\langle \mathcal{L}(\rho_k)^2 e_{2,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}H_2 + O\big(1+\sqrt{H_2}\big), \\ &\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(2)} - \gamma_k\Big)^2 = \frac{\log\log\log\log T + O\Big(\sqrt{\log\log\log\log T}\Big)}{\log^2 T}. \end{aligned} \tag{8.5}$$

Proof. Fourth moment of e_1. By (8.2) with $j = 1$ (that is (7.1)), $\mathcal{L}(\tau_k)e_{1,k} = u_k(X_1) + \mathfrak{t}_k(X_1) + \chi_k$. Lemma 8.2 with $r = 2$ gives $\langle u_k(X_1)^4\rangle \ll (1+\log(1/c_1))^2 \ll H_1^2$. For the transport, $|\mathfrak{t}_k| \le 2|\delta_k|g_1(\tau_k)$ with $g_1 = \mathcal{M}(\varphi S'_{X_1})$, so $\langle \mathfrak{t}_k^4\rangle \le 16\langle \delta_k^8\rangle^{1/2}\langle g_1(\tau_k)^8\rangle^{1/2}$; here $\langle\delta_k^8\rangle \ll H_0^4/\log^8 T$ by (3.3) with $j = 4$, and $\langle g_1(\tau_k)^8\rangle \ll (c_1\log T)^8$ by the sampling argument of Lemma 7.2 in L^8 (Sobolev on the intervals $\mathcal{U}_k$ with g^8, Theorem 3.17 with $p = 8$, Lemma 8.2 with $r = 2$). Hence $\langle \mathfrak{t}_k^4\rangle \ll (c_1^2 H_0)^2 = 1$, and $\chi_k \ll 1/T$. Dividing by $\mathcal{L}(\tau_k)^4 \asymp \log^4 T$,

$$\langle e_{1,k}^4\rangle_{\mathcal{I}_T} \ll \frac{H_1^2}{\log^4 T}. \tag{8.6}$$

Transport of the second step. The interval centred at ρ_k of radius $|e_{1,k}|$ contains γ_k and lies in $[T - 2T^{1/2}, 2T + 2T^{1/2}]$, so $|S_{X_2}(\gamma_k) - S_{X_2}(\rho_k)| \le 2|e_{1,k}|g_2(\rho_k)$ with $g_2 = \mathcal{M}(\varphi S'_{X_2})$, and by Cauchy–Schwarz, (8.6) and Lemma 8.4 (applicable since $X_2 \le T^{1/8}$),

$$\Big\langle \big(S_{X_2}(\gamma_k) - S_{X_2}(\rho_k)\big)^2\Big\rangle \le 4\langle e_{1,k}^4\rangle^{1/2}\langle g_2(\rho_k)^4\rangle^{1/2} \ll \frac{H_1}{\log^2 T}(c_2\log T)^2 = c_2^2 H_1 = 1.$$

Conclusion. By (8.2) with $j = 2$, $0 \le \chi_{2,k} \le e_{1,k}^2/(2\pi T)$, and $\langle \chi_{2,k}^2\rangle \ll T^{-2}$. Theorem 5.1 with $X = X_2$ (allowed, $2 \le X_2 \le T^{1/2}$) gives $\langle u_k(X_2)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{c_2} + O(1+\sqrt{\log(1/c_2)})$ with $\log\frac{1}{c_2} = \frac{1}{2}H_2$. Combining as in the proof of Theorem 7.3 gives the first statement of (8.5), and $\mathcal{L}(\rho_k) = \frac{\log T}{2\pi}(1 + O(T^{-1/2}))$ on $\mathcal{I}_T$ gives the second. □

8.5. All further steps: the exact identity, the local deformation, and the multiplicity

This subsection proves the Newton steps $m \geq 3$ with constants uniform in m, up to $m = \log^* T + O(1)$ steps, where the error becomes $O(1)$ in units of zeros.

Lemma 8.6 (The second-level identity)**.** *For $j \geq 1$, with $Y_k^{(j)} = \tau_k - x_k^{(j)}$ and $\chi''_{j,k} = \int_{x_k^{(j-1)}}^{\tau_k} (\mathcal{L}(\tau_k) - \mathcal{L}(y))\, dy$,*

$$Y_k^{(j)} = \Big(1 - \frac{\mathcal{L}(\tau_k)}{\mathcal{L}(x_k^{(j-1)})}\Big) Y_k^{(j-1)} + \frac{S_{X_j}(x_k^{(j-1)}) + \chi''_{j,k}}{\mathcal{L}(x_k^{(j-1)})}, \tag{8.7}$$
$$\text{hence} \qquad \mathcal{L}(\tau_k) Y_k^{(j)} = S_{X_j}(x_k^{(j-1)}) + R_{j,k},$$

with $|R_{j,k}| \ll T^{c_j - 1}$ if $|Y_k^{(j-1)}| \leq T^{c_{j-1}/2}$ and $c_{j-1} \leq c_j$ (true by (2.11) and induction): the three contributions are $\mathcal{L}(\tau_k)(1 - \mathcal{L}(\tau_k)/\mathcal{L}(x)) Y_k^{(j-1)} \ll (Y_k^{(j-1)})^2/T$, $(\mathcal{L}(\tau_k)/\mathcal{L}(x) - 1) S_{X_j}(x) \ll |Y_k^{(j-1)}| \sigma_{X_j}/(T \log T)$ and $\chi''_{j,k} \leq (Y_k^{(j-1)})^2/(4\pi(T - T^{1/2}))$. Newton's step does not accumulate the corrections of the previous levels: it replaces the previous one by the phase of the new cutoff evaluated at the previous point.

Proof. $N_0(x_k^{(j-1)}) - (k - \frac{1}{2}) = N_0(x_k^{(j-1)}) - N_0(\tau_k) = -\mathcal{L}(\tau_k) Y_k^{(j-1)} + \chi''_{j,k}$, with $0 \leq \chi''_{j,k} \leq (Y_k^{(j-1)})^2/(4\pi(T - T^{1/2}))$; insert in (8.1) and use $1 - \mathcal{L}(\tau_k)/\mathcal{L}(x) = O(|Y_k^{(j-1)}|/(T \log T))$ and $|S_{X_j}| \leq \sigma_{X_j}$. □

Proposition 8.7 (Local deformation of the second clock)**.** *Let $X_1 = T^{c_1}$, $X_2 = T^{c_2}$ with $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$, and $d_{j,k} = x_{k+1}^{(j)} - x_k^{(j)}$ ($d_{0,k} = \tau_{k+1} - \tau_k$). Then, for $T \geq T_0$,*

$$\Big\langle \Big| \frac{d_{2,k} - d_{0,k}}{d_{0,k}} \Big|^4 \Big\rangle_{\mathcal{I}_T} \ll c_2^4 = \frac{1}{H_1^2}, \qquad \frac{1}{\#\mathcal{I}_T} \#\Big\{ k \in \mathcal{I}_T : \Big| \frac{d_{2,k}}{d_{0,k}} - 1 \Big| > \alpha \Big\} \ll \frac{1}{\alpha^4 H_1^2} \quad (0 < \alpha \leq 1), \tag{8.8}$$

in particular $\#\{k \in \mathcal{I}_T : d_{2,k} \leq 0\} \ll \#\mathcal{I}_T/(\log\log\log T)^2$: the second clock preserves the order and a spacing comparable to that of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices.

Proof. By (8.7) with $j = 2$, $d_{2,k} - d_{0,k} = -\Delta_k\Big(S_{X_2}(x^{(1)})/\mathcal{L}(\tau)\Big) - \Delta_k(R_2/\mathcal{L}(\tau))$, where $\Delta_k a = a_{k+1} - a_k$; the last term is $\ll T^{c_1 - 1}$, negligible against $d_{0,k} \asymp 1/\log T$, and $\Delta_k(1/\mathcal{L}(\tau)) \ll 1/(T \log^3 T)$ is negligible too, so it suffices to bound $\Delta_k S_{X_2}(x^{(1)})/(d_{0,k}\mathcal{L}(\tau_k))$ in $\ell^4(\mathcal{I}_T)$. Let $g_j = \mathcal{M}(\varphi S'_{X_j})$ with $\varphi = 1$ on $[T - 2T^{1/2}, 2T + 2T^{1/2}]$. Two applications of $|S(a) - S(b)| \leq 2|a - b|\, \mathcal{M}(S')(a)$:

$$|x_{k+1}^{(1)} - x_k^{(1)}| \leq d_{0,k} + |y_{k+1} - y_k| \leq d_{0,k}\Big(1 + \frac{2g_1(\tau_k)}{\mathcal{L}(\tau_k)}\Big) + O\Big(T^{c_1/2 - 1}\Big),$$

$$|\Delta_k S_{X_2}(x^{(1)})| \leq 2|x_{k+1}^{(1)} - x_k^{(1)}|\, g_2(x_k^{(1)}).$$

Hence, with $\mathcal{L}(\tau_k) \asymp \log T$,

$$\Big\| \frac{\Delta S_{X_2}(x^{(1)})}{d_0\, \mathcal{L}(\tau)} \Big\|_{4, \mathcal{I}_T} \ll \Big\| \frac{g_2(x^{(1)})}{\log T} \Big\|_{4, \mathcal{I}_T} + \Big\| \frac{g_2(x^{(1)})}{\log T} \Big\|_{8, \mathcal{I}_T} \Big\| \frac{g_1(\tau)}{\log T} \Big\|_{8, \mathcal{I}_T} + o(1).$$

Sampling at the τ_k in L^8 (Sobolev on the disjoint intervals $\mathcal{U}_k$ as in Lemma 7.2, with Lemma 8.2 for $\int |S'_{X_1}|^8$, $\int |S''_{X_1}|^8$ and Theorem 3.17 with $p=8$) gives $\|g_1(\tau)/\log T\|_8 \ll c_1$. Sampling at the $\rho_k = x_k^{(1)}$ in L^8 is Lemma 8.4 with $F = g_2^8$: $\|F\|_2 = \|g_2\|_{16}^8$ and $\|F'\|_2 \le 8\|g_2\|_{16}^7\|g_2'\|_{16}$, which need Theorem 3.7 with products of 8 primes ($X_2^{16} \le T$, true for $T \ge T_0$ since $c_2 \to 0$) and Theorem 3.17 with $p=16$; the result is $\|g_2(x^{(1)})/\log T\|_8 \ll c_2$. The first inequality of (8.8) follows, and the second is Markov's inequality; the last statement is $\alpha = 1$. □

Proposition 8.7 controls consecutive differences; the multiplicity of the points $x_k^{(j)}$ at scale $\mathfrak{b}_T$, which Lemma 8.4 needs, is obtained for every j by comparing $x_k^{(j)}$ with γ_k through two unconditional counts of near pairs and near triples of ordinates. The ordinates enter the proof of the bound, as in Theorem 5.1, and not the construction of the points.

Lemma 8.8 (Near pairs and triples of ordinates)**.** *Let $b = \mathfrak{b}_T$, $M = \#\mathcal{I}_T$ and, for $\Lambda \ge 1$, $n_\Lambda(u) = \#\{l : |\gamma_l - u| \le \Lambda b\}$. There are absolute constants C_1, C_2 such that for $T \ge T_0$ and every $\Lambda \ge 1$,*

$$\#\{(k,l) \in \mathcal{I}_T^2 : |\gamma_k - \gamma_l| \le \Lambda b\} \le C_1 \Lambda M, \qquad \sum_{k \in \mathcal{I}_T} n_\Lambda(\gamma_k)^2 \le C_2 \Lambda^2 M. \tag{8.9}$$

Proof. If $4\Lambda b > 1$, then $N(u+h) - N(u) \le Ch\log T$ for $h \ge 1$ and $u \in [T-1, 2T]$, by (2.3) and Theorem 3.1; hence every γ_k has at most $C'\Lambda$ ordinates within Λb, and both bounds follow. Let $4\Lambda b \le 1$ and $h \in \{2\Lambda b, 4\Lambda b\}$. For $u \in [T-1, 2T]$, $N(u+h) - N(u) = h\mathcal{L}(u) + S(u+h) - S(u) + O(1/T)$ with $h\mathcal{L}(u) \le 5\Lambda$, so Theorem 3.6 with $k = 1, 2$ and $\log(2 + h\log T) \le \log(30\Lambda) \le 30\Lambda$ give

$$\int_{T-1}^{2T} \Big(N(u+h) - N(u)\Big)^{2k} du \le C\Lambda^{2k} T \qquad (k = 1, 2).$$

A pair (k,l) with $|\gamma_k - \gamma_l| \le \Lambda b$ lies in $[u, u+2\Lambda b]$ for every $u \in [\min(\gamma_k, \gamma_l) - \Lambda b, \min(\gamma_k, \gamma_l)]$, a set of measure Λb contained in $[T-1, 2T]$; hence the number of pairs is at most $\frac{1}{\Lambda b}\int_{T-1}^{2T} (N(u + 2\Lambda b) - N(u))^2 du \le C\Lambda T/b \le C_1 \Lambda M$, as $M \asymp T/b$. A triple (k,l,l') with $|\gamma_l - \gamma_k|, |\gamma_{l'} - \gamma_k| \le \Lambda b$ lies in $[u, u + 4\Lambda b]$ for u in a set of measure $2\Lambda b$; by Hölder's inequality the number of triples is at most $\frac{1}{2\Lambda b}\int_{T-1}^{2T} (N(u+4\Lambda b) - N(u))^3 du \le \frac{T^{1/4}(C\Lambda^4 T)^{3/4}}{2\Lambda b} \le C_2 \Lambda^2 M$. □

Theorem 8.9 (Multiplicity of the corrected clock points)**.** *Let $j \ge 0$ and let $x_k^{(j)}$ be given by* (8.1) *with cutoffs $X_i = T^{c_i}$, $c_i \le \frac{1}{2}$, so that $x_k^{(j)} \in [T - 2T^{1/2}, 2T + 2T^{1/2}]$ for $k \in \mathcal{I}_T$ and $T \ge T_0$ (Lemma 8.6). Let $\tilde{n}_j(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le x_k^{(j)} \le u\}$, and let $A \ge 1$, $H \ge 1$ be such that $\|\mathcal{L}(\tau_k) e_{j,k}\|_{4, \mathcal{I}_T} \le A\sqrt{H}$. Then, with an absolute constant C_3,*

$$\int_{\mathbb{R}} \tilde{n}_j(u)^2 \, du \le C_3 \, A\sqrt{H} \; \mathfrak{b}_T \, \#\mathcal{I}_T. \tag{8.10}$$

Proof. Write $b = \mathfrak{b}_T$, $M = \#\mathcal{I}_T$, $x_k = x_k^{(j)}$, $e_k = e_{j,k}$. As in Lemma 8.3, $\int \tilde{n}_j^2 = \sum_{k,l \in \mathcal{I}_T} (b - |x_k - x_l|)_+ \le b\,P$ with $P = \#\{(k,l) \in \mathcal{I}_T^2 : |x_k - x_l| < b\}$. Let $\Lambda_0 \ge 2$. The pairs with $|\gamma_k - \gamma_l| \le \Lambda_0 b$ number at most $C_1 \Lambda_0 M$ by (8.9). For the others, let $\Lambda = 2^i \Lambda_0$, $i \ge 0$, and consider the pairs with $\Lambda b < |\gamma_k - \gamma_l| \le 2\Lambda b$ and $|x_k - x_l| < b$. For them $|e_k - e_l| \ge (\Lambda - 1)b \ge \Lambda b/2$, and $b\mathcal{L}(\tau_k) \ge \frac{1}{2}$ for $T \ge T_0$, so $\mathcal{L}(\tau_k)|e_k - e_l| \ge \Lambda/4$ and

$$1 \le \Big(\frac{4\mathcal{L}(\tau_k)|e_k - e_l|}{\Lambda}\Big)^2 \le \frac{32}{\Lambda^2}\Big(\mathcal{L}(\tau_k)^2 e_k^2 + \mathcal{L}(\tau_l)^2 e_l^2\Big)\Big(1 + O(1/\log T)\Big).$$

Summing over the ordered pairs of the block and symmetrizing, their number P_Λ satisfies

$$P_\Lambda \le \frac{65}{\Lambda^2}\sum_{k\in\mathcal{I}_T}\mathcal{L}(\tau_k)^2 e_k^2\, n_{2\Lambda}(\gamma_k) \le \frac{65}{\Lambda^2}\Big(\sum_k \mathcal{L}(\tau_k)^4 e_k^4\Big)^{1/2}\Big(\sum_k n_{2\Lambda}(\gamma_k)^2\Big)^{1/2} \le \frac{130\sqrt{C_2}\,A^2 H}{\Lambda}\,M,$$

by Cauchy–Schwarz, the hypothesis and (8.9). Summing over $i \ge 0$ gives at most $260\sqrt{C_2}\,A^2HM/\Lambda_0$. With $\Lambda_0 = A\sqrt{H}\,(260\sqrt{C_2}/C_1)^{1/2}$ (which is ≥ 2 for C_2 large enough) the two parts are equal and $P \le 2\sqrt{260C_1\sqrt{C_2}}\,A\sqrt{H}\,M$. □

Only the fourth moment of the error of the j-th clock enters. For $j = 2$, with A bounded (shown below), the multiplicity of the second clock is $\ll \sqrt{H_2}\,\mathfrak{b}_T \#\mathcal{I}_T$. The factor $\sqrt{H}$ is not a loss in the iteration: it is absorbed by the cutoff of the next step.

All further steps

Let $H_0 = \log\log T$, $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$ with $H_1 = \log H_0$, and define recursively, with an absolute constant $0 < C_\star \le 1$ fixed in the proof of Theorem 8.10,

$$H_j = 2\log\frac{1}{c_j} \quad (j \ge 1), \qquad c_j = C_\star\, H_{j-1}^{-9/16} \quad (j \ge 3), \tag{8.11}$$

so that $H_2 = \log H_1$ as in Theorem 8.5, $H_j = \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$ for $j \ge 3$, and $\log H_{j-1} \le H_j$ for all j. All cutoffs depend only on T. Put

$$A_j(r) = \frac{\|\mathcal{L}(\tau_k)e_{j,k}\|_{2r,\mathcal{I}_T}}{\sqrt{H_j}} \quad (j \ge 1), \qquad A_0(r) = \frac{\|\mathcal{L}(\tau_k)\delta_k\|_{2r,\mathcal{I}_T}}{\sqrt{H_0}},$$

and let $m^* = m^*(T)$ be the first index m with $H_m < H^*$, where H^* is the absolute constant of the next theorem. Since $\log H_{j-1} \le H_j \le \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$, the sequence H_j decreases from $\log\log T$ to below H^* in $m^* = \log^* T + O(1)$ steps, where $\log^*$ is the iterated logarithm.

Theorem 8.10 (All Newton steps, uniformly)**.** *There are absolute constants $C_\star$, H^*, C and T_0 such that, with the cutoffs* (8.11)*, for every $T \ge T_0$ and every $1 \le m \le m^*$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{m,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_m}{4\pi^2} + \vartheta_m\, C\Big(1 + \sqrt{H_m}\Big), \qquad |\vartheta_m| \le 1, \tag{8.12}$$

and $\|\mathcal{L}(\tau_k)e_{j,k}\|_{4,\mathcal{I}_T} \le C\sqrt{H_j}$ for $1 \le j < m^*$. In particular, at $m = m^* = \log^* T + O(1)$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(m^*)} - \gamma_k\Big)^2 \le \frac{C'}{\log^2 T}: \tag{8.13}$$

the clock corrected $\log^* T + O(1)$ times with the Euler phases of the primes up to T^{c_j} locates the ordinates with a mean-square error of $O(1)$ mean spacings, with an absolute constant, using no ordinate in its construction.

Proof. Multiplicities. Let $\mathfrak{k}_j$ be the least $\kappa \ge 1$ with $\int \tilde{n}_j^2 \le \kappa \mathfrak{b}_T \# \mathcal{I}_T$. Then $\mathfrak{k}_0 = 1$ (the τ_k are separated by at least $\mathfrak{b}_T$, so only the diagonal contributes), $\mathfrak{k}_1 \le C$ (Lemma 8.3) and $\mathfrak{k}_j \le C_3 A_j(2)\sqrt{H_j}$ for $j \ge 2$ (Theorem 8.9 with $A = \max(A_j(2), 1)$).

Base. By Lemma 8.2 with $X = 2$, $|S_2| \le \sigma_2 < 1$ and $0 \le \chi_k \le 1/T$, $\|\mathcal{L}(\tau_k)\delta_k\|_{2r} \le \|u_k(2)\|_{2r} + 2 \le C_0(r^2 + \sqrt{rH_0})$, so $A_0(r) \le C_0 r^2$ for $r \ge 1$.

One step. Let $j \ge 1$ and $r \ge 1$ with $X_j \le T^{1/(160r)}$. By (8.3), $\mathcal{L}(x_k^{(j-1)}) = \mathcal{L}(\tau_k)(1 + O(T^{-1/2}))$, Lemma 8.2 (with $V \le \log \frac{1}{c_j} + O(1) \le H_j$), Lemma 8.4 at the points $x_k^{(j-1)}$ with multiplicity constant $\mathfrak{k}_{j-1}$, and $\|\chi_j\|_{2r} \ll T^{c_{j-1}-1}$,

$$\sqrt{H_j}\, A_j(r) \le C\Big(r^2 + \sqrt{rH_j}\Big) + 2\sqrt{H_{j-1}}\, A_{j-1}(2r) \cdot \frac{C\mathfrak{k}_{j-1}^{1/(8r)}\sqrt{r}\,(1 + \log X_j)}{\mathcal{L}(\tau_k)}\Big(1 + o(1)\Big) + o(1),$$

and $(1 + \log X_j)/\mathcal{L}(\tau_k) \le 2\pi c_j(1 + o(1))$, $\mathfrak{k}_{j-1}^{1/(8r)} \le \mathfrak{k}_{j-1}^{1/8}$. For $j \le 2$, $\sqrt{H_{j-1}}c_j = 1$ and $\mathfrak{k}_{j-1} \le C$; for $j \ge 3$, $\mathfrak{k}_{j-1}^{1/8}\sqrt{H_{j-1}}c_j \le (C_3 A_{j-1}(2))^{1/8} H_{j-1}^{1/16+1/2-9/16} C_\star = C_\star (C_3 A_{j-1}(2))^{1/8}$, and $A_{j-1}(2) \le A_{j-1}(2r)$ (monotonicity of the discrete norms). Fix $C_\star = \min(1, (4\pi C\, C_3^{1/8})^{-1})$. Then, for all $j \ge 1$ and $r \ge 1$,

$$A_j(r) \le C\Big(\sqrt{r} + \frac{r^2}{\sqrt{H_j}}\Big) + \frac{C_5\sqrt{r}}{\sqrt{H_j}} \max\Big(A_{j-1}(2r), A_{j-1}(2r)^{9/8}\Big), \tag{8.14}$$

with absolute C, C_5, provided $X_j \le T^{1/(160r)}$.

Induction. Let $B = \max(2C, C_0, 1)$ and $C_6 = \Big(4C/B + 4C_5(4B)^{9/8}/B\Big)^2$. We claim that for every $m \le m^*$,

$$A_j(r) \le Br^2 \qquad (0 \le j \le m-1,\ 1 \le r \le 2^{m-j}). \tag{8.15}$$

For $j = 0$ this is the base. Suppose it holds at $j-1$ for $r \le 2^{m-j+1}$, let $1 \le r \le 2^{m-j}$, and assume $H_j \ge C_6 4^{m-j}$ and $X_j \le T^{1/(160r)}$. Since $A_{j-1}(2r) \le 4Br^2$ and $4Br^2 \ge 1$, (8.14) gives

$$A_j(r) \le C\sqrt{r} + \frac{Cr^2 + C_5(4B)^{9/8} r^{11/4}}{\sqrt{H_j}} \le \frac{B}{2}r^2 + \frac{\big(C + C_5(4B)^{9/8}\big)r^3}{\sqrt{H_j}} \le Br^2,$$

because $r^3/\sqrt{H_j} \le r^2 \cdot 2^{m-j}/\sqrt{H_j} \le r^2/\sqrt{C_6}$ and $\sqrt{C_6} \ge 2(C + C_5(4B)^{9/8})/B$.

The two conditions. Let H^* be an absolute constant with $H^* \ge \max(4C_6, 160^2 \cdot 4)$ and so large that $e^{\frac{8}{9}(H - 2\log(1/C_\star))} \ge 4H$ and $e^H \ge 4H$ for $H \ge H^*$. If $m \le m^*$ then $H_{m-1} \ge H^*$ (by definition of m^*), and from $H_{j-1} \ge e^{H_j}$ $(j \le 2)$ or $H_{j-1} = e^{\frac{8}{9}(H_j - 2\log(1/C_\star))}$ $(j \ge 3)$ we get $H_{j-1} \ge 4H_j$ whenever $H_j \ge H^*$; by downward induction, $H_{m-1-i} \ge 4^i H^*$ for $0 \le i \le m-1$. Hence $H_j \ge 4^{m-1-j}H^* \ge C_6 4^{m-j}$ for $1 \le j \le m-1$, and $H_{j-1} \ge 4^{m-j}H^* \ge 160^2\, 4^{m-j}$ for $1 \le j \le m$, which gives $c_j \le H_{j-1}^{-1/2} \le 2^{-(m-j)}/160$, i.e. $X_j \le T^{1/(160r)}$ for $r \le 2^{m-j}$, and $X_j \ge 2$ because $c_j \ge C_\star(\log\log T)^{-9/16}$. So (8.15) holds for every $m \le m^*$, which is the statement $\|\mathcal{L}e_j\|_{4,\mathcal{I}_T} \le 4B\sqrt{H_j}$ for $j < m^*$.

The last step. For $m \le m^*$ and $r = 1$, (8.15) at $j = m-1$ gives $A_{m-1}(2) \le 4B$, so the transport term of the m-th step is $\|\mathcal{L}(\tau_k)\mathfrak{t}_k^{(m)}\|_{2,\mathcal{I}_T} \le C_5(4B)^{9/8}$, with $\mathfrak{t}_k^{(m)} = S_{X_m}(\gamma_k) -$

$S_{X_m}(x_k^{(m-1)})$. Theorem 5.1 at $X = X_m$ ($2 \le X_m \le T^{1/160}$) gives $\langle u_k(X_m)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{c_m} + O(1+\sqrt{H_m}) = \frac{H_m}{4\pi^2} + O(1+\sqrt{H_m})$, the cross term is at most $2\|u(X_m)\|_2\, C_5(4B)^{9/8} \ll 1+\sqrt{H_m}$, and $\chi_{m,k} \ll T^{-1/2}$; (8.2) gives (8.12). At $m = m^*$, $H_{m^*} < H^*$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1+O(1/\log T))$ give (8.13). □

Corollary 8.11 (Third step). *With $c_3 = C_\star H_2^{-9/16}$ and $H_3 = \frac{9}{8}\log H_2 + 2\log(1/C_\star)$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{3,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_3}{4\pi^2} + O\Big(1+\sqrt{H_3}\Big), \qquad \frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(3)} - \gamma_k\Big)^2 = \frac{H_3 + O(\sqrt{H_3})}{\log^2 T},$$

and the multiplicity of the second clock satisfies $\int \tilde n_2^2 \ll \sqrt{H_2}\,\mathfrak{b}_T \#\mathcal{I}_T$.

Proof. Theorem 8.10 with $m = 3$ (for $T \ge T_0$, $H_2 \ge H^*$), and Theorem 8.9 with $A = A_2(2) \le 4B$. □

The cutoff of the third step, T^{c_3} with $c_3 \asymp H_2^{-9/16}$ instead of $H_2^{-1/2}$, is what the multiplicity costs: the exponent $\frac{9}{16} = \frac{1}{2} + \frac{1}{16}$ comes from the factor $\sqrt{H_2}$ of (8.10) through the exponent $\frac{1}{8r} \le \frac{1}{8}$ of Lemma 8.4. The constants of Fujii's and Montgomery–Vaughan's bounds, which grow with the moment order 2^{m-j} used at level j, are absorbed by the normalization $H_j^{-1/2}$, because H_j grows as a tower in $m-j$ while the orders grow geometrically.

Table 2 illustrates the iteration at $T = 2^{19}$. The asymptotic cutoffs are not reachable there ($H_0 = 2.58$, $c_1 = 0.62$, $c_2 = 1.03$), so fixed exponents are used; the third step with $c_3 = c_2 = \frac{1}{2}$ is worse than the second, while $c_3 > c_2$ still improves. The multiplicity integral $\int \tilde n^2/(\mathfrak{b}_T \#\mathcal{I}_T)$ is 1 for the clock and grows slowly with the step and the cutoff; the proportion of inversions $d_{2,k} \le 0$ is below 0.4%.

Table 2: Iterated Newton steps over $2^{19} < \gamma_k \le 2^{20}$ with fixed cutoff exponents (c_1, c_2, c_3): mean squares of $\mathcal{L}e_j$ in units of zeros ($j = 0$ is the bare clock), fourth moment and root mean square of the relative deformation $(d_{2,k} - d_{0,k})/d_{0,k}$, proportion of inversions $d_{2,k} \le 0$, and the multiplicity integrals of $x^{(1)}$ and $x^{(2)}$ at scale $\mathfrak{b}_T$ (that of the clock is 1).

(c_1,c_2,c_3)	$\langle(\mathcal{L}e_0)^2\rangle$	$\langle(\mathcal{L}e_1)^2\rangle$	$\langle(\mathcal{L}e_2)^2\rangle$	$\langle(\mathcal{L}e_3)^2\rangle$	$\langle\mathrm{def}^4\rangle$	RMS def	inv.	mult $x^{(1)}$	mult $x^{(2)}$
$(\frac{1}{10},\frac{1}{5},\frac{1}{3})$	0.1128	0.0655	0.0384	0.0157	$8.8\cdot10^{-3}$	0.216	0	1.000	1.043
$(\frac{1}{5},\frac{1}{3},\frac{1}{2})$	0.1128	0.0406	0.0158	0.0079	$5.3\cdot10^{-2}$	0.329	0	1.049	1.114
$(\frac{1}{3},\frac{1}{2},\frac{1}{2})$	0.1128	0.0235	0.0099	0.0116	$1.6\cdot10^{-1}$	0.478	0.0039	1.134	1.254

A bounded error in units of zeros with a *fixed* final cutoff T^{c_0} is not obtained by this method: the Newton term has mean square of order $c^2\log\log T$ unless transport and defect cancel (Remark 7.7), and no estimate here proves such a cancellation.

8.6. The transport identity in a prime frequency

The transport between the clock and the zeros has an exact Fourier expression. For real w and $t > 2\pi$ not an ordinate put

$$\begin{aligned} I_T(w) &= \frac{1}{T}\int_T^{2T} \exp\Big(iwt + \frac{iw\,G(t)}{\mathcal{L}(t)}\Big)dt, \\ J_T(w) &= \frac{1}{T}\sum_{k:\,\gamma_k,\gamma_{k+1}\in(T,2T]} (\tau_{k+1}-\tau_k)\,e^{iwv_k}, \qquad v_k = \tau_{k+1/2}, \end{aligned} \tag{8.16}$$

where $\tau_{k+1/2} = N_0^{-1}(k)$ by (2.5). The phase $t + G(t)/\mathcal{L}(t)$ is the time t transported by the Newton step of N_0 towards the integer level $N(t)$; J_T depends only on the clock.

Theorem 8.12 (Transport identity)**.** *Uniformly for $0 < w \le \log T$ and $T \ge T_0$,*

$$I_T(w) = J_T(w) + \frac{2i}{T} \sum_{k \in \mathcal{I}_T} \sin\Big(\frac{w}{2\mathcal{L}(\tau_k)}\Big) \delta_k \, e^{iw\tau_k} + O\Big(\frac{1}{T}\Big). \tag{8.17}$$

If moreover $\log 2 \le w \le (1-\varepsilon)\log\frac{T}{2\pi}$, then $J_T(w) \ll_\varepsilon T^{-1}$, so the first term may be dropped (the lower bound on w is needed: $J_T(w) \to 1$ as $w \to 0$).

Proof. One gap. On (γ_k, γ_{k+1}), $N = k$ and $G = k - N_0$; the phase is $w\,\phi(t)$ with $\phi(t) = t - (N_0(t) - k)/N_0'(t)$, the Newton map of N_0 towards its root $v_k = N_0^{-1}(k)$. Since $N_0'' = \mathcal{L}' = 1/(2\pi t) > 0$, $\phi'(t) = (N_0(t) - k)N_0''(t)/N_0'(t)^2$, so for t in the gap $|\phi(t) - v_k| \le |t - v_k| \max|G| \,/(2\pi T \mathcal{L}(T)^2)$, and $|t - v_k| \le |G(t)|/\mathcal{L}(T)$ by the mean value theorem (v_k lies within $O(\log T/\log T) = O(1)$ of the gap). With $|G| \ll \log T$ this gives $\phi(t) = v_k + O(1/(T\log T))$ and $e^{iw\phi(t)} = e^{iwv_k} + O(1/T)$ for $w \le \log T$. Integrating over the gaps inside $(T, 2T]$ and over the two end pieces, of bounded length (Theorem 3.1),

$$I_T(w) = \frac{1}{T} \sum_k (\gamma_{k+1} - \gamma_k) e^{iwv_k} + O\Big(\frac{1}{T}\Big).$$

Summation by parts. Write $\gamma_{k+1} - \gamma_k = (\tau_{k+1} - \tau_k) - (\delta_{k+1} - \delta_k)$. The first part gives $J_T(w)$. For the second, $-\sum_k (\delta_{k+1} - \delta_k) e^{iwv_k} = \sum_k \delta_k (e^{iwv_k} - e^{iwv_{k-1}}) + O(1)$, since $|\delta_k| \le 1$. By (2.5), $x \mapsto \tau_x$ has $d\tau_x/dx = 1/\mathcal{L}(\tau_x)$ and $d^2\tau_x/dx^2 = -\mathcal{L}'(\tau_x)/\mathcal{L}(\tau_x)^3 \ll 1/(T\log^3 T)$ on the window; hence $v_k - v_{k-1} = \tau_{k+1/2} - \tau_{k-1/2} = 1/\mathcal{L}(\tau_k) + O(1/(T\log^3 T))$ and $\frac{1}{2}(v_k + v_{k-1}) = \tau_k + O(1/(T\log^3 T))$ (midpoint rule and second difference). Therefore

$$e^{iwv_k} - e^{iwv_{k-1}} = e^{iw(v_k+v_{k-1})/2} \cdot 2i \sin\Big(\frac{w(v_k - v_{k-1})}{2}\Big) = 2i \sin\Big(\frac{w}{2\mathcal{L}(\tau_k)}\Big) e^{iw\tau_k} + O\Big(\frac{w}{T\log^3 T}\Big),$$

and the error, summed over the $\asymp T\log T$ indices with $|\delta_k| \le 1$ and divided by T, is $O(1/(T\log T))$. The indices k of the gaps and the set $\mathcal{I}_T$ differ by $O(1)$ elements, each contributing $O(1/T)$.

The clock sum. For $w \le (1-\varepsilon)\log\frac{T}{2\pi}$ let $\Omega(x) = wv_x/2\pi$ with $v_x = \tau_{x+1/2}$; $\Omega'(x) = w/\log(\tau_{x+1/2}/2\pi)$ is monotone and lies in $[w/\log(T/\pi), 1-\varepsilon]$ on the window, so $\|\Omega'\| \ge \min(w/\log(T/\pi), \varepsilon)$. The weights $\tau_{k+1} - \tau_k = 1/\mathcal{L}(\tau_{k+1/2})$ are positive, decreasing and $\le 2\pi/\log(T/2\pi)$. By partial summation and Theorem 3.9, $\sum_k (\tau_{k+1} - \tau_k) e^{iwv_k} \ll \frac{1}{\log T}\Big(\varepsilon^{-1} + \log T/w\Big)$, and dividing by T gives $O_\varepsilon(1/T)$ since $w \ge \log 2$. □

Corollary 8.13 (The weighted response at the clock points)**.** *Let p be a prime with $\theta = \log p/L_T \in [\theta_0, \theta_1] \subset (0,1)$, $w = \log p$, and*

$$\mathfrak{w}_k = \frac{\#\mathcal{I}_T}{T\mathcal{L}(\tau_k)} \cdot \frac{\sin(w/2\mathcal{L}(\tau_k))}{\sin(\pi\theta)}, \qquad \widetilde{\mathcal{R}}_p^{\tau}(\mathcal{I}_T) = -\frac{2\pi\sqrt{p}}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} \mathfrak{w}_k \, s_k \sin(\tau_k \log p).$$

Then $\mathfrak{w}_k = 1 + O_{\theta_0,\theta_1}(1/\log T)$ and

$$\Re I_T(\log p) = \frac{\sin(\pi\theta)}{\pi\sqrt{p}} \, \widetilde{\mathcal{R}}_p^{\tau}(\mathcal{I}_T) + O\Big(\frac{1}{T}\Big). \tag{8.18}$$

Proof. By Theorem 6.1(a), $\delta_k = (s_k + \chi_k)/\mathcal{L}(\tau_k)$ with $0 \le \chi_k \ll 1/T$; insert in (8.17), take real parts ($\Re(2i\delta_k e^{iw\tau_k}) = -2\delta_k \sin(w\tau_k)$), and compare with the definition of $\mathfrak{w}_k$. The weights: $\frac{\#\mathcal{I}_T}{T\mathcal{L}(\tau_k)} = 1 + O(1/\log T)$ by (2.4) and $\mathcal{L}(\tau_k) = \mathcal{L}(T) + O(1)$; and $w/2\mathcal{L}(\tau_k) = \pi\theta(1 + O(1/\log T))$. □

Removing the weights, that is passing from $\widetilde{\mathcal{R}}_p^\tau$ to $\mathcal{R}_p^\tau$, requires control of the partial sums of $s_k \sin(\tau_k \log p)$: a small relative change of the weights does not by itself give a small change of a normalized oscillating sum. Here the weights are a smooth function of τ_k, and the partial sums are controlled by Theorem 11.3 in the range of slowly moving primes.

Corollary 8.14 (Removing the weights for slowly moving primes)**.** *Uniformly for primes $p \le T^{1/8}$,*

$$\left|\widetilde{\mathcal{R}}_p^\tau(\mathcal{I}_T) - \mathcal{R}_p^\tau(\mathcal{I}_T)\right| \ll \frac{1}{\log T} + \frac{\sqrt{p}\,\log p\,\log\log T}{\log^2 T} + \frac{p\log T}{T},$$

so that $\widetilde{\mathcal{R}}_p^\tau - \mathcal{R}_p^\tau \to 0$ uniformly for $p \le (\log T)^{4-\varepsilon}$; in particular, for $p \le (\log T)^{2-\varepsilon}$, $\Re I_T(\log p) = \frac{\sin(\pi\theta)}{\pi\sqrt{p}}(1 + o(1))$. For $p = T^\theta$ with fixed $\theta > 0$ the bound is useless.

Proof. Let $\epsilon(t) = \frac{\#\mathcal{I}_T}{T\mathcal{L}(t)} \frac{\sin(\log p/2\mathcal{L}(t))}{\sin\pi\theta} - 1$, so that $\mathfrak{w}_k = 1 + \epsilon(\tau_k)$, $|\epsilon| \ll 1/\log T$ and, differentiating with $\mathcal{L}'(t) = 1/(2\pi t)$, $|\epsilon'(t)| \ll 1/(T\log T)$ on $[T-1, 2T+1]$; hence $|\epsilon(\tau_{k+1}) - \epsilon(\tau_k)| \ll 1/(T\log^2 T)$. Let $K_0 < K_1$ be the extreme indices of $\mathcal{I}_T$ and $\Sigma_K = \sum_{K_0<k\le K} s_k \sin(\tau_k \log p)$. By partial summation,

$$\widetilde{\mathcal{R}}_p^\tau - \mathcal{R}_p^\tau = -\frac{2\pi\sqrt{p}}{\#\mathcal{I}_T}\Big(\epsilon(\tau_{K_1})\Sigma_{K_1} - \sum_{K_0<k<K_1}\big(\epsilon(\tau_{k+1}) - \epsilon(\tau_k)\big)\Sigma_k\Big).$$

The proof of Theorem 11.3, applied on $[T, t]$ with $t = \gamma_K$ instead of $[T, 2T]$ (Lemma 4.7 on $[t_1, t_2]$ with t_2 a non-ordinate within $1/T$ of γ_K; Theorem 11.1 at t; $\int_T^t G^2 \le \int_T^{2T} G^2$), gives for every $K \in \mathcal{I}_T$

$$\Sigma_K = \frac{a_p}{2}(K - K_0) + O\Big(T\log p\,\log\log T + \sqrt{p}\log^3 T\Big),$$

the error being absolute. Insert: the boundary term is $\ll \frac{\sqrt{p}}{\#\mathcal{I}_T\log T}(\#\mathcal{I}_T p^{-1/2} + T\log p\log\log T + \sqrt{p}\log^3 T)$ and the sum is $\ll \frac{\sqrt{p}}{\#\mathcal{I}_T} \cdot \frac{\#\mathcal{I}_T}{T\log^2 T} \cdot (\#\mathcal{I}_T p^{-1/2} + T\log p\log\log T + \sqrt{p}\log^3 T)$; with $\#\mathcal{I}_T \asymp T\log T$ both are bounded by the right side. □

Table 3 shows the two responses differing by 1–11% at $T = 2^{19}$ for θ between 0.1 and 0.8, where the corollary gives nothing.

Proposition 8.15 (Resonance of the clock sum)**.** *Let p be a prime with $T/2\pi \le p \le T/\pi$, and suppose $\min(2\pi p - T, 2T - 2\pi p) \ge \kappa T$ for a fixed $\kappa > 0$. Then*

$$J_T(\log p) = \frac{2\pi\sqrt{p}}{T} + O_\kappa\Big(T^{-3/5}\Big).$$

Proof. Let $w = \log p$ and let $\mathcal{K} = \{k : \gamma_k, \gamma_{k+1} \in (T, 2T]\}$, an integer interval $(k_1, k_2]$ with $v_{k_1} = T + O(1)$, $v_{k_2} = 2T + O(1)$. Since $\tau_{k+1} - \tau_k = \int_k^{k+1} dx/\mathcal{L}(\tau_x) = 1/\mathcal{L}(v_k) + O(1/(T\log^3 T))$ (midpoint rule, $d^2\tau_x/dx^2 \ll 1/(T\log^3 T)$), $TJ_T(w) = \sum_{k\in\mathcal{K}} \mathfrak{g}(k) e^{2\pi i f(k)} + O(1/\log^2 T)$ with $\mathfrak{g}(x) = 1/\mathcal{L}(v_x)$ and $f(x) = wv_x/2\pi$. Here $f'(x) = w/\log(v_x/2\pi)$ is continuous and decreasing, with values in $[\log p/\log(T/\pi), \log p/\log(T/2\pi)] \subset [1 - o(1), 1 + o(1)]$ for $T/2\pi \le p \le T/\pi$; $\mathfrak{g}$ is positive and decreasing, $\mathfrak{g}'(x) = -\mathcal{L}'(v_x)/\mathcal{L}(v_x)^3 < 0$ with $|\mathfrak{g}'|$ decreasing. Theorem 3.12 with $\eta = \frac12$ leaves only $\nu = 1$:

$$TJ_T(w) = \int_{k_1}^{k_2} \frac{e^{i(wv_x - 2\pi x)}}{\mathcal{L}(v_x)}\, dx + O\Big(\frac{1}{\log T}\Big) = \int_{v_{k_1}}^{v_{k_2}} e^{iF(t)} dt + O\Big(\frac{1}{\log T}\Big),$$

with $F(t) = wt - 2\pi N_0(t)$, after the substitution $t = v_x$, $x = N_0(t)$, $dx = \mathcal{L}(t)\, dt$. Now $F'(t) = w - \log(t/2\pi) = -\log(t/2\pi p)$ vanishes exactly at $t_p = 2\pi p$, which lies at distance $\ge \kappa T/2$ from both endpoints for $T \ge T_0$; $F''(t) = -1/t \in [-1/T, -1/3T]$ and $F'''(t) = 1/t^2 \le 1/T^2$, so Theorem 3.10 applies with $\lambda_2 \asymp 1/T$, $\lambda_3 \asymp 1/T^2$, $|F'| \gg_\kappa 1$ at the endpoints:

$$\int_{v_{k_1}}^{v_{k_2}} e^{iF} dt = (2\pi)^{1/2} e^{iF(t_p) - i\pi/4} (2\pi p)^{1/2} + O(T^{2/5}) + O_\kappa(1).$$

Finally $F(t_p) = 2\pi p \log p - 2\pi\big(p\log p - p + \frac78\big) = 2\pi p - \frac{7\pi}{4}$, so $e^{iF(t_p) - i\pi/4} = e^{2\pi i(p-1)} = 1$ and the leading term is $2\pi\sqrt{p}$. Divide by T. □

The constant $\frac78$ of the Riemann–von Mangoldt formula makes the stationary value real and positive. The same mechanism, with the frequency $\log(pq)$ of a pair of primes, is the onset of the cross term of Section 10.7.

Proposition 8.16 (The cross term of the primes at the clock points)**.** *Let I be a set of consecutive indices with $\tau_k \in [T_1, T_2]$ for $k \in I$, $T_1 \le T_2 \le 2T_1$, $T_1 \ge T_0$, and let $2 \le X \le (T_1/2\pi)^{1/2}$. For integers $n \ge 2$ put $\mathfrak{d}(n) = \min(|2\pi n - T_1|, |2\pi n - T_2|)$. Then for every $n \le T_2/\pi$,*

$$\sum_{k\in I} e^{i\tau_k \log n} = -\sqrt{n}\, \log n\, \mathbf{1}_{[T_1 < 2\pi n < T_2]} + O\Big(T_1^{2/5}\log T_1 + \log T_1 \min\Big(\frac{T_1}{\mathfrak{d}(n)}, T_1^{1/2}\Big)\Big), \tag{8.19}$$

and, for the prime part $S_X^{(1)}(t) = \sum_{p\le X} a_p \sin(t\log p)$ with $\mathrm{Cr}_X^{(1)}(I) = \langle S_X^{(1)}(\tau_k)^2\rangle_I - \frac12 \sum_{p\le X} a_p^2$,

$$\begin{aligned}\mathrm{Cr}_X^{(1)}(I) = {} & \frac{1}{2\pi^2 \# I} \sum_{\substack{p,q\le X \\ T_1 < 2\pi pq < T_2}} \log(pq) \\ & + O\Big(\frac{X^2 \log T_1}{\# I \log^2 X} + \frac{\sigma_X^2}{\# I}\Big(T_1^{2/5}\log T_1 + \log T_1 \max_{p,q\le X} \min\Big(\frac{T_1}{\mathfrak{d}(pq)}, T_1^{1/2}\Big)\Big)\Big),\end{aligned} \tag{8.20}$$

the sum running over ordered pairs, $p = q$ included.

Proof. (8.19). Let $w = \log n$, $I = (k_1, k_2]$ and $f(x) = w\tau_x/2\pi$; $f'(x) = w/\log(\tau_x/2\pi)$ is continuous and decreasing. If $n < (T_1/2\pi)^{1/2}$ then $f' < \frac12$ on I and Theorem 3.11 with $\eta = \frac12$ leaves only $\nu = 0$: $\int_{k_1}^{k_2} e^{2\pi i f} dx \ll 1/\min f' \ll \log T_1/w$ (first-derivative test, f' monotone),

which is covered by the error term. Otherwise $f' \in [w/\log(T_2/2\pi), w/\log(T_1/2\pi)] \subset (\frac{1}{2} - \frac{\log 2}{\log T_1}, \frac{3}{2})$ for $n \le T_2/\pi$ and $T_1 \ge T_0$, so Theorem 3.11 with $\eta = \frac{1}{2}$ leaves $\nu = 1$ and, when $f'(k_2) \le \frac{1}{2}$, also $\nu = 0$, whose integral is $\ll 1/\min f' \ll 1$ by the first-derivative test; with error $O(1)$:

$$\sum_{k\in I} e^{iw\tau_k} = \int_{k_1}^{k_2} e^{i\Phi(x)}dx + O(1), \qquad \Phi(x) = w\tau_x - 2\pi x.$$

Here $\Phi'(x) = 2\pi\Big(w/\log(\tau_x/2\pi) - 1\Big)$ vanishes exactly at the x_0 with $\tau_{x_0} = 2\pi n$, i.e. $x_0 = N_0(2\pi n) + \frac{1}{2}$; $\Phi''(x) = -w\mathcal{L}'(\tau_x)/\mathcal{L}(\tau_x)^3$, so $-\Phi'' \asymp 1/(T_1\log^2 T_1) =: \lambda_2$ with ratio bounded on I, and $|\Phi'''| \ll \lambda_2/(T_1\log T_1) =: \lambda_3$. If $T_1 < 2\pi n < T_2$, Theorem 3.10 gives the leading term $(2\pi)^{1/2}e^{i\Phi(x_0)-i\pi/4}|\Phi''(x_0)|^{-1/2}$ with $|\Phi''(x_0)|^{-1/2} = (2\pi n)^{1/2}\mathcal{L}(2\pi n)$ (since $w = 2\pi\mathcal{L}(\tau_{x_0})$) and

$$\Phi(x_0) = 2\pi n\log n - 2\pi\Big(n\log n - n + \tfrac{7}{8} + \tfrac{1}{2}\Big) = 2\pi n - \tfrac{11\pi}{4}, \qquad e^{i\Phi(x_0)-i\pi/4} = e^{2\pi i n}e^{-3\pi i} = -1,$$

so the leading term is $-(2\pi)^{1/2}(2\pi n)^{1/2}\mathcal{L}(2\pi n) = -\sqrt{n}\log n$; the error terms are $\lambda_2^{-4/5}\lambda_3^{1/5} \ll T_1^{2/5}\log T_1$ and $\min(1/|\Phi'(k_i)|, \lambda_2^{-1/2})$ at the two endpoints, where $|\Phi'(k_i)| \asymp \mathfrak{d}(n)/(T_1\log T_1)$ and $\lambda_2^{-1/2} \asymp T_1^{1/2}\log T_1$. If $2\pi n \notin (T_1, T_2)$, Φ' does not vanish on I and the last clause of Theorem 3.10 gives the same bound without the leading term.

(8.20). Expand $S_X^{(1)}(\tau_k)^2$ with $\sin a\sin b = \frac{1}{2}\cos(a-b) - \frac{1}{2}\cos(a+b)$. The terms $\cos(\tau_k\log(p/q))$, $p \ne q$: $\Omega(x) = |\log(p/q)|\tau_x/2\pi$ has $\Omega' \le \log X/\log(T_1/2\pi) \le \frac{1}{2}$ and $\|\Omega'\| \ge |\log(p/q)|/\log(T_2/\pi) \ge 1/(2X\log T_2)$, so Theorem 3.9 gives $\ll X\log T_1$ per pair, and $\sum_{p\ne q}|a_pa_q| \le \sigma_X^2 \ll X/\log^2 X$. The terms $-\frac{1}{2}\cos(\tau_k\log pq)$, $p = q$ included, are $-\frac{1}{2}\Re$ of (8.19) with $n = pq$: the main term is $\frac{1}{2}\sqrt{pq}\log(pq)$ for resonant pairs, which multiplied by $a_pa_q = 1/(\pi^2\sqrt{pq})$ and divided by $\#I$ gives $\log(pq)/(2\pi^2\#I)$; the errors are bounded by σ_X^2 times the error of (8.19). □

In the top window of Section 10.7 ($T_1/2\pi = 95\,536$, $T_2/2\pi = 180\,241$, $X = 1999$) the 7 807 resonant ordered pairs give $\frac{1}{2\pi^2\#I}\sum\log(pq) = 0.004663$, against $\mathrm{Cr}_{1999}^{(1)} = 0.004662$ measured. The error terms of (8.20) are not small at this height ($X^2\log T_1/(\#I\log^2 X) \approx 1$): the bound for the non-resonant pairs is crude, and the measured non-resonant contribution, $5\cdot 10^{-6}$, is far below it. The endpoint term is of the size of the main term only for the pairs with $\mathfrak{d}(pq) \ll T_1^{1/2}$, about 20 of the 7 807; so the agreement to 0.02% is better than what (8.20) guarantees, and the comparison of Section 10.7 is now a theorem for the resonant main term, with the non-resonant part still a numerical observation at this height.

The identity holds to $2\cdot 10^{-7}$, the size of $1/T$. For $\theta < 1$ the integral is the displacement sum; for $\theta > 1$ it is the clock sum, within 0.2% of $2\pi\sqrt{p}/T$. The integral $I_T(\log p)$ therefore characterizes the response at the clock points, not at the ordinates, and the two differ (Figure 8); the identity says nothing about $\mathcal{R}_p^\gamma$. It does show that G and the phase p^{it} cannot be treated as independent, and that the natural scale of $\Re I_T(\log p)$ is $p^{-1/2}$ rather than $p^{-1/2}(\log T)^{-\theta^2}$, which an independence heuristic would predict. Whether $\mathcal{R}_p^\tau$ has a limit for fixed θ, as the plateau of Figure 8 suggests, is open.

Almost orthogonality of the prime phases, spectral energy, and sorted steps

The responses $\mathcal{R}_p^\tau$ are the coordinates of the vector $(s_k)_{k\in\mathcal{I}_T}$ in the family $\sin(\tau_k\log p)$. Kusmin–Landau makes this family almost orthonormal for $p \le T^b$ with $b < \frac{1}{2}$; for $\frac{1}{2} \le b < 1$

Table 3: Transport identity (8.17) at $T = 2^{19}$ ($\#\mathcal{I}_T = 977\,804$). I: the twisted integral, computed gap by gap; D: the displacement sum $\frac{2i}{T}\sum \sin(\cdot)\delta_k p^{i\tau_k}$; J: the clock sum; $\mathcal{R}_p^\tau$ and $\widetilde{\mathcal{R}}_p^\tau$ as in (6.1) and Corollary 8.13. The last two primes are in the resonance range of Proposition 8.15, where $2\pi\sqrt{p}/T$ is $3.790\cdot10^{-3}$ and $4.339\cdot10^{-3}$.

p	θ	$\Re I$	$\Re D$	$\Re J$	$\lvert I-D-J\rvert$	$\mathcal{R}_p^\tau$	$\widetilde{\mathcal{R}}_p^\tau$
3	0.097	$5.1955\cdot10^{-2}$	$5.1954\cdot10^{-2}$	$9\cdot10^{-7}$	$1.9\cdot10^{-7}$	0.974	0.943
101	0.407	$2.4976\cdot10^{-2}$	$2.4975\cdot10^{-2}$	$9\cdot10^{-7}$	$2.1\cdot10^{-7}$	0.835	0.823
1009	0.610	$6.7591\cdot10^{-3}$	$6.7581\cdot10^{-3}$	$8\cdot10^{-7}$	$2.0\cdot10^{-7}$	0.703	0.717
10007	0.813	$1.2325\cdot10^{-3}$	$1.2330\cdot10^{-3}$	$-6\cdot10^{-7}$	$1.9\cdot10^{-7}$	0.626	0.699
100003	1.016	$3.7844\cdot10^{-3}$	$2.7\cdot10^{-6}$	$3.7816\cdot10^{-3}$	$1.9\cdot10^{-7}$	0.026	–
131101	1.040	$4.3330\cdot10^{-3}$	$1.1\cdot10^{-6}$	$4.3321\cdot10^{-3}$	$1.9\cdot10^{-7}$	−0.005	–

the frequencies $\log(pq)$ cross the stationary band of the clock, and it is the sparseness of the resonant pairs (Proposition 8.16) that keeps the Gram matrix close to the identity.

Proposition 8.17 (Almost orthogonality at the clock points)**.** *Let $0 < b < 1$, $M = \#\mathcal{I}_T$, $v_p(k) = \sqrt{2/M}\,\sin(\tau_k \log p)$ for primes $p \le T^b$, and $G = (\langle v_p, v_q\rangle)_{p,q\le T^b}$ with the inner product of $\mathbb{R}^M$. Then*

$$\|G - I\|_{\mathrm{op}} \ll_b (\log T)^{-1/4} + T^{b-1}\log T. \tag{8.21}$$

Proof. Let $C_T(y) = \frac{1}{M}\sum_{k\in\mathcal{I}_T} e^{i\tau_k \log y}$ for real $y > 0$. By $2\sin a \sin b = \cos(a-b) - \cos(a+b)$, $\langle v_p, v_q\rangle = \Re C_T(p/q) - \Re C_T(pq)$, so $G - I = D - H$ with $D_{pq} = \Re C_T(p/q)\mathbf{1}_{p\ne q}$ and $H_{pq} = \Re C_T(pq)$.

Quotients. For $p \ne q$ the phase $\Omega(x) = \tau_x \log(p/q)/2\pi$ has $\Omega'(x) = \log(p/q)/\log(\tau_x/2\pi)$, monotone with $0 < |\Omega'| \le b\log T/\log(T/2\pi) < 1$, so $\|\Omega'\| \ge |\log(p/q)|/\log(T/\pi)$ and Theorem 3.9 gives $|C_T(p/q)| \ll 1/(T|\log(p/q)|)$. Since $|\log(p/q)| \ge |p-q|/\max(p,q)$, the row sums satisfy $\sum_{q\ne p}|D_{pq}| \ll T^{-1}\sum_{q\le T^b, q\ne p}\max(p,q)/|p-q| \ll T^{b-1}\log T$, and Schur's test gives $\|D\|_{\mathrm{op}} \ll T^{b-1}\log T$.

Products in the stationary band. Let $\mathcal{S} = \{n : T/\sqrt{\log T} \le n \le T\sqrt{\log T}\}$. For $n \in \mathcal{S}$ the phase $f(x) = \tau_x \log n/2\pi$ has $f''(x) = -\log n\,\mathcal{L}'(\tau_x)/\mathcal{L}(\tau_x)^3$, so $-f'' \asymp 1/(T\log^2 T)$ on the window with bounded ratio, and Theorem 3.13 gives

$$\Big|\sum_{k\in\mathcal{I}_T} e^{i\tau_k\log n}\Big| \ll \frac{M}{(T\log^2 T)^{1/2}} + (T\log^2 T)^{1/2} \ll \sqrt{T}\log T, \qquad \text{i.e.} \quad |C_T(n)| \ll T^{-1/2}.$$

The number of ordered pairs (p,q) of primes $\le T^b$ with $pq \in \mathcal{S}$ is at most $\sum_p \#\{q \le T\sqrt{\log T}/p\}$ over $p \ge T^{1-b}/\sqrt{\log T}$, which is $\ll \frac{T\sqrt{\log T}}{\log T}\sum_{T^{1-b}/\sqrt{\log T}\le p\le T^b}\frac{1}{p} \ll_b T/\sqrt{\log T}$ by Theorem 3.8. Hence $\sum_{pq\in\mathcal{S}}|H_{pq}|^2 \ll_b (\log T)^{-1/2}$.

Products outside the band. If $pq < T/\sqrt{\log T}$ or $pq > T\sqrt{\log T}$, then $f'(x) = \log(pq)/\log(\tau_x/2\pi)$ differs from 1 by $\gg \log\log T/\log T$ and is $\le 2b < 2$, and it is $\ge \log 4/\log T$; so $\|f'\| \gg \log\log T/\log T$ and Theorem 3.9 gives $|C_T(pq)| \ll \log T/(M\log\log T) \ll 1/T$. Hence $\sum_{pq\notin\mathcal{S}}|H_{pq}|^2 \ll T^{2b-2}$.

Together, $\|H\|_{\mathrm{op}} \le \|H\|_{\mathrm{HS}} \ll_b (\log T)^{-1/4} + T^{b-1}$, and (8.21) follows. □

Proposition 8.18 (Spectral energy and collective transfer)**.** *Let $0 < b < 1$ and let $(r_k)_{k\in\mathcal{I}_T}$ be any real sequence. Then*

$$\sum_{p\le T^b}\frac{1}{p}\Big(2\pi\sqrt{p}\,\big\langle r_k\sin(\tau_k\log p)\big\rangle_{\mathcal{I}_T}\Big)^2 \le 2\pi^2\Big(1+o_b(1)\Big)\langle r_k^2\rangle_{\mathcal{I}_T}. \tag{8.22}$$

In particular:

(a) $\sum_{p\le T^b}|\mathcal{R}_p^\tau(\mathcal{I}_T)|^2/p \le 2\pi^2(1+o_b(1))\langle s_k^2\rangle_{\mathcal{I}_T} = \log\log T + O_b(\sqrt{\log\log T})$.

(b) For every sequence of points x_k computed from the clock and the primes, with $\mathcal{A}_p = -2\pi\sqrt{p}\langle\mathcal{L}(\tau_k)(\tau_k - x_k)\sin(\tau_k\log p)\rangle_{\mathcal{I}_T}$ (computable without ordinates) and $r_k = \mathcal{L}(\tau_k)(x_k-\gamma_k)$,

$$\sum_{p\le T^b}\frac{|\mathcal{R}_p^\tau(\mathcal{I}_T)-\mathcal{A}_p|^2}{p} \le 2\pi^2\Big(1+o_b(1)\Big)\langle r_k^2\rangle_{\mathcal{I}_T}, \qquad \sum_{\substack{p\le T^b\\ |\mathcal{R}_p^\tau-\mathcal{A}_p|>\varepsilon}}\frac{1}{p} \le \frac{2\pi^2+o_b(1)}{\varepsilon^2}\langle r_k^2\rangle_{\mathcal{I}_T}. \tag{8.23}$$

(c) With $x_k = x_k^{(m^*)}$ of Theorem 8.10, $\sum_{p\le T^b}|\mathcal{R}_p^\tau - \mathcal{A}_p^{(m^*)}|^2/p \le C_b$: the computable response of the final corrected clock reproduces $\mathcal{R}_p^\tau$ with a total error $O(1)$ in $L^2(dp/p)$, against the mass $\log(b/a)$ of the range $T^a \le p \le T^b$.

(d) If, uniformly for $T^a \le p \le T^b$, $\mathcal{R}_p^\tau - \mathcal{A}_p = h(\log p/\log T) + o(1)$ with h continuous, then $\liminf_T\langle r_k^2\rangle_{\mathcal{I}_T} \ge \frac{1}{2\pi^2}\int_a^b|h(\theta)|^2\theta^{-1}d\theta$.

Proof. For an almost orthonormal family, $\sum_p|\langle r,v_p\rangle|^2 \le \|G\|_{\mathrm{op}}\|r\|^2$ (expand $\|\sum_p c_pv_p\|^2 = c^{\mathsf{T}}Gc$ with $c_p = \langle r,v_p\rangle$ and apply Cauchy–Schwarz), and $\|G\|_{\mathrm{op}} \le 1+o_b(1)$ by Proposition 8.17. Since $\langle r,v_p\rangle = \sqrt{2M}\,\langle r_k\sin(\tau_k\log p)\rangle_{\mathcal{I}_T}$ and $\|r\|^2 = M\langle r_k^2\rangle$, this is $2\sum_p\langle r_k\sin(\tau_k\log p)\rangle^2 \le (1+o_b(1))\langle r_k^2\rangle$; multiply by $4\pi^2$. (a) is $r_k = s_k$ with (3.3); (b) is $r_k = \mathcal{L}(\tau_k)(x_k-\gamma_k)$, for which $\mathcal{R}_p^\tau - \mathcal{A}_p = -2\pi\sqrt{p}\langle r_k\sin(\tau_k\log p)\rangle$ by $s_k = \mathcal{L}(\tau_k)(\tau_k-x_k)+r_k+O(1/T)$, and Chebyshev's inequality; (c) is (b) with (8.12); (d) follows from (b) and $\sum_{T^a\le p\le T^b}|h(\log p/\log T)|^2/p \to \int_a^b|h|^2/\theta$ (Theorem 3.8 and partial summation). □

The inequality in (a) is almost saturated by the small primes, for which $\mathcal{R}_p^\tau \to 1$ (Theorem 11.3) and $\sum_{p\le T^b}1/p = \log\log T + \log b + O(1)$; it constrains the response of the moving primes only through the $O(\sqrt{\log\log T})$. Statement (c) is what the mean-square control of Theorem 8.10 says about the responses: an $o(1)$ in (8.12) would give $\mathcal{R}_p^\tau = \mathcal{A}_p^{(m^*)} + o(1)$ for almost all moving primes in harmonic density, by (b), and (d) shows that, conversely, a residual that is spectrally invisible in $[a,b]$ cannot have vanishing energy if the responses are reproduced there.

Lemma 8.19 (Sorted Newton steps)**.** *Let $\gamma_1 \le \cdots \le \gamma_M$ and let $z_1,\dots,z_M$ be arbitrary reals, with $z^\uparrow$ the same values in increasing order. Then $\sum_k|z_k^\uparrow - \gamma_k|^q \le \sum_k|z_k-\gamma_k|^q$ for every $q \ge 1$, and $\{z_k^\uparrow\} = \{z_k\}$ as multisets. Consequently, replacing $x_k^{(j)}$ by its sorted version at any level of* (8.1) *does not increase $\|e_j\|_{r,\mathcal{I}_T}$ for any r, does not change the multiplicity function $\tilde{n}_j$, and leaves Theorems 8.5, 8.9 and 8.10 unchanged; it removes the inversions $d_{j,k} \le 0$ of Proposition 8.7.*

Proof. For $a \le b$, $x \le y$ and φ convex, $\varphi(a-x)+\varphi(b-y) \le \varphi(a-y)+\varphi(b-x)$ (the points $a-y \le a-x, b-y \le b-x$ have the same sum in both pairings, and the outer pair majorizes the inner one). With $\varphi(t)=|t|^q$, each transposition of an inverted pair of z's does not increase the sum, and finitely many transpositions sort z. The recursion (8.3) uses only $\|e_{j-1}\|_{4r}$ and the sampling of g_j on the point set $\{x_k^{(j-1)}\}$, both unchanged or improved. □

9. Theorem E: the pointwise envelope

Let $\mathfrak{a} = \limsup_{t\to\infty} |S(t)|/\log t$. By Theorem 3.2, $\mathfrak{a} \le 0.10076$; whether $\mathfrak{a} = 0$ is not known.

Proposition 9.1 (The envelope of the clock is the envelope of S)**.** *(a)* $\limsup_{k\to\infty} |\tau_k - \gamma_k| = 2\pi\mathfrak{a}$. *(b) If* $X_k \ge 2$ *and* $\sigma_{X_k} = o(\log\tau_k)$*, in particular if* $X_k \le (\log\tau_k)^{2-\varepsilon}$*, then* $\limsup_{k\to\infty} |\tau_k^{X_k} - \gamma_k| = 2\pi\mathfrak{a}$.

Proof. (a) The values $S(\gamma_k^{\pm})$ are limits of values of S at non-ordinates. For $\varepsilon > 0$, (2.8) gives $|s_k| \le (\mathfrak{a}+\varepsilon)\log\gamma_k + O_\varepsilon(1)$, and $|\delta_k| = |s_k|/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log\gamma_k}{2\pi}(1+o(1))$ since $\delta_k = O(1)$; so $\limsup|\delta_k| \le 2\pi\mathfrak{a}$. Conversely, let $t > \gamma_1$ be a non-ordinate, let $\gamma < t < \gamma'$ be the nearest ordinates and $k = N(t)$, so that $\gamma_k = \gamma$ and $\gamma_{k+1} = \gamma'$. On (γ,γ') the function G is continuous and decreasing ($G' = -\mathcal{L}$), so $G(\gamma'^-) \le G(t) \le G(\gamma^+)$, while $s_k = k - \frac12 - N_0(\gamma) = G(\gamma^+) - \frac12$ and $s_{k+1} = k + \frac12 - N_0(\gamma') = G(\gamma'^-) + \frac12$. Hence $s_k \ge G(t) - \frac12$ and $s_{k+1} \le G(t) + \frac12$, so $\max(|s_k|, |s_{k+1}|) \ge |G(t)| - \frac12$, and $|G(t) - S(t)| = |\varpi(t)|/\pi \to 0$. Both ordinates are within bounded distance of t (Theorem 3.1). Choosing non-ordinates $t \to \infty$ with $|S(t)| \ge (\mathfrak{a}-\varepsilon)\log t$ gives indices $k \to \infty$ with $|\delta_k| = |s_k|/\mathcal{L}(\xi_k) \ge 2\pi(\mathfrak{a}-\varepsilon) + o(1)$.

(b) $|\tau_k^{X_k} - \tau_k| = |S_{X_k}(\tau_k)|/\mathcal{L}(\tau_k) \le \sigma_{X_k}/\mathcal{L}(\tau_k) \to 0$, and $\sigma_X \ll \sqrt{X}/\log X$. □

So the corrected clocks with slowly growing cutoffs have exactly the pointwise envelope of the bare clock, and for every cutoff the envelope of τ_k is that of S. A smaller coefficient for the clock is a smaller coefficient for $S(t)$.

Proposition 9.2 (Finite-height envelope)**.** *For every* $k \ge 1$ *let* $E_k = \mathrm{BW}_S(\gamma_k) - \frac12 + |\varpi(\gamma_k)|/\pi$*, and* $E_k = 2.5167 - \frac12 + |\varpi(\gamma_k)|/\pi$ *if* $\gamma_k \le 30\,610\,046\,000$*. Then* $|s_k| \le E_k$ *and*

$$-\frac{E_k}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{E_k}{\mathcal{L}(\gamma_k)}. \tag{9.1}$$

Proof. Theorem 3.2 at $\gamma_k^{\pm}$ (one-sided limits; the bounds are continuous in t) and (2.8) give $|s_k| \le E_k$. Since $\mathcal{L}$ is positive and increasing on $(2\pi,\infty)$ and $s_k = \int_{\gamma_k}^{\tau_k} \mathcal{L}$, $s_k \ge \mathcal{L}(\gamma_k)\delta_k$ if $\delta_k \ge 0$ and $s_k \le \mathcal{L}(\tau_k)\delta_k$ if $\delta_k < 0$. □

Since $|\varpi|/\pi < 0.01$, the numerator of the lower bound in (9.1) is at most $0.10076\log\gamma_k + 0.24460\log\log\gamma_k + 6.71844$, against $\ldots + 8.45844$ in (3.4); the upper bound of (3.4) is smaller by 0.01 for large γ_k, and the minimum of the two may be used. In the range of Odlyzko's table both are superseded by the bound from 2.5167: $|\delta_k| \le 2.0267/\mathcal{L}(\cdot)$, which is 1.047 at $k = K$ (Section 10.2).

Theorem 9.3 (Theorem E)**.** *Let $\mathfrak{e}$ be continuous on $[\frac12,1]$ with $\mathfrak{e}(1)=0$, such that for every $\varepsilon>0$, $\zeta(\sigma+it)\ll_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon}$ uniformly for $\frac12\le\sigma\le1$ and $|t|\ge2$, and put $\mathfrak{e}(\sigma)=\frac12-\sigma+\mathfrak{e}(1-\sigma)$ for $0\le\sigma<\frac12$. Then*

$$\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^1\frac{\mathfrak{e}(\sigma)}{\sqrt{\sigma(2-\sigma)}}\,d\sigma. \tag{9.2}$$

For the piecewise linear $\mathfrak{e}_{\mathrm B}$ through $(\frac12,\frac{13}{84})$, $(\frac{139}{194},\frac{13}{194})$, $(\frac{173}{207},\frac{13}{414})$, $(\frac{773}{854},\frac{13}{854})$, $(\frac{820}{867},\frac{13}{1734})$, $(\frac{3387}{3494},\frac{13}{3494})$, $(\frac{1149}{1169},\frac{13}{7014})$, $(\frac{13921}{14054},\frac{13}{14054})$, $(1,0)$, the right side is $0.0968160063\ldots$; hence

$$\limsup_{k\to\infty}|\tau_k-\gamma_k|=2\pi\mathfrak{a}\le0.6083129\ldots<0.60832. \tag{9.3}$$

Proof. Uniform bounds. Extend $\mathfrak{e}$ by 0 on $[1,\infty)$ and by $\frac12-\sigma$ on $(-\infty,0)$; the extension $\mathfrak{e}^*$ is continuous ($\mathfrak{e}(0)=\frac12+\mathfrak{e}(1)=\frac12$). For $\sigma\ge1$, $\zeta(\sigma+it)\ll\log|t|$ uniformly [33, Thm. 3.5]. For $-\frac{1}{50}\le\sigma<\frac12$, $\zeta(\sigma+it)=\chi(\sigma+it)\zeta(1-\sigma-it)$ with $|\chi(\sigma+it)|=(|t|/2\pi)^{1/2-\sigma}(1+O(1/|t|))$ uniformly [33, (4.12.3)], and $|\zeta(1-\sigma-it)|=|\zeta(1-\sigma+it)|$. Hence $\zeta(\sigma+it)\ll_\varepsilon|t|^{\mathfrak{e}^*(\sigma)+\varepsilon}$ uniformly for $-\frac{1}{50}\le\sigma\le4$, $|t|\ge2$.

Backlund's method. Let $0<\eta<\frac{1}{100}$, $\sigma_\circ=1+\eta$, $\varrho=1+2\eta$. Then $1<\sigma_\circ<\varrho$, $\sigma^\sharp=\frac54+\frac32\eta$, $\delta^\sharp=\frac14+\frac{\eta}{2}\in(0,\frac12)$, and $\varrho/(\sigma_\circ-\frac12)=2$. For $s=\sigma_\circ+\varrho e^{i\theta}$, $\Re s\in[-\eta,2+3\eta]$ and $|\Im s|\le2$, so for T large

$$\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(s)|\le\log\max_{\pm}|(s\pm iT-1)\zeta(s\pm iT)|\le\log(T+4)+(\mathfrak{e}^*(\Re s)+\varepsilon)\log(T+2)+C_\varepsilon=:F_\circ(\theta),$$

an even continuous function. By Theorem 3.21(a), along $\varkappa_m\to\infty$,

$$\limsup_m\mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho)\le\log T\cdot\frac{1}{\pi}\int_0^\pi\Big(\mathfrak{e}^*(\sigma_\circ+\varrho\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\eta}(1),$$

the terms $\log(T+4)$ and $-\log\sqrt{\eta^2+T^2}$ cancelling up to $O(1/T)$. The argument defining $S(T)$ runs along $2\to2+iT\to\sigma^\sharp+iT\to\frac12+iT$, and on the first two segments $\Re s>1$, where $\log\zeta$ is holomorphic; so $\pi S(T)=\arg\zeta(\sigma^\sharp+iT)+\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg((\sigma+iT-1)\zeta(\sigma+iT))-\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg(\sigma+iT-1)$, with $|\arg\zeta(\sigma^\sharp+iT)|\le\log\zeta(\sigma^\sharp)$ and the last term bounded by π. Theorem 3.21(b) with $\varkappa=\varkappa_m\to\infty$ gives, for T large,

$$|S(T)|\le\frac{\log T}{2\log2}\cdot\frac{1}{\pi}\int_0^\pi\Big(\mathfrak{e}^*(1+\eta+(1+2\eta)\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\eta}(1).$$

Limit. Divide by $\log T$, let $T\to\infty$, then $\varepsilon\to0$ and $\eta\to0$; $\mathfrak{e}^*$ is bounded and continuous, so by dominated convergence $\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^\pi\mathfrak{e}^*(1+\cos\theta)\,d\theta$. The integrand vanishes for $0\le\theta\le\frac{\pi}{2}$; on $[\frac{\pi}{2},\pi]$ substitute $\sigma=1+\cos\theta$, $d\sigma=-\sin\theta\,d\theta$, $\sin\theta=\sqrt{\sigma(2-\sigma)}$, which gives (9.2).

The profile. Let $\varepsilon'>0$, $(\mathsf k_0,\mathsf l_0)=(\frac{13}{84}+\varepsilon',\frac{55}{84}+\varepsilon')$ and $(\mathsf k_j,\mathsf l_j)=A^j(\mathsf k_0,\mathsf l_0)$ for $1\le j\le7$, exponent pairs by Theorem 3.19(a),(c), with $\mathsf l_j-\mathsf k_j\ge\frac12$; by (b), $|\zeta(\mathsf l_j-\mathsf k_j+it)|\ll|t|^{\mathsf k_j}\log|t|$, and $|\zeta(1+it)|\ll\log|t|$. The points $(\mathsf l_j-\mathsf k_j,\mathsf k_j)$ and $(1,0)$ have decreasing ordinates, and Theorem 3.20 between consecutive points gives the hypothesis of the theorem for the piecewise linear function through them. For $\varepsilon'=0$ these are the vertices listed in the statement; the right side of (9.2) depends continuously on ε', and its value at $\varepsilon'=0$, computed piece by piece with 40-digit quadrature by `opc_envelope_coefficient_v14.py`, is $0.0968160063238\ldots$. Multiply by 2π and use Proposition 9.1(a). □

Remark 9.4. (a) With the explicit exponents used by Bellotti and Wong ($\frac{27}{164}$ at $\sigma=\frac12$, Yang's lines $\sigma_j=1-j/(2^j-2)$ with exponents $1/(2^j-2)$, $4\le j\le 9$ [2]) and linear interpolation between $\sigma=0$ and $\sigma=\frac12$, as in [2, §3.1], the right side of (9.2) is 0.10074, the limit of their $C_1=0.10076$ as their parameters tend to $(1,1)$. The same right-hand profile reflected by the functional equation gives 0.09783, i.e. $2\pi\mathfrak{a}\le 0.6147$, with explicit inputs only; this uses that $\mathfrak{e}$ is convex, so that the reflected profile lies below the chord. (b) For the Lindelöf profile ($\mathfrak{e}=0$ on $[\frac12,1]$) the right side is 0.07863 ($2\pi\cdot$ it is 0.4940): Backlund's method with a disc cannot give less, whatever the exponents. (c) Theorem 9.3 is asymptotic: the explicit constants of the bounds for ζ do not enter, which is why non-explicit exponents may be used.

Proposition 9.5 (An explicit bound valid for all $T\ge e$)**.** *For $T\ge e$,*

$$\begin{aligned}&|S(T)|\le 0.09783\log T+0.722\log\log T+13,\\&\Big|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}\Big|\le 0.09783\log T+0.722\log\log T+14.\end{aligned}\tag{9.4}$$

Proof. Explicit profile. Let $\sigma_m=1-m/(2^m-2)$, $4\le m\le 9$. From the bounds collected in [2, (2.1), (2.3), (2.4)], for $|t|\ge 3$: $|\zeta(\frac12+it)|\le 2|t|^{27/164}\log|t|$ (below e^{105} from $0.618|t|^{1/6}\log|t|$, since $|t|^{1/492}\le 2/0.618$ there; above, $66.7|t|^{27/164}\le 2|t|^{27/164}\log|t|$), $|\zeta(\sigma_m+it)|\le 2|t|^{1/(2^m-2)}\log|t|$ and $|\zeta(1+it)|\le 2\log|t|$. Let $g_{\mathrm e}$ be the linear interpolation through $(\frac12,\frac{27}{164})$, $(\sigma_m,\frac{1}{2^m-2})$, $(1,0)$, extended by 0 on $[1,3]$. The Phragmén–Lindelöf principle [2, Lemma 2.5] for $(s-1)\zeta(s)$ with shift $Q=10$, the boundary values for $|t|\le 3$ being bounded by the elementary integral representation of ζ, gives $|\zeta(\sigma+it)|\le 10\,t^{g_{\mathrm e}(\sigma)}\log t$ for $\frac12\le\sigma\le 3$, $t\ge 10$. With $|\chi(\sigma+it)|\le 3t^{1/2-\sigma}$ for $-0.01\le\sigma\le\frac12$, $t\ge 10$ (Stirling's formula with explicit remainder), the reflected profile $h_{\mathrm e}$ ($h_{\mathrm e}=g_{\mathrm e}$ on $[\frac12,1]$, $\frac12-\sigma+g_{\mathrm e}(1-\sigma)$ on $[0,\frac12]$, $\frac12-\sigma$ below 0, 0 above 1) satisfies $|\zeta(\sigma+it)|\le 30\,t^{h_{\mathrm e}(\sigma)}\log t$ for $-0.01\le\sigma\le 3$, $t\ge 10$.

Fixed parameters. Take $\sigma_\circ=1.000002$, $\varrho=1.000005$ and, in the notation of [2, Thm. 1.2], $\eta=10^{-6}$; then $\sigma_\circ-\varrho<1-\sigma_\circ<-\eta$ and $\delta^\sharp\in(\frac14,\frac12)$. Put $D=\log(\varrho/(\sigma_\circ-\frac12))$ and $I=\int_0^\pi h_{\mathrm e}(\sigma_\circ+\varrho\cos\theta)\,d\theta$. On each linear piece $h_{\mathrm e}(\sigma)=\alpha\sigma+\beta$ a primitive is $(\alpha\sigma_\circ+\beta)\theta+\alpha\varrho\sin\theta$, the endpoints being arccosines of rationals, and interval arithmetic with directed rounding gives $0.097829231878517946338 76<I/(2\pi D)<0.097829231878517946338 77$ and $1/(2D)<0.721346479766175634597 98$.

Backlund's method. For $T\ge 100$ the bound for ζ gives $\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ+\varrho e^{i\theta})|\le(1+h_{\mathrm e}(\sigma_\circ+\varrho\cos\theta))\log(T+3)+\log\log(T+3)+\log 30$, and $\zeta(\sigma_\circ)/\zeta(2\sigma_\circ)\le 1+1/(\sigma_\circ-1)=500001$. Theorem 3.21 and the decomposition of the argument in the proof of Theorem 9.3 give $|S(T)|\le\frac{1}{2D}\big(-\log T+\log 500001+(1+\frac{I}{\pi})\log(T+3)+\log\log(T+3)+\log 30\big)+R(T)$, where $R(T)<1$ collects $\frac14$, the variation of $\arg(s-1)$, $E(T,\delta^\sharp)/2\pi$ and $\pi^{-1}\log\zeta(\sigma^\sharp)$. Since $\log(30\cdot 500001)/(2D)<11.92$ and the replacement of $\log(T+3)$ by $\log T$ costs less than 0.05, the constant is <13 for $T\ge 100$; for $e\le T\le 100$, $|S(T)|\le 2.5167$ (Theorem 3.2). Finally $N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}=\frac78+S(T)+\varpi(T)/\pi$ and $\frac78+|\varpi|/\pi<1$. □

The coefficient 0.09783 is the value of (9.2) for the explicit profile reflected by the functional equation (Remark 9.4(a)), now reached at fixed admissible parameters. The lower-order terms are larger than those of (3.1), and (9.4) is smaller than $\mathrm{BW}_S(T)$ only for $T\ge e^{4000}$; the minimum of the two bounds is valid for all $T\ge e$.

10. Computations

10.1. Data and methods

The ordinates are those of Odlyzko's table of the first $K = 2\,001\,052$ zeros of the Riemann zeta function, accurate to within $4 \cdot 10^{-9}$ [27], converted to IEEE double precision by correctly rounded decimal-to-binary conversion ($\gamma_K = 1\,132\,490.658714\ldots$). Spot checks against `mpmath.zetazero` at eight indices differ by at most $3.5 \cdot 10^{-10}$. The clock values τ_k are the double-precision numbers nearest to the exact roots of (2.5). They are computed by Newton's method in extended precision and certified for every k by evaluating $N_0 - (k - \frac{1}{2})$ at the rounded value and at both neighbouring floating-point numbers; the 20 679 indices at which the root lies within the evaluation error of a rounding midpoint are decided with `mpmath` at 40 significant digits, and an independent `mpmath` recomputation at 19 900 indices, including the first and last 100, agrees bit for bit. The Euler phase uses full local factors, evaluated either as $-\pi^{-1} \sum_{p \le X} \arg(1 - p^{-1/2-it})$ with the principal argument, or by (2.10) truncated where $p^{-m/2} \le 10^{-17}$. All means are over complete windows, without subsampling or fitted parameters. Windows of ordinals are $(0, 10^3]$, $(10^3, 10^4]$, $(10^4, 10^5]$, $(10^5, 5 \cdot 10^5]$, $(5 \cdot 10^5, 10^6]$, $(10^6, K]$; windows of height are $(T, 2T]$.

10.2. The clock and the ordinates

Figure 1 shows $\delta_k = \tau_k - \gamma_k$ for $k \le 2000$ together with the residual of the one-step clock with $\mathcal{P} = \{2, 3, 5, 7\}$. The root mean square of δ_k decreases from 0.3587 on $(0, 10^3]$ to 0.1789 on $(10^6, K]$, of the order of magnitude of $(2 \log\log T)^{1/2} / \log T$ (Remark 3.22). The maxima per window decrease only from 0.9937 to 0.7174 (Table 6); all of them exceed 0.6331, which bounds only the limit superior. The finite-height envelope (3.4) holds at every index and is used to at most 15% of its size. The envelope (9.1) with the bound 2.5167, valid for all ordinates of the table, gives $|s_k| \le 2.0267$ and is used to at most 66%: $\max_k |s_k| = 1.3325$, at $k = 1\,249\,694$ ($\gamma_k = 735\,839.17$, $\delta_k = -0.7174$).

10.3. Direct defects (Theorem B)

For $T \in \{2^{14}, 2^{16}, 2^{18}, 2^{19}\}$ and $c \in \{\frac{1}{10}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}\}$ we compute $u_k(X)$ at the true ordinates with $X = T^c$ constant on $(T, 2T]$; the windows contain 21 519, 100 536, 459 982 and 977 804 ordinates. Smaller exponents give $X < 2$ at these heights and are excluded, as in Theorem 5.1.

At fixed c the measured means are essentially independent of T, as Theorem 5.1 requires of a bounded quantity, and they decrease monotonically in c. Between $c = \frac{1}{2}$ and $c = \frac{1}{10}$ the increase of $\langle u_k^2 \rangle$ at $T = 2^{19}$ is 0.0534, against 0.0815 for the main term; at these heights the error term $O(1 + \sqrt{\log(1/c)})$ is not small compared with the main term, and the data do not determine a limit at fixed c. The one-step residual $r_k(X)$ agrees with $u_k(X)$ to within 3% for $c \le \frac{1}{5}$ and exceeds it at $c = \frac{1}{2}$ (0.0146 against 0.0122 at $T = 2^{19}$): this is the cost of evaluating S_X at τ_k instead of γ_k (Theorem 6.1(a)).

10.4. Exact accounting (Theorem C)

For $X \in \{7, 61, 401, 907, 1999\}$ and the five windows above $k = 1000$ we compute the gain $\langle s_k^2 - (s_k - S_X(\tau_k))^2 \rangle$, the diagonal term D_X and the cross term Cr_X of (6.5). Table 5 lists three windows; Figure 3 shows all five.

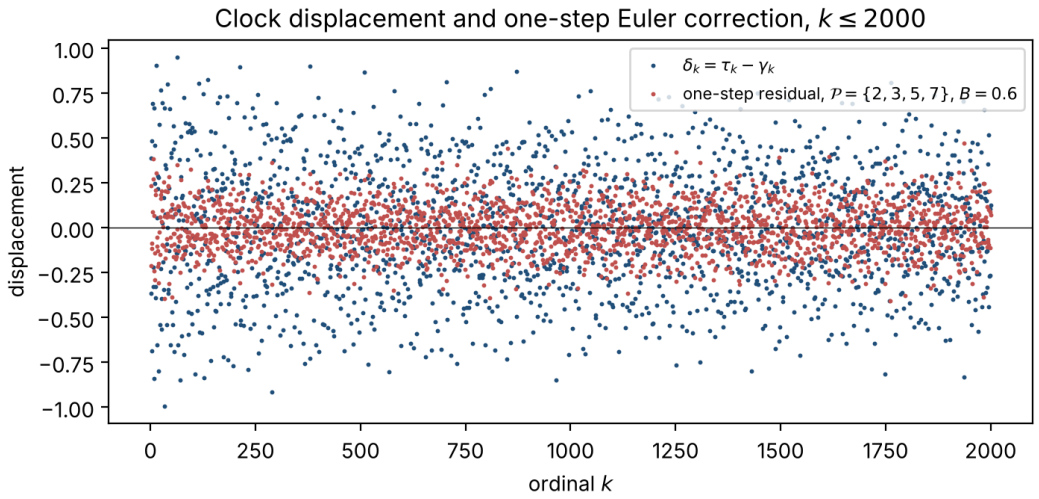


Figure 1: The displacement $\delta_k = \tau_k - \gamma_k$ of the RvM clock and the residual $\tau_k^{\mathcal{P}} - \gamma_k$ of the one-step clock with $\mathcal{P} = \{2, 3, 5, 7\}$, for $k \le 2000$; the correction $S_{\mathcal{P}}(\tau_k)/\mathcal{L}(\tau_k)$ is capped at $\pm B$ with $B = 0.6$ (saturation, Section 10.5).

Table 4: Mean squares of the defect $u_k(X)$ and of the one-step residual $r_k(X)$ over $T < \gamma_k \le 2T$, $X = T^c$.

T	c	X	$\#\{p \le X\}$	$\langle u_k(X)^2 \rangle$	$\langle r_k(X)^2 \rangle$	$\frac{1}{2\pi^2} \log \frac{1}{c}$
2^{14}	1/10	2.64	1	0.066738	0.066312	0.116650
2^{14}	1/5	6.96	3	0.039487	0.038425	0.081535
2^{14}	1/3	25.40	9	0.020174	0.018347	0.055656
2^{14}	1/2	128.00	31	0.010929	0.010701	0.035115
2^{16}	1/10	3.03	2	0.056337	0.055961	0.116650
2^{16}	1/5	9.19	4	0.039588	0.039045	0.081535
2^{16}	1/3	40.32	12	0.022282	0.021508	0.055656
2^{16}	1/2	256.00	54	0.011533	0.012316	0.035115
2^{18}	1/10	3.48	2	0.062696	0.062544	0.116650
2^{18}	1/5	12.13	5	0.041377	0.041265	0.081535
2^{18}	1/3	64.00	18	0.022737	0.022855	0.055656
2^{18}	1/2	512.00	97	0.011983	0.013848	0.035115
2^{19}	1/10	3.73	2	0.065599	0.065517	0.116650
2^{19}	1/5	13.93	6	0.040530	0.040623	0.081535
2^{19}	1/3	80.63	22	0.022911	0.023446	0.055656
2^{19}	1/2	724.08	128	0.012187	0.014567	0.035115

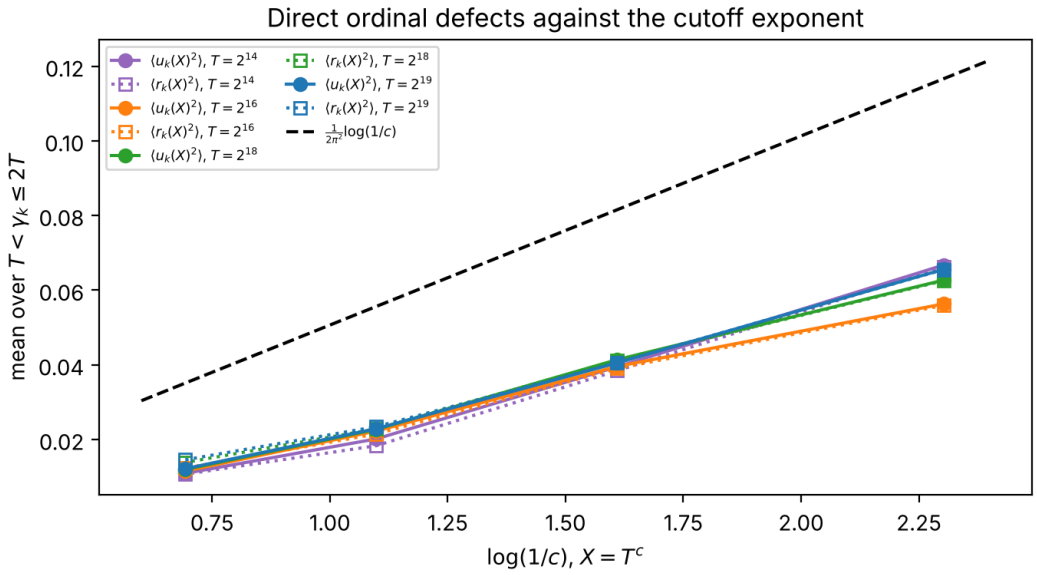


Figure 2: Mean square of $u_k(X)$ (filled markers) and of $r_k(X)$ (open markers) over $T < \gamma_k \le 2T$ against $\log(1/c)$, $X = T^c$, for the four heights of Table 4; dashed: $\frac{1}{2\pi^2}\log(1/c)$.

Table 5: Exact decomposition (6.5) of the one-step gain into the diagonal term D_X and the cross term Cr_X.

window of k	$\langle s_k^2\rangle$	X	gain	D_X	Cr_X	v_X
$(10^3, 10^4]$	0.085195	7	0.063224	0.063984	0.000760	0.066252
		61	0.075868	0.085258	0.009390	0.093853
		401	0.067945	0.097561	0.029616	0.111486
		907	0.062337	0.100483	0.038146	0.117730
		1999	0.055885	0.101829	0.045944	0.123202
$(10^5, 5\cdot 10^5]$	0.106325	7	0.064185	0.064201	0.000016	0.066252
		61	0.086367	0.086579	0.000212	0.093853
		401	0.093276	0.096932	0.003657	0.111486
		907	0.092779	0.099690	0.006911	0.117730
		1999	0.091379	0.101921	0.010542	0.123202
$(10^6, K]$	0.113224	7	0.064441	0.064444	0.000003	0.066252
		61	0.087434	0.087478	0.000044	0.093853
		401	0.098011	0.098495	0.000483	0.111486
		907	0.098607	0.101389	0.002782	0.117730
		1999	0.098325	0.103474	0.005149	0.123202

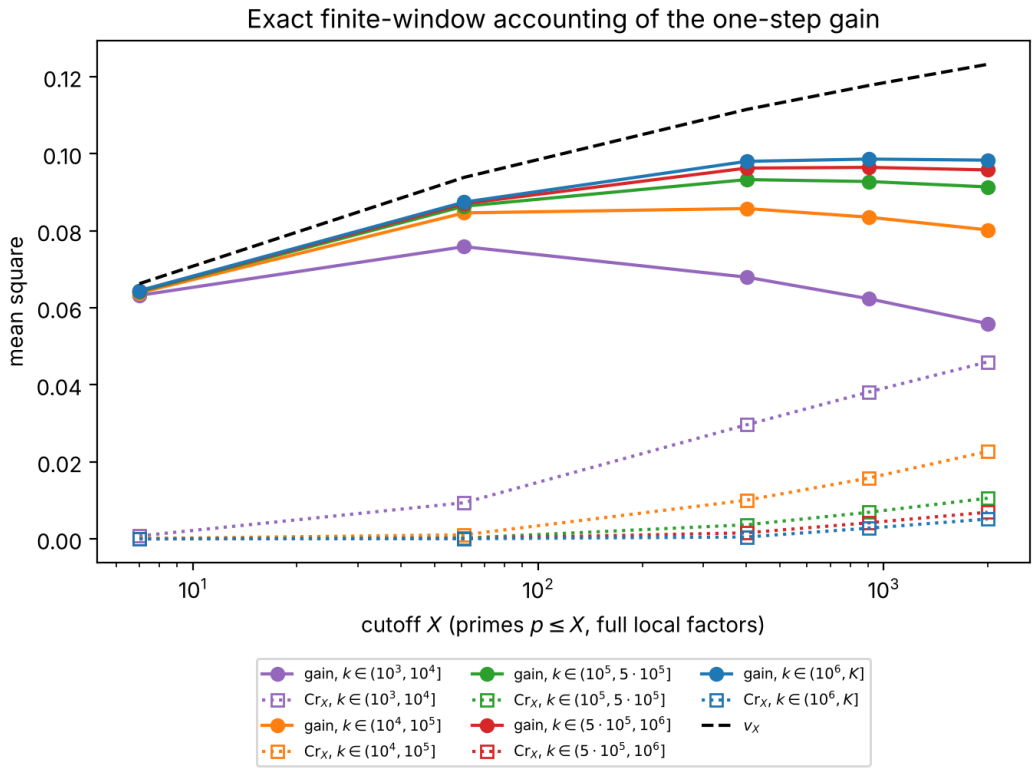


Figure 3: Gain of the one-step clock (filled markers) and cross term Cr_X (open markers) against the cutoff X, for five windows of ordinals; dashed: v_X.

The diagonal term D_X increases with X in every window and stays below v_X; the cross term Cr_X grows with X at fixed height and decreases with the height at fixed X. The gain is therefore maximal at a cutoff that increases with the height: $X = 61$ on $(10^3, 10^4]$, between 401 and 1999 on $(10^5, 5 \cdot 10^5]$, and near 907 on $(10^6, K]$. No parameter is fitted, and the identity gain $= D_X - \mathrm{Cr}_X$ is exact.

10.5. The one-step clock against the cutoff

For comparison with [9, 22] we also report the root mean square and the maximum of $\tau_k^X - \gamma_k$ for 40 cutoffs $2 \le X \le 1999$ (Figure 4) and for five rules, with saturation $\tau_k - \mathrm{sgn}(y_k)\min(|y_k|, B)$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, where indicated (Table 6, Figure 5). The minimizing cutoff increases from 7 on $(0, 10^3]$ to 907 on $(10^6, K]$, in agreement with the accounting above, and a least-squares fit $\log X_{\mathrm{opt}} = \theta_{\mathrm{fit}} \log(\tau/2\pi)$ over the five upper windows gives $\theta_{\mathrm{fit}} = 0.5744$. This exponent is exploratory: the curves are flat near the minimum (in the top window the root mean square is 0.06447 at $X = 701$, 0.06427 at $X = 907$ and 0.06428 at $X = 1009$), and it is not asserted to be asymptotically optimal. At these heights the minimizing cutoff lies beyond the onset $X \approx (\tau/2\pi)^{1/2}$ of the cross term of the primes (Section 10.7), where the gain of further primes is offset by the resonant pairs; θ_{fit} describes this finite-height balance. A large fixed cutoff is harmful at small heights: $\mathcal{P} = \{p \le 1999\}$ gives the residual 3.247 at $k = 1$. The largest residual for $\mathcal{P} = \{2, 3, 5, 7\}$, at $k = 258\,318$, follows the close pair $\gamma_{258317} = 175\,681.6491$, $\gamma_{258318} = 175\,681.7159$ (normalized gap 0.109), at which $S^*(\gamma_{258318}) = 1.039$, so that $S(\gamma^+_{258318}) = 1.539$, and $\delta_{258318} = 0.638$.

Table 6: Root mean square, maximum of $|\tau_k^X - \gamma_k|$ and number of residuals $\ge \frac{1}{2}$ over all K ordinates, and the same quantities in the top window $(10^6, K]$.

rule	RMS	max	$\# \ge \frac{1}{2}$	RMS, top	max, top	$\# \ge \frac{1}{2}$, top
bare clock τ_k	0.188066	0.993711	11589	0.178900	0.717359	2971
$\mathcal{P} = \{2, 3, 5, 7\}$, $B = 0.6$	0.120691	0.522347	1	0.117413	0.489293	0
$\mathcal{P} = \{p \le 61\}$, $B = 0.6$	0.085635	0.457259	0	0.085354	0.400144	0
$\mathcal{P} = \{p \le 1999\}$, no saturation	0.070757	3.247397	156	0.064897	0.528267	7
$X_k = \max(7, (\tau_k/2\pi)^{0.57})$, $B = 0.6$	0.066608	0.513886	2	0.064259	0.513886	1

10.6. Transport and the Newton term (Theorem D)

Table 7 decomposes the residual (7.1) on the window $T = 2^{19}$ of Table 4 (all 977 804 ordinates, full local factors). The identity $r_k = u_k + \mathfrak{t}_k + \chi_k$ holds to $2 \cdot 10^{-16}$, and $0 \le \chi_k \le 6.8 \cdot 10^{-8}$ up to rounding.

The transport is smaller than $c^2 \log\log T$ ($= 0.103$, 0.287, 0.644) by factors 21–35, and it is anticorrelated with $u_k(X)$, increasingly with c; this is why $\langle r_k^2 \rangle$ exceeds $\langle u_k^2 \rangle$ by only 0.0001–0.0024. The root mean square of $S'_X/\mathcal{L}$ is close to c, as in Remark 7.7, and the Newton term $y_k S'_X(\tau_k)$ has correlation 0.66–0.77 with $-\mathfrak{t}_k(X)$; at $c = \frac{1}{2}$ it overestimates the transport, and the denominator of (7.8) matters.

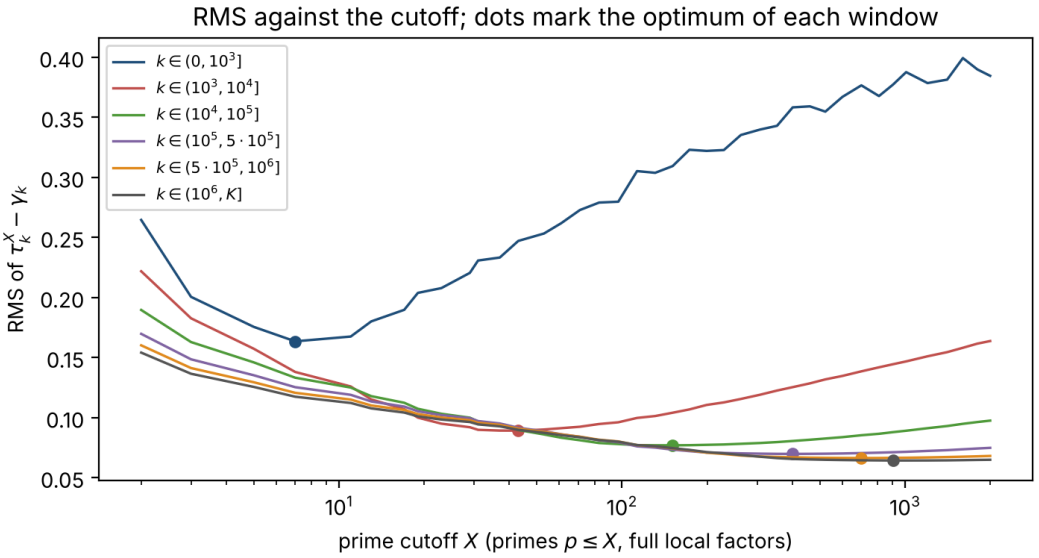


Figure 4: Root mean square of $\tau_k^X - \gamma_k$ (no saturation) against the cutoff X, by window of ordinals; dots mark the minimum in each window.

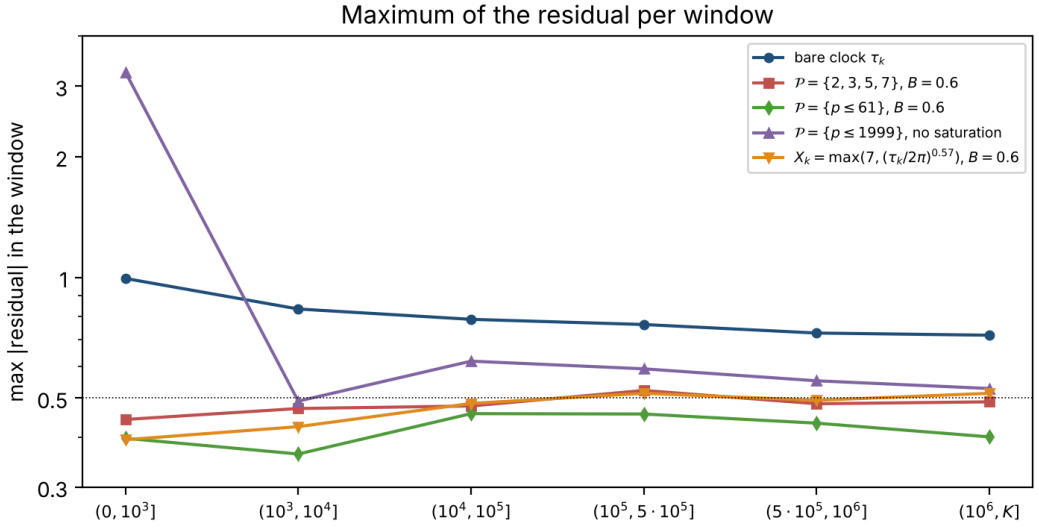


Figure 5: Maximum of $|\tau_k^X - \gamma_k|$ in each window of ordinals for the five rules of Table 6, logarithmic scale; dotted: $\frac{1}{2}$. The value 3.247 of the fixed set $\{p \le 1999\}$ in the first window occurs at $k = 1$.

Table 7: Transport term and Newton term of the one-step clock over $2^{19} < \gamma_k \le 2^{20}$, $X = T^c$: mean squares, the correlation of $u_k(X)$ with $\mathfrak{t}_k(X)$, the root mean square of $S'_X(\tau_k)/\mathcal{L}(\tau_k)$, and the proportion of indices with $F'_X(\tau_k) < 0$.

c	X	$\langle u_k^2 \rangle$	$\langle \mathfrak{t}_k^2 \rangle$	$\mathrm{corr}(u_k, \mathfrak{t}_k)$	$\langle r_k^2 \rangle$	$\langle (y_k S'_X(\tau_k))^2 \rangle$	RMS $\frac{S'_X(\tau_k)}{\mathcal{L}(\tau_k)}$	$F'_X(\tau_k) < 0$
1/5	13.93	0.040530	0.004905	−0.171	0.040623	0.003559	0.225	0.002%
1/3	80.63	0.022911	0.011391	−0.336	0.023446	0.014386	0.361	0.27%
1/2	724.08	0.012187	0.018283	−0.533	0.014567	0.046866	0.525	2.7%

Table 8: Cross term at the clock points on $(10^6, K]$; $(\tau_{\min}/2\pi)^{1/2} = 309.1$.

X	$\#\{p \le X\}$	Cr_X	$\mathrm{Cr}_X^{(1)}$
100	25	0.000083	0.000000
200	46	0.000149	0.000000
300	62	0.000186	0.000000
316	65	0.000195	0.000004
350	70	0.000248	0.000048
400	78	0.000449	0.000235
500	95	0.001107	0.000857
700	125	0.002056	0.001761
907	155	0.002782	0.002447
1200	196	0.003608	0.003210
1999	303	0.005149	0.004662

10.7. Onset of the cross term

Table 8 gives Cr_X on the top window $(10^6, K]$, where $\tau_k/2\pi$ runs over $[95\,536, 180\,241]$, for the full Euler phase and for its prime part $S_X^{(1)}(t) = -\frac{1}{\pi}\sum_{p\le X} p^{-1/2}\sin(t\log p)$, with $\mathrm{Cr}_X^{(1)} = \langle S_X^{(1)}(\tau_k)^2\rangle - \frac{1}{2}\sum_{p\le X} a_p^2$.

$\mathrm{Cr}_X^{(1)}$ vanishes to six decimals until X^2 reaches $\tau_{\min}/2\pi$ and then grows. For $X = 1999$ the 91 809 ordered pairs of primes split into the 7 807 pairs with $pq \in [\tau_{\min}/2\pi, \tau_{\max}/2\pi]$, which contribute 0.004657, and the others, which contribute 0.000005. The full Euler phase has earlier resonances, from products of prime powers in the same range, with smaller weights. This is the mechanism of Proposition 11.4: the phase $\omega\tau_x/2\pi$ with $\omega = \log n$ is stationary in the sense of the Poisson summation formula exactly at $\tau_x = 2\pi n$. A stationary-phase evaluation (heuristic: endpoint terms and the variation of the amplitude are neglected) at x_0 with $\tau_{x_0} = 2\pi n$, where $\omega\tau_{x_0}/2\pi - x_0 = n - \frac{11}{8}$ and the second derivative of the phase is $-2\pi/(\tau\log^2(\tau/2\pi))$, gives $\sum_{k\in I} e^{i\tau_k\log n} \approx -\sqrt{n}\log n$, hence a contribution $\log(pq)/(2\pi^2 \#I)$ of each resonant ordered pair to $\mathrm{Cr}_X^{(1)}$. Summed over the 7 807 pairs this is 0.004663, against 0.004657 measured.

11. Prime responses at the zeros and at the clock points

Subsections 11.1–11.4 are unconditional. Subsection 11.5 contains rigorous implications from two named hypotheses. Nothing heuristic is used; Theorems A–E do not depend on this section.

11.1. Fixed and slowly moving frequencies

Theorem 11.1 (Landau's formula on the critical line)**.** *Uniformly for $t \ge 2$ and integers $2 \le n \le t^{1/8}$,*

$$\sum_{0<\gamma\le t} n^{i\gamma} = -\frac{t}{2\pi}\frac{\Lambda(n)}{\sqrt{n}} + O\Big(\frac{t\log^2 n}{\log t} + \sqrt{n}\log^2 t\Big), \tag{11.1}$$

zeros counted with multiplicity. For fixed n the error is $O(t/\log t)$.

Proof. Let $\ell = \log n$. By the functional equation and $\zeta(\bar{s}) = \overline{\zeta(s)}$, $\rho \mapsto 1-\bar{\rho}$ maps the zeros with $0 < \gamma \le t$ onto themselves, preserving multiplicities and ordinates. Hence $\sum n^{\rho} = \sum n^{1-\bar{\rho}}$, and averaging the two expressions, $\sum_{0<\gamma\le t} n^{\rho} = \sqrt{n}\sum_{0<\gamma\le t} n^{i\gamma}\cosh((\beta-\frac12)\ell)$. Thus

$$\sum_{0<\gamma\le t} n^{i\gamma} = \frac{1}{\sqrt{n}}\sum_{0<\gamma\le t} n^{\rho} - \mathcal{K}_n(t), \qquad \mathcal{K}_n(t) = \sum_{0<\gamma\le t} n^{i\gamma}\Big(\cosh((\beta-\tfrac12)\ell)-1\Big).$$

The zeros with $\beta < \frac12$ correspond to those with $\beta > \frac12$, and those with $\beta = \frac12$ contribute 0, so $|\mathcal{K}_n(t)| \le 2\sum_{\beta>1/2}(\cosh((\beta-\frac12)\ell)-1)$. Writing $\cosh(x\ell)-1 = \int_0^x \ell\sinh(\ell y)\,dy$ and exchanging the finite sum with the integral,

$$\sum_{\substack{\beta>1/2\\0<\gamma\le t}}\Big(\cosh((\beta-\tfrac12)\ell)-1\Big) = \int_0^{1/2}\ell\sinh(\ell y)\,N^{+}(\tfrac12+y,t)\,dy \ll t\log t\cdot\ell\int_0^{\infty}\sinh(\ell y)e^{-ay}dy,$$

with $a = \frac14\log t$, by Theorem 3.15. For $\ell \le a/2$ the last integral is $\ell/(a^2-\ell^2) \le \frac43\ell/a^2$, so $\mathcal{K}_n(t) \ll t\ell^2/\log t$. Theorem 3.14 now gives $n^{-1/2}\sum n^{\rho} = -\frac{t}{2\pi}\Lambda(n)n^{-1/2} + O(\sqrt{n}\log(2nt)\log\log(3n))$, and $\log(2nt)\log\log(3n) \ll \log^2 t$ for $n \le t^{1/8}$. □

The symmetry removes the term linear in $\beta-\frac12$; Selberg's estimate then bounds the quadratic remainder. Without the symmetry one needs $\sum_{0<\gamma\le t}|\beta-\frac12| = o(t)$, which is not known.

Lemma 11.2. *For fixed real $\omega \ne 0$, $\sum_{k\in\mathcal{I}_T} e^{i\omega\tau_k} \ll_\omega \log T$. Hence for fixed integers $n_1, n_2 \ge 2$, $\langle \sin(\tau_k\log n_1)\sin(\tau_k\log n_2)\rangle_{\mathcal{I}_T} \to \frac12\mathbf{1}_{n_1=n_2}$.*

Proof. Take $\omega > 0$ (conjugate otherwise) and $\Omega(x) = \omega\tau_x/2\pi$ for real $x \ge 1$. By (2.5), $\Omega'(x) = \omega/\log(\tau_x/2\pi)$ is continuous, positive and decreasing, and for $x \in [\min\mathcal{I}_T, \max\mathcal{I}_T]$ and $T \ge T_0(\omega)$ it lies in $[\omega/\log(T/\pi), \frac12]$, so that $\|\Omega'\| \ge \omega/\log(T/\pi)$. $\mathcal{I}_T$ is a set of consecutive integers, and Theorem 3.9 gives the bound. For the second claim use $2\sin y_1\sin y_2 = \cos(y_1-y_2)-\cos(y_1+y_2)$ with $\omega = \log(n_1/n_2)$ (if $n_1 \ne n_2$) and $\omega = \log(n_1n_2)$, and $\#\mathcal{I}_T \asymp T\log T$. □

Theorem 11.3. *Uniformly for $T \ge T_0$ and integers $2 \le n \le T^{1/8}$,*

$$\langle s_k\sin(\gamma_k\log n)\rangle_{\mathcal{I}_T} = \frac{a_n}{2} + O\Big(\frac{\log n\,\log\log T}{\log T} + \frac{\sqrt{n}\log^2 T}{T\log n}\Big), \tag{11.2}$$

and the same with τ_k in place of γ_k. Hence for primes $p \le T^{1/8}$, $\mathcal{R}_p^{\gamma}(\mathcal{I}_T)$ and $\mathcal{R}_p^{\tau}(\mathcal{I}_T)$ are $1 + O(\frac{\sqrt{p}\log p\log\log T}{\log T} + \frac{p\log^2 T}{T\log p})$; they tend to 1 uniformly for $p \le (\log T)^{2-\varepsilon}$, and for every fixed prime power n.

Proof. For $t > 14$ not an ordinate let $\mathcal{H}_n(t) = \int_{14}^t S(y)n^{iy}dy$; the integrand is bounded and piecewise continuous. As measures on $(14,t]$, $dS = dN - (\mathcal{L}+\varpi'/\pi)dy$, and $y \mapsto n^{iy}/(i\ell_n)$ is a C^1 primitive of n^{iy}; integration by parts for Lebesgue–Stieltjes integrals gives

$$\mathcal{H}_n(t) = \Big[\frac{S(y)n^{iy}}{i\ell_n}\Big]_{14}^{t} - \frac{1}{i\ell_n}\sum_{14<\gamma_k\le t} n^{i\gamma_k} + \frac{1}{i\ell_n}\int_{14}^{t} n^{iy}\Big(\mathcal{L}+\frac{\varpi'}{\pi}\Big)dy = \frac{i}{\ell_n}\sum_{0<\gamma_k\le t} n^{i\gamma_k} + O(\log t), \tag{11.3}$$

uniformly in n (since $\ell_n \ge \log 2$), because $\gamma_1 > 14$, $S = O(\log t)$, $\int_{14}^t n^{iy}\mathcal{L}\,dy = O(\log t/\ell_n)$ by a second integration by parts ($\mathcal{L}'(y) = 1/(2\pi y)$), and $\varpi' = O(y^{-2})$. By Theorem 11.1, for $T \le t \le 2T+1$, $\mathcal{H}_n(t) = -i\frac{\Lambda(n)}{2\pi\sqrt{n}\,\ell_n}t + O(\frac{T\ell_n}{\log T} + \frac{\sqrt{n}\log^2 T}{\ell_n} + \log T)$. Integrating by parts once more, with $\frac{d}{dt}(t\mathcal{L}(t)) = \mathcal{L}(t) + \frac{1}{2\pi}$,

$$\int_T^{2T} S(t)n^{it}\mathcal{L}(t)\,dt = \Big[\mathcal{H}_n\mathcal{L}\Big]_T^{2T} - \int_T^{2T}\frac{\mathcal{H}_n(t)}{2\pi t}\,dt = -i\frac{\Lambda(n)}{2\pi\sqrt{n}\,\ell_n}\int_T^{2T}\mathcal{L}\,dt + O\Big(T\ell_n + \frac{\sqrt{n}\log^3 T}{\ell_n}\Big). \tag{11.4}$$

Taking imaginary parts, $\int_T^{2T} S\sin(\ell_n t)\mathcal{L}\,dt = \frac{a_n}{2}\int_T^{2T}\mathcal{L}\,dt + O(T\ell_n + \sqrt{n}\log^3 T/\ell_n)$. Apply Lemma 4.7 with $\psi(t) = \sin(\ell_n t)$ on $[t_1, t_2]$, where $t_1 \in [T, T+1/T]$ and $t_2 \in [2T, 2T+1/T]$ are non-ordinates; the change of endpoints alters both sides by $O(\log^2 T)$. The ϖ-term is $O(\log T)$, the boundary term is $O(\log^2 T)$, and $\frac12\int G^2|\psi'| \le \frac{\ell_n}{2}\int_T^{2T} G^2 \ll \ell_n T\log\log T$ by Theorem 3.3. Divide by $\#\mathcal{I}_T = \int_T^{2T}\mathcal{L}\,dt + O(\log T) \gg T\log T$. For τ_k: $|\sin(\ell_n\tau_k) - \sin(\ell_n\gamma_k)| \le \ell_n|\delta_k|$, and by (2.6) and (3.3), $\langle|s_k\delta_k|\rangle_{\mathcal{I}_T} \ll \langle s_k^2\rangle_{\mathcal{I}_T}/\log T \ll \log\log T/\log T$. For primes multiply by $2/|a_p| = 2\pi\sqrt{p}$. □

The same computation without the term $\frac12\int G^2\psi'$ gives, for the one-particle term of Section 11.2, $\mathcal{A}_{T,p} = 1 + O(\frac{\sqrt{p}\log p}{\log T} + \frac{p\log^2 T}{T\log p})$ uniformly for $p \le T^{1/8}$; so Hypothesis U below holds for $p \le (\log T)^2/(\log\log T)^{2+\varepsilon}$, where $\theta \to 0$, and not for fixed $\theta > 0$.

Proposition 11.4 (No cross term below $T^{1/4}$)**.** *For $\varepsilon > 0$ there is $T_0(\varepsilon)$ such that for $T \ge T_0(\varepsilon)$ and $2 \le X \le T^{1/4-\varepsilon}$, $|\operatorname{Cr}_X(\mathcal{I}_T)| \ll T^{-\varepsilon/2}\sqrt{\log\log T}$.*

Proof. Let $Y = T^{1/2-\varepsilon}$, $A' = \{n \in A_X : n \le Y\}$, $A'' = A_X \setminus A'$, and $S_X = S' + S''$ accordingly. For each $p \le X$ the first m with $p^m > Y$ has $p^{-m/2} < Y^{-1/2}$, so $|S''| \le \sum_{A''}|a_n| \le \frac{\#\{p\le X\}}{\pi(1-2^{-1/2})}Y^{-1/2} \le 2T^{-\varepsilon/2}$. Next, $2\langle S'(\tau_k)^2\rangle = \sum_{A'} a_n^2 + \sum_{n_1,n_2\in A'} a_{n_1}a_{n_2}\Big(\mathbf{1}_{n_1\ne n_2}\langle\cos(\tau_k\log\frac{n_1}{n_2})\rangle - \langle\cos(\tau_k\log n_1n_2)\rangle\Big)$, and every frequency ω that occurs satisfies $1/Y \le |\omega| \le 2\log Y = (1-2\varepsilon)\log T$. For $k \in \mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$, and as in Lemma 11.2 the phase $\Omega(x) = |\omega|\tau_x/2\pi$ has $\Omega'(x) = |\omega|/\log(\tau_x/2\pi) \in [\frac{1}{2Y\log T}, 1-\varepsilon]$ for $T \ge T_0(\varepsilon)$; Theorem 3.9 gives $|\sum_{k\in\mathcal{I}_T} e^{i\omega\tau_k}| \ll Y\log T + \varepsilon^{-1}$. Since $(\sum_{A_X}|a_n|)^2 \le \sigma_X^2 \ll X$, the off-diagonal part is $\ll X(Y\log T + \varepsilon^{-1})/(T\log T) \ll T^{-3\varepsilon}$, and $\frac12\sum_{A'} a_n^2 = v_X + O(T^{-\varepsilon})$. Finally $\operatorname{Cr}_X = \langle S'^2\rangle - v_X + 2\langle S'S''\rangle + \langle S''^2\rangle$, and $|\langle S'S''\rangle| \le \langle S'^2\rangle^{1/2}\,2T^{-\varepsilon/2}$ with $\langle S'^2\rangle \ll 1 + v_X \ll \log\log T$. □

Beyond this range the cross term is produced by products n_1n_2 of prime powers near $\tau_k/2\pi$, where the clock phase is stationary; Section 10.7 shows its onset for the primes at $X \approx (\tau/2\pi)^{1/2}$ and its size.

Corollary 11.5 (Gain at the clock points)**.** *(a) For a finite set A of prime powers and $\lambda \in \mathbb{R}$, $\langle s_k^2 - (s_k - \lambda S_A(\tau_k))^2\rangle_{\mathcal{I}_T} \to (2\lambda - \lambda^2)v_A$. (b) For fixed $\varepsilon > 0$ and λ, uniformly for $2 \le X \le (\log T)^{2-\varepsilon}$, with full local factors,*

$$\Big\langle s_k^2 - (s_k - \lambda S_X(\tau_k))^2\Big\rangle_{\mathcal{I}_T} = (2\lambda - \lambda^2)v_X + O_\lambda\Big(\frac{\sqrt{X}\log\log T}{\log T} + T^{-1/40}\Big).$$

Proof. (a) In (6.4), $\mathcal{R}_n^{\tau}(\mathcal{I}_T) \to 1$ by Theorem 11.3 and $\mathrm{Cr}_A(\mathcal{I}_T) \to 0$ by Lemma 11.2. (b) The left side is $2\lambda\langle s_k S_X(\tau_k)\rangle - \lambda^2\langle S_X(\tau_k)^2\rangle$, and $\langle S_X(\tau_k)^2\rangle = v_X + O(T^{-1/40})$ by Proposition 11.4. Let $Y = T^{1/8}$. For $n \in A_X$ with $n \le Y$, (11.2) multiplied by a_n and summed gives $\frac{1}{2}\sum a_n^2$ plus an error $\ll \frac{\log\log T}{\log T}\sum_{A_X}|a_n|\log n + \frac{\log^2 T}{T}\sum_{n\le Y}|a_n|\frac{\sqrt{n}}{\log n} \ll \frac{\sqrt{X}\log\log T}{\log T} + \frac{X\log^2 T}{T}$, since $\sum_{A_X}|a_n|\log n = \frac{1}{\pi}\sum_{p\le X}\frac{\log p}{\sqrt{p}-1} \ll \sqrt{X}$ and $|a_{p^m}|\sqrt{p^m}/\log p^m \le 1/(m^2\log p)$. For $n > Y$, $|\sum a_n\langle s_k \sin(\tau_k \log n)\rangle| \le \langle s_k^2\rangle^{1/2}\sum_{A_X, n>Y}|a_n| \ll \sqrt{\log\log T}\, XY^{-1/2}$, and $\frac{1}{2}\sum_{n>Y} a_n^2 \ll X/Y$. □

For $X = T^c$ the gain of the one-step clock is controlled by Theorem 7.3 instead.

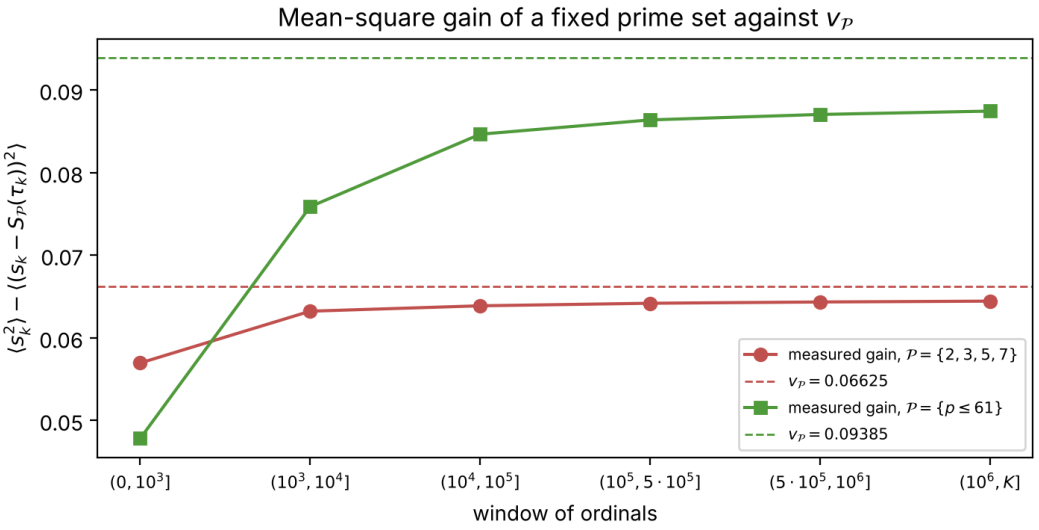


Figure 6: Mean-square gain $\langle s_k^2\rangle - \langle (s_k - S_{\mathcal{P}}(\tau_k))^2\rangle$ of two fixed prime sets $\mathcal{P}$ (full local factors) in each window of ordinals, and the constants $v_{\mathcal{P}}$ (dashed) of Corollary 11.5.

11.2. The response at the ordinates and its Fourier reduction

Let $T, 2T$ be non-ordinates, p a prime, and

$$\mathcal{A}_{T,p} = -\frac{2\pi\sqrt{p}}{\#\mathcal{I}_T}\int_T^{2T} G(t)\sin(t\log p)\mathcal{L}(t)\,dt, \qquad \mathcal{S}_{T,p} = \frac{1}{T}\int_T^{2T} S(t)^2 p^{it}\,dt. \tag{11.5}$$

Proposition 11.6 (Unconditional separation of one- and two-point terms)**.**

$$\begin{aligned}\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = \mathcal{A}_{T,p} &+ \frac{\pi\sqrt{p}\,T\log p}{\#\mathcal{I}_T}\Re\mathcal{S}_{T,p} - \frac{\pi\sqrt{p}}{\#\mathcal{I}_T}\Big[G^2\sin(t\log p)\Big]_T^{2T} \\ &+ \frac{\pi\sqrt{p}\log p}{\#\mathcal{I}_T}\int_T^{2T}(G^2 - S^2)\cos(t\log p)\,dt.\end{aligned} \tag{11.6}$$

If $\theta = \log p/L_T$ stays in a compact subset of $(0,1)$, the last two terms are $o(1)$ and

$$\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = \mathcal{A}_{T,p} + 2\pi^2\theta\sqrt{p}\,\big(1 + O(1/L_T)\big)\Re\mathcal{S}_{T,p} + o(1). \tag{11.7}$$

Proof. Lemma 4.7 with $\psi(t) = \sin(t\log p)$ on $[T,2T]$, multiplied by $-2\pi\sqrt{p}/\#\mathcal{I}_T$, gives (11.6), because $\int_T^{2T} G^2\cos(t\log p)\,dt = T\Re\mathcal{S}_{T,p} + \int_T^{2T}(G^2-S^2)\cos(t\log p)\,dt$. On $[T,2T]$, $|G^2-S^2| \le 2|S||\varpi|/\pi + \varpi^2/\pi^2 \ll \log T/T$, so the last term is $O(\sqrt{p}\log p/T)$, and the boundary term is $O(\sqrt{p}\log^2 T/(T\log T))$. Both are $o(1)$ for $p \le T^{\theta_1}$ with $\theta_1 < 1$. Finally $\#\mathcal{I}_T = \frac{TL_T}{2\pi}(1+O(1/L_T))$. □

Let $f_T = S\mathbf{1}_{[T,2T]}$, let $\mathfrak{m}_T$ be its distributional derivative (a finite signed measure, since f_T has compact support and bounded variation), and

$$Z_T(\omega) = \int_{\mathbb{R}} e^{i\omega t}\,d\mathfrak{m}_T(t), \qquad \widehat{f_T}(\omega) = \int_{\mathbb{R}} f_T(t)e^{i\omega t}dt. \tag{11.8}$$

Theorem 11.7 (Unconditional shifted two-point representation)**.** *For non-ordinates $T, 2T$,*

$$Z_T(\omega) = \sum_{T<\gamma<2T} \nu(\gamma)e^{i\omega\gamma} - \int_T^{2T}\Big(\mathcal{L}(t) + \frac{\varpi'(t)}{\pi}\Big)e^{i\omega t}\,dt + S(T)e^{i\omega T} - S(2T)e^{2i\omega T}, \tag{11.9}$$

$Z_T(\omega) = -i\omega\widehat{f_T}(\omega)$, so that $Z_T(0) = 0$ and $Z_T(\omega)/\omega$ extends continuously by $-i\widehat{f_T}$; and for every real ω_0

$$\int_T^{2T} S(t)^2 e^{i\omega_0 t}\,dt = -\frac{1}{2\pi}\int_{\mathbb{R}} \frac{Z_T(\omega)Z_T(\omega_0-\omega)}{\omega(\omega_0-\omega)}\,d\omega, \tag{11.10}$$

the integral converging absolutely.

Proof. On $(T,2T)$, $dS = dN - (\mathcal{L}+\varpi'/\pi)dt$ as measures, with atoms $\nu(\gamma)\delta_\gamma$; the cutoff adds $S(T)\delta_T$ and $-S(2T)\delta_{2T}$. This gives $\mathfrak{m}_T$ and (11.9). Since f_T is of bounded variation with compact support, integration by parts gives $\int e^{i\omega t}d\mathfrak{m}_T = -i\omega\int f_T e^{i\omega t}dt$, the boundary terms vanishing. As $f_T \in L^1\cap L^2$ is real, Plancherel's identity applied to f_T and $f_T e^{-i\omega_0 t}$ gives $\int f_T^2 e^{i\omega_0 t}dt = \frac{1}{2\pi}\int \widehat{f_T}(\omega)\widehat{f_T}(\omega_0-\omega)\,d\omega$, absolutely convergent since $\int|\widehat{f_T}(\omega)\widehat{f_T}(\omega_0-\omega)|d\omega \le \|\widehat{f_T}\|_2^2 = 2\pi\|f_T\|_2^2$. Substitute $\widehat{f_T} = iZ_T/\omega$. □

With $\omega_0 = \theta L_T = \log p$ and $\omega = \alpha L_T$ put

$$\mathcal{B}_T(\theta) = \int_{\mathbb{R}} \frac{Z_T(\alpha L_T)Z_T((\theta-\alpha)L_T)}{\alpha(\theta-\alpha)}\,d\alpha, \qquad \text{so that} \qquad \mathcal{S}_{T,p} = -\frac{\mathcal{B}_T(\theta)}{2\pi T L_T}. \tag{11.11}$$

Expanding the zero sums gives the weights $e^{i\alpha L_T\gamma}e^{i(\theta-\alpha)L_T\gamma'} = p^{i\gamma'}e^{i\alpha L_T(\gamma-\gamma')}$: $\mathcal{S}_{T,p}$ is a pair correlation of the zeros of the Riemann zeta function *shifted by a prime frequency*, which a stationary pair correlation (a function of $\gamma-\gamma'$ only) does not determine.

11.3. The kernel $\mathfrak{K}_\theta$

For $0<\theta<1$ define the continuous, piecewise linear function

$$\mathfrak{K}_\theta(\alpha) = \begin{cases} \theta, & \alpha \le -1, \\ \theta-1-\alpha, & -1 \le \alpha \le \theta-1, \\ 0, & \theta-1 \le \alpha \le 1, \\ \alpha-1, & 1 \le \alpha \le 1+\theta, \\ \theta, & \alpha \ge 1+\theta, \end{cases} \tag{11.12}$$

equivalently, with $h(q) = \max(1 - |q|, 0)$,

$$\mathfrak{K}_\theta(\alpha) = 1 - h(\alpha) - h(\theta - \alpha) - h(\theta) + 2\max\Big(1 - \max(|\alpha|, |\theta - \alpha|, \theta), 0\Big), \tag{11.13}$$

as one checks on each of the five intervals.

Proposition 11.8 (Integral and support of the kernel)**.** *For $0 < \theta < 1$,*

$$\int_{\mathbb{R}} \frac{\mathfrak{K}_\theta(\alpha)}{\alpha(\theta - \alpha)}\, d\alpha = -\frac{2\Phi(\theta)}{\theta}, \tag{11.14}$$

the integral being absolutely convergent, and $\mathfrak{K}_\theta(\alpha) = 0$ whenever $|\alpha| + |\theta - \alpha| + \theta < 2$.

Proof. $\mathfrak{K}_\theta$ is bounded and vanishes on $[\theta - 1, 1]$, which contains 0 and θ; the integrand is bounded, continuous outside $[\theta - 1, 1]$ and $O(\alpha^{-2})$ as $|\alpha| \to \infty$, hence absolutely integrable. It is invariant under $\alpha \mapsto \theta - \alpha$, which maps $[1, \infty)$ onto $(-\infty, \theta - 1]$, so the integral is twice that over $[1, \infty)$. On $[1 + \theta, \infty)$, $\frac{\theta}{\alpha(\theta-\alpha)} = \frac{1}{\alpha} - \frac{1}{\alpha-\theta}$, whose integral is $\lim_{y\to\infty}\Big[\log\frac{\alpha}{\alpha-\theta}\Big]_{1+\theta}^{y} = -\log(1+\theta)$. On $[1, 1+\theta]$, $\frac{\alpha-1}{\alpha(\theta-\alpha)} = -\frac{1}{\theta\alpha} + \frac{\theta-1}{\theta(\theta-\alpha)}$, whose integral is $-\frac{\log(1+\theta)}{\theta} + \frac{\theta-1}{\theta}\log(1-\theta)$. The sum is $-\Phi(\theta)/\theta$; doubling gives (11.14). If $\alpha > 1$ then $|\alpha|+|\theta-\alpha|+\theta \geq \alpha+(\alpha-\theta)+\theta = 2\alpha > 2$, and if $\alpha < \theta - 1$ the same follows by the symmetry; elsewhere $\mathfrak{K}_\theta = 0$. □

Proposition 11.9 (The square of the prime polynomial has no prime frequency)**.** *For $Y \geq 2$, $T \geq 2$ and every prime p, $\big|\frac{1}{T}\int_T^{2T} S_Y^{\mathrm{pr}}(t)^2 p^{it}\, dt\big| \ll Y^3/T$, with an absolute constant. In particular, with $E_0 = S - S_{Y_0}^{\mathrm{pr}}$, uniformly for primes $p \leq T^{\theta_1}$ with fixed $\theta_1 < 1$,*

$$\mathcal{S}_{T,p} = \frac{1}{T}\int_T^{2T} \Big(2S_{Y_0}^{\mathrm{pr}}(t)E_0(t) + E_0(t)^2\Big)p^{it}\, dt + o(p^{-1/2}).$$

Proof. Expanding the product of sines, $S_Y^{\mathrm{pr}}(t)^2 p^{it}$ is a combination, with coefficients of modulus at most $\frac{1}{4\pi^2}(qr)^{-1/2}$, of $e^{i\lambda t}$ with $\lambda = \log(pq^{\pm1}r^{\pm1})$, $q, r < Y$ primes. No λ vanishes: $p = qr$, $pq = r$, $pr = q$ and $pqr = 1$ are impossible for primes. Write $e^\lambda = A/B$ with coprime positive integers; then $|\lambda| \geq \log(1 + 1/\min(A, B)) \geq 1/(2\min(A, B))$, and $\min(A, B) \leq \max(r, qr) < Y^2$ in all cases, or $|\lambda| \geq \log 8$. Hence $|\int_T^{2T} e^{i\lambda t} dt| \leq 2/|\lambda| \leq 4Y^2$, and the sum of the coefficients is $\ll (\sum_{q<Y} q^{-1/2})^2 \ll Y$. For the second statement take $Y = Y_0 = T^{1/160}$: $Y_0^3/T = T^{-157/160} = o(T^{-\theta_1/2})$. □

Thus the moving-prime law below is a statement about the terms that contain Selberg's remainder E_0. The bound $\int_T^{2T} E_0^4 \ll T$ of Theorem 3.5 gives only $|\frac{1}{T}\int E_0^2 p^{it}| \ll 1$ and $|\frac{1}{T}\int S_{Y_0}^{\mathrm{pr}} E_0 p^{it}| \ll \sqrt{\log\log T}$, without the oscillation of size $p^{-1/2}$ that the law requires.

11.4. The arithmetic of the Newton nonlinearity

The corrected clocks of Section 8 are functions of the clock and of the phases $t\log q$ of the primes $q \leq X_j$; the second step contains $S_{X_2}(x_k^{(1)})$ with $x_k^{(1)} = \tau_k - S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$, that is, compositions of prime phases. This subsection determines, without hypotheses, which frequencies such compositions can produce and which they cannot, and why the error of the corrected clocks in mean square and the response to a prime $p = T^\theta$ are different questions. Throughout, $M = \#\mathcal{I}_T$.

Proposition 11.10 (Spectral barrier at the clock points)**.** *Let $0<\theta<1$, let $p=p(T)$ be a prime with $\log p/\log T\to\theta$, let $\epsilon_T\to 0$, and let*

$$Q_T(t)=\sum_{a,b\le T^{\epsilon_T}} c_{a,b}\,e^{it\log(a/b)},\qquad \sum_{a,b}|c_{a,b}|\le T^{o(1)},$$

with integers $a,b\ge 1$ and complex coefficients. Then

$$\frac{\sqrt{p}}{M}\sum_{k\in\mathcal{I}_T} Q_T(\tau_k)\,p^{\,i\tau_k}=o(1).$$

Proof. For each pair (a,b) let $\Omega(x)=\tau_x\log(pa/b)/2\pi$; by (2.5), $\Omega'(x)=\log(pa/b)/\log(\tau_x/2\pi)$ is continuous and monotone on the window, and since $|\log(a/b)|\le\epsilon_T\log T$, $\Omega'(x)=\theta+o(1)$ uniformly in (a,b) and x. For $T\ge T_0$ therefore $\|\Omega'\|\ge\frac12\min(\theta,1-\theta)$ and Theorem 3.9 gives $\sum_{k\in\mathcal{I}_T}e^{i\tau_k\log(pa/b)}=O_\theta(1)$. Summing with the coefficients, $|\sum_k Q_T(\tau_k)p^{i\tau_k}|\ll_\theta T^{o(1)}$, and $\sqrt{p}/M=T^{\theta/2-1+o(1)}$. □

A combination of prime frequencies of multiplicative complexity $T^{o(1)}$ cannot respond, at the clock points, to a prime of scale T^θ: the frequencies $\log(a/b)$ that could resonate with $\log p$ have a or b of size $T^{\theta+o(1)}$, which a polynomial in the phases of primes $\le T^c$ reaches only in degree $D\ge\min(\theta,1-\theta)/c$. For the first Newton step, $c=c_1$, this critical degree is $\asymp\min(\theta,1-\theta)\sqrt{\log\log T}$. The next lemma shows where the nonlinearity of the Newton step puts its spectral mass.

Lemma 11.11 (Bessel expansion of a Newton phase)**.** *Let $X\ge 2$ and $\lambda\in\mathbb{R}$, and let $P_X=S_X^{(1)}$ be the prime part of S_X,*

$$P_X(t)=-\frac{1}{\pi}\sum_{q\le X}\frac{\sin(t\log q)}{\sqrt{q}}.$$

With $z_q=\lambda/(\pi\sqrt{q})$ and integer vectors $\mathbf{j}=(j_q)_{q\le X}$ indexed by the primes $q\le X$,

$$e^{-i\lambda P_X(t)}=\sum_{\mathbf{j}}A_{\mathbf{j}}\,e^{it\Lambda_{\mathbf{j}}},\qquad A_{\mathbf{j}}=\prod_{q\le X}J_{j_q}(z_q),\qquad \Lambda_{\mathbf{j}}=\sum_{q\le X}j_q\log q,\tag{11.15}$$

the series converging absolutely, J_j the Bessel functions of integer order. Moreover

$$\sum_{\mathbf{j}}A_{\mathbf{j}}^2=1,\qquad \sum_{\mathbf{j}}A_{\mathbf{j}}^2\Lambda_{\mathbf{j}}=0,\qquad \sum_{\mathbf{j}}A_{\mathbf{j}}^2\Lambda_{\mathbf{j}}^2=\frac{\lambda^2}{2\pi^2}\sum_{q\le X}\frac{\log^2 q}{q},\tag{11.16}$$

and, with $D(\mathbf{j})=\sum_q|j_q|$ and $|z_q|\le 1$ for all q, for every $s>0$ and $D\ge 0$,

$$\sum_{D(\mathbf{j})\ge D}A_{\mathbf{j}}^2\le e^{-sD}\exp\Big(\frac{C_s\lambda^2}{\pi^2}\sum_{q\le X}\frac{1}{q}\Big),\qquad C_s=e^s\exp(e^s/4+1/2).\tag{11.17}$$

Proof. $e^{-i\lambda P_X(t)}=\prod_{q\le X}\exp(iz_q\sin(t\log q))$ and Jacobi's expansion $e^{iz\sin x}=\sum_{j\in\mathbb{Z}}J_j(z)e^{ijx}$ [36, §2.22], absolutely convergent since $|J_j(z)|\le(|z|/2)^{|j|}e^{z^2/4}/|j|!$ for real z (from the power series of J_j); the product of finitely many absolutely convergent series is (11.15). Parseval's

identity for $x \mapsto e^{iz\sin x}$ on $[0,2\pi]$ gives $\sum_j J_j(z)^2 = 1$; for its derivative $iz\cos x\, e^{iz\sin x}$, whose Fourier coefficients are $ijJ_j(z)$, it gives $\sum_j j^2 J_j(z)^2 = \frac{1}{2\pi}\int_0^{2\pi} z^2\cos^2 x\,dx = \frac{z^2}{2}$, and the inner product of the function with its derivative gives $\sum_j jJ_j(z)^2 = \frac{1}{2\pi}\int_0^{2\pi} z\cos x\,dx = 0$. Since $A_{\mathbf{j}}^2 = \prod_q J_{j_q}(z_q)^2$ and $\Lambda_{\mathbf{j}}$ is linear in $\mathbf{j}$, the three identities (11.16) follow by expanding the products. For (11.17), $J_0(z)^2 \le 1$ and, for $|z| \le 1$,

$$\sum_{j\neq 0} e^{s|j|} J_j(z)^2 \le 2e^{z^2/2} \sum_{j\ge 1} \frac{(e^s z^2/4)^j}{(j!)^2} \le 2e^{z^2/2}\cdot \frac{e^s z^2}{4}\exp\Big(\frac{e^s z^2}{4}\Big) \le C_s z^2,$$

so $\sum_{\mathbf{j}} e^{sD(\mathbf{j})} A_{\mathbf{j}}^2 = \prod_q \sum_j e^{s|j|} J_j(z_q)^2 \le \prod_q (1 + C_s z_q^2) \le \exp(C_s \sum_q z_q^2)$, and Chebyshev's inequality in D gives (11.17). □

Proposition 11.12 (Spectral concentration and dispersion of the Newton phases)**.** *Let $X_1 = T^{c_1}$, $c_1 = H_0^{-1/2}$, and for a prime power n and $t > 2\pi$ let $\lambda_n(t) = \log n/\mathcal{L}(t)$, so that $\sin(x_k^{(1)}\log n) = \Im\big(e^{i\tau_k \log n} e^{-i\lambda_n(\tau_k) S_{X_1}(\tau_k)}\big)$; the prime-power part $S_{X_1} - P_{X_1}$ contributes to $e^{-i\lambda S_{X_1}}$ a further factor of the form* (11.15)*, with $z_{p^m} = \lambda/(\pi m p^{m/2})$ at the frequencies $m\log p$, which changes nothing below and is omitted.*

(a) (First step: concentration in low degrees.) For $n \le X_1$ and $t \in [T, 2T]$, $|\lambda_n(t)| \le 2\pi c_1(1+o(1))$, and in (11.17) *with $X = X_1$, $\lambda = \lambda_n(t)$, the exponent $\frac{C_s\lambda^2}{\pi^2}\sum_{q\le X_1}\frac{1}{q} \le 4C_s c_1^2 H_0(1+o(1)) \le 5C_s$ is bounded: the mass of $e^{-i\lambda_n P_{X_1}}$ in degrees $\ge D$ is $\le C_s' e^{-sD}$, uniformly in $n \le X_1$ and t.*

(b) (Second step: dispersion.) For $n \le X_2 = T^{c_2}$, $c_2 = H_1^{-1/2}$, the mass in degrees $\ge D$ is only bounded by $e^{-sD}\exp(C_s'' H_0/H_1)$, which is not uniform, but the frequencies $\Lambda_{\mathbf{j}}$ of $e^{-i\lambda_n(t)P_{X_1}}$ have, by (11.16) *and Theorem 3.8,*

$$\Big(\sum_{\mathbf{j}} A_{\mathbf{j}}^2 \Lambda_{\mathbf{j}}^2\Big)^{1/2} = \frac{|\lambda_n(t)|}{2\pi}\log X_1\,(1+o(1)) \le c_1 c_2 \log T\,(1+o(1)),$$

hence, for every $0 < \theta < 1$,

$$\sum_{|\Lambda_{\mathbf{j}}| \ge \theta\log T} A_{\mathbf{j}}^2 \le \frac{c_1^2 c_2^2}{\theta^2}(1+o(1)) = \frac{1+o(1)}{\theta^2 H_0 H_1}. \tag{11.18}$$

Proof. (a) $\lambda_n(t) = \log n/\mathcal{L}(t) \le c_1 \log T/\mathcal{L}(T) = 2\pi c_1(1 + O(1/\log T))$ and $\sum_{q\le X_1} 1/q = \log\log X_1 + O(1) \le H_0$; $|z_q| \le |\lambda|/\pi \le 1$. (b) The same with c_2, $\sum_{q\le X_1}\log^2 q/q = \frac{1}{2}\log^2 X_1(1+o(1))$ by partial summation from Theorem 3.8, and Chebyshev's inequality for the probability measure $A_{\mathbf{j}}^2$ on the vectors $\mathbf{j}$. □

The second Newton step therefore generates frequencies of arbitrarily high multiplicative degree, but its spectral mass at the frequencies $|\Lambda| \ge \theta\log T$ of the moving primes $p = T^\theta$ is $\ll (H_0H_1)^{-1}$: the reduction of the mean-square error (Theorem 8.10) and the reconstruction of the response $1 - \Phi(\theta)$ at $p = T^\theta$ are different objectives, the second living in a spectral tail that the first does not see. For a fixed cutoff the compositions produce no exact resonance either.

Proposition 11.13 (Exact orthogonality for a fixed cutoff)**.** *Let $X \geq 2$ be fixed, let F be a continuous function on the torus $\mathbb{T}^{\#\{q\leq X\}}$, and let $F_X(t) = F\Big((t\log q)_{q\leq X}\Big)$. Then for every prime $p > X$,*

$$\lim_{U\to\infty} \frac{1}{U}\int_U^{2U} e^{it\log p} F_X(t)\,dt = 0.$$

Proof. Given $\varepsilon > 0$, choose a trigonometric polynomial $\Pi(\mathbf{v}) = \sum_{\mathbf{m}} d_{\mathbf{m}} e^{i\mathbf{m}\cdot\mathbf{v}}$ on the torus with $\sup|F - \Pi| \leq \varepsilon$ (Stone–Weierstrass). Then $e^{it\log p}\Pi((t\log q)_q) = \sum_{\mathbf{m}} d_{\mathbf{m}} e^{it\log(p\prod_q q^{m_q})}$, and each frequency $\log(p\prod_q q^{m_q})$ is nonzero, by unique factorization (p does not occur among the q); so each term has mean $O_{\mathbf{m}}(1/U)$ over $[U, 2U]$, and the limit superior of the modulus of the mean of $e^{it\log p}F_X$ is at most ε. □

What a corrected clock with growing cutoff can create is not an exact resonance but frequencies near $\log p$ in finite windows, through the degrees above the critical one; their uniform control is what a proof of the moving-prime law at the corrected clock points would need, and it is not given here. The estimate that would connect the two problems is

$$\sup_{T^a\leq p\leq T^b} \sqrt{p}\,\Big|\frac{1}{M}\sum_{k\in\mathcal{I}_T} \mathcal{L}(\tau_k)\Big(x_k^{(m)} - \gamma_k\Big)e^{i\tau_k\log p}\Big| \longrightarrow 0 \qquad (0 < a < b < 1), \tag{11.19}$$

for some $m \leq m^*$: it would transfer the response at the ordinates to the response at the m-th corrected clock, which is a function of the clock and the primes alone. Nothing of the kind is proved for a single prime; Theorem 8.10 together with Proposition 8.18(c) controls the left side only in $L^2(dp/p)$, with total error $O(1)$.

The computable part of the response

For the corrected clocks of Section 8, the response $\mathcal{R}_p^\tau$ splits exactly into a part computable from the clock and the primes and a part carried by the error. Write, for $m \geq 1$ and a prime p,

$$\begin{aligned}
\mathcal{A}_p^{(m)} &= -2\pi\sqrt{p}\Big\langle \mathcal{L}(\tau_k)(\tau_k - x_k^{(m)})\sin(\tau_k\log p)\Big\rangle_{\mathcal{I}_T},\\
\mathcal{E}_p^{(m)} &= -2\pi\sqrt{p}\Big\langle \mathcal{L}(\tau_k)(x_k^{(m)} - \gamma_k)\sin(\tau_k\log p)\Big\rangle_{\mathcal{I}_T},\\
\mathcal{D}_p^{(m)} &= -2\pi\sqrt{p}\Big\langle S_{X_m}(\tau_k)\sin(\tau_k\log p)\Big\rangle_{\mathcal{I}_T},\\
\mathcal{B}_p^{(m)} &= -2\pi\sqrt{p}\Big\langle B_k^{(m)}\sin(\tau_k\log p)\Big\rangle_{\mathcal{I}_T}, \qquad B_k^{(m)} = S_{X_m}(x_k^{(m-1)}) - S_{X_m}(\tau_k),
\end{aligned} \tag{11.20}$$

so that $\mathcal{A}^{(m)}$ is the response of the correction applied by the m steps, $\mathcal{D}^{(m)}$ that of the prime polynomial evaluated on the clock, $\mathcal{B}^{(m)}$ that of its transport to the previous points, and $\mathcal{E}^{(m)}$ that of the residual error.

Theorem 11.14 (Computable and residual responses)**.** *Let $X_m = T^{c_m}$ with $c_m \leq \frac{1}{2}$ and $x_k^{(m)}$ as in* (8.1)*. Then:*

(i) $\mathcal{R}_p^\tau(\mathcal{I}_T) = \mathcal{A}_p^{(m)} + \mathcal{E}_p^{(m)} + O(\sqrt{p}/T)$ and $\mathcal{A}_p^{(m)} = \mathcal{D}_p^{(m)} + \mathcal{B}_p^{(m)} + O(\sqrt{p}\,T^{c_m-1})$, for every $m \geq 1$ and every prime p; $\mathcal{B}_p^{(1)} = 0$.

(ii) For $\varepsilon>0$, uniformly for primes p with $pX_m\le T^{1-\varepsilon}$,

$$\mathcal{D}_p^{(m)}=\mathbf{1}_{[p\le X_m]}+O_\varepsilon\Big(\frac{(p+X_m)\log T}{T}\Big). \tag{11.21}$$

(iii) Hence $\mathcal{A}_p^{(1)}=\mathbf{1}_{[p\le X_1]}+o(1)$ for $pX_1\le T^{1-\varepsilon}$: the first step has no response at the moving primes $p=T^\theta$, $0<\theta<1$. For every m and $X_m<p$, $pX_m\le T^{1-\varepsilon}$,

$$\mathcal{R}_p^\tau(\mathcal{I}_T)=\mathcal{B}_p^{(m)}+\mathcal{E}_p^{(m)}+o(1)\,: \tag{11.22}$$

the response of the ordinates to a moving prime is the response of the transport plus that of the residual.

(iv) For $0<b<1$, $\sum_{p\le T^b}|\mathcal{B}_p^{(m)}|^2/p\le 2\pi^2(1+o_b(1))\langle (B_k^{(m)})^2\rangle_{\mathcal{I}_T}$, and $\langle (B_k^{(2)})^2\rangle\ll H_0/H_1$.

Proof. (i) $s_k=\mathcal{L}(\tau_k)\delta_k-\chi_k=\mathcal{L}(\tau_k)(\tau_k-x_k^{(m)})+\mathcal{L}(\tau_k)(x_k^{(m)}-\gamma_k)-\chi_k$ with $0\le\chi_k\le 1/T$, and $\mathcal{L}(\tau_k)(\tau_k-x_k^{(m)})=S_{X_m}(x_k^{(m-1)})+R_{m,k}$ with $|R_{m,k}|\ll T^{c_m-1}$ (Lemma 8.6); $S_{X_m}(x_k^{(m-1)})=S_{X_m}(\tau_k)+B_k^{(m)}$.

(ii) $S_{X_m}(\tau_k)\sin(\tau_k\log p)=\sum_{n\in A_{X_m}}a_n\big[\frac12\cos(\tau_k\log\frac np)-\frac12\cos(\tau_k\log np)\big]$. The term $n=p$ exists exactly when $p\le X_m$ and equals $\frac{a_p}{2}(1-\cos(2\tau_k\log p))$; since $-2\pi\sqrt p\cdot\frac{a_p}{2}=1$, it contributes $1-\langle\cos(2\tau_k\log p)\rangle$. Every other frequency $\omega\in\{\log(n/p),\log(np),2\log p\}$ satisfies $|\omega|\le\log(X_mp)\le(1-\varepsilon)\log T$, so $\Omega(x)=\tau_x|\omega|/2\pi$ has $\Omega'\le 1-\varepsilon$ and $\|\Omega'\|\ge\min(|\omega|/\log T,\varepsilon)$ on the window; Theorem 3.9 gives $|\langle e^{i\tau_k\omega}\rangle_{\mathcal{I}_T}|\ll_\varepsilon\frac{1}{T|\omega|}+\frac{1}{T\log T}$. For $|\log(n/p)|$: if $n\notin[p/2,2p]$ then $|\omega|\ge\log 2$; if $n\in[p/2,2p]$, $n\ne p$, then $|\omega|\ge|n-p|/(2p)$ and $\sum_{n\in[p/2,2p]}|a_n|/|\omega|\ll p^{-1/2}\sum_{0<|n-p|\le p}p/|n-p|\ll\sqrt p\log p$. Summing with $\sum_{A_{X_m}}|a_n|\ll\sqrt{X_m}$ and multiplying by $2\pi\sqrt p$ gives (11.21).

(iii) is (i) and (ii). (iv) is Proposition 8.18 with $r_k=B_k^{(m)}$; for $m=2$, $|B_k^{(2)}|\le 2|y_k|g_2(\tau_k)$ with $y_k=\tau_k-x_k^{(1)}$ and $g_2=\mathcal{M}(\varphi S'_{X_2})$ (as in Proposition 8.7), and the sampling of g_2 at the clock points in L^8 (proof of Theorem 8.5) and $\langle y_k^8\rangle\ll H_0^4/\log^8 T$ give $\langle (B^{(2)})^2\rangle\le 4\langle y^4\rangle^{1/2}\langle g_2(\tau)^4\rangle^{1/2}\ll\frac{H_0}{\log^2 T}(c_2\log T)^2=H_0/H_1$. □

By (11.22), a limit of $\mathcal{R}_p^\tau$ at $p=T^\theta$ is a statement about $\mathcal{B}_p^{(m)}+\mathcal{E}_p^{(m)}$. For the transport, the Bessel expansion of Lemma 11.11 gives every frequency of degree below the critical one as a non-resonant mean $O(T^{-\varepsilon})$, but their number has ℓ^1 mass $\exp(c\,c_2\sqrt{X_1}/\log X_1)$ while only the ℓ^2 mass is bounded; and the factor $\sqrt p$ in (11.20) makes any ℓ^2 or energy bound useless for a single prime. Whether $\mathcal{B}_p^{(m)}\to 0$ (the response of the moving primes then living entirely in the residual) or $\mathcal{B}_p^{(m)}$ tends to an arithmetic function of θ is not decided by the estimates of this paper.

Computation. Table 9 lists $\mathcal{R}_p^\tau$, $\mathcal{D}_p^{(m)}$, $\mathcal{B}_p^{(m)}$ and $\mathcal{E}_p^{(m)}$ at $T=2^{19}$ for the cutoff sets of Table 2 and for primes from $\theta=0.10$ to $\theta=1.04$ (`opc_computable_response_v14.py`). At this height the stationary band of the clock is $83\,443\lesssim n\lesssim 166\,886$, so (11.21) applies only when $pX_m<83\,443$; where pX_m enters the band, $\mathcal{D}_p^{(m)}$ and $\mathcal{B}_p^{(m)}$ take values of order one through the pair resonance of Proposition 8.16 (for instance $\mathcal{D}_{1009}^{(3)}=0.207$ with $X_3=724$, $\mathcal{D}_{50021}^{(1)}=1.199$ with $X_1=3$), which is a finite-height effect and not a response at θ. In the clean range $X_m<p$, $pX_m<83\,443$, the computable response is small: $|\mathcal{B}_p^{(2)}|\le 0.07$ and

$|\mathcal{B}_p^{(3)}| \le 0.08$ at $p = 101$ and 211, against $\mathcal{R}_p^\tau = 0.84$ and 0.79, and the residual carries the rest. This is descriptive, at one height, and consistent with $\mathcal{B}_p^{(m)} \to 0$ for fixed m; it says nothing about $m = m^*(T)$.

Table 9: Computable and residual responses at $T = 2^{19}$ ($\#\mathcal{I}_T = 977\,804$) for the three cutoff sets of Table 2: $\mathcal{R}_p^\tau$ and, for $m = 1, 2, 3$, the diagonal part $\mathcal{D}_p^{(m)}$, the transport part $\mathcal{B}_p^{(m)}$ and the residual $\mathcal{E}_p^{(m)}$ of (11.20). A dagger marks $pX_m \ge 83\,443$ (pair resonance); prime powers $n = q^j$ with $q \le X_m$ and the width of the stationary band blur this boundary, e.g. $\mathcal{D}_{10007}^{(1)} = 0.090$ with $X_1 = 3$ comes from $n = 8$, $8p = 80\,056$. Values from `outputs/table_L_computable_response.csv`.

p	θ	$\mathcal{R}_p^\tau$	$\mathcal{D}^{(1)}$	$\mathcal{B}^{(1)}$	$\mathcal{E}^{(1)}$	$\mathcal{D}^{(2)}$	$\mathcal{B}^{(2)}$	$\mathcal{E}^{(2)}$	$\mathcal{D}^{(3)}$	$\mathcal{B}^{(3)}$	$\mathcal{E}^{(3)}$
$(c_1, c_2, c_3) = (\frac{1}{10}, \frac{1}{5}, \frac{1}{3})$, $X = (3, 13, 80)$											
3	0.097	0.974	1.000	0	−0.026	1.000	−0.033	0.007	1.000	−0.026	−0.001
7	0.172	0.957	0.000	0	0.957	1.000	−0.026	−0.017	1.000	−0.084	0.041
31	0.303	0.900	0.000	0	0.900	0.000	0.001	0.898	1.001	−0.097	−0.005
101	0.407	0.835	0.000	0	0.835	0.001	0.003	0.831	0.002	0.047	0.785
211	0.472	0.791	0.001	0	0.790	0.001	0.006	0.784	0.003	0.079	0.709
1009	0.610	0.703	0.002	0	0.701	0.012	0.019	0.672	0.012	0.186	0.504
2003	0.671	0.678	0.010	0	0.668	0.022	0.038	0.618	0.238†	0.147	0.293
10007	0.813	0.626	0.090	0	0.536	0.330†	0.090	0.206	0.332†	0.246	0.048
50021	0.955	1.238	1.199†	0	0.039	1.199†	0.035	0.004	1.199†	0.077	−0.038
100003	1.016	0.026	0.000	0	0.026	0.001	0.030	−0.005	0.001	0.003	0.021
$(c_1, c_2, c_3) = (\frac{1}{5}, \frac{1}{3}, \frac{1}{2})$, $X = (13, 80, 724)$											
3	0.097	0.974	1.000	0	−0.026	1.000	−0.067	0.041	1.000	−0.002	−0.024
31	0.303	0.900	0.000	0	0.900	1.001	−0.105	0.004	1.001	−0.077	−0.024
101	0.407	0.835	0.001	0	0.834	0.002	0.035	0.797	1.002	0.044	−0.212
211	0.472	0.791	0.001	0	0.790	0.003	0.065	0.723	1.131†	−0.063	−0.277
1009	0.610	0.703	0.012	0	0.691	0.012	0.170	0.521	0.207†	0.347	0.149
2003	0.671	0.678	0.022	0	0.656	0.238†	0.131	0.309	0.264†	0.311	0.103
10007	0.813	0.626	0.330†	0	0.296	0.332†	0.254	0.039	0.332†	0.249	0.045
$(c_1, c_2, c_3) = (\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, $X = (80, 724, 724)$											
3	0.097	0.974	1.000	0	−0.026	1.000	−0.096	0.069	1.000	0.028	−0.055
31	0.303	0.900	1.001	0	−0.102	1.001	−0.122	0.020	1.001	−0.040	−0.062
101	0.407	0.835	0.002	0	0.832	1.002	−0.008	−0.160	1.002	−0.283	0.115
211	0.472	0.791	0.003	0	0.788	1.131†	−0.114	−0.226	1.131†	−0.416	0.076
1009	0.610	0.703	0.012	0	0.691	0.207†	0.350	0.146	0.207†	0.174	0.322
2003	0.671	0.678	0.238†	0	0.440	0.264†	0.320	0.094	0.264†	0.174	0.240

11.5. The moving-prime law under Hypotheses U and W

Hypothesis 11.15 (Hypothesis U: uniform one-particle input)**.** For every $0 < \theta_0 < \theta_1 < 1$, uniformly for primes with $\theta_0 \le \log p / L_T \le \theta_1$, $\mathcal{A}_{T,p} = 1 + o(1)$.

A sufficient condition is the uniform estimate $\sup_{T \le t \le 2T} |\sum_{T < \gamma_k \le t} p^{i\gamma_k} + \frac{(t-T)\log p}{2\pi\sqrt{p}}| = o(T \log p/\sqrt{p})$ in that range, by two integrations by parts as in the proof of Theorem 11.3. Theorem 11.1 gives it only for $p \le T^{o(1)}$, through the error term $t \log^2 p / \log t$.

Hypothesis 11.16 (Hypothesis W: weighted arithmetic bridge)**.** Uniformly for primes with θ in any compact subset of $(0, 1)$,

$$\int_{\mathbb{R}} \frac{Z_T(\alpha L_T) Z_T((\theta - \alpha) L_T) + \mathcal{N}_T p^{-1/2} \mathfrak{K}_\theta(\alpha)}{\alpha(\theta - \alpha)} \, d\alpha = o\Big(\frac{T L_T}{\sqrt{p}}\Big), \tag{11.23}$$

with the continuous extensions of $Z_T(\omega)/\omega$ at $\omega = 0$.

Proposition 11.17 (Moving-prime law under Hypotheses U and W)**.** *Under Hypothesis W,*

$$\mathcal{S}_{T,p} = -\frac{1}{2\pi^2\sqrt{p}}\frac{\Phi(\theta)}{\theta} + o(p^{-1/2}). \tag{11.24}$$

Under Hypotheses U and W, or U and the real part of (11.24)*, uniformly for θ in compact subsets of $(0,1)$,*

$$\mathcal{R}_p^{\gamma}(\mathcal{I}_T) = 1 - \Phi(\theta) + o(1). \tag{11.25}$$

Conversely, under Hypothesis U, (11.25) *is equivalent to the real part of* (11.24)*. Nothing is asserted about $\mathcal{R}_p^{\tau}$.*

Proof. The integral of the kernel term in (11.23) converges absolutely (Proposition 11.8); hence so does that of the first term, which is $\mathcal{B}_T(\theta)$, and by (11.14) $\mathcal{B}_T(\theta) = \frac{\mathcal{N}_T}{\sqrt{p}}\frac{2\Phi(\theta)}{\theta} + o(TL_T/\sqrt{p}) = \frac{TL_T}{\pi\sqrt{p}}\frac{\Phi(\theta)}{\theta} + o(TL_T/\sqrt{p})$; (11.11) gives (11.24). Then $\sqrt{p}\,\Re\mathcal{S}_{T,p} = O(1)$, so in (11.7) the factor $1 + O(1/L_T)$ contributes $o(1)$ and $2\pi^2\theta\sqrt{p}\,\Re\mathcal{S}_{T,p} = -\Phi(\theta) + o(1)$; with $\mathcal{A}_{T,p} = 1 + o(1)$ this is (11.25). Conversely, solve (11.7) for $\Re\mathcal{S}_{T,p}$. □

Remark 11.18. $\Phi(\theta) = 2\sum_{j\ge1}\frac{\theta^{2j}}{(2j-1)2j} = \theta^2 + \frac{\theta^4}{6} + \frac{\theta^6}{15} + \cdots$, so $1 - \theta^2$ is only the quadratic truncation of (11.25), and $1 - \Phi(\theta)$ vanishes at $\theta = 0.9036\ldots$. Hypotheses U and W are not consequences of Theorem 11.1, which reaches only $\theta \to 0$. By Proposition 11.9, (11.24) is a statement about $\frac{1}{T}\int(2S_{Y_0}^{\mathrm{pr}}E_0 + E_0^2)p^{it}$, with required error $o(p^{-1/2}) = T^{-\theta/2+o(1)}$.

12. Heuristic support and numerical evidence

This section contains no statement about the Riemann zeta function that is used elsewhere. Subsection 12.1 proves a theorem about a random-matrix model; its relevance to the zeros of the Riemann zeta function is a heuristic.

12.1. A determinantal model whose connected kernel is $\mathfrak{K}_\theta$

On a circle of length R_0 let $\Pi_{\mathfrak{n}}$ be the orthogonal projection onto $\mathfrak{n}$ consecutive Fourier modes, and consider its determinantal point process, which has exactly $\mathfrak{n}$ points. For $q \in R_0^{-1}\mathbb{Z}$ let Ξ_q be the centred linear statistic of $y \mapsto e^{2\pi iqy}$, so that $\Xi_0 = 0$ and $\overline{\Xi_q} = \Xi_{-q}$. Let $R_0, \mathfrak{n} \to \infty$ with $\mathfrak{n}/R_0 \to 1$, so that the local kernel tends to the sine kernel. For lattice $\theta > 0$ and real ι_1, ι_2 the tilt

$$d\mathbb{P}_{\iota_1,\iota_2} = \frac{\exp\{-\iota_1\Re\Xi_{-\theta} - \iota_2\Im\Xi_{-\theta}\}}{\mathbb{E}\exp\{-\iota_1\Re\Xi_{-\theta} - \iota_2\Im\Xi_{-\theta}\}}\,d\mathbb{P}_{0,0} \tag{12.1}$$

is a probability measure (the statistics are bounded), and differentiation at $(0,0)$, justified by boundedness, gives $(\partial_{\iota_1} + i\partial_{\iota_2})\operatorname{Cov}_{\iota_1,\iota_2}(\Xi_\alpha, \Xi_{\theta-\alpha})|_{(0,0)} = -\operatorname{cum}_3(\Xi_\alpha, \Xi_{\theta-\alpha}, \Xi_{-\theta})$, the third joint cumulant.

Theorem 12.1 (Connected kernel of the model)**.** *Let lattice frequencies tend to $\alpha, \theta - \alpha, -\theta$ with $0 < \theta < 1$. Then $R_0^{-1}\operatorname{cum}_3(\Xi_\alpha, \Xi_{\theta-\alpha}, \Xi_{-\theta}) \to \mathfrak{K}_\theta(\alpha)$ and $R_0^{-1}\operatorname{cum}_2(\Xi_\theta, \Xi_{-\theta}) \to \theta$.*

Proof. For commuting multiplication operators $\mathsf{m}_1, \mathsf{m}_2, \mathsf{m}_3$, the third joint cumulant of the corresponding linear statistics of a determinantal process with projection kernel is [30]

$$\begin{aligned}\operatorname{cum}_3 = {} & \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_2\mathsf{m}_3\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_2\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_1\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}) - \operatorname{Tr}(\mathsf{m}_2\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_1\Pi_{\mathfrak{n}}) \\ & + \operatorname{Tr}(\mathsf{m}_1\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}) + \operatorname{Tr}(\mathsf{m}_1\Pi_{\mathfrak{n}}\mathsf{m}_3\Pi_{\mathfrak{n}}\mathsf{m}_2\Pi_{\mathfrak{n}}),\end{aligned} \tag{12.2}$$

the third-order term of the expansion of $\log\det(\mathrm{Id} + \Pi_{\mathfrak{n}}(e^{\mathsf{m}} - 1)\Pi_{\mathfrak{n}})$. Multiplication by $e^{2\pi i q y}$ shifts a mode by $R_0 q$; a trace with two projections counts the overlap of two integer intervals of length $\mathfrak{n}$, and a trace with three their triple overlap. For the shifts $R_0\alpha$, $R_0(\theta - \alpha)$, $-R_0\theta$, whose sum is 0, the triple overlap has length $\max(\mathfrak{n} - R_0\max(|\alpha|, |\theta - \alpha|, \theta), 0)$ for both cyclic orders, and the pair overlaps have lengths $R_0 h(\cdot) + o(R_0)$. Dividing by R_0 gives the right side of (11.13). The same count for two operators gives $R_0^{-1}\operatorname{cum}_2(\Xi_\theta, \Xi_{-\theta}) \to 1 - h(\theta) = \theta$. □

Why Hypothesis W is plausible. A window $[T, 2T]$ contains about $\mathcal{N}_T = TL_T/2\pi$ zeros of the Riemann zeta function at unit mean spacing after unfolding, and the one-particle prime amplitude in Hypothesis U is $-\mathcal{N}_T\theta/\sqrt{p}$. This equals the linear response of $\mathbb{E}\,\Xi_\theta$ to the tilt (12.1) with $\iota_1 = 2/\sqrt{p}$, $\iota_2 = 0$ and $R_0 = \mathcal{N}_T$, namely $-\frac{\iota_1}{2}\operatorname{cum}_2(\Xi_\theta, \Xi_{-\theta}) \sim -\mathcal{N}_T\theta/\sqrt{p}$, by the susceptibility θ of Theorem 12.1. If the zeros responded to the prime p like the model, the connected product in (11.11) would be $-\mathcal{N}_T\mathfrak{K}_\theta(\alpha)/\sqrt{p}$, which is the content of Hypothesis W. This is an analogy, not an argument: stationary sine-kernel statistics do not determine the shifted correlation (11.11), and convergence on compact sets would not control the singular weight and the tails. Note that $\mathfrak{K}_\theta$ vanishes where $|\alpha| + |\theta - \alpha| + \theta < 2$ (Proposition 11.8), so the whole value $-2\Phi(\theta)/\theta$ comes from frequencies outside that region.

12.2. Numerical evidence

Conjecture 12.2. *For primes with $\theta = \log p/L_T$ in compact subsets of $(0, 1)$, $\mathcal{R}_p^\gamma(\mathcal{I}_T) = 1 - \Phi(\theta) + o(1)$.*

For all 2262 primes $p \le 20\,000$ we compute $\mathcal{R}_p^\gamma$ and $\mathcal{R}_p^\tau$ on the five windows of ordinals above $k = 1000$, with $\theta = \log p/\operatorname{median}_k \log(\tau_k/2\pi)$, and compare the primes with $0 < \theta < 1$ with $1 - \Phi(\theta)$. Table 10 separates individual errors from bin means over θ-bins of width 0.05 containing at least five primes; Table 11 lists the bins of the two top windows, and Figure 7 all bins.

Table 10: Errors of $\mathcal{R}_p^\gamma - (1 - \Phi(\theta))$ over the primes $p \le 20\,000$ with $\theta < 1$.

window of k	primes	max individual error	RMS error	max |bin-mean error|
$(10^3, 10^4]$	159	0.34557	0.09425	0.02637
$(10^4, 10^5]$	901	0.44085	0.08511	0.02371
$(10^5, 5\cdot 10^5]$	2262	0.27857	0.06429	0.02972
$(5\cdot 10^5, 10^6]$	2262	0.31682	0.06086	0.03531
$(10^6, K]$	2262	0.16425	0.03309	0.01951

The bin means follow $1 - \Phi(\theta)$ and separate from $1 - \theta^2$ for $\theta \ge 0.6$. The residuals are not noise: in the two top windows every bin mean exceeds the candidate, by about 0.004

Table 11: Bin means of $\mathcal{R}_p^\gamma$ and $\mathcal{R}_p^\tau$ in the two top windows; "law" is the mean of $1-\Phi(\theta)$ over the same primes, s.e. the standard error of the mean difference.

window of k	θ-bin	primes	$\mathcal{R}_p^\gamma$	law	difference	s.e.	$\mathcal{R}_p^\tau$
$(5\cdot 10^5, 10^6]$	$[0.30, 0.35)$	6	0.8985	0.8933	+0.0052	0.0003	0.8845
	$[0.35, 0.40)$	8	0.8592	0.8549	+0.0043	0.0001	0.8515
	$[0.40, 0.45)$	13	0.8174	0.8127	+0.0047	0.0002	0.8180
	$[0.45, 0.50)$	22	0.7671	0.7619	+0.0052	0.0004	0.7829
	$[0.50, 0.55)$	33	0.7114	0.7059	+0.0054	0.0004	0.7496
	$[0.55, 0.60)$	54	0.6513	0.6443	+0.0070	0.0007	0.7205
	$[0.60, 0.65)$	86	0.5849	0.5753	+0.0096	0.0008	0.6960
	$[0.65, 0.70)$	142	0.5090	0.4979	+0.0111	0.0009	0.6771
	$[0.70, 0.75)$	234	0.4239	0.4103	+0.0136	0.0012	0.6689
	$[0.75, 0.80)$	380	0.3305	0.3123	+0.0181	0.0017	0.6905
	$[0.80, 0.85)$	630	0.2225	0.2018	+0.0207	0.0021	0.7548
	$[0.85, 0.90)$	645	0.1355	0.1002	+0.0353	0.0034	0.8411
$(10^6, K]$	$[0.30, 0.35)$	7	0.8952	0.8909	+0.0043	0.0001	0.8779
	$[0.35, 0.40)$	12	0.8556	0.8518	+0.0038	0.0002	0.8426
	$[0.40, 0.45)$	16	0.8126	0.8085	+0.0041	0.0003	0.8076
	$[0.45, 0.50)$	27	0.7669	0.7629	+0.0040	0.0002	0.7749
	$[0.50, 0.55)$	48	0.7126	0.7076	+0.0050	0.0003	0.7406
	$[0.55, 0.60)$	77	0.6518	0.6453	+0.0065	0.0004	0.7085
	$[0.60, 0.65)$	129	0.5845	0.5761	+0.0084	0.0006	0.6815
	$[0.65, 0.70)$	222	0.5074	0.4974	+0.0100	0.0007	0.6640
	$[0.70, 0.75)$	368	0.4218	0.4099	+0.0119	0.0009	0.6524
	$[0.75, 0.80)$	627	0.3271	0.3120	+0.0151	0.0011	0.6642
	$[0.80, 0.85)$	718	0.2393	0.2198	+0.0195	0.0016	0.7015

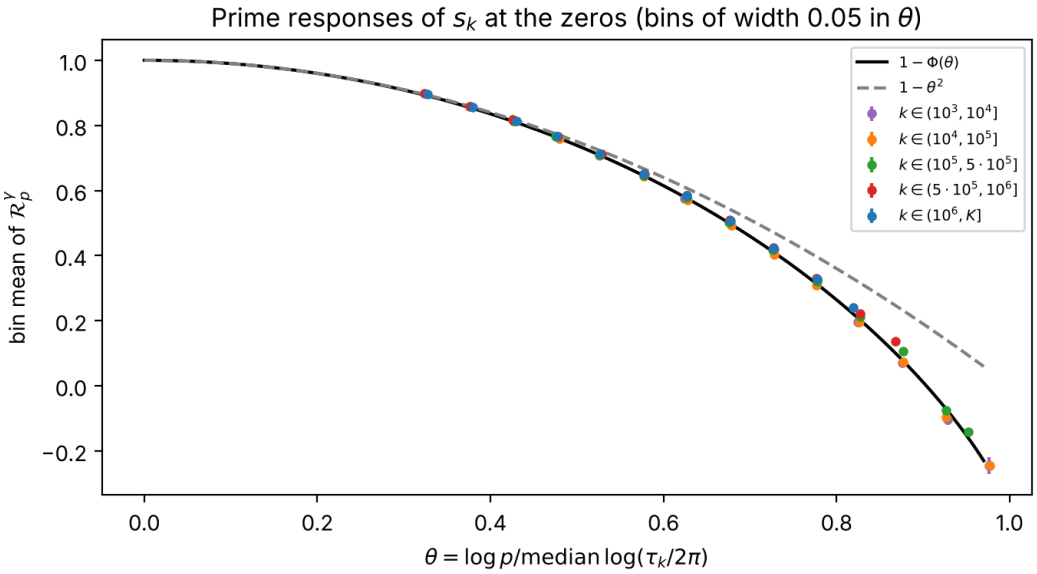


Figure 7: Bin means of $\mathcal{R}_p^\gamma$ (bars: descriptive standard error, see the text) against $1-\Phi(\theta)$ (solid) and $1-\theta^2$ (dashed), for five windows of ordinals.

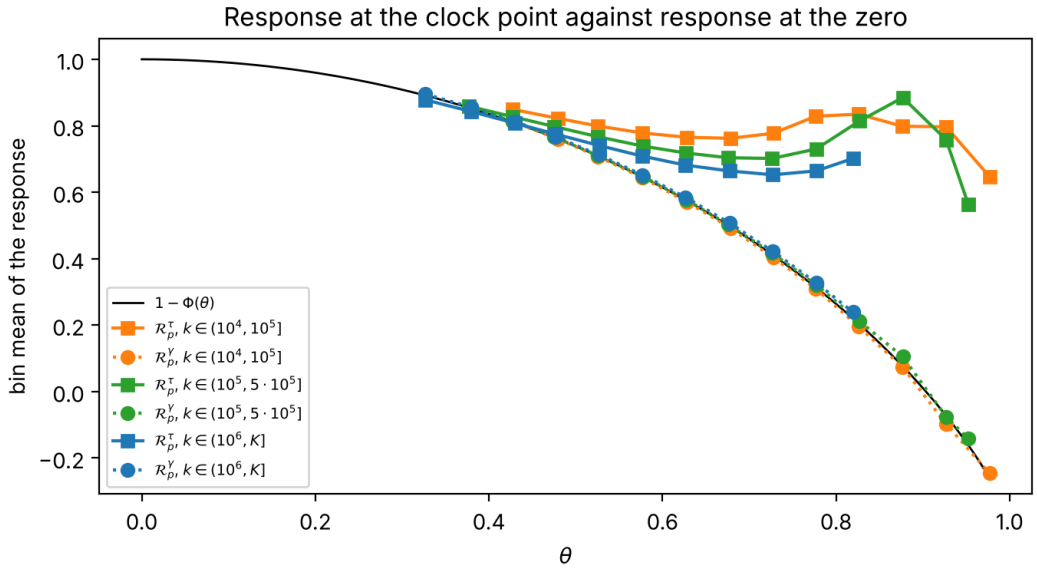


Figure 8: Bin means of the response at the clock points, $\mathcal{R}_p^\tau$ (squares), and at the zeros, $\mathcal{R}_p^\gamma$ (circles), for three windows of ordinals; solid: $1-\Phi(\theta)$.

at $\theta \approx 0.3$ and 0.02–0.035 at $\theta \approx 0.8$–0.9, many standard errors. In every bin the excess is smaller in the top window than in the preceding one, by 6–23%; since θ is computed with the median height of each window, a bin contains different primes in the two windows, and two windows do not determine a rate of decay. The data are consistent with Conjecture 12.2 only together with a finite-height correction of a few hundredths. The standard errors treat the primes of a bin as independent; since all responses are computed on the same ordinates, they are descriptive only. The response at the clock points follows a different curve, with a plateau near 0.65–0.70 for $\theta \in [0.65, 0.80]$ in the top window (Figure 8), as Remark 6.4 allows.

13. Conclusions

13.1. What was known

Before this work, the relation between the RvM clock and the ordinates of the zeros of the Riemann zeta function was known at two levels. Pointwise, $\delta_k = \tau_k - \gamma_k$ is bounded, with $\limsup |\delta_k| \le 0.6331$ as an immediate consequence of the bound of Bellotti and Wong [2]. On average, s_k is asymptotically Gaussian with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae for the Gram displacement proved by Korolev [20, 19], so $\langle \delta_k^2 \rangle \sim 2\log\log T/\log^2 T$: the clock is close to the zeros in height, but it misplaces almost every zero by an unbounded number of positions. Selberg's approximation [28, 12] replaces $S(t)$ by short prime sums in t-mean, and França–LeClair and LeClair [10, 22, 11] write the equation of an individual zero with an Euler phase and a growing number of primes, with computations at a fixed number of primes.

13.2. What this paper adds

1. *An exact dictionary between clock-type models and the zeros* (Theorems 4.1 and 4.2). For any C^1 model F and any continuous $\mathfrak{g}$, sums of $\mathfrak{g}$-increments of $k - \frac{1}{2} - F(\gamma_k)$ over the ordinates are exactly dF-integrals of $\mathfrak{g}(N - F)$ up to two boundary terms. For powers this gives the even moments with the constant $\frac{1}{12}$ per zero; for exponentials it transfers the signed law of $S(t)$ to s_k (Corollary 4.4).

2. *The prime information needed by the clock* (Theorems 5.1 and 5.12). The corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ holds at the true ordinates with mean-square ordinal defect $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log(\log T/\log X)})$, also with a cutoff varying at every index. In the main term each factor e in $\log X$ removes $\frac{1}{2\pi^2}$ from the mean square; the error term is not resolved at this scale. The defect is bounded in mean square if and only if the primes reach a fixed power of the height.

3. *A computable clock that is closer to the zeros by an order of magnitude* (Theorem 7.3, Corollaries 7.4 and 7.5, Theorem 8.5). The one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$ uses only τ_k and the primes. With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log \tau_k}}$, its mean-square error in height is $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$, against $(2 + o(1))\log\log T/\log^2 T$ for the bare clock; in units of zeros the mean square drops from $\frac{1}{2\pi^2}\log\log T$ to $\frac{1}{4\pi^2}\log\log\log T$. This is the main finding: the information in the primes up to a slowly decreasing power of the height cancels all but a $\log\log\log T/(2\log\log T)$ part of the mean square of the error of the clock. The restriction on c comes from the transport between τ_k and γ_k (Lemma 7.2); a root of the corrected equation gains nothing where it

can be selected (Proposition 7.6). A second Newton step, which is possible because the first error is smaller than δ_k, lowers the mean square to $\frac{1}{4\pi^2}\log\log\log\log T$ in units of zeros, and the m-th step to $\frac{H_m}{4\pi^2}+O(1+\sqrt{H_m})$ with absolute constants (Theorem 8.10); after $\log^* T+O(1)$ steps the error is $O(1)$ in units of zeros. The proof needs an exact moment recursion, a multiplicity bound for the corrected points of every level, which do not inherit the separation of the τ_k (Theorem 8.9), and a cutoff law that absorbs the multiplicity.

4. *The pointwise envelope* (Proposition 9.1, Theorem 9.3, Proposition 9.5). $\limsup|\tau_k-\gamma_k| = 2\pi\limsup|S(t)|/\log t$ exactly, and no slowly growing prime correction changes it. Backlund's method with the reflected exponent profile gives $\limsup|S(t)|/\log t \le \frac{1}{2\pi\log 2}\int_0^1 \mathfrak{e}(\sigma)\,d\sigma/\sqrt{\sigma(2-\sigma)}$; Bourgain's exponent pair gives $0.0968160\ldots$, hence $\limsup|\tau_k-\gamma_k|<0.60832$, and the explicit inputs alone give $|S(T)|\le 0.09783\log T+0.722\log\log T+13$ for all $T\ge e$.

5. *An exact finite-height account and the prime responses* (Theorem 6.1, Section 11). The gain of τ_k^X is the diagonal response minus a cross term, with the exact nonnegative term χ_k. The responses tend to 1 for primes up to $(\log T)^{2-\varepsilon}$ unconditionally, the cross term vanishes for $X\le T^{1/4-\varepsilon}$, and on the data it switches on at $X\approx(\tau/2\pi)^{1/2}$ through resonant pairs $pq\approx\tau_k/2\pi$. For $p=T^\theta$ the response at the zeros is reduced exactly to a pair correlation shifted by a prime frequency; under two named hypotheses it is $1-\Phi(\theta)$, which the data follow up to a systematic excess of a few hundredths. The compositions of prime phases produced by the Newton steps are expanded exactly in Bessel coefficients (Section 11.4): they cannot respond to $p=T^\theta$ below a critical multiplicative degree, their spectral mass at the frequencies of the moving primes is $O((H_0H_1)^{-1})$, and for a fixed cutoff they are exactly orthogonal to p^{it}.

6. *The transport identity in a prime frequency* (Theorem 8.12, Corollary 8.13, Proposition 8.15). The twisted integral $\frac{1}{T}\int_T^{2T} p^{i(t+G(t)/\mathcal{L}(t))}dt$ equals, up to $O(1/T)$, a sum over the displacements δ_k at the clock points plus a clock sum that vanishes for $p\le T^{1-\varepsilon}$ and equals $2\pi\sqrt{p}/T$ at the resonance $p\approx T/2\pi$, the constant $\frac{7}{8}$ of the Riemann–von Mangoldt formula making the stationary value real and positive. The identity is unconditional, uses no correlation between zeros, and shows that the phase p^{it} and the displacement cannot be treated as independent; it characterizes the response at the clock, in a weighted form that coincides with $\mathcal{R}_p^\tau$ for $p\le(\log T)^{4-\varepsilon}$ (Corollary 8.14), and not the response at the ordinates. The pair version (Proposition 8.16) makes the resonant main term of the cross term a theorem, and the sparseness of the resonant pairs makes the prime phases at the clock points almost orthonormal for all $p\le T^{1-\varepsilon}$ (Proposition 8.17), whence a spectral energy inequality: the responses of a computable approximation differ from $\mathcal{R}_p^\tau$ by a total $L^2(dp/p)$ error bounded by $2\pi^2$ times its mean-square error (Proposition 8.18); for the final corrected clock this error is $O(1)$. The response of the moving primes is carried by the transport of the prime phases and by the residual, not by the prime polynomials on the clock (Theorem 11.14).

13.3. Status of the proofs and pending verification

Every statement called Theorem, Proposition, Lemma or Corollary in Sections 4–9, 8 and 11.1–11.4 is proved in the text from the external results of Section 3. The following points are recorded so that a reader can check them independently.

1. *External inputs read in the original.* Theorems 3.5 and 3.6 (Fujii, pp. 225 and 232–233: the exponent $1/(80k)$, the constants $k^{1/2}, k^2, k^2, k^{1/2}, k^{3/2}$ of the five terms, and the Main Theorem for $S(t+h)-S(t)$), 3.4 (Korolev, Thm. 4.2, with his convention for the Gram points), 3.17 (Kinnunen, Thm. 1.4, and Stein, Thm. I.1 as quoted there), 3.19 (Bourgain, Thm. 6 of arXiv:1408.5794v2; Ivić, Lemma 2.8 and (7.57)), 3.10, 3.11, 3.12 and 3.13 (Titchmarsh, Lemmas 4.6, 4.7, 4.10 and Theorem 5.9), 3.15 (Titchmarsh, Thm. 9.19(C)), 3.9 (Graham–Kolesnik, Thm. 2.1), 3.14 (Gonek's formula as stated in Aryan, (1.1)), 3.21 and 3.2 (Bellotti–Wong, numbering of arXiv:2412.15470v2). Theorem 3.18 is taken from Lerner's statement for the real line; Carleson's and Hunt's papers, Billingsley's Theorems 26.3 and 30.2, Folland's §3.5, Soshnikov's cumulant formula, Jacobi's expansion in Watson's treatise and Tsang's thesis (through [7]) are cited as stated in those sources.

2. *Constants in the moment order.* Lemma 8.2 and Lemma 8.4 carry explicit dependence on the moment order r; Theorem 8.10 uses them with r up to 2^m, $m \le \log^* T + O(1)$, and the only requirement on T is $T^{1/(160r)} \ge 2$. The constants $C_\star$, H^*, B, C_6 are absolute and not computed; the statement is asymptotic in T.

3. *Proposition 9.5.* The explicit constants 10, 30, $|\chi(\sigma+it)| \le 3t^{1/2-\sigma}$, $R(T) < 1$ and the interval-arithmetic values of $I/(2\pi D)$ and $1/(2D)$ come from a computation communicated to the author and have not been recomputed; the coefficient 0.09783 agrees with `opc_envelope_coefficient_v14.py` (profile (b)). This is the only numerical input of a proved statement that is not reproduced by the programs of Appendix Appendix A.

4. *Propositions 8.15 and 8.16.* In (8.20) the non-resonant error term is crude, and the agreement of Section 10.7 to 0.02% is better than what is proved at that height.

5. *Corollaries 8.13 and 8.14.* The passage from the weighted response to $\mathcal{R}_p^\tau$ is proved for $p \le (\log T)^{4-\varepsilon}$ and not for $p = T^\theta$; Table 3 shows the difference at one height in the second range.

6. *What is not proved.* The moving-prime law $1-\Phi(\theta)$ (Section 11.5) rests on Hypotheses U and W; the estimate (11.19) that would transfer it to the corrected clocks is open; a bounded error with a fixed cutoff (Remark 7.7) is open; the cutoff law (8.11) is not claimed to be optimal.

7. *Numerical statements.* All printed numbers are produced by the programs listed in Appendix Appendix A from the two input files whose SHA-256 identifiers are given below. The following are descriptive, with no asymptotic statement behind them: the exponent $\theta_{\mathrm{fit}} = 0.5744$ and the minimizing cutoff per window (Table 6, Figure 4); the standard errors of Figures 7–8 and Table 11, since the primes of a bin share the ordinates; the comparison of the bin means with $1-\Phi(\theta)$ and the excess of 0.004–0.035; the plateau of $\mathcal{R}_p^\tau$ near 0.65–0.70; the correlations and the shortfall of the transport in Table 7; the cutoffs and all columns

of Table 2, at a height where the cutoffs (8.11) are not reachable; the values of Table 9, where the stationary band contaminates $pX_m \gtrsim 83\,443$ and the clean range contains three primes per cutoff set. The following are comparisons with a proved statement: the resonant main term of Section 10.7 (Proposition 8.16); the two resonance values of Table 3 (Proposition 8.15); the identity $|I - D - J| \le 2 \cdot 10^{-7}$ (Theorem 8.12); the envelope used to 66% and 15% (Proposition 9.2); the exact decomposition of Table 5 (Theorem 6.1). The following are data checks: the eight spot checks against `mpmath.zetazero`, the 19 900 bit-for-bit indices and the 20 679 midpoint decisions of the certificate; $|\delta_{34}| = 0.9937$ and $\max_k |s_k| = 1.3325$ at $k = 1\,249\,694$.

13.4. What is missing for a complete prediction of the zeros from the clock

Theorem 8.10 locates the ordinates from τ_k and the primes with a mean-square error of $O(1)$ mean spacings. A complete prediction, assigning to each zero its index and position, requires the following, in increasing order of difficulty; none is proved here.

1. *From $O(1)$ to $o(1)$ in mean square* (Question 13.2). A correction h_k computable from the clock and the primes with $2\langle r_k h_k\rangle - \langle h_k^2\rangle = \langle r_k^2\rangle - o(1)$, $r_k = \mathcal{L}(\tau_k)(x_k^{(m^*)} - \gamma_k)$. By Proposition 8.18 this is equivalent to reproducing the responses $\mathcal{R}_p^\tau$ of the moving primes in $L^2(dp/p)$, and by Theorem 11.14 and Table 9 those responses are carried by the residual, not by the prime polynomials on the clock. The two routes known are the cancellation between transport and defect of Remark 7.7 and the transfer estimate (11.19); neither is proved.

2. *The constant of Theorem 5.1* (Question 13.1). The term $O(1)$ is Selberg's remainder, $\|S - S^{\mathrm{pr}}_{T^{1/160}}\|_{L^4} \le C$ with C unknown. Assigning indices needs $|\mathcal{L}(\tau_k)e_k| < \frac{1}{2}$ for almost all k, hence a final mean square below $\frac{1}{4}$; Proposition 5.5 shows that rounding the corrected count fails on a positive proportion for small cutoffs, and for cutoffs near $T^{1/2}$ the question is exactly the value of C.

3. *Cutoffs $X > T^{1/2}$.* Theorem 5.1, Theorem 3.7 and the moment lemmas require $X^2 \le T$; the pair resonance at $pX \approx T/2\pi$ marks where this regime ends. Without a version of Theorem 5.1 for $\frac{1}{2} < c < 1$, the mean square cannot be driven to zero by enlarging the cutoff.

4. *Pointwise control.* Everything proved is in mean square. Proposition 9.1 gives $\limsup |\tau_k - \gamma_k| = 2\pi\mathfrak{a}$ for the bare clock and shows that cutoffs $\le (\log T)^{2-\varepsilon}$ do not change it; for the corrected clocks with $X = T^c$ no bound for $\sup_k |e_{m,k}|$ is known, and S_X has extreme values $\gg \exp(c'\sqrt{\log X/\log\log X})$. A zero-by-zero prediction needs at least $|\mathcal{L}(\tau_k)e_k| < \frac{1}{2}$ outside $o(\#\mathcal{I}_T)$ indices, and the moment method does not reach it.

5. *The moving-prime law.* Not needed for the prediction itself, but by item 1 it is the same arithmetic: if the residual carries the response of the moving primes, the function $\rho^\tau(\theta)$, if it exists, is the spectrum of what is missing.

13.5. Why a finer knowledge of $\tau_k - \gamma_k$ matters

The clock is the deterministic part of the ordinates, and δ_k carries all the arithmetic. Statistics of δ_k and s_k are statistics of the zeros of the Riemann zeta function that do not depend on an unfolding of the ordinates or on Gram intervals. Theorem A turns questions about the ordinates into questions about $N - F$ in t-mean, so any improvement of the t-mean description of $S(t)$ (sharper forms of Selberg's approximation, longer prime polynomials, mixed moments with prime powers) passes to the zeros. Theorem D shows that the size of $\tau_k^X - \gamma_k$ is governed by two separate quantities: the prime information missing below X, which Theorem B measures, and the transport $S_X(\gamma_k) - S_X(\tau_k)$, whose observed anticorrelation with the defect (Table 7) is not explained by any proved estimate. Pointwise, Proposition 9.1 identifies the extreme displacement of the clock with the extreme size of $S(t)$, so that every bound for one is a bound for the other. The measured responses $\mathcal{R}_n^\tau$, $\mathcal{R}_n^\gamma$ and the cross terms Cr_X are exact finite-height invariants of the zeros, which any model of the zeros must reproduce; the response $1 - \Phi(\theta)$ is a function that the stationary pair correlation does not determine.

13.6. Future work

Question 13.1. For fixed $c \in (0, \frac{1}{2}]$, do the bounded means of $u_k(T^c)^2$ converge, and to $(2\pi^2)^{-1}\log(1/c)$ plus which constant?

Question 13.2. Does $\langle r_k(T^c)^2\rangle_{\mathcal{I}_T}$ stay bounded for fixed c, that is, does the transport cancel against the defect as in Table 7? Is the error of the corrected clock after $\log^* T + O(1)$ steps bounded in units of zeros for *almost every* index, and not only in mean square? Can Theorems 5.1 and 7.3 be extended to $c > \frac{1}{2}$?

Question 13.3. Can Hypotheses U and W be proved for $p = T^\theta$, that is, can the terms of Proposition 11.9 that contain Selberg's remainder be evaluated to $o(p^{-1/2})$? What is the finite-height correction to $1 - \Phi(\theta)$ seen in Table 11?

Theorem 4.1 applies to any counting staircase and any C^1 model, in particular to the zeros of a primitive Dirichlet L-function; an analogue of Theorem 5.1 would additionally need Selberg's approximation for $L(s, \chi)$ [29, 12] and uniformity in the conductor.

Declaration of competing interest

The author declares no competing interests.

Data availability

All programs, the input-generation script with its certificates, the numerical outputs (CSV and JSON) and the figures are in the public GitHub repository `ordinal_prime_cutoff_v14` (link to be inserted). The ordinates are Odlyzko's table [27]; the two input files (arrays of 2 001 052 little-endian double-precision numbers) are regenerated from that table by `make_inputs_v14.py`. Their SHA-256 identifiers are
`fafb9436ce6a506de58c96c9bc9765d3942de3f15c921e99e0015b3e945fe0ac` (ordinates)
and

`b4c6758f0aef3724d3345f894ff82d5bbcb0f6139025b8b266091f713faa9504` (clock). The computations used Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2 and mpmath 1.3.0.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI assistant (Claude, Anthropic) to edit the text, check proofs and write analysis programs. The author reviewed and edited the content and takes full responsibility for it.

References

[1] F. Aryan, On an extension of the Landau–Gonek formula, J. Number Theory 233 (2022) 389–404, https://doi.org/10.1016/j.jnt.2021.06.015. (Cited on p. 13.)

[2] C. Bellotti, P.-J. Wong, Improved estimates for the argument and zero-counting function of the Riemann zeta-function, Math. Comp. (2025) (statements numbered as in arXiv:2412.15470v2), https://doi.org/10.1090/mcom/4133. (Cited on pp. 4, 11, 14, 15, 45, and 67.)

[3] P. Billingsley, Probability and Measure, third ed., Wiley, New York, 1995. (Cited on pp. 14 and 16.)

[4] J. Bourgain, Decoupling, exponential sums and the Riemann zeta function, J. Amer. Math. Soc. 30 (1) (2017) 205–224 (theorems numbered as in arXiv:1408.5794v2), https://doi.org/10.1090/jams/860. (Cited on p. 13.)

[5] L. Carleson, On convergence and growth of partial sums of Fourier series, Acta Math. 116 (1966) 135–157, https://doi.org/10.1007/BF02392815. (Cited on p. 13.)

[6] R.M. Corless, G.H. Gonnet, D.E.G. Hare, D.J. Jeffrey, D.E. Knuth, On the Lambert W function, Adv. Comput. Math. 5 (1996) 329–359, https://doi.org/10.1007/BF02124750. (Cited on p. 10.)

[7] A. Fazzari, M. Gerspach, P. Minelli, Selberg's central limit theorem weighted by linear statistics of zeta zeros, preprint, arXiv:2507.04150, 2025, https://doi.org/10.48550/arXiv.2507.04150. (Cited on pp. 13 and 69.)

[8] G.B. Folland, Real Analysis: Modern Techniques and Their Applications, second ed., Wiley, New York, 1999. (Cited on pp. 17 and 22.)

[9] G. França, A. LeClair, Statistical and other properties of Riemann zeros based on an explicit equation for the n-th zero on the critical line, preprint, arXiv:1307.8395, 2013, https://doi.org/10.48550/arXiv.1307.8395. (Cited on pp. 4 and 50.)

[10] G. França, A. LeClair, Transcendental equations satisfied by the individual zeros of Riemann ζ, Dirichlet and modular L-functions, Commun. Number Theory Phys. 9 (1) (2015) 1–50, https://doi.org/10.4310/CNTP.2015.v9.n1.a1. (Cited on pp. 4 and 67.)

[11] G. França, A. LeClair, A theory for the zeros of Riemann ζ and other L-functions (updated), preprint, arXiv:1407.4358v2, 2024, https://doi.org/10.48550/arXiv.1407.4358. (Cited on pp. 4, 18, and 67.)

[12] A. Fujii, On the zeros of Dirichlet L-functions. I, Trans. Amer. Math. Soc. 196 (1974) 225–235, https://doi.org/10.1090/S0002-9947-1974-0349603-2. (Cited on pp. 4, 11, 12, 67, and 71.)

[13] S.M. Gonek, An explicit formula of Landau and its applications to the theory of the zeta-function, in: A Tribute to Emil Grosswald: Number Theory and Related Analysis, Contemp. Math., vol. 143, Amer. Math. Soc., Providence, RI, 1993, pp. 395–413, https://doi.org/10.1090/conm/143/1210528. (Cited on pp. 4 and 13.)

[14] S.W. Graham, G. Kolesnik, Van der Corput's Method of Exponential Sums, London Math. Soc. Lecture Note Ser., vol. 126, Cambridge University Press, Cambridge, 1991, https://doi.org/10.1017/CBO9780511661976. (Cited on p. 12.)

[15] E. Hasanalizade, Q. Shen, P.-J. Wong, Counting zeros of the Riemann zeta function, J. Number Theory 235 (2022) 219–241, https://doi.org/10.1016/j.jnt.2021.06.032. (Cited on pp. 4, 11, and 14.)

[16] R.A. Hunt, On the convergence of Fourier series, in: Orthogonal Expansions and their Continuous Analogues (Proc. Conf., Edwardsville, Ill., 1967), Southern Illinois Univ. Press, Carbondale, 1968, pp. 235–255. (Cited on p. 13.)

[17] A. Ivić, The Riemann Zeta-Function, Wiley, New York, 1985 (reprinted by Dover, Mineola, NY, 2003). (Cited on p. 13.)

[18] J. Kinnunen, The Hardy–Littlewood maximal function of a Sobolev function, Israel J. Math. 100 (1997) 117–124, https://doi.org/10.1007/BF02773636. (Cited on p. 13.)

[19] M. Korolev, Gram's law and the argument of the Riemann zeta function, Publ. Inst. Math. (Beograd) (N.S.) 92(106) (2012) 53–78, https://doi.org/10.2298/PIM1206053K. (Cited on pp. 4, 11, and 67.)

[20] M.A. Korolev, On Gram's law in the theory of the Riemann zeta function, Izv. Math. 76 (2) (2012) 275–309, https://doi.org/10.1070/IM2012v076n02ABEH002583. (Cited on pp. 4, 11, and 67.)

[21] E. Landau, Über die Nullstellen der Zetafunktion, Math. Ann. 71 (1912) 548–564, https://doi.org/10.1007/BF01456808. (Cited on pp. 4 and 13.)

[22] A. LeClair, Riemann hypothesis and random walks: the zeta case, Symmetry 13 (11) (2021) 2014, https://doi.org/10.3390/sym13112014. (Cited on pp. 4, 5, 18, 50, and 67.)

[23] A.K. Lerner, On weighted norm inequalities for the Carleson operator, preprint, arXiv:1310.3352, 2013, https://doi.org/10.48550/arXiv.1310.3352. (Cited on p. 13.)

[24] J.E. Littlewood, On the zeros of the Riemann zeta-function, Proc. Cambridge Philos. Soc. 22 (1924) 295–318, https://doi.org/10.1017/S0305004100014225. (Cited on p. 11.)

[25] F. Mertens, Ein Beitrag zur analytischen Zahlentheorie, J. Reine Angew. Math. 78 (1874) 46–62, https://doi.org/10.1515/crll.1874.78.46. (Cited on p. 12.)

[26] H.L. Montgomery, R.C. Vaughan, Hilbert's inequality, J. Lond. Math. Soc. (2) 8 (1974) 73–82, https://doi.org/10.1112/jlms/s2-8.1.73. (Cited on p. 12.)

[27] A.M. Odlyzko, Tables of zeros of the Riemann zeta function: the first 2,001,052 zeros, accurate to within $4 \cdot 10^{-9}$, `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`, accessed 17 October 2026. (Cited on pp. 46 and 71.)

[28] A. Selberg, Contributions to the theory of the Riemann zeta-function, Arch. Math. Naturvid. 48 (5) (1946) 89–155. (Cited on pp. 4, 11, 13, and 67.)

[29] A. Selberg, Contributions to the theory of Dirichlet's L-functions, Skr. Norske Vid.-Akad. Oslo I 1946 (3) (1946) 2–62. (Cited on pp. 4 and 71.)

[30] A. Soshnikov, Gaussian limit for determinantal random point fields, Ann. Probab. 30 (1) (2002) 171–187, https://doi.org/10.1214/aop/1020107764. (Cited on p. 64.)

[31] E.M. Stein, Singular Integrals and Differentiability Properties of Functions, Princeton University Press, Princeton, NJ, 1970. (Cited on p. 13.)

[32] G. Tenenbaum, Introduction to Analytic and Probabilistic Number Theory, third ed., Grad. Stud. Math., vol. 163, Amer. Math. Soc., Providence, RI, 2015, https://doi.org/10.1090/gsm/163. (Cited on p. 12.)

[33] E.C. Titchmarsh, The Theory of the Riemann Zeta-Function, second ed., revised by D.R. Heath-Brown, Clarendon Press, Oxford, 1986. (Cited on pp. 4, 9, 11, 12, 13, and 44.)

[34] T.S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line II, J. Number Theory 134 (2014) 280–292, https://doi.org/10.1016/j.jnt.2013.07.017. (Cited on pp. 4, 11, and 14.)

[35] K.-M. Tsang, The distribution of the values of the Riemann zeta-function, Ph.D. thesis, Princeton University, 1984. (Cited on pp. 4 and 13.)

[36] G.N. Watson, A Treatise on the Theory of Bessel Functions, second ed., Cambridge University Press, Cambridge, 1944. (Cited on p. 58.)

Appendix A. Programs

The programs below are those of the repository (`code/` and `tools/`); each is reproduced verbatim. They run under Python 3 with NumPy, SciPy, Matplotlib and mpmath, from the repository root, in the order of the `README`. The input files `zeross.bin` and `zLamb.bin` are produced by the first program from Odlyzko's table.

Appendix A.1. `make_inputs_v14.py`

```
"""Build the two input files of ordinal_prime_cutoff_v14 from Odlyzko's table, with certificates.

Usage:
    python3 make_inputs_v14.py ZEROS6 OUT_DIR [--legacy-clock data/zLamb.bin]

ZEROS6 is A. Odlyzko's table "zeros6" (first 2,001,052 zeros of the Riemann zeta function, accurate
to within 4e-9), as downloaded from https://www-users.cse.umn.edu/~odlyzko/zeta_tables/ ; plain
text or gzip, one ordinate per line.

Writes
    OUT_DIR/zeross.bin     ordinates gamma_k, k = 1..2,001,052, little-endian float64
    OUT_DIR/zLamb.bin      RvM clock tau_k, N_0(tau_k) = k - 1/2, correctly rounded float64
    OUT_DIR/make_inputs_certificate.json

Certificates
 1. zeross.bin: each decimal string is converted by Python's correctly rounded float(); the count,
    finiteness and monotonicity are checked, and the SHA-256 is compared with the identifier
    fafb9436...fe0ac printed in the paper.  Equality of the hashes proves that the file used in
    the paper is exactly the correctly rounded Odlyzko table.
 2. zLamb.bin: N_0(t) = t/(2 pi) log(t/(2 pi e)) + 7/8 is strictly increasing on (2 pi, inf).  For
    each k we compute a high-precision root by Newton's method in extended precision, round it to
    float64, and then evaluate the residual F(t) = N_0(t) - (k - 1/2) at the rounded value t and at
    its two float64 neighbours, in extended precision (64-bit significand).  The stored value is the
    neighbour with the smallest |F|; because N_0 is increasing, this is the float64 nearest to the
    exact root (correct rounding) whenever the decision margin exceeds the evaluation error, which
    is checked for every k; the few k at which the root lies within the evaluation error of a
    float64 midpoint are decided by mpmath at 40 significant digits.  In addition, an independent check with
    mpmath at 40 significant digits (mpmath.lambertw) is made at 20,000 indices including both ends.
 3. If a legacy clock file is given, the maximal difference to it is reported (the paper's runs
    used the legacy file; its differences are at the level of 1-2 units in the last place).
"""
import argparse
import gzip
import hashlib
import json
from pathlib import Path

import numpy as np
import mpmath as mp

K = 2001052
SHA_ZEROS = "fafb9436ce6a506de58c96c9bc9765d3942de3f15c921e99e0015b3e945fe0ac"
SHA_LEGACY_CLOCK = "b6f2a8d64600951f34ee29c701caff57ffb3c62805d1e1d3be8fa393f24b1252"


def sha256(path):
    h = hashlib.sha256()
    with open(path, "rb") as f:
        for block in iter(lambda: f.read(1 << 20), b""):
            h.update(block)
    return h.hexdigest()


def read_table(path):
    opener = gzip.open if str(path).endswith(".gz") else open
    vals = []
    with opener(path, "rt") as f:
        for line in f:
            line = line.strip()
            if line:
                vals.append(float(line))          # correctly rounded decimal -> binary64
    g = np.array(vals, dtype="<f8")
    if len(g) != K:
        raise ValueError("expected %d ordinates, found %d" % (K, len(g)))
    if not np.isfinite(g).all() or np.any(np.diff(g) < 0) or g[0] <= 14:
        raise ValueError("ordinates must be finite, nondecreasing and start above 14")
```

```
    return g


def clock(K):
    """Correctly rounded tau_k for k = 1..K and the minimal decision margin."""
    LD = np.longdouble
    two_pi = LD("6.28318530717958647692528676655900")
    e_ld = LD("2.71828182845904523536028747135266")
    k = np.arange(1, K + 1, dtype=LD)
    target = k - LD(1) / LD(2)

    def F(t):
        return t / two_pi * np.log(t / (two_pi * e_ld)) + LD(7) / LD(8) - target

    from scipy.special import lambertw
    a = np.arange(1, K + 1, dtype=float) - 11 / 8
    t = (2 * np.pi * a / np.real(lambertw(a / np.e))).astype(LD)
    for _ in range(4):                                  # Newton in extended precision
        t = t - F(t) / (np.log(t / two_pi) / two_pi)
    t64 = t.astype(np.float64)
    cand = np.stack([np.nextafter(t64, -np.inf), t64, np.nextafter(t64, np.inf)])
    res = np.abs(np.stack([F(c.astype(LD)) for c in cand]))
    best = np.argmin(res, axis=0)
    out = cand[best, np.arange(K)]
    srt = np.sort(res, axis=0)
    margin = srt[1] - srt[0]                            # gap between best and second-best |F|
    # evaluation error of F in extended precision: |F| terms are <= ~ k, eps_ld ~ 1.1e-19
    eval_err = 8 * np.finfo(LD).eps * (np.arange(1, K + 1, dtype=LD) + 10)
    amb = np.flatnonzero(~(margin > 2 * eval_err))
    # undecided in extended precision (root within the evaluation error of a midpoint):
    # decide with mpmath at 40 significant digits, which rounds correctly to binary64
    mp.mp.dps = 40
    for j in amb:
        a = mp.mpf(int(j) + 1) - mp.mpf(11) / 8
        out[j] = float(mp.re(2 * mp.pi * a / mp.lambertw(a / mp.e)))
    decided = margin > 2 * eval_err
    ok = True                                           # every k decided by one of the two methods
    ratio = float(np.min(margin[decided] / (2 * eval_err[decided])))
    return out, ratio, float(np.max(eval_err)), ok, int(len(amb))


def mp_check(tau, idx):
    mp.mp.dps = 40
    worst = mp.mpf(0)
    for k in idx:
        a = mp.mpf(int(k) + 1) - mp.mpf(11) / 8             # index k+1 (1-based)
        exact = 2 * mp.pi * a / mp.lambertw(a / mp.e) if a != 0 else 2 * mp.pi * mp.e
        exact = mp.re(exact)
        x = float(exact)                                    # correctly rounded by mpmath
        if x != tau[k]:
            raise AssertionError("not correctly rounded at k=%d: %r vs %r" % (k + 1, tau[k], x))
        worst = max(worst, abs(mp.mpf(tau[k]) - exact))
    return float(worst)


def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("zeros6", type=Path)
    ap.add_argument("out_dir", type=Path)
    ap.add_argument("--legacy-clock", type=Path, default=None)
    args = ap.parse_args()
    args.out_dir.mkdir(parents=True, exist_ok=True)
    cert = {}

    g = read_table(args.zeros6)
    zpath = args.out_dir / "zeross.bin"
    g.tofile(zpath)
    cert["zeross.bin"] = {"count": K, "sha256": sha256(zpath),
```

```
                          "matches_paper": sha256(zpath) == SHA_ZEROS}
    print("zeross.bin sha256", cert["zeross.bin"]["sha256"], "matches paper:", cert["zeross.bin"]["
     ↪ matches_paper"])

    tau, margin, err, ok, namb = clock(K)
    rng = np.random.default_rng(20261017)
    idx = np.unique(np.concatenate([np.arange(0, 100), np.arange(K - 100, K),
                                    rng.integers(0, K, 19800)]))
    mpmax = mp_check(tau, idx)
    tpath = args.out_dir / "zLamb.bin"
    tau.astype("<f8").tofile(tpath)
    cert["zLamb.bin"] = {"sha256": sha256(tpath), "correct_rounding_certified_for_all_k": ok,
                         "min_ratio_margin_to_twice_evaluation_error": margin, "max_evaluation_error_bound"
     ↪ : err,
                         "indices_decided_by_mpmath_40_digits": namb,
                         "mpmath_checked_indices": int(len(idx)),
                         "mpmath_max_abs_error": mpmax}
    print("zLamb.bin sha256", cert["zLamb.bin"]["sha256"], "correct rounding certified:", ok,
          "min margin/(2 x eval error) = %.3f > 1, max eval error %.2e (%d near-midpoint cases decided by
     ↪ mpmath)" % (margin, err, namb), "mpmath max |error| %.3e" % mpmax)
    if args.legacy_clock is not None:
        old = np.fromfile(args.legacy_clock, dtype="<f8")
        d = np.abs(old - tau)
        ulp = np.spacing(tau)
        cert["legacy_clock"] = {"sha256": sha256(args.legacy_clock),
                                "is_paper_file": sha256(args.legacy_clock) == SHA_LEGACY_CLOCK,
                                "max_abs_difference": float(d.max()),
                                "max_difference_in_ulps": float(np.max(d / ulp)),
                                "fraction_bitwise_equal": float(np.mean(d == 0))}
        print("legacy clock: max |difference| %.3e = %.1f ulp, bitwise equal fraction %.4f"
              % (d.max(), np.max(d / ulp), np.mean(d == 0)))
    (args.out_dir / "make_inputs_certificate.json").write_text(json.dumps(cert, indent=2) + "\n")


if __name__ == "__main__":
    main()
```

Appendix A.2. `opc_analysis_v14.py`

```
"""Ordinal-dependent prime cutoffs for Riemann zeta zero ordinates.

Numerical companion of the manuscript on ordinal prime cutoffs.

Data (not included in the repository, see README):
  data/zeross.bin : ordinates gamma_k of the zeros of zeta, k = 1..2,001,052 (little-endian float64)
  data/zLamb.bin  : Lambert clock tau_k = 2*pi*(k-11/8)/W((k-11/8)/e), same length (little-endian float64)

Notation (as in the manuscript):
  delta_k   = tau_k - gamma_k                    (displacement of the clock)
  T'(k)     = 2*pi / log(tau_k/(2*pi))           (derivative of the clock in the ordinal)
  S_P(t)    = -(1/pi) sum_{p in P} arg(1 - p^{-1/2-it})     (Euler phase, full local factors)
  tau_k^P   = tau_k - T'(k) S_P(tau_k)           (one-step prime-corrected clock)
  delta_k^P = tau_k^P - gamma_k = delta_k - T'(k) S_P(tau_k)
  saturation with level B: the correction T'(k) S_P(tau_k) is replaced by sgn(.) min(|.|, B).

Outputs: ../outputs/*.csv, *.txt, *.tex and ../figures/*.png
Usage:   python3 opc_analysis_v14.py [path_to_data_dir]
"""
import os
import sys
import time
import hashlib
import json
import csv
import platform
import scipy
```

```
28
29 import numpy as np
30 import matplotlib
31 import matplotlib.ticker
32 matplotlib.use("Agg")
33 import matplotlib.pyplot as plt
34 import mpmath as mp
35 from scipy.special import lambertw
36
37 T0 = time.time()
38 HERE = os.path.dirname(os.path.abspath(__file__))
39 DATA = sys.argv[1] if len(sys.argv) > 1 else os.path.join(HERE, "..", "data")
40 OUT = os.path.join(HERE, "..", "outputs")
41 FIG = os.path.join(HERE, "..", "figures")
42 os.makedirs(OUT, exist_ok=True)
43 os.makedirs(FIG, exist_ok=True)
44 LOG = open(os.path.join(OUT, "opc_analysis_log.txt"), "w")
45
46
47 def log(*a):
48     s = " ".join(str(x) for x in a)
49     print(s)
50     LOG.write(s + "\n")
51     LOG.flush()
52
53
54 # ------------------------------------------------------------------------------ data
55 tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), dtype="<f8")
56 gam = np.fromfile(os.path.join(DATA, "zeross.bin"), dtype="<f8")
57 K = 2001052
58 if len(tau) != K or len(gam) != K:
59     raise ValueError("Expected exactly 2,001,052 values in each input; no truncation is allowed")
60 if not np.isfinite(tau).all() or not np.isfinite(gam).all():
61     raise ValueError("Input contains a non-finite ordinate")
62 if np.any(np.diff(tau) <= 0) or np.any(np.diff(gam) < 0):
63     raise ValueError("Clock must be increasing and ordinates nondecreasing")
64 metadata = {"python": platform.python_version(), "numpy": np.__version__,
65             "scipy": scipy.__version__, "matplotlib": matplotlib.__version__,
66             "mpmath": mp.__version__, "files": {}}
67 for filename in ("zeross.bin", "zLamb.bin"):
68     path = os.path.join(DATA, filename)
69     h = hashlib.sha256()
70     with open(path, "rb") as stream:
71         for block in iter(lambda: stream.read(1024 * 1024), b""):
72             h.update(block)
73     metadata["files"][filename] = {"size_bytes": os.path.getsize(path), "sha256": h.hexdigest()}
74 with open(os.path.join(OUT, "analysis_metadata.json"), "w") as stream:
75     json.dump(metadata, stream, indent=2)
76
77 k = np.arange(1, K + 1, dtype=float)
78 Lk = np.log(tau / (2 * np.pi))          # log(tau_k / 2pi) = 1 + W_k
79 Tp = 2 * np.pi / Lk                     # T'(k)
80 delta = tau - gam                       # delta_k = tau_k - gamma_k
81 s_ord = delta / Tp                      # ordinal-normalized displacement
82 log("K =", K, " tau_K =", tau[-1], " gamma_K =", gam[-1])
83
84 # check of the clock formula on a sample of indices
85 idx = np.array([0, 1, 9, 99, 999, 99999, K - 1])
86 a = idx + 1 - 11.0 / 8.0
87 Wv = np.real(lambertw(a / np.e))
88 tau_chk = 2 * np.pi * a / Wv
89 log("clock formula check, max |tau_file - 2pi(k-11/8)/W((k-11/8)/e)| on sample:",
90     float(np.max(np.abs(tau_chk - tau[idx]))))
91
92 EDGES = [0, 1000, 10**4, 10**5, 5 * 10**5, 10**6, K]
93 WIN = list(zip(EDGES[:-1], EDGES[1:]))
94 WLAB = ["(0,1e3]", "(1e3,1e4]", "(1e4,1e5]", "(1e5,5e5]", "(5e5,1e6]", "(1e6,%d]" % K]
95 WTEX = [r"$(0,10^3]$", r"$(10^3,10^4]$", r"$(10^4,10^5]$", r"$(10^5,5\cdot10^5]$", r"$(5\cdot10^5,10^6]$",
```

```
        r"$(10^6,K]$"]   # display labels used in the figures


def rms(x):
    return float(np.sqrt(np.mean(x * x)))


def primes_upto(n):
    sv = np.ones(n + 1, bool)
    sv[:2] = False
    for q in range(2, int(n ** 0.5) + 1):
        if sv[q]:
            sv[q * q::q] = False
    return np.nonzero(sv)[0]


def euler_term(p, t):
    """-(1/pi) arg(1 - p^{-1/2-it}): contribution of the prime p to S_P(t), all powers p^m"""
    return -np.angle(1.0 - p ** -0.5 * np.exp(-1j * t * np.log(p))) / np.pi


def saturate(c, B):
    return np.sign(c) * np.minimum(np.abs(c), B) if B is not None else c


def li2(x, terms=200):
    m = np.arange(1, terms + 1)
    return float(np.sum(x ** m / m ** 2))


def cP(plist):
    """(1/2pi^2) sum_{p in P} Li_2(1/p): asymptotic mean-square gain of the full Euler phase over P"""
    return sum(li2(1.0 / p) for p in plist) / (2 * np.pi ** 2)


PR = primes_upto(2000)
LADDER = [p for p in PR if p in (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 43, 53, 61, 71, 83, 97, 113,
     ↪ 131,
                                 151, 173, 199, 229, 263, 307, 353, 401, 461, 523, 601, 701, 809, 907,
     ↪ 1009, 1201,
                                 1409, 1601, 1801, 1999)]
SCAN = [p for p in PR if 11 <= p <= 71]
THETA = 0.57                                    # variable rule X_k = max(7, (tau_k/2pi)^THETA)
Xvar = np.maximum(7.0, (tau / (2 * np.pi)) ** THETA)
B0 = 0.6
parameters = {"ordinal_windows": WIN, "window_labels": WLAB,
              "prime_max": 2000, "cutoff_ladder": [int(p) for p in LADDER],
              "saturation_scan": [int(p) for p in SCAN],
              "variable_exponent": THETA, "variable_minimum_cutoff": 7,
              "method_saturation": B0, "saturation_levels": [0.5, 0.6, 0.7]}
with open(os.path.join(OUT, "analysis_parameters.json"), "w") as stream:
    json.dump(parameters, stream, indent=2)

# ------------------------------------------------------------------------- one pass over the primes
S = np.zeros(K)          # S_P(tau_k) with P = primes <= current p
Svar = np.zeros(K)       # variable cutoff: prime p used for k with X_k >= p
snap = {}
rms_ladder = {}          # p -> list of RMS per window (no saturation)
max_ladder = {}
scan_rows = []
for p in PR:
    tp = euler_term(p, tau)
    S += tp
    a0 = int(np.searchsorted(Xvar >= p, True)) if Xvar[-1] >= p else K
    if a0 < K:
        Svar[a0:] += tp[a0:]
    if p in LADDER:
        r = delta - Tp * S
```

```
        rms_ladder[p] = [rms(r[a:b]) for a, b in WIN]
        max_ladder[p] = [float(np.max(np.abs(r[a:b]))) for a, b in WIN]
    if p in SCAN:
        for B in (0.5, 0.6, 0.7):
            r = delta - saturate(Tp * S, B)
            ar = np.abs(r)
            j = int(np.argmax(ar))
            scan_rows.append((p, B, float(ar[j]), j + 1, int(np.sum(ar >= 0.5)), rms(r), float(r[258317])))
    if p in (7, 61, 1999):
        snap[p] = S.copy()
log("prime loop done, %.0f s" % (time.time() - T0))

# ---------------------------------------------------------------------------- Table A: methods by window
methods = [("bare clock", np.zeros(K), None),
           ("P={2,3,5,7}, B=0.6", Tp * snap[7], B0),
           ("P={p<=61}, B=0.6", Tp * snap[61], B0),
           ("P={p<=1999} (fixed)", Tp * snap[1999], None),
           ("X_k=max(7,(tau_k/2pi)^0.57), B=0.6", Tp * Svar, B0)]
resid = {}
with open(os.path.join(OUT, "table_A_methods_by_window.csv"), "w") as f:
    f.write("method,window,RMS,max_abs,count_ge_half\n")
    for name, c, B in methods:
        r = delta - saturate(c, B)
        resid[name] = r
        for (a, b), wl in zip(WIN, WLAB):
            f.write('"%s","%s",%.6f,%.6f,%d\n' % (name, wl, rms(r[a:b]), np.max(np.abs(r[a:b])),
                                                  np.sum(np.abs(r[a:b]) >= 0.5)))
        j = int(np.argmax(np.abs(r)))
        log("A| %-38s global RMS %.6f  max %.6f at k=%d  #>=1/2: %d" %
            (name, rms(r), abs(r[j]), j + 1, int(np.sum(np.abs(r) >= 0.5))))
        log("   per-window RMS:", " ".join("%.4f" % rms(r[a:b]) for a, b in WIN))
        log("   per-window max:", " ".join("%.4f" % np.max(np.abs(r[a:b])) for a, b in WIN))

# ---------------------------------------------------------------------------- Table B: RMS against the
    ↪ cutoff
with open(os.path.join(OUT, "table_B_rms_vs_cutoff.csv"), "w") as f:
    csv.writer(f).writerow(["X"] + ["RMS_"+w for w in WLAB] + ["max_"+w for w in WLAB])
    for p in LADDER:
        f.write("%d," % p + ",".join("%.6f" % v for v in rms_ladder[p]) + "," +
                ",".join("%.6f" % v for v in max_ladder[p]) + "\n")
opt = []
for j, ((a, b), wl) in enumerate(zip(WIN, WLAB)):
    pbest = min(LADDER, key=lambda p: rms_ladder[p][j])
    Lmed = float(np.median(Lk[a:b]))
    opt.append((wl, pbest, rms_ladder[pbest][j], Lmed, rms(delta[a:b])))
    log("B| window %-16s optimal X = %5d  RMS = %.4f   median log(tau/2pi) = %.3f  bare RMS = %.4f" %
        (wl, pbest, rms_ladder[pbest][j], Lmed, rms(delta[a:b])))
xs = np.array([o[3] for o in opt[1:]])
ys = np.log(np.array([o[1] for o in opt[1:]], float))
theta_fit = float(np.sum(xs * ys) / np.sum(xs * xs))
log("B| least-squares fit log X_opt = theta * log(tau/2pi) on windows 2-6: theta = %.4f" % theta_fit)
with open(os.path.join(OUT, "table_B_optimal_cutoff.csv"), "w") as f:
    f.write("window,X_opt,RMS_at_opt,median_log_tau_over_2pi,bare_RMS\n")
    for o in opt:
        f.write('"%s",%d,%.6f,%.4f,%.6f\n' % o)
    f.write("theta_fit,%.6f\n" % theta_fit)

# ---------------------------------------------------------------------------- Table C: fixed cutoff and
    ↪ saturation, global
with open(os.path.join(OUT, "table_C_fixed_cutoff_saturation.csv"), "w") as f:
    f.write("X,B,global_max,k_at_max,count_ge_half,RMS,residual_at_k258318\n")
    for row in scan_rows:
        f.write("%d,%.1f,%.6f,%d,%d,%.6f,%.6f\n" % row)
for row in scan_rows:
    if row[0] in (11, 13, 31, 61, 71):
        log("C| X=%3d B=%.1f  max %.4f at k=%8d  #>=1/2 %d  RMS %.4f  r(258318) %+.4f" % row)

```

```
# ------------------------------------------------------------------------------ Table D: measured mean-square gain against theory
PP = {7: [2, 3, 5, 7], 61: [int(p) for p in PR if p <= 61], 1999: [int(p) for p in PR if p <= 1999]}
with open(os.path.join(OUT, "table_D_gain_vs_theory.csv"), "w") as f:
    f.write("window,mean_s2,gain_P7,c_P7,gain_P61,c_P61\n")
    for (a, b), wl in zip(WIN, WLAB):
        s2 = float(np.mean(s_ord[a:b] ** 2))
        g7 = s2 - float(np.mean((s_ord[a:b] - snap[7][a:b]) ** 2))
        g61 = s2 - float(np.mean((s_ord[a:b] - snap[61][a:b]) ** 2))
        f.write('"%s",%.6f,%.6f,%.6f,%.6f,%.6f\n' % (wl, s2, g7, cP(PP[7]), g61, cP(PP[61])))
        log("D| %-16s <s^2> %.4f  gain P7 %.4f (c=%.4f)  gain P61 %.4f (c=%.4f)" %
            (wl, s2, g7, cP(PP[7]), g61, cP(PP[61])))

# ------------------------------------------------------------------------------ Table E: projection coefficients
a, b = 10**6, K
def Lambda(n):
    for p in PR:
        if p > n:
            break
        if n % p == 0:
            m = n
            while m % p == 0:
                m //= p
            return float(np.log(p)) if m == 1 else 0.0
    return 0.0

Lmed_top = float(np.median(Lk[a:b]))


def factor(n):
    out = {}
    for p in PR:
        if p > n:
            break
        while n % p == 0:
            out[p] = out.get(p, 0) + 1
            n //= p
    return out


def second_order(n):
    """second-order prediction log(n)/(a*b*log(tau/2pi)) for n = p^a q^b with two distinct primes, else nan """
    fz = factor(n)
    if len(fz) != 2:
        return float("nan")
    e = list(fz.values())
    return float(np.log(n)) / (e[0] * e[1] * Lmed_top)


rows_E = []
for n in range(2, 65):
    proj = float(np.mean(s_ord[a:b] * np.sin(tau[a:b] * np.log(n))))
    rho = -2 * np.pi * np.sqrt(n) * proj
    theo = Lambda(n) / np.log(n)
    rows_E.append((n, proj, rho, theo, second_order(n)))
with open(os.path.join(OUT, "table_E_projection_coefficients.csv"), "w") as f:
    f.write("n,projection,rho_n=-2pi*sqrt(n)*projection,Lambda(n)/log(n),second_order_log(n)/(ab*L)\n")
    for r_ in rows_E:
        f.write("%d,%.6e,%.6f,%.6f,%.6f\n" % r_)
so_err = [abs(r_[2] - r_[4]) for r_ in rows_E if r_[4] == r_[4]]
log("E| median log(tau/2pi) in the window: %.4f ; max |rho_n - log n/(ab L)| over n = p^a q^b <= 64: %.4f" %
    (Lmed_top, max(so_err)))
log("E| n, rho_n, first-order, second-order:", " ".join("%d:%.3f/%.3f/%.3f" % (r_[0], r_[2], r_[3], r_[4])
                                                        for r_ in rows_E))
pp_err = [abs(r_[2] - r_[3]) for r_ in rows_E if r_[3] > 0]
```

```
np_err = [abs(r_[2]) for r_ in rows_E if r_[3] == 0]
log("E| window k in (1e6, K]: max |rho_n - Lambda(n)/log n| over prime powers n<=64: %.4f ; max |rho_n| over"
    " non-prime-powers: %.4f" % (max(pp_err), max(np_err)))
log("E| rho_2..rho_12:", " ".join("%d:%.3f" % (r_[0], r_[2]) for r_ in rows_E[:11]))

# ---------------------------------------------------------------------------- the extreme residual: a close pair
kk = 258318
i0 = kk - 1
ng = (gam[1:] - gam[:-1]) * np.log(gam[:-1] / (2 * np.pi)) / (2 * np.pi)
rank = int(np.sum(ng < ng[i0 - 1])) + 1
mp.mp.dps = 25
with open(os.path.join(OUT, "close_pair_k258318.txt"), "w") as f:
    f.write("k, gamma_k (zeross.bin), tau_k, delta_k=tau-gamma, gap to next, normalized gap to next\n")
    for j in range(i0 - 3, i0 + 4):
        f.write("%d %.9f %.9f %+.4f %.4f %.4f\n" % (j + 1, gam[j], tau[j], delta[j], gam[j + 1] - gam[j], ng[j]))
    for n in (kk - 1, kk, kk + 1):
        z = mp.zetazero(n)
        f.write("mpmath zetazero(%d) = %s ; difference with zeross.bin = %.3e\n" %
                (n, mp.nstr(z.imag, 18), float(z.imag) - gam[n - 1]))
    f.write("normalized gap (k-1,k) = %.4f, rank %d among %d gaps\n" % (ng[i0 - 1], rank, len(ng)))
    for n in (1, 1000, 100000, 1000000, K):
        z = mp.zetazero(n)
        f.write("data check: mpmath zetazero(%d) - zeross.bin = %.3e\n" % (n, float(z.imag) - gam[n - 1]))
    f.write("S(gamma_k) (midpoint) = %s\n" % mp.nstr(kk - mp.mpf(1) / 2 - mp.siegeltheta(mp.mpf(gam[i0])) / mp.pi - 1, 10))
    for name in resid:
        f.write("residual %-38s %+.4f\n" % (name, resid[name][i0]))
log(open(os.path.join(OUT, "close_pair_k258318.txt")).read())

# ---------------------------------------------------------------------------- figures
plt.rcParams.update({"font.size": 10, "figure.dpi": 200, "savefig.dpi": 200})
C0, C1, C2, C3, C4 = "#1f4e79", "#c0504d", "#4f9a3a", "#8064a2", "#e08a1e"
# display labels in the notation of the paper (calligraphic P = finite set of primes)
DISP = {"bare clock": r"bare clock $\tau_k$",
        "P={2,3,5,7}, B=0.6": r"$\mathcal{P}=\{2,3,5,7\}$, $B=0.6$",
        "P={p<=61}, B=0.6": r"$\mathcal{P}=\{p\leq 61\}$, $B=0.6$",
        "P={p<=1999} (fixed)": r"$\mathcal{P}=\{p\leq 1999\}$, no saturation",
        "X_k=max(7,(tau_k/2pi)^0.57), B=0.6": r"$X_k=\max(7,(\tau_k/2\pi)^{0.57})$, $B=0.6$"}

# figure: displacement for k <= 2000
fig, ax = plt.subplots(figsize=(7.2, 3.6))
n0 = 2000
ax.plot(k[:n0], delta[:n0], ".", ms=2.2, color=C0, label=r"$\delta_k=\tau_k-\gamma_k$")
ax.plot(k[:n0], resid["P={2,3,5,7}, B=0.6"][:n0], ".", ms=2.2, color=C1,
        label=r"one-step residual, $\mathcal{P}=\{2,3,5,7\}$, $B=0.6$")
ax.axhline(0, color="k", lw=0.5)
ax.set_xlabel("ordinal $k$")
ax.set_ylabel("displacement")
ax.legend(loc="upper right", fontsize=8)
ax.set_title("Clock displacement and one-step Euler correction, $k\\leq 2000$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig01_displacement_first2000.png"))

# figure: binned RMS against log10 k
bins = np.unique(np.round(np.logspace(1, np.log10(K), 36)).astype(int))
mid, curves = [], {nm: [] for nm in resid}
for a_, b_ in zip(bins[:-1], bins[1:]):
    mid.append(np.sqrt(a_ * b_))
    for nm in resid:
        curves[nm].append(rms(resid[nm][a_:b_]))
fig, ax = plt.subplots(figsize=(7.2, 4.0))
for (nm, col) in zip(resid, (C0, C1, C2, C3, C4)):
    ax.plot(mid, curves[nm], "-o", ms=2.5, lw=1.2, color=col, label=DISP[nm])
ax.set_xscale("log")
ax.set_xlabel("ordinal $k$ (log scale)")
```

```
ax.set_ylabel("RMS of the residual in the bin")
ax.legend(fontsize=7.5)
ax.set_title("Root mean square of the residual, logarithmic bins of $k$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "figS1_rms_by_height.png"))

# figure: RMS against the cutoff, per window
fig, ax = plt.subplots(figsize=(7.2, 4.0))
cols = (C0, C1, C2, C3, C4, "#555555")
for j, wl in enumerate(WTEX):
    yv = [rms_ladder[p][j] for p in LADDER]
    ax.plot(LADDER, yv, "-", lw=1.2, color=cols[j], label="$k\\in$" + wl)
    pb = opt[j][1]
    ax.plot([pb], [opt[j][2]], "o", color=cols[j], ms=5)
ax.set_xscale("log")
ax.set_xlabel("prime cutoff $X$ (primes $p\\leq X$, full local factors)")
ax.set_ylabel("RMS of $\\tau_k^X-\\gamma_k$")
ax.legend(fontsize=7.5)
ax.set_title("RMS against the cutoff; dots mark the optimum of each window")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig04_rms_vs_cutoff.png"))

# figure: optimal cutoff against log(tau/2pi)
fig, ax = plt.subplots(figsize=(6.0, 4.0))
xo = np.array([o[3] for o in opt])
yo = np.log(np.array([o[1] for o in opt], float))
ax.plot(xo, yo, "o", color=C0, label="RMS-optimal $\\log X$ per window")
xx = np.linspace(4, 12.5, 50)
ax.plot(xx, theta_fit * xx, "-", color=C1, label="$\\log X=%.3f\\,\\log(\\tau/2\\pi)$" % theta_fit)
ax.set_xlabel("median of $\\log(\\tau_k/2\\pi)$ in the window")
ax.set_ylabel("$\\log X_{\\rm opt}$")
ax.legend(fontsize=8)
ax.set_title("The optimal prime cutoff grows with the height")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "figS2_optimal_cutoff.png"))

# figure: close pair
fig, ax = plt.subplots(figsize=(7.2, 2.8))
js = np.arange(i0 - 4, i0 + 5)
ax.plot(gam[js], np.zeros_like(js), "o", color="k", ms=6, label="$\\gamma_k$ (zeros)")
ax.plot(tau[js], 0.6 * np.ones_like(js), "s", color=C0, ms=5, label="$\\tau_k$ (clock)")
ax.plot(tau[js] - (Tp * snap[7])[js], 1.2 * np.ones_like(js), "^", color=C1, ms=5, label="$\\tau_k^{\\
    ↪ mathcal{P}}$, $\\mathcal{P}=\\{2,3,5,7\\}$")
ax.plot(tau[js] - (Tp * snap[61])[js], 1.8 * np.ones_like(js), "v", color=C2, ms=5, label="$\\tau_k^{\\
    ↪ mathcal{P}}$, $\\mathcal{P}=\\{p\\leq61\\}$")
for j in js:
    ax.plot([gam[j], tau[j]], [0, 0.6], "-", color="#999999", lw=0.6)
ax.annotate("$k=258317,258318$", xy=(gam[i0 - 1], 0), xytext=(gam[i0 - 1] - 1.2, -0.5), fontsize=8,
            arrowprops=dict(arrowstyle="->", lw=0.6))
ax.set_yticks([])
ax.set_ylim(-0.8, 2.3)
ax.set_xlabel("$t$")
ax.legend(fontsize=7, ncol=4, loc="upper center")
ax.set_title("The extreme residual: a close pair of zeros followed by a large gap")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "figS3_close_pair.png"))

# figure: measured gain against theory
g7s, g61s, s2s = [], [], []
for a_, b_ in WIN:
    s2 = float(np.mean(s_ord[a_:b_] ** 2))
    s2s.append(s2)
    g7s.append(s2 - float(np.mean((s_ord[a_:b_] - snap[7][a_:b_]) ** 2)))
    g61s.append(s2 - float(np.mean((s_ord[a_:b_] - snap[61][a_:b_]) ** 2)))
fig, ax = plt.subplots(figsize=(7.2, 3.8))
xw = np.arange(len(WLAB))
ax.plot(xw, g7s, "o-", color=C1, label="measured gain, $\\mathcal{P}=\\{2,3,5,7\\}$")
ax.axhline(cP(PP[7]), color=C1, ls="--", lw=1, label="$v_{\\mathcal{P}}=%.5f$" % cP(PP[7]))
```

```
ax.plot(xw, g61s, "s-", color=C2, label="measured gain, $\\mathcal{P}=\\{p\\leq 61\\}$")
ax.axhline(cP(PP[61]), color=C2, ls="--", lw=1, label="$v_{\\mathcal{P}}=%.5f$" % cP(PP[61]))
ax.set_xticks(xw)
ax.set_xticklabels(WTEX, fontsize=8)
ax.set_xlabel("window of ordinals")
ax.set_ylabel("$\\langle s_k^2\\rangle-\\langle (s_k-S_{\\mathcal{P}}(\\tau_k))^2\\rangle$")
ax.legend(fontsize=7.5)
ax.set_title("Mean-square gain of a fixed prime set against $v_{\\mathcal{P}}$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig06_gain_fixed_sets.png"))

# figure: maximum per window
fig, ax = plt.subplots(figsize=(7.2, 3.8))
for (nm, col, mk) in zip(resid, (C0, C1, C2, C3, C4), "osd^v"):
    ax.plot(xw, [np.max(np.abs(resid[nm][a_:b_])) for a_, b_ in WIN], "-" + mk, color=col, ms=4, label=DISP
    ↪ [nm])
ax.axhline(0.5, color="k", lw=0.6, ls=":")
ax.set_xticks(xw)
ax.set_xticklabels(WTEX, fontsize=8)
ax.set_ylabel("max $|$residual$|$ in the window")
ax.set_yscale("log")
ax.set_ylim(0.3, 4.0)
ax.set_yticks([0.3, 0.5, 1.0, 2.0, 3.0])
ax.set_yticklabels(["0.3", "0.5", "1", "2", "3"])
ax.yaxis.set_minor_formatter(matplotlib.ticker.NullFormatter())
ax.legend(fontsize=7)
ax.set_title("Maximum of the residual per window")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig05_max_by_window.png"))

log("total time %.0f s" % (time.time() - T0))
LOG.close()
```

Appendix A.3. `opc_checks_v14.py`

```
"""Tests of the finite-height and finite-window statements of ordinal_prime_cutoff_v14.

Usage: python3 opc_checks_v14.py DATA_DIR

Inputs : DATA_DIR/zeross.bin (ordinates gamma_k), DATA_DIR/zLamb.bin (clock tau_k), each exactly
         2,001,052 little-endian float64 values; ../outputs/table_F_direct_defects.csv and
         ../outputs/table_G_prime_responses.csv written by opc_extensions_v14.py.
Outputs: ../outputs/check_envelope.csv          finite-height envelope (Remark on Bellotti--Wong)
         ../outputs/check_accounting.csv        exact finite-window gain = diagonal - cross term
         ../outputs/check_response_bins.csv     grouped moving-prime responses (bin width 0.05, >= 5 primes)
         ../outputs/check_log.txt               everything printed
         ../figures/fig02_direct_defects.png    direct ordinal defects against the cutoff exponent
         ../figures/fig03_accounting.png        gain, diagonal part and cross term against the cutoff
         ../figures/fig07_response_law.png      grouped responses R^gamma against 1 - Phi(theta)
         ../figures/fig08_response_clock_vs_zero.png   R^tau against R^gamma

Notation (as in the paper): N_0(t) = t/(2 pi) log(t/(2 pi e)) + 7/8, L(t) = log(t/2pi)/(2pi),
s_k = k - 1/2 - N_0(gamma_k) (exact), delta_k = tau_k - gamma_k,
BW(t) = 0.10076 log t + 0.24460 log log t + 8.08344,
a_n = -Lambda(n)/(pi sqrt(n) log n), S_X(t) = sum over prime powers p^m, p <= X, of a_{p^m} sin(t log p^m),
v_X = (1/2) sum a_n^2, Cr_X = <S_X(tau_k)^2> - v_X, D_X = 2 <s_k S_X(tau_k)> - v_X.
No parameter is fitted.
"""
import csv
import os
import sys

import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
```

```

HERE = os.path.dirname(os.path.abspath(__file__))
DATA = sys.argv[1] if len(sys.argv) > 1 else os.path.join(HERE, "..", "data")
OUT = os.path.join(HERE, "..", "outputs")
FIG = os.path.join(HERE, "..", "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)
LOG = open(os.path.join(OUT, "check_log.txt"), "w")
K = 2001052


def log(*a):
    s = " ".join(str(x) for x in a)
    print(s, flush=True)
    LOG.write(s + "\n")
    LOG.flush()


def load(name):
    path = os.path.join(DATA, name)
    if os.path.getsize(path) != 8 * K:
        raise ValueError("%s: expected exactly %d float64 values" % (path, K))
    x = np.fromfile(path, dtype="<f8")
    if not np.isfinite(x).all() or np.any(np.diff(x) < 0):
        raise ValueError("%s: values must be finite and nondecreasing" % path)
    return x


def primes_upto(n):
    sieve = np.ones(n + 1, dtype=bool)
    sieve[:2] = False
    for p in range(2, int(n ** 0.5) + 1):
        if sieve[p]:
            sieve[p * p::p] = False
    return np.flatnonzero(sieve)


def Phi(th):
    th = np.asarray(th, dtype=float)
    return (1 + th) * np.log1p(th) + (1 - th) * np.log1p(-th)


gam = load("zeross.bin")
tau = load("zLamb.bin")
k = np.arange(1, K + 1, dtype=float)
N0g = gam / (2 * np.pi) * np.log(gam / (2 * np.pi * np.e)) + 7 / 8
s = k - 0.5 - N0g
delta = tau - gam
Lg = np.log(gam / (2 * np.pi)) / (2 * np.pi)
Lt = np.log(tau / (2 * np.pi)) / (2 * np.pi)
WINDOWS = [(0, 1000), (1000, 10000), (10000, 100000), (100000, 500000), (500000, 1000000), (1000000, K)]
LAB = ["(0,1e3]", "(1e3,1e4]", "(1e4,1e5]", "(1e5,5e5]", "(5e5,1e6]", "(1e6,K]"]

# ------------------------------------------------------------- 1. finite-height envelope
BW = 0.10076 * np.log(gam) + 0.24460 * np.log(np.log(gam)) + 8.08344
upper = (BW - 11 / 8) / Lg
lower = -(BW + 3 / 8) / Lt
ok_s = bool(np.all((s >= -BW - 3 / 8) & (s <= BW - 11 / 8)))
ok_d = bool(np.all((delta >= lower) & (delta <= upper)))
log("envelope: ordinal sandwich holds for all k:", ok_s, "; displacement envelope holds for all k:", ok_d)
rows = []
for (a, b), lab in zip(WINDOWS, LAB):
    d, u, l = delta[a:b], upper[a:b], lower[a:b]
    rows.append((lab, b - a, float(gam[a]), float(gam[b - 1]), float(d.max()), float(d.min()),
                 float(u.min()), float(l.max()), float(np.max(d / u)), float(np.max(-d / np.abs(l)))))
    log("envelope %-10s max delta=%.6f min delta=%.6f  min upper=%.4f  max lower=%.4f  max delta/upper=%.4f
     ↪   max(-delta)/|lower|=%.4f"
        % (lab, rows[-1][4], rows[-1][5], rows[-1][6], rows[-1][7], rows[-1][8], rows[-1][9]))
```

```
with open(os.path.join(OUT, "check_envelope.csv"), "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(("window", "zero_count", "gamma_first", "gamma_last", "max_delta", "min_delta",
                "min_upper_envelope", "max_lower_envelope", "max_delta_over_upper", "
     ↪ max_minus_delta_over_abs_lower"))
    w.writerows(rows)
j = int(np.argmax(np.abs(delta)))
log("global max |delta_k| = %.6f at k = %d (gamma = %.6f)" % (abs(delta[j]), j + 1, gam[j]))

# ------------------------------------------------------------------ 2. exact finite-window accounting
PR = primes_upto(2000)
XS = [7, 61, 401, 907, 1999]
ACC_W = [1, 2, 3, 4, 5]                       # indices into WINDOWS: all windows above k = 1000
acc = []
for wi in ACC_W:
    a, b = WINDOWS[wi]
    t, sw = tau[a:b], s[a:b]
    S = np.zeros(b - a)
    vX = 0.0
    done = 0
    for X in XS:
        for p in PR[(PR > done) & (PR <= X)]:
            p = int(p)
            lp = np.log(p)
            m = 1
            while p ** (-m / 2) > 1e-17:
                S += -np.sin(m * lp * t) / (np.pi * m * p ** (m / 2))
                vX += 1 / (2 * np.pi ** 2 * m * m * p ** m)
                m += 1
        done = X
        gain = float(np.mean(sw * sw) - np.mean((sw - S) ** 2))
        D = float(2 * np.mean(sw * S) - vX)
        Cr = float(np.mean(S * S) - vX)
        acc.append((LAB[wi], X, b - a, float(np.mean(sw * sw)), gain, D, Cr, vX))
        log("accounting %-10s X=%5d  <s^2>=%.6f  gain=%.6f  D=%.6f  Cr=%.6f  D-Cr-gain=%.1e  v_X=%.6f"
            % (LAB[wi], X, float(np.mean(sw * sw)), gain, D, Cr, D - Cr - gain, vX))
with open(os.path.join(OUT, "check_accounting.csv"), "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(("window", "X", "zero_count", "mean_s2", "gain", "diagonal_D_X", "cross_Cr_X", "v_X"))
    w.writerows(acc)

# ------------------------------------------------------------------ 3. grouped moving-prime responses
G = []
with open(os.path.join(OUT, "table_G_prime_responses.csv")) as f:
    for r in csv.DictReader(f):
        G.append((int(r["window"]), int(r["prime"]), float(r["theta_median"]), float(r["rho_gamma"]),
                  float(r["rho_tau"])))
G = np.array(G)
bins = []
edges = np.round(np.arange(0.0, 1.0001, 0.05), 2)
for wnd in range(1, 6):
    Gw = G[G[:, 0] == wnd]
    for lo, hi in zip(edges[:-1], edges[1:]):
        sel = Gw[(Gw[:, 2] >= lo) & (Gw[:, 2] < hi)]
        if len(sel) < 5:
            continue
        law = 1 - Phi(sel[:, 2])
        bins.append((wnd, lo, hi, len(sel), float(np.mean(sel[:, 2])), float(np.mean(sel[:, 3])),
                     float(np.mean(law)), float(np.mean(sel[:, 3]) - np.mean(law)),
                     float(np.std(sel[:, 3] - law) / np.sqrt(len(sel))), float(np.mean(sel[:, 4]))))
with open(os.path.join(OUT, "check_response_bins.csv"), "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(("window", "theta_lo", "theta_hi", "prime_count", "mean_theta", "mean_R_gamma",
                "mean_law", "difference", "standard_error_of_difference", "mean_R_tau"))
    w.writerows(bins)
for r in bins:
    if r[0] == 5:
        log("bins top window [%.2f,%.2f) n=%4d  R^g=%.6f  law=%.6f  diff=%+.6f (se %.6f)  R^t=%.6f"
```

```
            % (r[1], r[2], r[3], r[5], r[6], r[7], r[8], r[9]))

# ---------------------------------------------------------------- figures
plt.rcParams.update({"font.size": 10, "figure.dpi": 200, "savefig.dpi": 200})
F = []
with open(os.path.join(OUT, "table_F_direct_defects.csv")) as f:
    for r in csv.DictReader(f):
        F.append((int(r["r"]), float(r["c"]), float(r["mean_u2"]), float(r["mean_one_step_r2"])))
F = np.array(F)
fig, ax = plt.subplots(figsize=(7, 4))
cols = {14: "tab:purple", 16: "tab:orange", 18: "tab:green", 19: "tab:blue"}
for r in (14, 16, 18, 19):
    sel = F[F[:, 0] == r]
    x = np.log(1 / sel[:, 1])
    ax.plot(x, sel[:, 2], "o-", color=cols[r], label=r"$\langle u_k(X)^2\rangle$, $T=2^{%d}$" % r)
    ax.plot(x, sel[:, 3], "s:", mfc="none", color=cols[r], label=r"$\langle r_k(X)^2\rangle$, $T=2^{%d}$" %
     ↪  r)
xx = np.linspace(0.6, 2.4, 50)
ax.plot(xx, xx / (2 * np.pi ** 2), "k--", label=r"$\frac{1}{2\pi^2}\log(1/c)$")
ax.set_xlabel(r"$\log(1/c)$, $X=T^c$")
ax.set_ylabel("mean over $T<\\gamma_k\\leq 2T$")
ax.set_title("Direct ordinal defects against the cutoff exponent")
ax.legend(fontsize=6.5, ncol=2)
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig02_direct_defects.png"))

fig, ax = plt.subplots(figsize=(7, 5.4))
wcols = {"(1e3,1e4]": "tab:purple", "(1e4,1e5]": "tab:orange", "(1e5,5e5]": "tab:green",
         "(5e5,1e6]": "tab:red", "(1e6,K]": "tab:blue"}
TEXW = {"(1e3,1e4]": r"$(10^3,10^4]$", "(1e4,1e5]": r"$(10^4,10^5]$", "(1e5,5e5]": r"$(10^5,5\cdot10^5]$",
        "(5e5,1e6]": r"$(5\cdot10^5,10^6]$", "(1e6,K]": r"$(10^6,K]$"}
for lab in wcols:
    rr = [x for x in acc if x[0] == lab]
    X = [x[1] for x in rr]
    ax.plot(X, [x[4] for x in rr], "o-", color=wcols[lab], label=r"gain, $k\in$" + TEXW[lab])
    ax.plot(X, [x[6] for x in rr], "s:", mfc="none", color=wcols[lab], label=r"$\mathrm{Cr}_X$, $k\in$" +
     ↪ TEXW[lab])
ax.plot(XS, [acc[i][7] for i in range(len(XS))], "k--", label=r"$v_X$")
ax.set_xscale("log")
ax.set_xlabel("cutoff $X$ (primes $p\\leq X$, full local factors)")
ax.set_ylabel("mean square")
ax.set_title("Exact finite-window accounting of the one-step gain")
ax.legend(fontsize=7, ncol=3, loc="upper center", bbox_to_anchor=(0.5, -0.17))
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig03_accounting.png"))

B = np.array([b for b in bins])
fig, ax = plt.subplots(figsize=(7, 4))
th = np.linspace(0, 0.97, 200)
ax.plot(th, 1 - Phi(th), "k-", label=r"$1-\Phi(\theta)$")
ax.plot(th, 1 - th ** 2, color="gray", ls="--", label=r"$1-\theta^2$")
wl = {1: r"$(10^3,10^4]$", 2: r"$(10^4,10^5]$", 3: r"$(10^5,5\cdot10^5]$", 4: r"$(5\cdot10^5,10^6]$", 5: r"
     ↪ $(10^6,K]$"}
wc = {1: "tab:purple", 2: "tab:orange", 3: "tab:green", 4: "tab:red", 5: "tab:blue"}
for wnd in range(1, 6):
    sel = B[B[:, 0] == wnd]
    ax.errorbar(sel[:, 4], sel[:, 5], yerr=sel[:, 8], fmt="o", ms=3.5, color=wc[wnd], label=r"$k\in$" + wl[
     ↪ wnd])
ax.set_xlabel(r"$\theta=\log p/\mathrm{median}\,\log(\tau_k/2\pi)$")
ax.set_ylabel(r"bin mean of $\mathcal{R}^{\gamma}_p$")
ax.set_title(r"Prime responses of $s_k$ at the zeros (bins of width 0.05 in $\theta$)")
ax.legend(fontsize=7)
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig07_response_law.png"))

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(th, 1 - Phi(th), "k-", lw=1, label=r"$1-\Phi(\theta)$")
for wnd in (2, 3, 5):
```

```
    sel = B[B[:, 0] == wnd]
    ax.plot(sel[:, 4], sel[:, 9], "s-", color=wc[wnd], label=r"$\mathcal{R}^{\tau}_p$, $k\in$" + wl[wnd])
    ax.plot(sel[:, 4], sel[:, 5], "o:", color=wc[wnd], label=r"$\mathcal{R}^{\gamma}_p$, $k\in$" + wl[wnd])
ax.set_xlabel(r"$\theta$")
ax.set_ylabel("bin mean of the response")
ax.set_title("Response at the clock point against response at the zero")
ax.legend(fontsize=7)
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig08_response_clock_vs_zero.png"))
log("done")
LOG.close()
```

Appendix A.4. opc_computable_response_v14.py

```
#!/usr/bin/env python3
"""Computable and residual parts of the prime responses at the clock points (ordinal_prime_cutoff_v14,
    ↪ Section 11.4).

Usage:  python3 opc_computable_response_v14.py DATA_DIR [OUT_DIR]

Window T = 2^19 < gamma_k <= 2T.  For cutoff exponents (c_1, c_2, c_3) and the Newton points x^(m) of
opc_iteration_v14.py, and for primes p:
  R_tau   = -2 pi sqrt(p) < s_k sin(tau_k log p) >,                  s_k = L(tau_k)(tau_k - gamma_k);
  A_m     = -2 pi sqrt(p) < L(tau_k)(tau_k - x^(m)_k) sin(tau_k log p) >   (computable, no zeros);
  D_m     = -2 pi sqrt(p) < S_{X_m}(tau_k) sin(tau_k log p) >            (diagonal part);
  B_m     = A_m - D_m   (response of the transport S_{X_m}(x^(m-1)) - S_{X_m}(tau_k));
  E_m     = R_tau - A_m (response of the residual L(tau_k)(x^(m)_k - gamma_k)).
The column 'res' marks p X_m >= T/(2 pi), where the frequencies log(np), n <= X_m, reach the stationary band
of the clock (Proposition 8.15) and D_m is no longer small.  Output: OUT_DIR/table_L_computable_response.csv.
    ↪
No parameter is fitted.
"""
import csv
import os
import sys

import numpy as np

DATA = sys.argv[1] if len(sys.argv) > 1 else "data"
OUT = sys.argv[2] if len(sys.argv) > 2 else "outputs"
os.makedirs(OUT, exist_ok=True)
g = np.fromfile(os.path.join(DATA, "zeross.bin"), "<f8")
tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), "<f8")
K = len(g)
k = np.arange(1, K + 1, dtype=float)
N0 = lambda x: x / (2 * np.pi) * np.log(x / (2 * np.pi * np.e)) + 7 / 8
L = lambda x: np.log(x / (2 * np.pi)) / (2 * np.pi)


def primes(n):
    sv = np.ones(n + 1, bool)
    sv[:2] = False
    for p in range(2, int(n ** 0.5) + 1):
        if sv[p]:
            sv[p * p::p] = False
    return np.flatnonzero(sv)


def phase(x, P):
    out = np.zeros_like(x)
    for p in P:
        lp = np.log(p)
        m = 1
        while p ** (-m / 2) > 1e-17:
            out += -np.sin(m * lp * x) / (np.pi * m * p ** (m / 2))
            m += 1
```

```
    return out


T = 2.0 ** 19
sel = (g > T) & (g <= 2 * T)
gg, tt, kk = g[sel], tau[sel], k[sel]
M = len(gg)
LT = np.log(T / (2 * np.pi))
s = L(tt) * (tt - gg)
plist = [3, 7, 31, 101, 211, 1009, 2003, 10007, 20011, 50021, 100003, 131101]


def resp(v, p):
    return -2 * np.pi * np.sqrt(p) * np.mean(v * np.sin(tt * np.log(p)))


rows = []
for cs in ((1 / 10, 1 / 5, 1 / 3), (1 / 5, 1 / 3, 1 / 2), (1 / 3, 1 / 2, 1 / 2)):
    x = [tt.copy()]
    Xs, ph_tau = [], []
    for c in cs:
        X = int(T ** c)
        P = primes(X)
        Xs.append(X)
        ph_tau.append(phase(tt, P))
        xp = x[-1]
        x.append(xp - (N0(xp) + phase(xp, P) - (kk - 0.5)) / L(xp))
    Y = [L(tt) * (tt - x[m]) for m in (1, 2, 3)]
    for p in plist:
        row = dict(c1=f"{cs[0]:.4f}", c2=f"{cs[1]:.4f}", c3=f"{cs[2]:.4f}", X1=Xs[0], X2=Xs[1], X3=Xs[2],
                   p=p, theta=f"{np.log(p) / LT:.4f}", R_tau=f"{resp(s, p):.4f}")
        for m in (1, 2, 3):
            A = resp(Y[m - 1], p)
            D = resp(ph_tau[m - 1], p)
            row[f"A{m}"] = f"{A:.4f}"
            row[f"D{m}"] = f"{D:.4f}"
            row[f"B{m}"] = f"{A - D:.4f}"
            row[f"E{m}"] = f"{resp(s, p) - A:.4f}"
            row[f"res{m}"] = int(p * Xs[m - 1] >= T / (2 * np.pi))
        rows.append(row)
        print(row)
with open(os.path.join(OUT, "table_L_computable_response.csv"), "w", newline="") as fh:
    wr = csv.DictWriter(fh, fieldnames=list(rows[0].keys()))
    wr.writeheader()
    wr.writerows(rows)
```

Appendix A.5. opc_envelope_check_v14.py

```
#!/usr/bin/env python3
"""Finite-height envelopes of delta_k = tau_k - gamma_k (ordinal_prime_cutoff_v14, Proposition 8.x).

Usage:  python3 opc_envelope_check_v14.py DATA_DIR [OUT_DIR]

For every index k:
  (i)  |s_k - varpi(gamma_k)/pi| <= max |S(gamma_k^+-)| - 1/2, and Bellotti-Wong (Cor. 1.5, eq. (1.2)) give
       |S(t)| <= B_S(t) for t >= e and |S(t)| <= 2.5167 for 0 <= t <= 30 610 046 000;
  (ii) -E/L(tau_k) <= delta_k <= E/L(gamma_k) with E = B - 1/2 + |varpi(gamma_k)|/pi.
varpi(t) = 1/(48 t) + 7/(5760 t^3) + O(t^-5) is evaluated by this expansion (|error| < 1e-12 for t > 14).
Output: OUT_DIR/check_envelope_S.csv.  No parameter is fitted.
"""
import csv
import os
import sys

import numpy as np

```

```
DATA = sys.argv[1] if len(sys.argv) > 1 else "data"
OUT = sys.argv[2] if len(sys.argv) > 2 else "outputs"
g = np.fromfile(os.path.join(DATA, "zeross.bin"), "<f8")
tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), "<f8")
K = len(g)
k = np.arange(1, K + 1, dtype=float)
N0 = g / (2 * np.pi) * np.log(g / (2 * np.pi * np.e)) + 7 / 8
s = k - 0.5 - N0
d = tau - g
Lg = np.log(g / (2 * np.pi)) / (2 * np.pi)
Lt = np.log(tau / (2 * np.pi)) / (2 * np.pi)
varpi = 1 / (48 * g) + 7 / (5760 * g ** 3)
BS = 0.10076 * np.log(g) + np.minimum(0.24460 * np.log(np.log(g)) + 7.20844, 1.68845 * np.log(np.log(g)) +
    ↪ 1.50956)
assert g[-1] <= 30610046000
BP = np.full(K, 2.5167)
BWN = 0.10076 * np.log(g) + 0.24460 * np.log(np.log(g)) + 8.08344
rows = []
for name, B in (("Platt range 2.5167", BP), ("Bellotti-Wong Cor 1.5", BS)):
    E = B - 0.5 + np.abs(varpi) / np.pi
    up, lo = E / Lg, -E / Lt
    ok_s = bool(np.all(np.abs(s - varpi / np.pi) <= B - 0.5))
    ok_d = bool(np.all((d <= up) & (d >= lo)))
    rows.append(dict(envelope=name, sandwich_ok=ok_s, displacement_ok=ok_d,
                     max_abs_s=f"{np.max(np.abs(s)):.6f}", max_ratio_abs_s_over_E=f"{np.max(np.abs(s) / E)
    ↪ :.4f}",
                     max_delta_over_upper=f"{np.max(d / up):.4f}", max_minus_delta_over_lower=f"{np.max(d /
    ↪  lo):.4f}",
                     upper_at_k34=f"{up[33]:.4f}", lower_at_k34=f"{lo[33]:.4f}",
                     upper_at_K=f"{up[-1]:.4f}", lower_at_K=f"{lo[-1]:.4f}"))
old_up, old_lo = (BWN - 11 / 8) / Lg, -(BWN + 3 / 8) / Lt
rows.append(dict(envelope="v8 (N-bound) for comparison", sandwich_ok=bool(np.all((s >= -BWN - 3 / 8) & (s
    ↪ <= BWN - 11 / 8))),
                 displacement_ok=bool(np.all((d <= old_up) & (d >= old_lo))), max_abs_s=f"{np.max(np.abs(s)
    ↪ ):.6f}",
                 max_ratio_abs_s_over_E="", max_delta_over_upper=f"{np.max(d / old_up):.4f}",
                 max_minus_delta_over_lower=f"{np.max(d / old_lo):.4f}", upper_at_k34=f"{old_up[33]:.4f}",
                 lower_at_k34=f"{old_lo[33]:.4f}", upper_at_K=f"{old_up[-1]:.4f}", lower_at_K=f"{old_lo
    ↪ [-1]:.4f}"))
with open(os.path.join(OUT, "check_envelope_S.csv"), "w", newline="") as fh:
    w = csv.DictWriter(fh, fieldnames=list(rows[0].keys()))
    w.writeheader()
    w.writerows(rows)
for r in rows:
    print(r)
j = int(np.argmax(np.abs(s)))
print(f"max |s_k| = {abs(s[j]):.6f} at k = {j + 1}, gamma = {g[j]:.6f}, delta = {d[j]:.6f}")
```

Appendix A.6. `opc_envelope_coefficient_v14.py`

```
#!/usr/bin/env python3
"""Coefficient of the asymptotic envelope limsup |S(t)|/log t (Theorem E of ordinal_prime_cutoff_v14).

Usage:  python3 opc_envelope_coefficient_v14.py OUT_DIR

For an exponent profile e(sigma) on [0,1] (zeta(sigma+it) << |t|^{e(sigma)+eps} uniformly), the
Backlund-Jensen bound of Bellotti-Wong (Lemmas 2.1 and 2.3 of arXiv:2412.15470), with the disc centred
at b and radius rho tending to (1,1), gives

    limsup |S(t)|/log t <= C(e) = 1/(2 pi log 2) * int_0^1 e(sigma) / sqrt(sigma (2 - sigma)) d sigma .

The script evaluates C(e) for four profiles, with 40-digit quadrature on each linear piece:
  (a) the profile of Bellotti-Wong: 27/164 at sigma=1/2, Yang's lines sigma_k=1-k/(2^k-2) (k=4..9) with
      exponent 1/(2^k-2), 0 at sigma=1, linear interpolation between 0 and 1/2 (no reflection);
  (b) the same right-hand profile, reflected to [0,1/2] by the functional equation;
  (c) Bourgain's exponent pair (13/84,55/84) and its images A^j (j=1..7) under van der Corput's
```

```
      A-process, lower convex hull, reflected to [0,1/2];
  (d) the Lindelof profile (floor of the method).
It also evaluates the general disc (b, rho) for profile (a) as a check against 0.10076.
No data are read.
"""
import json
import sys
from fractions import Fraction as Fr

import mpmath as mp

mp.mp.dps = 40


def lower_hull(points):
    pts = sorted(set(points))
    hull = []
    for p in pts:
        while len(hull) >= 2:
            (x1, y1), (x2, y2) = hull[-2], hull[-1]
            if (y2 - y1) * (p[0] - x1) >= (p[1] - y1) * (x2 - x1):
                hull.pop()
            else:
                break
        hull.append(p)
    return hull


def piecewise(points):
    xs = [mp.mpf(x.numerator) / x.denominator if isinstance(x, Fr) else mp.mpf(x) for x, _ in points]
    ys = [mp.mpf(y.numerator) / y.denominator if isinstance(y, Fr) else mp.mpf(y) for _, y in points]

    def f(s):
        for i in range(len(xs) - 1):
            if xs[i] <= s <= xs[i + 1]:
                return ys[i] + (ys[i + 1] - ys[i]) * (s - xs[i]) / (xs[i + 1] - xs[i])
        raise ValueError(s)
    return f, xs


def profile(right_points, reflect, left_value_at_half=None):
    """Return e(sigma) on [0,1] and the list of breakpoints."""
    right, xs = piecewise(right_points)
    half = mp.mpf(1) / 2

    def e(s):
        if s >= 1:
            return mp.mpf(0)
        if s >= half:
            return right(s)
        if reflect:
            return half - s + right(1 - s)
        # linear interpolation between e(0)=1/2 and e(1/2)
        return (1 - 2 * s) * half + 2 * s * left_value_at_half
    bps = sorted(set([mp.mpf(0), half, mp.mpf(1)] + xs + ([1 - x for x in xs] if reflect else [])))
    return e, bps


def coefficient(e, bps):
    w = lambda s: e(s) / mp.sqrt(s * (2 - s))
    total = mp.mpf(0)
    for a, b in zip(bps[:-1], bps[1:]):
        if b > a:
            total += mp.quad(w, [a, b])
    return total / (2 * mp.pi * mp.log(2))


def disc_coefficient(e, b, rho):
```

```
    """(1/pi) int_0^pi e(b + rho cos th) dth / (2 log(rho/(b-1/2))), e = 0 for sigma >= 1."""
    g = lambda th: e(min(max(b + rho * mp.cos(th), mp.mpf(0)), mp.mpf(1))) if b + rho * mp.cos(th) >= 0
     ↪ else mp.mpf(1) / 2 - (b + rho * mp.cos(th))
    val = mp.quad(g, mp.linspace(0, mp.pi, 64)) / mp.pi
    return val / (2 * mp.log(rho / (b - mp.mpf(1) / 2)))


def main(out_dir):
    half = Fr(1, 2)
    yang = [(Fr(1) - Fr(k, 2 ** k - 2), Fr(1, 2 ** k - 2)) for k in range(4, 10)]
    bw_right = lower_hull([(half, Fr(27, 164))] + yang + [(Fr(1), Fr(0))])
    e_a, b_a = profile(bw_right, reflect=False, left_value_at_half=mp.mpf(27) / 164)
    e_b, b_b = profile(bw_right, reflect=True)

    pairs = []
    k, l = Fr(13, 84), Fr(55, 84)
    pairs.append((k, l))
    for _ in range(7):
        k, l = k / (2 * k + 2), half + l / (2 * k + 2)
        pairs.append((k, l))
    bourgain_points = [(1 - k, k) for k, l in pairs] + [(Fr(1), Fr(0))]
    bourgain_right = lower_hull(bourgain_points)
    e_c, b_c = profile(bourgain_right, reflect=True)

    lindelof_right = [(half, Fr(0)), (Fr(1), Fr(0))]
    e_d, b_d = profile(lindelof_right, reflect=True)

    res = {
        "formula": "C(e) = (1/(2 pi log 2)) * int_0^1 e(sigma)/sqrt(sigma(2-sigma)) dsigma",
        "a_BellottiWong_profile_no_reflection": float(coefficient(e_a, b_a)),
        "b_same_profile_with_reflection": float(coefficient(e_b, b_b)),
        "c_Bourgain_Aimages_with_reflection": float(coefficient(e_c, b_c)),
        "d_Lindelof_floor": float(coefficient(e_d, b_d)),
        "check_disc_b1.0001_rho1.0002_profile_a": float(disc_coefficient(e_a, mp.mpf("1.0001"), mp.mpf("
     ↪ 1.0002"))),
        "Bourgain_hull_points": [[str(x), str(y), float(x), float(y)] for x, y in bourgain_right],
        "BW_hull_points": [[str(x), str(y), float(x), float(y)] for x, y in bw_right],
        "exponent_pairs_used": [[str(k), str(l)] for k, l in pairs],
    }
    for key in ("a_BellottiWong_profile_no_reflection", "b_same_profile_with_reflection",
                "c_Bourgain_Aimages_with_reflection", "d_Lindelof_floor"):
        res[key + "_times_2pi"] = float(2 * mp.pi * res[key])
    res["c_exact_40_digits"] = mp.nstr(coefficient(e_c, b_c), 30)
    res["c_times_2pi_40_digits"] = mp.nstr(2 * mp.pi * coefficient(e_c, b_c), 30)
    with open(f"{out_dir}/envelope_coefficient.json", "w") as fh:
        json.dump(res, fh, indent=1)
    for key, val in res.items():
        if not isinstance(val, list):
            print(f"{key}: {val}")


if __name__ == "__main__":
    main(sys.argv[1] if len(sys.argv) > 1 else "outputs")
```

Appendix A.7. `opc_extensions_v14.py`

```
"""Direct ordinal defects and finite-window prime responses; no fitted parameters.

Usage: python opc_extensions_v14.py DATA_DIR [--skip-responses]
Writes tables and diagnostics beside the original companion outputs.
"""
import argparse
import csv
import hashlib
import json
import platform
```

```
import sys
from pathlib import Path

import numpy as np
import scipy


def primes_upto(n):
    sieve = np.ones(n + 1, dtype=bool)
    sieve[:2] = False
    for p in range(2, int(n**0.5) + 1):
        if sieve[p]:
            sieve[p*p::p] = False
    return np.flatnonzero(sieve)


def digest(path):
    h = hashlib.sha256()
    with path.open('rb') as f:
        for block in iter(lambda: f.read(1024*1024), b''):
            h.update(block)
    return h.hexdigest()


def phase(p, t):
    return -np.angle(1 - p**(-0.5)*np.exp(-1j*t*np.log(p)))/np.pi


def main():
    parser = argparse.ArgumentParser()
    parser.add_argument('data_dir', type=Path)
    parser.add_argument('--skip-responses', action='store_true')
    args = parser.parse_args()
    root = Path(__file__).resolve().parents[1]
    out = root/'outputs'
    out.mkdir(exist_ok=True)
    expected = 2001052
    arrays = {}
    metadata = {'python': platform.python_version(), 'numpy': np.__version__,
                'scipy': scipy.__version__, 'files': {}}
    for name in ('zeross.bin', 'zLamb.bin'):
        path = args.data_dir/name
        if not path.is_file() or path.stat().st_size != 8*expected:
            raise ValueError(f'{path}: expected exactly {expected} little-endian float64 values')
        values = np.fromfile(path, dtype='<f8')
        if not np.isfinite(values).all() or np.any(np.diff(values) < 0):
            raise ValueError(f'{path}: values must be finite and nondecreasing')
        arrays[name] = values
        metadata['files'][name] = {'size_bytes': path.stat().st_size, 'sha256': digest(path)}
    gam, tau = arrays['zeross.bin'], arrays['zLamb.bin']
    k = np.arange(1, expected+1, dtype=float)
    Ltau = np.log(tau/(2*np.pi))/(2*np.pi)
    N0gam = gam/(2*np.pi)*np.log(gam/(2*np.pi*np.e)) + 7/8
    s = k - 0.5 - N0gam
    q = Ltau*(tau-gam)
    metadata['max_clock_equation_error'] = float(np.max(np.abs(
        tau/(2*np.pi)*np.log(tau/(2*np.pi*np.e))+7/8-k+0.5)))
    metadata['max_abs_q_minus_s'] = float(np.max(np.abs(q-s)))
    PR = primes_upto(20000)
    C = (1/160, 1/80, 1/40, 1/20, 1/10, 1/5, 1/3, 1/2)
    metadata['parameters'] = {'expected_count':expected, 'prime_max':20000,
        'defect_dyadic_exponents':[14,16,18,19], 'defect_exponents':list(C),
        'defect_cutoff_is_constant_on_each_height_window':True,
        'cutoffs_below_two':'excluded', 'Euler_phase':'full local factors',
        'prime_response_ordinal_windows':[[1000,10000],[10000,100000],
            [100000,500000],[500000,1000000],[1000000,expected]],
        'response_target':'exact s_k=k-1/2-N0(gamma_k)',
        'theta':'log(p)/median(log(tau_k/(2*pi)))',
```

```
        'response_bin_width':0.05, 'minimum_primes_per_bin':5}
    defect_rows = []
    for r in (14, 16, 18, 19):
        T = float(2**r)
        mask = (gam > T) & (gam <= 2*T)
        t, z, st = gam[mask], tau[mask], s[mask]
        for c in C:
            X = T**c
            if X < 2:
                continue
            P = PR[PR <= X]
            Sg = np.zeros(len(t))
            St = np.zeros(len(t))
            for p in P:
                Sg += phase(int(p), t)
                St += phase(int(p), z)
            u = st-Sg
            # Exact ordinal residual of the one-step clock, without saturation.
            rk = q[mask]-St
            lead = np.log(1/c)/(2*np.pi**2)
            defect_rows.append((r, int(T), len(t), c, X, len(P),
                                float(np.mean(u*u)), lead,
                                float(np.mean(u*u))-lead,
                                float(np.mean(rk*rk))))
        print(f'direct defects: T=2^{r}, zeros={len(t)}', flush=True)
    with (out/'table_F_direct_defects.csv').open('w', newline='') as f:
        w=csv.writer(f)
        w.writerow(('r','T','zero_count','c','X','prime_count','mean_u2',
                    'log_1_over_c_over_2pi2','difference','mean_one_step_r2'))
        w.writerows(defect_rows)

    if not args.skip_responses:
        windows = [(1000,10000),(10000,100000),(100000,500000),
                   (500000,1000000),(1000000,expected)]
        medians = [float(np.median(np.log(tau[a:b]/(2*np.pi)))) for a,b in windows]
        rows = []
        for ip,p0 in enumerate(PR):
            p = int(p0)
            lg = np.log(p)
            sn_g = np.sin(gam*lg)
            sn_t = np.sin(tau*lg)
            for j,((a,b),L) in enumerate(zip(windows,medians)):
                rg = -2*np.pi*np.sqrt(p)*np.mean(s[a:b]*sn_g[a:b])
                rt = -2*np.pi*np.sqrt(p)*np.mean(s[a:b]*sn_t[a:b])
                th = lg/L
                law = (1-(1+th)*np.log1p(th)-(1-th)*np.log1p(-th)
                       if 0 < th < 1 else float('nan'))
                rows.append((j+1,a,b,p,th,float(rg),float(rt),float(law)))
            if ip % 200 == 0:
                print(f'prime responses: {ip+1}/{len(PR)} (p={p})',flush=True)
        with (out/'table_G_prime_responses.csv').open('w',newline='') as f:
            w=csv.writer(f)
            w.writerow(('window','k_lower_exclusive','k_upper_inclusive','prime',
                        'theta_median','rho_gamma','rho_tau','sine_response_prediction'))
            w.writerows(rows)
        diagnostics=[]
        rr=np.array(rows)
        for j in range(1,6):
            selected=rr[(rr[:,0]==j)&(rr[:,4]>0)&(rr[:,4]<1)]
            central=selected[(selected[:,4]>=0.1)&(selected[:,4]<=0.8)]
            diagnostics.append({'window':j,'prime_count_total':len(PR),
              'prime_count_theta_below_one':len(selected),
              'max_abs_gamma_minus_prediction':float(np.max(np.abs(selected[:,5]-selected[:,7]))),
              'rms_gamma_minus_prediction':float(np.sqrt(np.mean((selected[:,5]-selected[:,7])**2))),
              'prime_count_theta_01_to_08':len(central),
              'max_abs_gamma_minus_prediction_theta_01_to_08':float(np.max(np.abs(central[:,5]-central
    ↪ [:,7])))})
        metadata['prime_response_diagnostics']=diagnostics
```

```
    (out/'extension_metadata.json').write_text(json.dumps(metadata,indent=2)+'\n')
    print(json.dumps(metadata,indent=2),flush=True)


if __name__ == '__main__':
    main()
```

Appendix A.8. `opc_resonance_v14.py`

```
#!/usr/bin/env python3
"""Onset of the cross term Cr_X at the clock points (ordinal_prime_cutoff_v14, Section 9).

Usage:  python3 opc_resonance_v14.py DATA_DIR [OUT_DIR]

Top window of ordinals (10^6, K].  (1) Cr_X = <S_X(tau_k)^2> - v_X with full local factors, and the same for the
prime part S^(1) alone, for a grid of cutoffs X.  (2) For the prime part S^(1)(t) = sum_{p<=1999} a_p sin(t log p), a_p = -1/(pi sqrt p), the cross
term Cr^(1) = sum_{p,q} a_p a_q (<sin(tau log p) sin(tau log q)> - [p=q]/2) is split into the ordered pairs with
p q in [tau_min/2pi, tau_max/2pi] (resonant) and the others; the stationary-phase value
sum_{resonant ordered pairs} log(pq) / (2 pi^2 #I) is printed for comparison (heuristic).
Output: OUT_DIR/table_I_cross_term_onset.csv, table_I_resonance.json.  No parameter is fitted.
"""
import csv
import json
import os
import sys

import numpy as np

DATA = sys.argv[1] if len(sys.argv) > 1 else "data"
OUT = sys.argv[2] if len(sys.argv) > 2 else "outputs"
os.makedirs(OUT, exist_ok=True)
tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), "<f8")
K = len(tau)
tt = tau[10 ** 6:]                      # indices k = 10^6+1, ..., K
nI = len(tt)


def primes(n):
    sv = np.ones(n + 1, bool)
    sv[:2] = False
    for p in range(2, int(n ** 0.5) + 1):
        if sv[p]:
            sv[p * p::p] = False
    return np.flatnonzero(sv)


P = primes(1999)
grid = [100, 200, 250, 300, 316, 350, 400, 500, 700, 907, 1200, 1999]
S = np.zeros(nI)
S1 = np.zeros(nI)
v = 0.0
v1 = 0.0
rows = []
gi = 0
for idx, p in enumerate(P):
    lp = np.log(p)
    m = 1
    while p ** (-m / 2) > 1e-17:
        a = -1.0 / (np.pi * m * p ** (m / 2))
        term = a * np.sin(m * lp * tt)
        S += term
        v += 0.5 * a * a
        if m == 1:
```

```
            S1 += term
            v1 += 0.5 * a * a
        m += 1
    nxt = P[idx + 1] if idx + 1 < len(P) else np.inf
    while gi < len(grid) and grid[gi] < nxt:
        rows.append(dict(X=grid[gi], n_primes=int(np.sum(P <= grid[gi])), mean_S2=f"{np.mean(S * S):.6f}",
                         v_X=f"{v:.6f}", Cr_X=f"{np.mean(S * S) - v:.6f}",
                         Cr1_X_primes_only=f"{np.mean(S1 * S1) - v1:.6f}"))
        gi += 1
with open(os.path.join(OUT, "table_I_cross_term_onset.csv"), "w", newline="") as fh:
    w = csv.DictWriter(fh, fieldnames=list(rows[0].keys()))
    w.writeheader()
    w.writerows(rows)
for r in rows:
    print(r)

# (2) prime part, pair decomposition through the Gram matrix of the sine vectors
lp = np.log(P.astype(float))
G = np.zeros((len(P), len(P)))
step = 50000
for i0 in range(0, nI, step):
    blk = np.sin(np.outer(lp, tt[i0:i0 + step]))
    G += blk @ blk.T
G /= nI
a = -1.0 / (np.pi * np.sqrt(P.astype(float)))
M = np.outer(a, a) * (G - 0.5 * np.eye(len(P)))
U1, U2 = tt.min() / (2 * np.pi), tt.max() / (2 * np.pi)
PQ = np.outer(P.astype(float), P.astype(float))
res = (PQ >= U1) & (PQ <= U2)
pred = float(np.sum(np.log(PQ[res])) / (2 * np.pi ** 2 * nI))
out = dict(window="(10^6, K]", n_indices=nI, tau_over_2pi_range=[float(U1), float(U2)], X=1999, n_primes=
    ↪ len(P),
          Cr1_total=float(M.sum()), Cr1_resonant_pairs=float(M[res].sum()), Cr1_nonresonant=float(M[~res].
    ↪ sum()),
          n_resonant_ordered_pairs=int(res.sum()), n_nonresonant_ordered_pairs=int((~res).sum()),
          stationary_phase_value_heuristic=pred, sqrt_tau_min_over_2pi=float(np.sqrt(U1)))
with open(os.path.join(OUT, "table_I_resonance.json"), "w") as fh:
    json.dump(out, fh, indent=1)
print(json.dumps(out, indent=1))
```

Appendix A.9. `opc_transport_v14.py`

```
#!/usr/bin/env python3
"""Transport term and Newton second-order term of the one-step clock (ordinal_prime_cutoff_v14, Section 7).

Usage:  python3 opc_transport_v14.py DATA_DIR [OUT_DIR]

Window: T = 2^19, indices with T < gamma_k <= 2T.  For X = T^c, c in {1/5, 1/3, 1/2}, full local factors:
  u_k        = s_k - S_X(gamma_k)                        (ordinal defect)
  t_k        = S_X(gamma_k) - S_X(tau_k)                 (transport term)
  chi_k      = L(tau_k) delta_k - s_k                    (exact second-order term, >= 0)
  r_k        = L(tau_k) (tau_k^X - gamma_k) = u_k + t_k + chi_k
  newton_k   = S_X(tau_k) S_X'(tau_k) / L(tau_k)         (second-order Newton term)
Output: OUT_DIR/table_H_transport.csv and transport_log.txt.  No parameter is fitted.
"""
import csv
import os
import sys

import numpy as np

DATA = sys.argv[1] if len(sys.argv) > 1 else "data"
OUT = sys.argv[2] if len(sys.argv) > 2 else "outputs"
os.makedirs(OUT, exist_ok=True)

g = np.fromfile(os.path.join(DATA, "zeross.bin"), "<f8")
```

```
tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), "<f8")
K = len(g)
k = np.arange(1, K + 1, dtype=float)
N0 = lambda x: x / (2 * np.pi) * np.log(x / (2 * np.pi * np.e)) + 7 / 8
L = lambda x: np.log(x / (2 * np.pi)) / (2 * np.pi)
s = k - 0.5 - N0(g)
d = tau - g


def primes(n):
    sv = np.ones(n + 1, bool)
    sv[:2] = False
    for p in range(2, int(n ** 0.5) + 1):
        if sv[p]:
            sv[p * p::p] = False
    return np.flatnonzero(sv)


def phase(x, P, deriv=0):
    out = np.zeros_like(x)
    for p in P:
        lp = np.log(p)
        m = 1
        while p ** (-m / 2) > 1e-17:
            if deriv == 0:
                out += -np.sin(m * lp * x) / (np.pi * m * p ** (m / 2))
            else:
                out += -lp * np.cos(m * lp * x) / (np.pi * p ** (m / 2))
            m += 1
    return out


T = 2.0 ** 19
sel = (g > T) & (g <= 2 * T)
gg, tt, ss, dd = g[sel], tau[sel], s[sel], d[sel]
Lt = L(tt)
rows = []
lines = [f"T=2^19, window size {sel.sum()}"]
for c in (1 / 5, 1 / 3, 1 / 2):
    X = T ** c
    P = primes(int(X))
    Sg, St, Spt = phase(gg, P), phase(tt, P), phase(tt, P, 1)
    u = ss - Sg
    tr = Sg - St
    chi = Lt * dd - ss
    r = Lt * dd - St
    newton = St * Spt / Lt
    ms = lambda v: float(np.mean(v * v))
    row = dict(c=f"{c:.6f}", X=f"{X:.2f}", n_primes=len(P), ms_u=f"{ms(u):.6f}", ms_transport=f"{ms(tr):.6f
    ↪ }",
               corr_u_transport=f"{np.corrcoef(u, tr)[0, 1]:+.4f}", ms_r=f"{ms(r):.6f}",
               max_chi=f"{chi.max():.3e}", min_chi=f"{chi.min():.3e}",
               rms_SXprime_over_L=f"{np.sqrt(ms(Spt / Lt)):.4f}", frac_FXprime_negative=f"{np.mean(Lt + Spt
    ↪  < 0):.5f}",
               ms_newton=f"{ms(newton):.6f}", corr_transport_minus_newton=f"{np.corrcoef(tr, -newton)[0,
    ↪ 1]:+.4f}",
               check_identity_max_abs=f"{np.max(np.abs(r - (u + tr + chi))):.2e}")
    rows.append(row)
    lines.append(str(row))
with open(os.path.join(OUT, "table_H_transport.csv"), "w", newline="") as fh:
    w = csv.DictWriter(fh, fieldnames=list(rows[0].keys()))
    w.writeheader()
    w.writerows(rows)
with open(os.path.join(OUT, "transport_log.txt"), "w") as fh:
    fh.write("\n".join(lines) + "\n")
print("\n".join(lines))
```

Appendix A.10. `opc_twist_v14.py`

```
#!/usr/bin/env python3
"""Transport identity in a prime frequency (ordinal_prime_cutoff_v14, Section on the twisted integral).

Usage:  python3 opc_twist_v14.py DATA_DIR [OUT_DIR]

Window T = 2^19 < t <= 2T.  For primes p and w = log p:
  I      = (1/T) int_T^{2T} exp(i w t + i w G(t)/L(t)) dt, computed exactly on every gap of consecutive
           ordinates (G = N - N_0 is k - N_0(t) on the gap (gamma_k, gamma_{k+1}); the integrand is a smooth
           function there and is integrated by adaptive Simpson to relative accuracy 1e-10);
  D      = (2i/T) sum_{k in I_T} sin(w/(2 L(tau_k))) delta_k p^{i tau_k}   (delta_k = tau_k - gamma_k);
  J      = (1/T) sum_k (tau_{k+1} - tau_k) exp(i w v_k), v_k = N_0^{-1}(k) = tau_{k+1/2}  (clock sum, no zeros);
  Rtau   = -2 pi sqrt(p) <s_k sin(tau_k log p)>_{I_T};
  Rw     = weighted response: -2 pi sqrt(p) <b_k s_k sin(tau_k log p)>, b_k = (M/(T L(tau_k))) sin(w/(2L(tau_k)))/sin(pi theta).
Output: OUT_DIR/table_J_twist.csv and twist_log.txt.  No parameter is fitted.
"""
import csv
import os
import sys

import numpy as np
from scipy.special import lambertw

DATA = sys.argv[1] if len(sys.argv) > 1 else "data"
OUT = sys.argv[2] if len(sys.argv) > 2 else "outputs"
os.makedirs(OUT, exist_ok=True)
g = np.fromfile(os.path.join(DATA, "zeross.bin"), "<f8")
tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), "<f8")
K = len(g)
k = np.arange(1, K + 1, dtype=float)
N0 = lambda x: x / (2 * np.pi) * np.log(x / (2 * np.pi * np.e)) + 7 / 8
L = lambda x: np.log(x / (2 * np.pi)) / (2 * np.pi)
s = k - 0.5 - N0(g)
d = tau - g
T = 2.0 ** 19
i0, i1 = int(np.searchsorted(g, T)), int(np.searchsorted(g, 2 * T))   # gamma[i0] > T, gamma[i1-1] <= 2T
LT = np.log(T / (2 * np.pi))
sel = slice(i0, i1)
M = i1 - i0


def tau_x(x):
    y = x - 11 / 8
    return 2 * np.pi * y / lambertw(y / np.e).real


def gap_integral(a, b, Nv, w):
    """int_a^b exp(i(w t + w (Nv - N0(t))/L(t))) dt on each gap, Simpson with 9 nodes (the phase varies by O(1/T))."""
    nodes = np.linspace(0, 1, 9)
    wts = np.array([1, 4, 2, 4, 2, 4, 2, 4, 1], dtype=float) / 24
    tot = np.zeros(len(a), dtype=complex)
    for u, wt in zip(nodes, wts):
        t = a + (b - a) * u
        tot += wt * np.exp(1j * (w * t + w * (Nv - N0(t)) / L(t)))
    return (b - a) * tot


rows = []
lines = [f"T=2^19, M=#I_T={M}, gaps used {i1 - 1 - i0}"]
a, b = g[i0:i1 - 1], g[i0 + 1:i1]
Nv = np.arange(i0 + 1, i1, dtype=float)             # N(t) on the gap (gamma_{i0+1} ... ) : index i -> N = i+1 (1-based)
vk = tau_x(Nv + 0.5)                                 # v_k = N_0^{-1}(k) = tau_{k+1/2}
for p in (3, 101, 1009, 10007, 100003, 131101):
```

```
    w = np.log(p)
    th = w / LT
    I = gap_integral(a, b, Nv, w).sum() / T
    D = 2j * np.sum(np.sin(w / (2 * L(tau[sel]))) * d[sel] * np.exp(1j * w * tau[sel])) / T
    J = np.sum((tau[i0 + 1:i1] - tau[i0:i1 - 1]) * np.exp(1j * w * vk)) / T
    Rt = -2 * np.pi * np.sqrt(p) * np.mean(s[sel] * np.sin(w * tau[sel]))
    bk = (M / (T * L(tau[sel]))) * np.sin(w / (2 * L(tau[sel]))) / np.sin(np.pi * th) if th < 1 else np.
    ↪ full(M, np.nan)
    Rw = -2 * np.pi * np.sqrt(p) * np.mean(bk * s[sel] * np.sin(w * tau[sel]))
    res = 2 * np.pi * np.sqrt(p) / T if T / (2 * np.pi) <= p <= T / np.pi else 0.0
    row = dict(p=p, theta=f"{th:.4f}", Re_I=f"{I.real:+.6e}", Im_I=f"{I.imag:+.6e}", Re_D=f"{D.real:+.6e}",
    ↪  Im_D=f"{D.imag:+.6e}",
               Re_J_clock=f"{J.real:+.6e}", Im_J_clock=f"{J.imag:+.6e}", resonance_2pi_sqrtp_over_T=f"{res
    ↪ :.6e}",
               I_minus_D_minus_J=f"{abs(I - D - J):.2e}", R_tau=f"{Rt:.4f}", R_tau_weighted=f"{Rw:.4f}",
               sin_pi_theta_Rw_over_pi_sqrtp=f"{np.sin(np.pi * th) * Rw / (np.pi * np.sqrt(p)):+.6e}")
    rows.append(row)
    lines.append(str(row))
with open(os.path.join(OUT, "table_J_twist.csv"), "w", newline="") as fh:
    wr = csv.DictWriter(fh, fieldnames=list(rows[0].keys()))
    wr.writeheader()
    wr.writerows(rows)
with open(os.path.join(OUT, "twist_log.txt"), "w") as fh:
    fh.write("\n".join(lines) + "\n")
print("\n".join(lines))
```

Appendix A.11. `opc_validate_v14.py`

```
"""Independent numerical checks of the two exact identities added to the paper."""
import json
from pathlib import Path
import numpy as np
from scipy.integrate import quad


def main():
    checks = {}
    # A repeated ordinate and a nonmonotone model exercise both delicate cases.
    gamma=np.array([1.0,2.0,2.0,3.5,4.2])
    F=lambda t: np.sin(3*t)+0.2*t
    dF=lambda t: 3*np.cos(3*t)+0.2
    u=np.arange(1,len(gamma)+1)-0.5-F(gamma)
    for power in (2,4,6):
        left=sum(quad(lambda t:(k+1-F(t))**power*dF(t),
                      gamma[k],gamma[k+1],epsabs=1e-10)[0]
                 for k in range(len(gamma)-1))
        P=lambda x: ((0.5+x)**(power+1)+(0.5-x)**(power+1))/(power+1)
        right=sum(P(x) for x in u[1:-1]) + (
            (0.5+u[0])**(power+1)+(0.5-u[-1])**(power+1))/(power+1)
        err=abs(left-right)
        if err>1e-8*max(1,abs(right)):
            raise AssertionError(('jump identity',power,left,right))
        checks[f'jump_identity_power_{power}_absolute_error']=float(err)
    # Integrate the positive-part overlap formula, not the antiderivative used
    # in the manuscript, and compare with the closed logarithmic expression.
    for th in (0.1,0.3,0.5,0.8,0.95):
        h=lambda x:max(1-abs(x),0)
        def f(a):
            if th-1 <= a <= 1:
                return 0.0
            kernel=1-h(a)-h(th-a)-h(th)+2*max(1-max(abs(a),abs(th-a),th),0)
            return kernel/(a*(th-a))
        edges=[-np.inf,-1,th-1,1,1+th,np.inf]
        value=sum(quad(f,a,b,epsabs=1e-11)[0] for a,b in zip(edges,edges[1:]))
        exact=-2*((1+th)*np.log1p(th)+(1-th)*np.log1p(-th))/th
        err=abs(value-exact)
```

```
        if err>1e-9:
            raise AssertionError(('kernel integral',th,value,exact))
        checks[f'kernel_integral_theta_{th}_absolute_error']=float(err)
    path=Path(__file__).resolve().parents[1]/'outputs'/'identity_checks.json'
    path.write_text(json.dumps(checks,indent=2)+'\n')
    print(json.dumps(checks,indent=2))


if __name__=='__main__':
    main()
```

Appendix A.12. `tools/make_tips_v14.py`

```
"""Generate the hover texts (\\CGTip@<label>) shown by \\CGref and \\CGeqref in PDF viewers.

usage: python3 make_tips.py main.tex main.aux tips.tex

For every \\label the text is the kind and number of the object, its page and a plain-text
rendering of its statement or equation.  The LaTeX source is converted token by token, as TeX
reads it (a control word is a backslash followed by letters), so that no symbol is lost by the
gluing of a converted macro to the following letters; \\frac, \\sqrt and \\binom are converted
with their arguments; \\CGref and \\CGeqref inside the text are replaced by the numbers they
print; optional arguments and section titles are read with balanced delimiters.
"""
import os
import re
import sys

tex = open(sys.argv[1]).read()
aux = sys.argv[2]
out = sys.argv[3]
nums = {}
if os.path.exists(aux):
    for m in re.finditer(r"\\newlabel\{([^}]*)\}\{\{(\{[^{}]*\}|[^{}]*)\}\{([^}]*)\}", open(aux).read()):
        nums[m.group(1)] = (m.group(2).strip("{}").replace("$", ""), m.group(3))

GREEK = set(("alpha beta gamma delta epsilon varepsilon zeta eta theta vartheta iota kappa varkappa lambda mu nu xi "
             "pi varpi rho varrho sigma varsigma tau upsilon phi varphi chi psi omega Gamma Delta Theta Lambda Xi "
             "Pi Sigma Upsilon Phi Psi Omega").split())
MAP = {"langle": "<", "rangle": ">", "iff": "<=>", "Rightarrow": "=>", "Leftarrow": "<=", "Longrightarrow": "=>",
       "Leftrightarrow": "<=>", "Longleftrightarrow": "<=>", "setminus": "\\", "subset": "in", "subseteq": "in",
       "Subset": "in", "supset": "contains", "cup": "u", "cap": "n", "le": "<=", "leq": "<=", "ge": ">=", "geq": ">=",
       "ne": "!=", "neq": "!=", "to": "->", "mapsto": "->", "longrightarrow": "->", "rightarrow": "->",
       "downarrow": "decreasing to", "uparrow": "increasing to", "infty": "inf", "cdot": "*", "pm": "+-",
       "mp": "-+", "times": "x", "ll": "<<", "gg": ">>", "lesssim": "<~", "gtrsim": ">~", "asymp": "~", "sim": "~",
       "approx": "~", "simeq": "~", "equiv": "==", "in": "in", "notin": "not in", "mid": "|", "nmid": "does not divide",
       "vert": "|", "Vert": "||", "lvert": "|", "rvert": "|", "lVert": "||", "rVert": "||", "ast": "*", "star": "*",
       "circ": "o", "vee": "v", "wedge": "^", "neg": "not", "forall": "for all", "exists": "exists",
       "emptyset": "empty", "varnothing": "empty", "partial": "d", "nabla": "grad", "prime": "'", "dots": "...",
       "cdots": "...", "ldots": "...", "lfloor": "floor(", "rfloor": ")", "lceil": "ceil(", "rceil": ")",
       "NR": "N_RVM", "RVM": "RVM", "crit": "crit", "vpi": "varpi_RVM", "eps": "eps", "Z": "Z", "C": "C",
       "R": "R", "N": "N", "Q": "Q", "Re": "Re", "Im": "Im", "Log": "Log", "Mell": "*Mel", "hbar": "hbar",
       "ell": "l", "imath": "i", "jmath": "j", "aleph": "aleph", "Pr": "Pr", "pmod": "mod", "bmod": "mod",
       "widehat": "hat", "hat": "hat", "tilde": "tilde", "widetilde": "tilde", "overline": "conj", "bar": "conj",
       "dot": "dot", "ddot": "ddot", "vec": "", "xrightarrow": "->", "overset": "", "underset": "",
       "S": "S", "NO": "N_0", "IT": "I_T", "lL": "L", "sgn": "sgn", "Cr": "Cr", "BW": "BW", "Rg": "R^gamma", "Rt": "R^tau"}
```

```
DROP = set(("mathcal mathscr mathfrak mathbf mathrm mathsf mathbb mathit mathtt boldsymbol bm operatorname text "
            "textup textrm textit textbf textsf texttt emph mbox hbox left right bigl bigr Bigl Bigr biggl biggr "
            "Biggl Biggr big Big bigg Bigg middle displaystyle textstyle scriptstyle boxed quad qquad label tag "
            "notag nonumber allowbreak nobreak hfill hfil vfill noindent par limits nolimits phantom hphantom "
            "vphantom smash strut relax protect leavevmode medskip smallskip bigskip centering raggedright "
            "footnotesize small normalsize").split())


def _arg(s, i):
    """one TeX argument at s[i:] (after spaces): braced group, control sequence or single character"""
    while i < len(s) and s[i] in " \n\t":
        i += 1
    if i >= len(s):
        return "", i
    if s[i] == "{":
        depth = 0
        for j in range(i, len(s)):
            if s[j] == "{":
                depth += 1
            elif s[j] == "}":
                depth -= 1
                if depth == 0:
                    return s[i + 1:j], j + 1
        return s[i + 1:], len(s)
    if s[i] == "\\":
        m = re.match(r"\\[a-zA-Z]+|\\.", s[i:])
        return m.group(0), i + len(m.group(0))
    return s[i], i + 1


def _optarg(s, i):
    """optional [ ... ] argument with balanced brackets and braces; returns (content or None, index)"""
    j = i
    while j < len(s) and s[j] in " \n\t":
        j += 1
    if j >= len(s) or s[j] != "[":
        return None, i
    depth_sq, depth_br = 0, 0
    for k in range(j, len(s)):
        c = s[k]
        if c == "{":
            depth_br += 1
        elif c == "}":
            depth_br -= 1
        elif c == "[" and depth_br == 0:
            depth_sq += 1
        elif c == "]" and depth_br == 0:
            depth_sq -= 1
            if depth_sq == 0:
                return s[j + 1:k], k + 1
    return s[j + 1:], len(s)


def _refs(s):
    """\\CGref{lab} -> printed number, \\CGeqref{lab} -> (number); \\cite[..]{..} -> [ref, ..]"""
    s = re.sub(r"\\CGeqref\{([^}]*)\}", lambda m: "(" + nums.get(m.group(1), (m.group(1), ""))[0] + ")", s)
    s = re.sub(r"\\(?:CGref|ref|eqref)\{([^}]*)\}", lambda m: nums.get(m.group(1), (m.group(1), ""))[0], s)
    out_s, i = [], 0
    while True:
        k = s.find("\\cite", i)
        if k < 0:
            out_s.append(s[i:])
            break
```

```
        out_s.append(s[i:k])
        j = k + 5
        opt, j = _optarg(s, j)
        _, j = _arg(s, j)
        out_s.append("[ref" + (", " + opt if opt else "") + "]")
        i = j
    return "".join(out_s)


def _structures(s):
    """\\texorpdfstring (its PDF-string argument), \\frac, \\tfrac, \\dfrac, \\sqrt, \\binom, innermost
    ↪ first"""
    s = re.sub(r"\\[td]frac(?![a-zA-Z])", r"\\frac", s)
    for name, fmt, nargs in (("\\texorpdfstring", "{1}", 2), ("\\frac", "({0})/({1})", 2),
                             ("\\binom", "binom({0},{1})", 2), ("\\sqrt", "sqrt({0})", 1)):
        for _ in range(60):
            k = s.rfind(name)
            if k < 0:
                break
            if s[k + len(name):k + len(name) + 1].isalpha():       # a longer control word
                break
            i = k + len(name)
            if name == "\\sqrt":
                _, i = _optarg(s, i)
            args = []
            for _ in range(nargs):
                a, i = _arg(s, i)
                args.append(a)
            s = s[:k] + fmt.format(*args) + s[i:]
    return s


def _macros(s):
    def rep(m):
        name = m.group(1)
        nxt = s_[m.end():m.end() + 1]
        if name in DROP:
            r = ""
        elif name in MAP:
            r = MAP[name]
        else:
            r = name                                    # Greek letters and operator names keep their name
        if r and r[-1].isalnum() and (nxt.isalnum() or nxt in "\\{"):
            r += " "
        return r
    s_ = s
    return re.sub(r"\\([a-zA-Z]+)\*?", rep, s)


def clean(s, limit=320):
    s = re.sub(r"(?<!\\)%.*", "", s)
    s = re.sub(r"\\label\{[^}]*\}", "", s)
    s = re.sub(r"\\begin\{[^}]*\}(\[[^]]*\])?|\\end\{[^}]*\}", " ", s)
    s = s.replace("\\~", "")
    s = re.sub(r"\\c\{([a-zA-Z])\}", r"\1", s)                          # cedilla
    s = re.sub(r"\\[\"'‘^]\{?([a-zA-Z])\}?", r"\1", s)                  # accents: \"o -> o
    s = re.sub(r"(?<!\\)~", " ", s)                 # ties
    s = _refs(s)
    s = _structures(s)
    s = re.sub(r"\\[,;:! ]", " ", s)
    s = s.replace("\\{", "{").replace("\\}", "}").replace("\\|", "||").replace("\\#", "#").replace("\\%",
    ↪ "%")
    s = s.replace("\\&", "&").replace("\\_", "_")
    s = _macros(s)
    s = s.replace("&", " ").replace("\\\\", "; ").replace("$", "")
    s = s.replace("{", "").replace("}", "").replace("\\", "")
    s = re.sub(r"\s+", " ", s).strip()
    s = re.sub(r"\(\s+", "(", s)
```

```
    s = re.sub(r"\s+\)", ")", s)
    if len(s) > limit:
        s = s[:limit - 3].rstrip() + "..."
    s = s.replace("#", "no.").replace("%", " pct")
    s = s.replace("_", r"\_").replace("^", r"\textasciicircum{}").replace("~", r"\textasciitilde{}")
    return s


ENVS = {"theorem": "Theorem", "proposition": "Proposition", "lemma": "Lemma", "corollary": "Corollary",
        "definition": "Definition", "hypothesis": "Hypothesis", "remark": "Remark", "experiment": "
    ↪ Experiment",
        "conjecture": "Conjecture", "heuristic": "Heuristic", "example": "Example", "proof": "Proof"}
tips = {}
for m in re.finditer(r"\\label\{([^}]*)\}", tex):
    lab = m.group(1)
    pos = m.start()
    num, page = nums.get(lab, ("", ""))
    before = tex[:pos]
    if lab.startswith(("sec:", "subsec:", "app:")):
        hs = list(re.finditer(r"\\(section|subsection|subsubsection)\*?(?=[\[{])", before[-2000:]))
        title = lab
        if hs:
            seg = before[-2000:][hs[-1].end():]
            _, i = _optarg(seg, 0)
            t, _ = _arg(seg, i)
            title = clean(t)
        tips[lab] = "Section %s: %s (p. %s)" % (num, title, page)
    elif lab.startswith("eq:"):
        b = max(before.rfind(r"\begin{equation}"), before.rfind(r"\begin{align}"),
                before.rfind(r"\begin{multline}"), before.rfind(r"\begin{gather}"))
        e = tex.find(r"\end{", pos)
        body = tex[b:e]
        body = re.sub(r"\\begin\{(equation|align|multline|gather)\}", "", body)
        if r"\begin{align}" in tex[b:b + 15]:
            for sg in body.split("\\\\"):
                if lab in sg:
                    body = sg
        tg = re.search(r"\\tag\{([^}]*)\}", body)
        if tg and not num:
            num = clean(tg.group(1))
        tips[lab] = "Eq. (%s), p. %s: %s" % (num, page, clean(body))
    elif lab.startswith(("fig:", "tab:")):
        c = tex.rfind(r"\caption{", 0, pos)
        cap, _ = _arg(tex, c + 8)
        tips[lab] = "%s %s (p. %s): %s" % ("Figure" if lab.startswith("fig") else "Table", num, page, clean
    ↪ (cap))
    else:
        mm = list(re.finditer(r"\\begin\{(\w+)\}", before[-3000:]))
        if mm:
            env = mm[-1].group(1)
            seg = before[-3000:][mm[-1].end():]
            title, _ = _optarg(seg, 0)
            e = tex.find(r"\end{" + env + "}", pos)
            body = clean(tex[pos + len(lab) + 8:e])
            tips[lab] = "%s %s%s (p. %s): %s" % (ENVS.get(env, env.capitalize()), num,
                                                 (" (" + clean(title) + ")") if title else "", page, body)
with open(out, "w") as f:
    f.write("\\makeatletter\n")
    for k, v in tips.items():
        f.write("\\expandafter\\gdef\\csname CGTip@%s\\endcsname{%s}\n" % (k, v))
    f.write("\\makeatother\n")
print(len(tips), "tips")
```

Appendix A.13. `tools/replace_tips_v14.py`

```
"""Replace the inlined tooltip block of the paper by a freshly generated one.
```

```
2
3 Usage: python3 replace_tips.py PAPER.tex TIPS.tex
4 The block starts at the line after '% Tooltip texts shown when hovering over internal references.'
5 and ends at the first line '\\makeatother' after it (both lines of the block included)."""
6 import sys
7
8 paper, tips = sys.argv[1], sys.argv[2]
9 lines = open(paper).read().split("\n")
10 start = lines.index("% Tooltip texts shown when hovering over internal references.") + 1
11 end = start + lines[start:].index("\\makeatother")
12 new = open(tips).read().rstrip("\n").split("\n")
13 assert lines[start] == "\\makeatletter" and new[0] == "\\makeatletter" and new[-1] == "\\makeatother"
14 lines[start:end + 1] = new
15 open(paper, "w").write("\n".join(lines))
16 print("replaced lines %d-%d with %d lines" % (start + 1, end + 1, len(new)))
```